Research Paper Writing and Research Tools

A Complete Practical Guide from Research Idea to Journal Publication

A Training and Reference Handbook for PhD Scholars, Research Scholars, Postgraduate Students, Faculty Members, and Early-Career Researchers

Edition 1.0

Prepared as an independent-study and workshop manual

Preface

This handbook exists because of a gap between two things that are both widely available and rarely connected.

The first is instruction in research *methods*: how to train a model, run a regression, design an interview protocol, compute a statistic. Most postgraduate programmes teach this reasonably well.

The second is instruction in research *practice*: how to decide what is worth studying, how to establish that nobody has established it already, how to construct an argument that a sceptical expert will accept, how to choose where to publish, and how to respond when three anonymous specialists tell you that your work is inadequate. This is usually transmitted by apprenticeship — absorbed from a supervisor, a lab culture, or a reading group — and when that apprenticeship is weak, absent, or overloaded, the researcher is left to reconstruct it alone. The visible symptoms are familiar: a scholar who has built an impressive system and cannot explain what is new about it; a manuscript desk-rejected in four days; a literature review that lists forty papers and concludes nothing; a "research gap" that amounts to *nobody has tried this combination yet*; three years of work published in a venue that no committee recognises.

This handbook is an attempt to write that apprenticeship down.

What this handbook assumes

It assumes you can already do the technical work of your field, or are learning to. It does not teach machine learning, statistics, or qualitative coding. It teaches what to do with those skills so that the result becomes a defensible contribution to a permanent record.

It assumes you are working on something real. Every chapter ends with exercises that operate on *your* topic, not on a toy example. The handbook is designed to be worked through with a notebook and a laptop open, not read passively.

It assumes good faith and rewards it. A recurring theme is that the shortcuts available to a researcher under pressure — an unverified citation, a favourable subset of results, a paraphrase that is really a copy, a similarity score massaged below a threshold — are not merely unethical but *ineffective*, because they fail precisely at the point where the work is examined most closely.

What this handbook is not

It is not a substitute for a supervisor, an ethics committee, or your institution's regulations. Where it describes general practice, your specific programme may impose stricter or simply different requirements, and those requirements win.

It is not a guide to gaming metrics. There is no chapter on raising your h-index. The premise throughout is that the durable strategy is to do work that is worth citing and to describe it accurately.

It is not current forever. Chapters 9, 16, 41–46 and 50–52 describe software platforms, database features, journal metrics, and publisher policies, all of which change — some of them yearly. Every such chapter therefore tells you *how to verify the current state of affairs from an authoritative source* rather than asking you to trust a printed description. Where this handbook and a publisher's own current documentation disagree, the publisher is right.

A note on examples and on honesty

Two categories of example appear in this handbook, and they are deliberately distinguished.

Real works are cited normally and appear in the References. These are established, verifiable publications — foundational methods, well-known datasets, reporting guidelines, and published studies of research practice. You are encouraged to read them.

Hypothetical examples are constructed for teaching. The extended case study in Part XVII, the running example of cross-hospital chest-radiograph classification, the method named CLUSTER-DG, the placeholder papers labelled P1–P15, and every number attached to them are **invented for instructional purposes**. They are realistic — the underlying phenomenon is documented in real literature, cited where relevant — but they are not findings, and nothing in them may be cited. Wherever such material appears it is marked **[HYPOTHETICAL]**.

This distinction is not pedantry. A significant fraction of the integrity failures this handbook warns against begin with a plausible-looking example being copied out of a teaching document and into a manuscript. If you take one habit from this book, take the habit of opening every source you cite.

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How to Use This Guide

Four ways in

1. Sequentially, as a course. Parts I–VII cover everything that happens before you run an experiment; Parts VIII–IX cover producing trustworthy evidence; Parts X–XIII cover writing and tooling; Parts XIV–XVI cover ethics and publication. Worked in order, with the exercises, this is roughly a semester of part-time study and will take you from an unfocused interest to a submitted manuscript.

2. By current stage. Use the table below.

If you are here right now	Start at
I have an interest but no topic	Chapter 4
I have a topic but it feels too broad	Chapter 4, §4.3; Chapter 5
I have read some papers but cannot find a gap	Chapters 13, 15, 17–19
I have a gap; I do not know what to promise	Chapters 6, 7, 20, 21
I am designing experiments	Chapters 22–26
I have results and cannot start writing	Chapter 30 onwards; and §29.1
I have a draft and it feels weak	Chapters 31–37; Part XVIII checklists
I am choosing a journal	Chapters 50–52
I have reviewer comments	Chapter 56

3. As a reference. The chapters on evaluation metrics (27), research gaps (17), search strategies (10), and journal verification (51–52) are written to be consulted in isolation. Each defines its own terms.

4. As teaching material. Every chapter contains exercises, common-mistake tables, and checklists that can be lifted into a workshop or a lab meeting. Part XVIII and the Appendices are reusable templates.

The structure of a chapter

Most substantive sections follow a fixed pattern, so you can navigate to the part you need:

> **Definition □ Purpose □ Detailed explanation □ Step-by-step procedure □ Example □ Common mistakes □ Best practices □ Tools □ Verification checklist**

Not every section needs all nine elements, and they are omitted where they would be padding.

Conventions

Convention	Meaning
[HYPOTHETICAL]	Invented for teaching. Not a finding. Do not cite.
[VERIFY]	Platform features, metrics, or policies that change. Check the authoritative source named.
□ / □	A weak example followed by an improved version of the same thing
Exercise n.n	Work to do on your own topic
Fact vs **Recommendation**	Where a claim could be mistaken for consensus, the handbook states which it is

The single most important idea in this handbook

If you read nothing else, read this.

A research paper is not a report of what you did. It is an **argument** that something is now known which was not known before, supported by evidence a sceptic would accept. Every section of a paper exists to defend one link in that argument. Work that cannot be expressed as such an argument may still be valuable — as software, as a service, as a thesis chapter — but it will not survive peer review at a serious venue, because there is nothing for the reviewer to be convinced *of*.

Most of this handbook is the practical consequence of that one sentence.

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PART I — UNDERSTANDING RESEARCH

Part I establishes the conceptual foundation on which everything else in this handbook depends. Chapter 1 defines research and, more importantly, distinguishes it from the activities it is most often confused with — building systems and solving problems. Chapter 2 surveys the types of research and shows how the type is dictated by the question rather than chosen for convenience. Chapter 3 lays out the complete research lifecycle, stage by stage, with the decisions and failure modes attached to each.

Readers who are impatient to begin searching the literature may be tempted to skip to Part III. Resist this. The single most expensive error in research is not a badly executed experiment; it is a well-executed experiment that answers a question nobody was asking. Parts I and II exist to prevent that.

Chapter 1 — What Is Research?

1.1 Definition

Definition. Research is a systematic, documented, and critically evaluated process of inquiry that produces knowledge which did not previously exist in the accessible record, in a form that others can independently verify.

Every clause in that sentence carries weight, and it is worth unpacking each before going further.

Systematic. The inquiry follows a procedure that was chosen in advance for reasons that can be stated. The opposite is not disorder but *opportunism* — trying things until something works and then reporting the thing that worked as though it had been the plan. The test of whether your process is systematic is not whether it felt organised but whether a competent stranger could repeat it from your description.

Documented. The procedure, the data, the analysis, and the reasoning are recorded in sufficient detail that they can be examined. Undocumented research is indistinguishable from assertion.

Critically evaluated. The researcher actively attempts to find reasons why the conclusion might be wrong — alternative explanations, confounds, limits of applicability — and reports what that attempt found. This is the hardest of the four and the one most often skipped, because it requires arguing against your own result.

Produces knowledge that did not previously exist in the accessible record. Two qualifications matter here. *Accessible* means published, indexed, and findable — knowledge that exists inside your own head, or inside your company, or in an unpublished thesis nobody can obtain, has not yet entered the record. *Did not previously exist* is a claim about the literature, not about your personal awareness; the fact that you did not know something is not evidence that the field did not know it. Establishing this claim is the entire purpose of Parts III to VI of this handbook.

In a form that others can independently verify. Verification is what separates research from expert opinion. A result that only you can obtain, because only you have the data or only you know the undocumented preprocessing step, is a claim rather than a finding.

1.2 The purpose of research

It is worth being concrete about *why* this activity is organised the way it is, because the conventions of research writing — which can appear arbitrary and bureaucratic to a newcomer — are almost all consequences of its purpose.

Research exists to build a **cumulative, correctable body of knowledge**. Cumulative means each contribution is designed to be built upon by strangers; correctable means each contribution is designed to be *checked*, and overturned if wrong. Nearly every convention that frustrates early-career researchers follows from these two requirements:

Convention	Purpose it serves
Citing prior work	Makes the increment visible; locates your claim in the cumulative structure
Describing method in tedious detail	Enables independent verification and reuse
Comparing against baselines	Distinguishes your contribution from the difficulty of the task
Reporting limitations	Tells later researchers where the claim stops being safe
Peer review	Filters claims before they enter the record others will build on
Reporting variance and statistics	Allows a reader to judge whether the effect is real
Publishing negative and null results	Prevents the field repeating your dead end

Once you see the conventions as engineering solutions to the problem of building shared, checkable knowledge, they stop being hoops to jump through. A reviewer who demands a missing baseline is not being obstructive; they are protecting the cumulative property. A reviewer who asks for confidence intervals is protecting the correctable property.

1.3 Research versus project development

This is the distinction that determines whether most early-career work becomes publishable, and it causes more confusion than any other topic in this handbook — particularly in computer science, where the two activities look almost identical from the outside and use the same tools.

Table 1.1 — Research versus project development

Dimension	Project / development	Research
Goal	Deliver a working artefact	Produce verifiable new knowledge
Success criterion	It works, on time, to specification	A claim is novel, significant, and validated
Question form	"How do I build X?"	"Under what conditions does X hold, and why?"
Comparison	Against requirements	Against baselines and the state of the art
Evaluation	Testing, user acceptance	Controlled experiment, statistics, ablation
Meaning of failure	A defect to be fixed	A *finding* to be reported and explained
Reuse of existing work	Encouraged and sufficient	Necessary but insufficient — must be exceeded or examined
Generalisation	Usually irrelevant	Usually the whole point
Primary output	Software, report, demonstration	Paper, dataset, theorem, protocol
Audience	Users, client, institution	The field, indefinitely

The relationship between the two is not opposition. Most computational research *contains* a development project. The project is the **instrument**; the knowledge claim is the **product**. A telescope is an engineering achievement; "Jupiter has moons" is the discovery. Papers are cited for discoveries, not for instruments — which is why a paper that reports only the instrument reads to a reviewer as a technical report that has been submitted to the wrong kind of venue.

1.3.1 The diagnostic question

Here is the fastest way to classify your own work. Ask:

> **If my system works exactly as intended, what will the field know that it did not know before?**

Three kinds of answer are possible.

- **A specific, transferable statement** — "that cross-institutional degradation in this task is approximately 0.12 AUC under leakage-free evaluation", or "that this class of method fails when the input distribution has property P". This is research.
- **"That my system works"** — this is development. It may be excellent, useful, fundable development. It is not yet a research contribution.
- **"I am not sure"** — this is the most common and most rescuable answer. It means you have an instrument and have not yet chosen a question. Chapters 4 and 5 exist for exactly this situation.

1.3.2 Worked rescue: three development projects turned into research

The following transformations are the single most useful skill in Part I. Notice that in each case the *engineering work barely changes* — what changes is what is measured, what it is compared against, and what is claimed.

[HYPOTHETICAL] Example A. *Development framing:* "Build a web system that detects plagiarism in student assignments for our department."

This is bounded by a local need, solvable by integrating existing tools, and successful when deployed. Generalisation is irrelevant. No baseline comparison is required.

Research framing: "Sentence-embedding plagiarism detectors are reported at high F_1 on English benchmarks. Their behaviour on **code-mixed** student text (for example Hindi–English) is unmeasured, and there is reason to expect degradation: the subword tokenisers used by these models are trained predominantly on monolingual corpora, so transliterated and code-switched text fragments into unfamiliar subword sequences. We quantify the degradation across four levels of obfuscation and test whether script normalisation recovers it."

What changed: a *mechanism hypothesis* (tokeniser fragmentation) was added, a *measurable outcome* was specified, and the claim became transferable to anyone working on multilingual text.

[HYPOTHETICAL] Example B. *Development framing:* "Apply YOLO to helmet detection for traffic enforcement."

Research framing: "Object detectors are typically benchmarked on datasets whose images are well-lit and lightly occluded. We characterise how detection performance on two-wheeler helmet compliance degrades across illumination and occlusion strata in naturalistic traffic imagery, and measure whether a lightweight image-enhancement front-end recovers performance within a fixed edge-inference latency budget."

What changed: the study now measures a *degradation curve* across conditions rather than a single accuracy number, and introduces a *constraint* (latency budget) that makes the trade-off scientifically interesting rather than merely engineering-adjacent.

Example C, grounded in real literature. *Development framing:* "Train a CNN to detect pneumonia on chest radiographs."

This has been done many times and is not a contribution. But the *real* published literature provides the rescue. Zech et al. (2018) showed that convolutional networks trained on chest radiographs can learn to exploit hospital-specific confounders — such as differences in equipment or department-level disease prevalence — with the consequence that performance measured within one institution substantially overstates performance at a new one. That observation converts a saturated engineering task into a live research area, because it implies that a large body of reported performance numbers may not describe what they appear to describe.

Research framing built on it: "We re-evaluate published architectures under institution-disjoint protocols and quantify the degradation, then test whether it can be mitigated without access to institutional metadata."

The general lesson: **an engineering task becomes a research question when you find a reason to doubt what the reported numbers mean.**

1.4 Research versus problem solving

Problem solving and research overlap but are not the same, and the difference is instructive.

Solving a problem means moving a specific situation from an undesired state to a desired one. It is complete when the situation is resolved. Research means producing a statement about a *class* of situations, and it is complete when the statement is established and its limits are known.

An engineer who fixes a memory leak has solved a problem. A researcher who characterises the class of allocation patterns under which this family of runtimes leaks, and demonstrates the boundary conditions, has produced knowledge. The first is worth doing and stops mattering when the software is retired. The second remains true afterwards.

This distinction has a practical consequence for how you frame work in a paper. Reviewers react badly to papers that read as case reports — “we had this problem, we did these things, it improved” — not because the work is worthless but because there is no statement whose generality can be assessed. The fix is to identify what class your specific case belongs to, and to say something about the class.

1.5 Scientific investigation and evidence-based research

Scientific investigation is the practice of subjecting a candidate explanation to a test that could have refuted it. The critical phrase is *could have refuted it*. A test that would produce the same outcome whether or not your explanation is correct provides no evidence, however elaborate it is.

This has a sharp practical implication for computational research. Suppose your hypothesis is that a new attention mechanism improves performance because it captures long-range dependencies. Running your model and observing higher accuracy does not test that hypothesis, because higher accuracy is also consistent with a dozen other explanations: more parameters, more effective regularisation, a longer training schedule, a luckier initialisation, or unequal hyperparameter tuning effort. To *test* the hypothesis you need a comparison in which only the mechanism differs — which is what an ablation study is for (§26.6), and why an ablation is not an optional extra but the core evidential instrument for a mechanistic claim.

Evidence-based research means that each claim in your paper is traceable to something specific: a measurement, a proof, a cited result, or a documented observation. In practice this means being able to answer, for any sentence in your paper, the question *how do you know that?* The three acceptable answers are:

1. **We measured it** — and here is the experiment, the variance, and the statistical test.
2. **It is established in the literature** — and here is the citation, which we have read.
3. **It is a definition or a logical consequence** — and here is the derivation.

A fourth answer — *it is generally believed* or *it is well known* — is acceptable only for genuine textbook material, and is a frequent hiding place for unexamined assumptions. If you find yourself writing “it is well known that”, stop and check whether it is known at all.

1.6 Originality and novelty

These terms are used loosely in everyday speech and need to be separated.

Originality is a property of the *work*: it was produced by you, not copied. Originality is a baseline requirement and its absence is misconduct (Part XIV).

Novelty is a property of the *contribution relative to the literature*: the knowledge is new to the field. Novelty is what makes work publishable and its absence is not misconduct — it is simply redundancy.

The two are independent. A study can be entirely original and completely unnovel: you may have honestly and independently discovered something that four other groups published in 2021. This is an ordinary experience and the appropriate response is not embarrassment but reframing — usually by asking what your version does that theirs does not, or by reporting the independent replication as such, which is a genuine contribution in its own right.

Novelty is discussed in depth in Chapter 20. Two points belong here because they shape how you should think from the beginning:

Novelty is not a binary. It is a spectrum from *incremental* (a better result on an established problem by an established route) through *substantive* (a new method, a new resource, a new evaluation protocol) to *paradigmatic* (a reframing of what the problem is). Almost all published research, including almost all *good* published research, is in the first two categories. The expectation that a PhD must be paradigmatic is a major and unnecessary source of paralysis.

Novelty is not the same as difficulty. Work can be extremely laborious and add nothing, and it can be conceptually simple and change a field. Reviewers assess what is *learned*, not what was *endured*. This is counter-intuitive and worth internalising early, because a great deal of effort is routinely spent on work whose difficulty is not converting into knowledge.

1.7 Four properties of trustworthy findings

Four terms are used constantly in research writing and are frequently confused with one another. They are not interchangeable, and reviewers use them precisely.

Table 1.2 — Validity, reliability, reproducibility, generalizability

Property	Question it answers	Threat if absent	Concrete example of failure
Validity	Are we measuring what we claim to measure?	The result is about something other than what you say	A model that appears to detect disease is detecting scanner artefacts
Reliability	Would repeating the measurement give a consistent answer?	The result is noise	Reported gain of 0.4 points where seed-to-seed variation is 1.5 points
Reproducibility	Can others obtain this result from your materials?	The result cannot be checked	Numbers unobtainable from the released code
Generalizability	Does the result hold beyond the studied sample?	The result is true but useless	Accuracy holds on one hospital's data and collapses elsewhere

1.7.1 Validity, and the vocabulary of validity threats

Validity is subdivided in the methodological literature, and the vocabulary is worth knowing because reviewers use it.

- **Construct validity** — does your measurement correspond to the concept you claim? If you claim to measure "code readability" using cyclomatic complexity, a reviewer may reasonably dispute the construct.
- **Internal validity** — within your study, is the observed relationship attributable to the factor you manipulated, rather than to a confound? Unequal hyperparameter tuning between your method and the baseline is an internal validity threat. So is data leakage.
- **External validity** — does the result transfer to other populations, domains, and settings? Single-dataset studies are weak here by construction.
- **Conclusion validity** (sometimes *statistical conclusion validity*) — do your statistical procedures support the inference you draw? Reporting a gain without variance, or selecting the one comparison that reached significance, are threats here.

These four categories, which originate in the experimental design literature (Campbell and Stanley, 1963; Shadish, Cook and Campbell, 2002) and are widely applied in empirical software engineering (Wohlin et al., 2012), give you the standard structure for a "Threats to Validity" section — and, more usefully, a checklist to run against your own design *before* you collect data.

1.7.2 Reproducibility: terminology and why it matters

Terminology in this area is genuinely inconsistent across communities, which is worth knowing so you are not confused by conflicting definitions. A widely used convention distinguishes:

- **Repeatability** — the same team, the same setup, obtains the same result.
- **Reproducibility** — a *different* team, the same artefacts (code, data), obtains the same result.
- **Replicability** — a different team, *independently implemented*, obtains a consistent result.

Some communities swap the last two. When precision matters, state what you mean rather than relying on the label.

The practical importance is not philosophical. Large-scale surveys and structured re-evaluations across several fields have repeatedly found that a substantial fraction of published results are hard to obtain independently, and that apparent improvements sometimes disappear when comparisons are made under equal conditions. Baker (2016) reported survey evidence of widespread concern about reproducibility across scientific disciplines. In machine learning specifically, methodological critiques and re-evaluation studies — for example Sculley et al. (2018) on empirical rigour, Lipton and Steinhardt (2019) on failure modes of scholarship, Dacrema et al. (2019) on recommender-system baselines, Melis et al. (2018) on language-model baselines, and Lucic et al. (2018) on generative adversarial network comparisons — have documented cases in which reported gains narrowed or vanished once baselines were tuned with comparable effort.

Two conclusions follow, and they are among the most actionable in this handbook:

1. **Tune your baselines as hard as you tune your own method, and say so in the paper.** One sentence stating that all methods received an identical search budget removes the most common reason reviewers distrust an improvement claim.
2. **The reproducibility gap is a research opportunity.** A rigorous re-evaluation requires careful experimental work rather than new theory or large compute, which makes it achievable for a researcher with modest resources, and such studies are frequently well cited because they change practice. This is developed in §17.10.

1.7.3 Reliability in practice: the single-run problem

A very common defect in student work is reporting a single run to four decimal places. Stochastic training procedures produce a *distribution* of outcomes; a single sample from that distribution is not an estimate of the method's quality, and the difference between two single samples is not evidence of a difference between two methods.

The remedy is mechanical and cheap: run each configuration with at least five different random seeds — ten is better — and report the mean with a standard deviation or a confidence interval. `0.822 ± 0.009` is more informative than `0.8221`, and it is the form that allows a reader to judge whether your improvement of `0.014` means anything. Statistical testing is covered in §26.8 and §29.5.

1.8 What makes a contribution rather than an implementation

We can now state the distinction precisely. Consider three levels:

Figure 1.1 — The relationship between activity, evidence, and contribution

```
ACTIVITY                 EVIDENCE                   CONTRIBUTION
what you did             what you measured          what the field gains
────────────             ─────────────────          ────────────────────

"We trained a      →     "It reached 0.94      →    ??? nothing, if this
 DenseNet on              accuracy on the            number is already
 CheXpert."               official test split."      in the literature

"We re-split       →     "Macro AUC falls        →  "Reported CXR
 three public             from 0.897 to 0.781       performance overstates
 datasets by              (95% CI 0.104–0.128        cross-hospital
 institution and          reduction) across five     behaviour by roughly
 re-evaluated five        architectures, over        0.12 AUC under
 published models."       10 seeds."                 leakage-free
                                                     evaluation."
```

[HYPOTHETICAL — numbers illustrative]

The activity in the second row is *less* technically demanding than in the first: no new architecture, no novel loss. Yet it produces a contribution, because it produces a statement about the world that other researchers must now take into account.

This is the mechanism by which researchers with limited computational resources produce influential work. You do not need to out-train a large laboratory. You need to ask a question whose answer changes what people should do.

1.8.1 The seven categories of contribution

It helps enormously to know explicitly which kind of contribution you are making, because each requires a different kind of evidence.

Contribution type	What you claim	Evidence required
New method	A procedure that achieves something previously unachieved or achieves it better	Fair comparison, ablation isolating the novel component, statistics
New theory or analysis	A proof, bound, complexity result, or formal characterisation	Correct derivation; assumptions stated; ideally empirical corroboration
New empirical knowledge	A measured fact about the world or about existing methods	Careful measurement, controls, variance, replication across settings
New resource	A dataset, benchmark, corpus, or tool others will use	Construction protocol, quality assessment, annotation agreement, licence, baselines
New evaluation	A protocol or metric that measures something better	Demonstration that existing evaluation is inadequate; validation of the new one
Synthesis	An organised, critical account of a body of work	Reproducible search protocol, systematic screening, comparative analysis
Reproduction or refutation	An existing result does or does not hold	Faithful re-implementation, documented discrepancies, diagnosis of cause

Most early-career researchers assume “new method” by reflex. In practice the strongest achievable first papers are frequently in the *new empirical knowledge*, *new resource*, *new evaluation*, and *reproduction* categories, because they are completable within a year, require modest compute, and are difficult for a reviewer to dispute when done rigorously.

1.9 Weak and strong research, weak and strong contributions

The following contrasts are deliberately concrete. All are hypothetical formulations constructed to illustrate the pattern.

1.9.1 Weak research

[HYPOTHETICAL] □ *“We propose a novel hybrid CNN-LSTM framework for sentiment classification. Experiments on a benchmark dataset show that our model achieves 92.4% accuracy, outperforming existing state-of-the-art methods and demonstrating the effectiveness of the proposed approach.”*

What is wrong, in order of severity:

1. **No question.** Nothing was unknown at the start; nothing is known at the end beyond one number.
2. **No mechanism.** Why should combining these components help? Without a stated reason, there is nothing to test and no explanation to transfer.
3. **“A benchmark dataset.”** One dataset supports no generalisation claim, and it is not even named.
4. **“Existing state-of-the-art methods”** — unnamed, so the comparison is unauditable; and probably not tuned with equal effort.
5. **No variance, no statistical test.** 92.4% from how many runs?
6. **“Demonstrating the effectiveness”** — a conclusion restating the result, not interpreting it.
7. **Unfalsifiable framing.** There is no outcome of this study that would have been reported as informative other than “we won”.

1.9.2 Strong research on the same topic

[HYPOTHETICAL] □ *“Reported gains from recurrent components in sentiment classification are typically measured on datasets whose documents are short. We hypothesise that these gains are a function of document length and largely vanish once transformer context windows exceed typical document length. We test this by evaluating four architectures across five corpora stratified into four document-length bands, with an identical 50-trial tuning budget per architecture per band and ten seeds each. Recurrent components yield a mean improvement of 2.1 points macro-F_1 in the shortest band (95% CI 1.4–2.8) but no detectable improvement in the two longest bands (95% CI includes zero). This suggests that a portion of previously reported architectural gains is attributable to corpus composition rather than to modelling capacity.”*

Why this is strong:

- It states a **hypothesis with a mechanism** (document length) that could have been refuted.
- The design **isolates the factor** by stratifying on it.
- **Fairness is controlled** and stated (equal tuning budget).
- **Uncertainty is quantified**, and a null result in two bands is *reported rather than hidden*.
- The conclusion is a **transferable statement about the literature**, not about the authors’ model.
- The finding would have been publishable in either direction.

1.9.3 Weak and strong contribution statements

[HYPOTHETICAL] □ Weak	Why it fails	[HYPOTHETICAL] □ Strong
"We propose a novel deep learning framework."	"Framework" conceals what the thing is; "novel" is asserted, not demonstrated	"We propose a site-clustering regulariser that requires no institutional metadata at training time."
"Our method outperforms state-of-the-art methods."	Unbounded, unquantified, invites a counterexample	"Our method improves worst-institution AUC by 0.041 (95% CI 0.032–0.050) over the strongest of four baselines under an identical tuning budget."
"We achieve high accuracy."	Relative to what threshold, on what data?	"We recover approximately 82% of the benefit obtained by methods that require privileged site labels, without using them."
"This is the first work in this area."	Almost always falsifiable by a specialist	"To our knowledge this is the first leakage-free multi-institution evaluation of these five architectures; the closest prior study [ref] evaluates two of them on a single external set."
"We solve the problem of domain shift."	Overclaim	"We reduce, but do not eliminate, cross-institution degradation; approximately 0.075 AUC of the observed 0.116 gap remains."

Note the pattern: strong statements are **longer**, contain **numbers with uncertainty**, name their **scope**, and **concede** what they do not achieve. Calibrated claims are not weaker than bold ones — they are harder to attack, and reviewers read calibration as competence.

1.10 Common mistakes in conceptualising research

Mistake	Why it happens	Correction
Treating the implementation as the contribution	Engineering training rewards working artefacts	Ask: what will the field *know*?
Choosing the method before the question	Methods are exciting; questions are hard	Derive the method from the gap (Part VI)
Assuming novelty requires a new architecture	Visibility of high-profile architecture papers	Review the seven contribution types (§1.8.1)
Equating effort with contribution	Effort is what the researcher experiences	Reviewers assess what is learned
Reporting only favourable outcomes	Fear that null results are failures	Design so that either outcome is informative
Believing a single run is a result	Deadline pressure; compute cost	Five seeds minimum; report variance
Using "it is well known that"	Convenient; avoids searching	Either cite it or measure it
Confusing "nobody has done this" with a gap	Superficially resembles novelty	Add a mechanism and a measurement (§19.3)

1.11 Verification checklist for Chapter 1

Before proceeding, confirm that you can answer each of these about your own work.

- [ ] I can state, in one sentence, what the field will know if my study succeeds.
- [ ] That sentence is about a class of situations, not only about my system.
- [ ] I can name which of the seven contribution types (§1.8.1) I am attempting.
- [ ] I can name the evidence that contribution type requires.
- [ ] I can state a mechanism — a reason why my approach should work — that could turn out to be wrong.
- [ ] I have identified at least one outcome, other than "my method wins", that would be worth reporting.

- [ ] I can name the principal threat to internal validity in my planned design.
- [ ] I can name the principal threat to external validity.
- [ ] I know how many seeds or repetitions I will run, and how I will report variance.
- [ ] I have not used the words “novel”, “framework”, or “state of the art” in my own description without being able to justify each.

Exercises

Exercise 1.1 — Classification. Write your current work in one sentence. Classify it as project development, research, or unclear. If project or unclear, complete this sentence: *“The field does not currently know whether ______, and this study would show it.”*

Exercise 1.2 — The mechanism test. State why your approach should work, in the form *“Because [property of the problem], [property of the method] should produce [effect].”* If you cannot fill all three slots, you have an implementation plan and not yet a hypothesis.

Exercise 1.3 — The falsifiability test. Describe the result you expect. Then describe the opposite result. Write one sentence explaining why the opposite result would still be worth publishing. If you cannot, revisit §1.9.

Exercise 1.4 — Contribution typing. Identify which of the seven contribution types your work targets. Then list the specific evidence that type demands, and mark each item as *already planned* or *missing*. The missing items are your experimental to-do list.

Exercise 1.5 — Critical reading. Take any paper in your area. Identify (a) the activity, (b) the evidence, © the contribution, in the sense of Figure 1.1. If you cannot find ©, you have learned something important about the standards of the venue that published it.

Chapter 2 — Types of Research

2.1 Why classification matters practically

Research typologies can feel like taxonomy for its own sake. They are not. The type of research you are conducting determines four concrete things:

1. **What counts as adequate evidence** — a proof, a controlled experiment, a saturated set of interviews, or a reproducible synthesis.
2. **What your validity threats are** — and therefore what a reviewer will attack.
3. **What reporting standard applies** — for example PRISMA for systematic reviews (Page et al., 2021), or the reporting conventions of your subfield.
4. **Where it can be published** — venues differ sharply in the types they accept.

Choosing a type by convenience, and then discovering that your evidence does not match your claim, is a common and expensive error. The type should be *derived from the question*.

2.2 By purpose: fundamental, applied, and translational

Fundamental (basic) research seeks to understand a phenomenon, without a specified application. *Why do heavily overparameterised networks generalise at all? What is the sample complexity of this learning problem?* Evidence is typically theoretical or carefully controlled empirical work. Success is explanatory power. The risk is irrelevance to practice; the reward is durability — foundational results remain citable for decades.

Applied research addresses a specified practical problem. *Can retinopathy be screened reliably using low-cost fundus cameras in primary care?* Evidence must include performance under realistic conditions. Success is usefulness. The risk is that the result is tied to a context that changes.

Translational research deliberately bridges the two: it takes a method established in idealised conditions and studies what happens when deployment constraints are imposed. Deployment constraints — latency, memory, privacy, missing metadata, distribution shift, human workflow — become the *independent variables* rather than nuisances. This is an under-occupied and highly productive space for early-career researchers, because the constraints are real, the questions are genuinely open, and the compute requirements are often modest.

2.3 By approach: experimental, theoretical, computational, observational

Experimental research manipulates one or more variables and measures the effect, holding others constant. Most machine-learning papers are experimental in form, though frequently weak in execution because the “holding others constant” requirement is not met (unequal tuning, different data pipelines, different training budgets).

Theoretical research derives consequences from stated assumptions: proofs, bounds, convergence guarantees, complexity results, impossibility results. Evidence is the correctness of the derivation. The characteristic failure is a result that is technically correct but whose assumptions exclude every case of practical interest — which is why strong theoretical papers state explicitly where their assumptions bite.

Computational or simulation research studies systems through models when direct experimentation is impractical. The characteristic threat is that conclusions describe the simulator rather than the world, which is why validation of the model against reality is essential.

Observational or empirical research measures what already exists without intervening: mining a hundred thousand public repositories, analysing published papers, characterising a corpus, measuring the behaviour of deployed systems. It cannot establish causation on its own, and the discipline of the field is to be careful with causal language. Its great advantage is scale.

2.4 By data type: quantitative, qualitative, mixed

Quantitative research measures, counts, and analyses numerically, aiming at generalisation. Its instruments are metrics, statistical tests, effect sizes, confidence intervals. Its blind spot is that it can only answer questions you thought to measure.

Qualitative research investigates meaning, mechanism, and experience: interviews, observation, document analysis, open-ended survey responses. It is not "soft" — rigorous qualitative work has explicit standards, including systematic coding, intercoder agreement, negative-case analysis, saturation, reflexivity, and audit trails. Reporting guidance such as COREQ and SRQR is used in several fields.

Its relevance to computational researchers is often underestimated. If your contribution concerns tools, explanations, interfaces, developer practice, or clinical adoption, then *how people actually interpret and use the system* is an empirical question that numbers alone cannot answer.

Mixed-method research integrates both deliberately — not merely running both, but designing so that one informs the other. A common and powerful pattern: a quantitative evaluation establishes that a model performs well, and a qualitative study establishes that intended users do not trust or cannot act on its output. The combination is a much stronger contribution than either half, and the tension between the two results is usually the most interesting thing in the paper.

2.5 By stage of knowledge: exploratory, descriptive, comparative, explanatory

This axis is about what is already known, and it governs how strongly you may phrase your conclusions.

Stage	Question form	Typical design	Permissible claim strength
Exploratory	What is going on here?	Open-ended, small-scale, hypothesis-generating	Suggestive only; explicitly preliminary
Descriptive	What are the characteristics of X?	Measurement, characterisation, corpus analysis	Factual about the studied sample
Comparative	Does A differ from B, and by how much?	Controlled comparison with matched conditions	Difference, with effect size and uncertainty
Explanatory / causal	Why does A produce B?	Ablation, intervention, mediation analysis	Mechanism, if the design isolates it

A very common reviewer objection is a mismatch between design and claim: an exploratory study whose abstract asserts a general causal mechanism. Matching the two is largely a matter of writing discipline, and Chapter 36 addresses it directly.

2.6 Review research: narrative reviews, systematic reviews, and surveys

Because reviews are often a researcher's first publication, they deserve specific treatment.

Narrative review. An expert account of a body of work, organised by the author's judgement. Strength: interpretation, taxonomy, agenda-setting. Weakness: selection is not reproducible, so a sceptical reader cannot tell whether inconvenient work was omitted. Usually written by established researchers, often by invitation.

Systematic literature review (SLR). A review conducted according to a pre-specified, reported protocol: databases searched, exact search strings, dates, inclusion and exclusion criteria, screening procedure, quality appraisal, and extraction fields. The defining property is that *another team could repeat it*. In medicine and increasingly elsewhere, PRISMA (Page et al., 2021) is the standard reporting framework; in software engineering, Kitchenham's guidelines (Kitchenham and Charters, 2007) are widely used.

Systematic mapping study. A lighter-weight relative of the SLR that characterises the shape of a field — what is studied, by whom, with what methods, in what venues — without appraising outcomes in depth. Useful when a field is too large or too heterogeneous for meta-analysis.

Meta-analysis. Statistical pooling of quantitative results across studies to estimate an overall effect. Requires comparable outcome measures, which in computational fields is often the binding obstacle: results measured on different datasets with different splits and different metrics generally cannot be pooled, and attempting it produces a meaningless average.

Survey (in the computer-science sense). A structured, tutorial-oriented account of methods in an area, usually with a taxonomy and comparison tables. Note the terminological collision: in social science a "survey" is a questionnaire study of human respondents, an entirely different thing. State which you mean.

2.6.1 Recommendation: the systematic review as a first paper

This is a recommendation, not a fact. For a researcher in their first or second year, a rigorous systematic review is often the highest-expected-value first publication, for four reasons:

1. It forces genuine mastery of the literature, which is required anyway.
2. Its by-products *are* the artefacts you need for your own empirical work — the literature matrix (Chapter 15) and the research gap (Part VI) fall out of it.
3. It requires no compute and no data access.
4. It remains citable for years.

The honest counterweight: it is more work than newcomers expect. Screening several hundred to a few thousand records is normal, and the protocol must be executed faithfully. It also does not demonstrate experimental skill, so it should not be your *only* output.

2.7 Choosing a type from your question

Figure 2.1 — Choosing a research type from a research question

```
What does your question ask for?
 │
 ├─ "Is it true that…?" / "Does A differ from B?"
 │      → COMPARATIVE EXPERIMENTAL
 │        evidence: controlled comparison, equal budgets, seeds, statistics
 │
 ├─ "How much / how large / under what conditions?"
 │      → DESCRIPTIVE or COMPARATIVE EMPIRICAL
 │        evidence: measurement across strata, confidence intervals
 │
 ├─ "Why does it happen?"
 │      → EXPLANATORY (ablation / intervention / mediation)
 │        evidence: designs isolating one factor at a time
 │
 ├─ "Must it always be so?" / "What is the limit?"
 │      → THEORETICAL
 │        evidence: proof; assumptions stated and their bite acknowledged
 │
 ├─ "What do people do / think / experience?"
 │      → QUALITATIVE or MIXED
 │        evidence: systematic coding, agreement, saturation, reflexivity
 │
 ├─ "What is already known, and where does it disagree?"
 │      → SYSTEMATIC REVIEW
 │        evidence: reproducible protocol, screening funnel, appraisal
 │
 └─ "Does the published result hold?"
        → REPRODUCTION / REPLICATION
          evidence: faithful re-implementation, documented discrepancies
```

Table 2.1 — Research types, questions they answer, and typical evidence

Type	Answers	Core evidence	Principal threat
Fundamental	Why does this phenomenon occur?	Theory plus controlled empirics	Practical irrelevance
Applied	Does this work for this purpose?	Realistic-condition evaluation	Context dependence
Translational	What happens under deployment constraints?	Constrained evaluation, cost measurement	Constraint chosen unrealistically
Experimental	Does the manipulation change the outcome?	Controlled comparison	Confounds; unequal effort
Theoretical	What follows from these assumptions?	Proof	Vacuous assumptions
Computational	How does the modelled system behave?	Simulation plus validation	Simulator ≠ world
Observational	What exists at scale?	Large-sample measurement	Causal overreach
Quantitative	How much?	Statistics	Measuring the wrong construct
Qualitative	What does it mean to participants?	Coded data, agreement, saturation	Researcher bias; unwarranted generalisation
Mixed	Both, integrated	Both, plus integration argument	Two studies stapled together
Systematic review	What is the state of evidence?	Reproducible protocol	Protocol not actually reproducible
Reproduction	Does the result hold?	Re-implementation, diagnosis	Unfair re-implementation

2.8 Common mistakes

Mistake	Correction
Calling a narrative review a “systematic review” without a protocol	Either follow a protocol and report it, or call it a review
Averaging accuracies across incomparable datasets as “meta-analysis”	Pool only genuinely comparable outcomes; otherwise present a structured comparison
Causal language (“improves”, “causes”, “leads to”) on observational data	Use associational language, or add an intervention
Claiming generality from a single dataset	Add datasets, or bound the claim explicitly
Running a questionnaire and calling it qualitative	Open-text responses need systematic coding to count as qualitative analysis
Choosing the type from available skills rather than from the question	Derive the type from the question; acquire the method or change the question

Exercises

Exercise 2.1 Write your research question. Trace it through Figure 2.1 and record the type it implies. Then list the evidence that type requires and mark what you currently have.

Exercise 2.2 Identify the strongest validity threat for your type from Table 2.1, and write one sentence describing how your design will mitigate it.

Exercise 2.3 If your question implies a type you have never used, decide now whether to acquire the method, collaborate, or reframe the question. Record the decision and the date.

Chapter 3 — The Research Lifecycle

3.1 The lifecycle as a whole

The stages below are presented linearly because they must be described in some order. They are not executed linearly, and pretending otherwise is a source of real damage: researchers who believe the process is a pipeline treat every backward step as a failure, when in fact iteration is the normal mode of competent work.

Figure 3.1 — The research lifecycle, with feedback paths

```
┌───────────────────────── PHASE 1: FRAMING ─────────────────────────┐
│  1  Research area                                                  │
│  2  Research problem                                               │
│  3  Research question(s)                                           │
│  4  Literature search                                              │
│  5  Paper reading and extraction                                   │
│  6  Literature matrix                                              │
│  7  Critical review / synthesis                                    │
│  8  Research gap                                                   │
│  9  Objectives, hypotheses, contributions                          │
└────────────────────────────────────────────────────────────────────┘
                                 │
┌───────────────────── PHASE 2: EVIDENCE ────────────────────────────┐
│ 10  Methodology design                                             │
│ 11  Dataset selection and preparation                              │
│ 12  Experimental design (baselines, splits, ablations, seeds)      │
│ 13  Execution                                                      │
│ 14  Evaluation and analysis                                        │
│ 15  Results                                                        │
│ 16  Discussion and interpretation                                  │
└────────────────────────────────────────────────────────────────────┘
                                 │
┌───────────────────── PHASE 3: COMMUNICATION ───────────────────────┐
│ 17  Figures and tables                                            │
│ 18  Manuscript writing                                            │
│ 19  References                                                    │
│ 20  Ethics and similarity check                                   │
│ 21  Journal selection                                             │
│ 22  Submission                                                    │
│ 23  Peer review                                                   │
│ 24  Revision and response                                         │
│ 25  Publication, archiving, dissemination                         │
└────────────────────────────────────────────────────────────────────┘

FEEDBACK PATHS THAT THE LINEAR DIAGRAM HIDES
 • Gap not defensible under scrutiny ............. return to stage 4
 • Objectives not measurable ...................... return to stage 8
 • Baselines unavailable or unrunnable ........... return to stage 9 or 12
 • Results contradict the hypothesis ............. return to stage 10
                                                   (NEVER to the data — see §3.4)
 • Reviewer identifies a missing comparison ....... return to stage 12
 • Scope mismatch at desk rejection .............. return to stage 21
 • Result already published while you worked ...... return to stage 8 and reframe
```

3.2 Stage-by-stage specification

Table 3.1 — Lifecycle stages: inputs, outputs, decisions, mistakes, tools

#	Stage	Input	Output	Key decision	Characteristic mistake	Tools
1	Research area	Interest, supervisor capability, resources	A named sub-area with a readable literature	Breadth versus competition	Choosing a domain and calling it a topic	Scopus/WoS trend analysis, arXiv, conference programmes
2	Research problem	Sub-area	A statement of a specific unknown	Is the answer actually unknown?	Restating an engineering task	Recent papers' limitations sections
3	Research question	Problem	1–3 answerable questions	Question type (Fig. 2.1)	Questions no experiment can settle	Chapter 6 frameworks
4	Literature search	Questions	A logged, reproducible record set	Databases, strings, inclusion window	One database, one spelling	Scopus, WoS, IEEE, ACM, Semantic Scholar
5	Reading and extraction	Record set	Filled extraction templates	Which papers deserve depth	Highlighting instead of extracting	Zotero, three-pass method
6	Literature matrix	Extractions	A sortable table of 10–25 studies	Which columns	A single "notes" column	Sheets/Excel, Zotero export
7	Critical synthesis	Matrix	Grouped, evaluated account	Organising principle	Paper-by-paper listing	Chapter 14
8	Research gap	Synthesis	An evidenced gap statement	Which gap type	"Room for improvement"	Chapter 18
9	Objectives	Gap	Aim, objectives, hypotheses, contributions	Measurability	Activity titles ("Study of…")	Chapter 7
10	Methodology	Objectives	A reproducible design	Design that isolates the factor	Environment mistaken for methodology	Chapter 22
11	Dataset	Methodology	Prepared, documented, licensed data	Split unit	Random splits on grouped data	Chapter 23
12	Experimental design	Methodology, data	Protocol: baselines, splits, seeds, ablations	Tuning budget parity	Only comparing your own model	Chapter 26
13	Execution	Protocol	Logged runs, artefacts	Configuration control	Tuning by editing source	MLflow, W&B, config files
14	Evaluation	Runs	Metrics, tests, error analysis	Metric choice	Accuracy on imbalanced data	Chapter 27
15	Results	Analysis	Tables, figures, observations	What to report	Reporting only favourable findings	Chapters 28, 35
16	Discussion	Results	Interpretation, limits	Mechanism versus speculation	Restating results	Chapters 29, 36
17	Figures and tables	Results	Publication-quality floats	One message per figure	Screenshots; unreadable fonts	Chapters 38–39
18	Writing	All above	Manuscript	Writing order	Introduction promising more than delivered	Chapters 30–37
19	References	Reading	Verified, styled bibliography	Manager and style	Unopened or fabricated citations	Chapters 43–44
20	Ethics and similarity	Manuscript	Clean report, declarations	Interpretation of matches	Chasing a percentage	Chapters 47–49
21	Journal selection	Manuscript	Target plus two backups	Scope fit	Choosing after writing	Chapters 50–52

#	Stage	Input	Output	Key decision	Characteristic mistake	Tools
22	Submission	Manuscript, letter	Submission record	Template compliance	Unproofed system PDF	Chapters 53–54
23	Peer review	Submission	Reviews and decision	Nothing — you wait	Panicking at “major revision”	Chapter 55
24	Revision	Reviews	Revised manuscript, response letter	Where to concede, where to argue	Undisclosed changes	Chapter 56
25	Publication	Accepted paper	DOI, archive, dissemination	Where to deposit	Neglecting the proofs	Chapter 57

3.3 Where time actually goes

Beginners consistently misallocate effort, and the misallocation has a predictable shape.

Figure 3.2 — Where time is actually spent versus where beginners expect to spend it

```
                    EXPECTED              REALISTIC (first paper)
Framing (1–9)       ███ 10%               ████████████ 30–35%
Evidence (10–16)    ████████████████ 70%  ████████████████ 40–45%
Communication       ██ 15%                ████████ 20–25%
 (17–22)
Review (23–25)      █ 5%                  (mostly waiting: 3–12 months)
```

In calendar terms, a realistic first journal paper in a computational field spans roughly:

Phase	Realistic duration	Note
Framing	4–10 weeks	Almost always compressed, and this is the root cause of most later rejections
Evidence	8–20 weeks	Budget three times your first estimate for compute; baselines and ablations dominate
Communication	3–6 weeks	Underestimated; the abstract and response letter take longer than expected
Review to publication	3–12 months	Largely outside your control

The single most valuable planning insight in this chapter: **the framing phase is cheap in resources and decisive in outcome.** Four extra weeks spent establishing a defensible gap routinely saves six months of experiments that answer nothing.

3.4 One feedback path that is not permitted

Every backward arrow in Figure 3.1 is legitimate except one distinction that must be stated explicitly, because it is the point at which ordinary pressure turns into misconduct.

When results contradict your hypothesis, you may legitimately:

- Return to the **methodology** and ask whether the design actually tested the hypothesis;
- Discover and fix a genuine **bug**, then re-run everything, including the baselines;
- Add experiments that **probe why** the prediction failed;
- **Report the disconfirmation** and revise the hypothesis, clearly labelling the revision as post hoc;
- Conclude that the effect is absent, and publish that.

You may not:

- Adjust, exclude, or “clean” data until the hypothesis is supported;

- Try many analyses and report only the one that reached significance (p-hacking);
- Present a hypothesis formed after seeing the results as though it had been predicted (HARKing);
- Drop the datasets, seeds, or metrics on which the method lost, without disclosure;
- Report the best of many runs as the result.

The first list is science; the second is falsification or misrepresentation, discussed in Chapter 49. The boundary is not always obvious in the moment, which is why the practical safeguard is **pre-specification**: write down, before you look at test results, what you predict, what you will measure, which test you will use, and what would count as a disconfirmation. A dated entry in a research journal costs two minutes and settles the question later.

3.5 Recommended practice: the research journal

This is a recommendation. Keep a single dated, append-only log for each project, containing:

- Decisions and their reasons ("using site-wise splits because §Zech et al. suggests site confounding; decided 12 March")
- Search strings and the dates you ran them
- Pre-specified predictions and analysis plans
- Every time you evaluated on the test set, and why
- Discrepancies found when reproducing others' work
- Ideas deferred, with enough detail to resume

Three payoffs: your methods section is half-written; your response letters can cite what you did and when; and if your integrity is ever questioned, a contemporaneous record is the strongest evidence available. Plain text under version control, Obsidian, Notion, or a paper notebook all work — the tool matters far less than the habit.

3.6 Verification checklist for Part I

- [ ] I can state my contribution type and the evidence it requires (§1.8.1).
- [ ] I can state a falsifiable mechanism (§1.7, Exercise 1.2).
- [ ] I know my research type and its principal validity threat (Chapter 2).
- [ ] I know which lifecycle stage I am actually at (Table 3.1).
- [ ] I have written down what I predict, before seeing test results (§3.4).
- [ ] I have started a dated research journal (§3.5).
- [ ] I have allocated at least four weeks to framing before beginning experiments.

Exercises

Exercise 3.1 Locate yourself in Table 3.1. Write down the stage number, its required output, and whether you actually hold that output. If not, you have found your immediate task.

Exercise 3.2 Draw your own version of Figure 3.1 for your project, and mark every backward arrow you have already traversed. Most researchers are surprised by how many there are — and reassured.

Exercise 3.3 Write a pre-specification entry: your prediction, the measurement, the statistical test, the significance level, and what result would disconfirm you. Date it. Do not modify it later; add new entries instead.

Exercise 3.4 Estimate your compute requirement as *configurations* × *seeds* × *datasets*, then multiply by three. Compare with the resources you actually have. If the result is infeasible, return to Chapter 4 now rather than in six months.

PART II — SELECTING A RESEARCH PROBLEM

> Part II covers the four decisions that determine whether a research project is completable and worth completing: the area you work in (Chapter 4), the problem you address within it (Chapter 5), the questions that make that problem answerable (Chapter 6), and the objectives that commit you to specific, checkable work (Chapter 7).
>
> These decisions are cheap to make and expensive to unmake. A poorly chosen area wastes years; a poorly specified problem produces experiments that answer nothing; unmeasurable objectives produce a thesis that cannot be examined. The material here is deliberately procedural, because “choose a good topic” is advice that helps nobody.

Chapter 4 — Selecting a Research Area

4.1 The problem with "follow your interest"

The standard advice is to work on what interests you. This is necessary and radically insufficient. Interest sustains you through year three; it does not tell you whether the work is possible, supervisable, or publishable. A research area must satisfy five constraints simultaneously, and the failure of any one of them is sufficient to end a project.

Table 4.1 — Five constraints on the choice of a research area

Constraint	Question to answer honestly	Failure mode if ignored
Personal endurance	Can I read this literature for three to five years without resentment?	Abandonment in year two, after sunk cost
Supervisory capability	Can my supervisor critique my technical work here, not merely approve it?	Unsupervised drift; weak papers that pass internally and fail externally
Resource availability	Do I have the data, compute, subjects, licences, and ethics route?	Stalled experiments; a year lost to access negotiations
Field momentum	Is publication volume in this area rising, flat, or collapsing?	Working in a dead area, or arriving late at a saturated peak
Career relevance	Do job advertisements and funding calls name this area?	Employability mismatch at the end

The second constraint is the one least often said aloud. If your supervisor cannot critique your work — because the area is outside their expertise, or because they supervise thirty students — you are not badly supervised so much as *unsupervised*, and you must construct a substitute critique network deliberately: a reading group, participation in peer review, correspondence with the authors of papers you read, and preprint feedback. This is achievable, but only if you recognise the need early.

4.2 Assessing field momentum

Purpose. To distinguish an emerging area (good entry point), a saturated area (high competition, fast obsolescence), and a declining area (easy novelty, few readers).

Procedure.

1. In Scopus or Web of Science, search your candidate area's core terms restricted to title, abstract, and keywords.
2. Use the platform's analysis view to plot **documents by year**. **[VERIFY]** The exact name of this feature changes; in Scopus it has been presented as "Analyze search results" and in Web of Science as "Analyze Results". Confirm on the platform.
3. Read the shape of the curve.

Curve shape	Interpretation	Strategic implication
Rising steadily	Healthy, growing area	Good entry point; readers exist; questions remain open
Rising near-vertically	Hype phase	High visibility, brutal competition, results obsolete in 18 months
Flat and high	Mature, saturated	Novelty is hard; incremental gains are contested
Flat and low	Niche	Novelty is easy; few citers; harder to place in strong venues
Declining	Superseded	Establish *why* it declined before entering

4. In the same analysis view, inspect **top source titles** (these are your candidate target journals, Chapter 51), **top authors** (these are your likely reviewers), and **top affiliations** (these are your competitors and

potential collaborators).

That fifteen-minute exercise yields your venue list, your reviewer pool, and your competitive landscape simultaneously. It is among the highest-return activities in this handbook and is skipped by most beginners.

4.3 Narrowing: the funnel

Figure 4.1 — The narrowing funnel: from research area to research objective

```
LEVEL 0  DISCIPLINE          Computer Science
   |                          (hundreds of thousands of papers/year — meaningless as a topic)
   ▼
LEVEL 1  BROAD AREA          Machine learning for medical image analysis
   |                          (~10,000+ papers/year — still unreadable)
   ▼
LEVEL 2  SUB-AREA            Domain generalisation in chest radiograph classification
   |                          (a readable literature: roughly 100–300 papers)
   ▼
LEVEL 3  SPECIFIC PROBLEM    Published CXR classifiers are evaluated with random splits
   |                          that mix institutions across train and test, so reported
   |                          performance may not describe cross-hospital behaviour
   ▼
LEVEL 4  RESEARCH QUESTION   RQ1: How large is the performance drop under
   |                          institution-disjoint evaluation?
   |                         RQ2: Can it be mitigated without institutional metadata?
   ▼
LEVEL 5  OBJECTIVE           To re-evaluate five published architectures on three public
                             datasets under both random and institution-disjoint protocols,
                             over ten seeds, reporting macro AUC, worst-institution AUC,
                             and calibration error with 95% confidence intervals.
```

[HYPOTHETICAL as a worked example, though grounded in the real finding of Zech et al. (2018) that CXR models can exploit institution-specific confounders.]

4.3.1 The one-breath test

You have narrowed sufficiently when you can say this sentence without hesitation:

> *"I compare **[methods]** on **[datasets]** under **[condition]**, measured by **[metrics]**, to find out whether **[question]**."*

If you cannot fill all five slots with specifics, you are at least one level too broad. This test is more useful than any abstract discussion of scope because it fails loudly.

4.3.2 Narrowing operators

Apply these until the one-breath test passes. Each operator reduces scope along a different axis, and using two or three in combination is normal.

Operator	Question	Example narrowing
Population	Which subjects, domains, or corpora?	All images → paediatric chest radiographs
Task	Which exact output?	"Understanding" → multi-label finding classification
Condition	Under what circumstance?	→ under low-light and occlusion
Constraint	Subject to what limit?	→ within 50 ms on an edge device
Comparison	Against what?	→ versus four representative domain-generalisation methods
Outcome	Measured how?	→ by worst-group AUC and calibration error

4.3.3 Over-narrowing

The opposite error is rarer but real: a topic so specific that no one outside your laboratory can use the result. The diagnostic is that you cannot name anyone who would cite it.

The remedy is *not* to broaden the study. It is to raise the level of the **claim** while keeping the study specific. A study on one proprietary dataset with 200 images is weak. The same study, framed as “we characterise how detection performance degrades as annotation density falls below N examples per class, using X as a case”, makes a transferable claim while performing the same experiments. The study is narrow; the knowledge is not.

4.4 Four routes into a topic

Different researchers arrive from different directions, and each route has a characteristic weakness worth knowing in advance.

Technology-driven. You are interested in a technique — diffusion models, graph neural networks, retrieval-augmented generation — and seek a place to apply it. *Characteristic weakness:* produces “apply X to Y” work with no reason why Y should be interesting for X. *Remedy:* identify a property of Y that violates an assumption of X, and measure what happens. That converts application into science (§5.5).

Problem-driven. You know a real difficulty — clinicians distrust model output; farmers cannot afford connectivity — and seek methods. *Characteristic weakness:* the problem may not be tractable, or may require data you cannot obtain. *Remedy:* run the feasibility audit (§4.6) before committing.

Dataset-driven. A dataset exists or you can build one. *Characteristic weakness:* dataset availability is not a research question; “we applied five models to this new dataset” is a technical report. *Remedy:* the dataset itself can be the contribution, if constructed and documented to publishable standards (§1.8.1, resource contribution), or ask what the data allows you to measure that was previously unmeasurable.

Literature-driven. You read carefully and find a contradiction, an admitted limitation, or an untested assumption. *Characteristic weakness:* slow to start; requires substantial reading before the topic appears. *Advantage:* this route produces the most defensible gaps, because the gap arrives already supported by citations. Part VI is built on it.

4.5 Finding emerging topics and unresolved problems

Purpose. To locate questions that are open, recognised as open, and not yet crowded.

4.5.1 Sources of emerging topics

Source	What to look for	Cadence	Access
Preprint servers (arXiv and equivalents)	New listings in your categories; set keyword alerts	Weekly	Free
Semantic Scholar	Alerts, citation velocity, influential citations	Weekly	Free
Citation-graph tools (Connected Papers, Litmaps, ResearchRabbit)	Prominent nodes you have not read	Per topic	Free tiers
Scopus / Web of Science analysis views	Trend, top sources, top authors	Per topic	Subscription
Journal special-issue calls	Editors publicly declaring which gaps they want filled	Monthly	Free
Workshop titles at major conferences	The field’s twelve-month agenda	Yearly	Free
Benchmark leaderboards	Metrics that have *stopped improving*	Per topic	Free
Funding-agency calls	What is considered strategically important	Per cycle	Free

Two of these deserve emphasis because they are consistently underused.

Special-issue calls for papers are the most explicit statement of demand available anywhere. An editor writing a call is saying, in public, "we want papers on this and will handle them." That is a gap with a receptive gatekeeper and a deadline.

Flattening leaderboard curves are a research opportunity in either of two ways: progress is genuinely saturating, which is interesting, or the benchmark has ceased to measure what matters, which is more interesting.

4.5.2 Where unresolved problems are stated explicitly

Researchers routinely publish lists of open problems, and beginners routinely do not read them.

1. **Limitations sections** of recent papers.
2. **Future work** in conclusions.
3. **Open-challenge sections** of survey papers.
4. **Public peer review**, where available. Some venues publish reviews and author responses. **[VERIFY]** Availability varies by venue and year; OpenReview hosts public reviews for several major machine-learning conferences. Reading expert criticism of accepted *and rejected* papers teaches you the field's actual standards of evidence in the field's own words — and each stated weakness is a candidate gap that already carries an expert endorsement.
5. **Rebuttals**, where authors concede what they could not do.
6. **Reproducibility reports** and replication studies.
7. **Registered reports and pre-registrations**, which state predictions before results.

4.6 The feasibility audit

Purpose. To discover that a project is impossible *now*, in an afternoon, rather than in eight months.

Procedure. Complete every row with a specific answer. "We will find a dataset" is not an answer.

Dimension	What to verify concretely	Red flag
Data exists	Named dataset, size, label type, access path	"We will collect our own" with no protocol, budget, or timeline
Licence permits use	Terms for research use, redistribution, derivative works	A data-use agreement requiring institutional signature you have not begun
Ethics route	Whether committee approval is needed; who applies; typical duration	Human or patient data with no identified approval route
Compute	GPU-hours for *your model* × *baselines* × *ablations* × *seeds* × *datasets*	An estimate that assumes one run per configuration
Baselines runnable	Official code located; does it execute on your machine?	No public implementation and no reimplementation plan
Evaluation defined	Metrics, splits, statistical test chosen in advance	"We will look at the accuracy"
Skills	Methods you must learn; who will teach you	A method central to the design that nobody available understands
Time	Backwards from a real deadline, with 40% slack	A plan with no slack for failure

4.6.1 Compute arithmetic, done honestly

Beginners estimate compute as *one training run*. The real figure is a product:

```
configurations   =  1 proposed method  +  4 baselines          =   5
× seeds          =  × 10                                      =  50
× datasets       =  × 3                                       = 150 runs
× hours per run  =  × 2 GPU-hours                             = 300 GPU-hours

plus hyperparameter search   (often 2–5× the above)
plus the ablation study      (typically 5–15 further configurations)
plus re-runs after a bug     (assume at least one full re-run)
plus experiments reviewers demand in revision   (budget 20%)

REALISTIC TOTAL:  roughly 3× your first estimate
```

Doing this arithmetic before committing is the single most effective protection against a stalled project. If the number exceeds your resources, you have three honest options: reduce the scope (fewer datasets, smaller models), change the question to one your resources can answer (evaluation and reproducibility questions are often far cheaper — §17.10), or secure more resources before starting.

4.6.2 A note on discovering infeasibility

If this audit shows your project is impossible, that is a *success*, and it should be treated as one. Discovering infeasibility in week one costs an afternoon; discovering it in month eight costs a year. Researchers who feel embarrassed by this outcome tend to press on, and pressing on is how projects consume a doctorate without producing a paper.

4.7 Common mistakes

Mistake	Correction
Choosing a topic from titles rather than from reading	Read fifteen papers properly first (Part IV)
Confusing a domain (“AI in healthcare”) with a problem	Apply the funnel to Level 3 and the one-breath test
Ignoring licences and ethics until submission	Audit them in week one (§4.6)
Ignoring baseline availability until the final month	Clone and run one baseline early (§4.6)
Entering a hype peak with no durable core	Balance one visible topic with one durable contribution
Assuming your supervisor’s approval means the topic is sound	Approval is not critique; seek external criticism
Estimating compute as one run per configuration	Use the arithmetic in §4.6.1

4.8 Verification checklist

- [ ] My area passes all five constraints in Table 4.1, including supervisory capability.
- [ ] I have plotted documents-by-year and can describe the curve shape.
- [ ] I have listed the top five source titles and top five authors in my area.
- [ ] My topic passes the one-breath test with all five slots filled.
- [ ] I can name at least one real dataset, with its licence and access route.
- [ ] I have located at least one baseline implementation and confirmed it runs.
- [ ] I have completed the compute arithmetic and compared it with my actual resources.
- [ ] I have identified my single greatest risk and written one sentence on mitigation.

Exercises

Exercise 4.1 — The funnel. Write your topic at all six levels of Figure 4.1. Most researchers discover they are working at Level 1 or 2 and believe they are at Level 3.

Exercise 4.2 — Momentum. Produce the documents-by-year curve for your area. Record the shape, the top five journals, and the top five authors.

Exercise 4.3 — The feasibility audit. Complete every row of §4.6 with specifics: dataset names with licences, baseline repository URLs you have actually opened, a GPU-hour figure from §4.6.1.

Exercise 4.4 — Limitations harvest. Take ten recent papers. Copy every sentence from their limitations and future-work sections into a spreadsheet, verbatim, with paper and page. You will use this in Chapter 18; it is also the fastest route to a topic.

Chapter 5 — Identifying a Research Problem

5.1 Definition

Definition. A research problem is a specific, identified absence or inadequacy in current knowledge, whose resolution would change what the field knows or can do, and which is answerable by evidence you can obtain.

Note what this excludes. A research problem is not a *difficulty* (something being hard), not a *task* (something needing to be built), and not a *topic* (something being interesting). It is a specified gap between what is known and what needs to be known.

5.2 Research problem versus general problem

A general problem is a state of the world that is unsatisfactory. A research problem is a state of *knowledge* that is incomplete. The two are related but the conversion is not automatic, and doing it explicitly is the core skill of this chapter.

Table 5.1 — General problems transformed into research problems

General problem	Why it is not yet a research problem	Research problem derived from it
Farmers lose crops to disease	A state of the world; no knowledge claim	*[HYPOTHETICAL]* Leaf-disease classifiers are trained on curated images with uniform backgrounds; their degradation under field illumination and occlusion is unquantified, and it is unknown whether few-shot adaptation recovers it within an edge-device inference budget
Our university needs plagiarism detection	A local procurement need	*[HYPOTHETICAL]* Sentence-embedding similarity detectors are validated on monolingual English corpora; their behaviour on code-mixed text is unmeasured, and tokeniser fragmentation is a plausible mechanism of failure
Hospitals want automated triage	An aspiration	Model performance reported within one institution may not transfer across institutions (Zech et al., 2018); the magnitude of that gap under leakage-free protocols, and whether it can be mitigated without institutional metadata, is not established
Large models are expensive	An observation	*[HYPOTHETICAL]* The accuracy–latency frontier for this task across four model scales on commodity edge hardware has not been characterised, so practitioners cannot select a model for a given latency budget
Students find our course difficult	An observation	*[HYPOTHETICAL]* It is unknown which specific misconception accounts for failure on this topic, and whether a targeted intervention addressing it improves outcomes relative to additional practice

Observe the grammatical signature of the right-hand column. Each contains a *gap marker*: **unquantified**, **unmeasured**, **not established**, **has not been characterised**, **it is unknown which**. If your problem statement contains no such marker, it is probably not yet a research problem.

5.3 The four-part test

Apply all four. A statement that fails any one of them is not yet usable.

1. **Is the answer currently unknown in the accessible literature?** Not unknown to you — unknown to the field. Establishing this requires the documented search of Part III. If five papers already answer it, you have found a reading gap in yourself, which is useful but different.
2. **Can it be measured or proven?** Name the measurement or the proof strategy now. If you cannot, the problem is not yet operational.
3. **Would a negative result still be publishable?** This is the sharpest and least-known diagnostic. If the only reportable outcome is “my method wins”, you have an advocacy exercise. If both outcomes teach

the field something, you have science. Applied to the hospital example: if degradation turns out to be *small*, that is genuinely important news, because it would mean that the published literature can be trusted more than feared. Either way the study is worth doing.

4. **Does it matter to anyone outside your institution?** Name the population who would act differently if they knew the answer: practitioners, other researchers, tool builders, policy makers.

5.4 Anatomy of a problem statement

A problem statement is a short piece of formal writing — typically three to six sentences — that appears in your proposal, your introduction, and your funding applications. It has a fixed structure.

Figure 5.1 — Anatomy of a problem statement

```
┌─ CONTEXT ──────────────────────────────────────────────────────┐
│ What domain, why anyone should care. One or two sentences, concrete, │
│ ideally with a number. No history of the discipline.             │
└────────────────────────────────────────────────────────────────┘

┌─ CURRENT STATE ────────────────────────────────────────────────┐
│ What is done now and what it achieves. Cited. Fair to prior work. │
└────────────────────────────────────────────────────────────────┘

┌─ THE INADEQUACY ───────────────────────────────────────────────┐
│ What specifically is missing, wrong, unmeasured, or assumed. This  │
│ is the pivot of the whole statement. Cite evidence for the claim.  │
└────────────────────────────────────────────────────────────────┘

┌─ CONSEQUENCE ──────────────────────────────────────────────────┐
│ What cannot currently be done, decided, or trusted as a result.    │
│ This is where significance lives — not in adjectives.              │
└────────────────────────────────────────────────────────────────┘

┌─ SCOPE AND CONSTRAINTS ────────────────────────────────────────┐
│ What you will and will not address. Stating limits early is a mark │
│ of competence, not weakness.                                     │
└────────────────────────────────────────────────────────────────┘
```

5.4.1 Worked example

[HYPOTHETICAL] □ Weak problem statement. *"Deep learning has revolutionised medical imaging. Many researchers have proposed models for chest X-ray classification. However, there are still some challenges and limitations. There is a need for a more accurate and efficient model. This research proposes a novel framework to address these challenges."*

Diagnosis: the context is a cliché; "some challenges and limitations" identifies nothing; "a need for a more accurate model" is true of every task forever; no consequence is stated; there is no scope. Nothing here could be false.

[HYPOTHETICAL, grounded in Zech et al. (2018)] □ Improved problem statement.

> *"Chest radiography is among the most frequently performed diagnostic imaging examinations, and automated triage of abnormal studies has been proposed as a response to radiologist workload [context]. Convolutional and transformer-based classifiers report high discriminative performance on public benchmarks such as CheXpert and MIMIC-CXR [current state]. However, published evidence indicates that such models can learn institution-specific confounders rather than pathology (Zech et al., 2018), and in the studies we surveyed, evaluation is predominantly conducted on random splits in which radiographs from the same institution appear in both training and test partitions [the inadequacy]. Consequently, reported performance figures cannot be interpreted as estimates of cross-institutional performance, and a hospital considering deployment has no basis on which to estimate the accuracy it would actually obtain [consequence]. This study addresses the magnitude of that discrepancy and its mitigation in the setting where institutional metadata is unavailable at training time; it does not address prospective clinical validation or the effect on clinician decision-making [scope]."*

Each clause does one job, the inadequacy is supported by a real citation, the consequence names an affected decision-maker, and the scope pre-empts two obvious reviewer objections.

5.5 The mechanism requirement

This section contains the most transferable single technique in Part II.

A very large fraction of weak research problems have the form **"X has not been applied to Y."** This is an absence of *activity*, not an absence of *knowledge*, and it is not a research problem — because the fact that nobody has done something is not evidence that doing it would teach anyone anything.

The conversion is always the same. Ask: **which assumption of X is violated by Y, and what should happen as a result?**

Weak: absence of activity	Strong: mechanism plus measurement
"Transformers have not been applied to language L."	"The subword tokenisers of pretrained transformers are fitted on corpora in which L is scarce; we predict that morphologically rich L suffers excess fragmentation, that fragmentation correlates with error, and we measure both across four language families."
"No one has used method M on our sensor data."	"M assumes stationarity; our sensor exhibits documented drift over deployment months. We measure the rate at which M degrades with drift magnitude and test one recalibration strategy."
"Nobody has combined A and B."	"A fails on long inputs, B fails on noisy inputs, and the two failure modes are hypothesised to be independent; if so, their combination should be robust to both, which is testable by crossing input length with noise level."

The mechanism converts a combination into a *prediction*, and a prediction is something an experiment can refute. This is the difference between engineering novelty and scientific contribution.

5.6 Common mistakes

Mistake	Correction
A problem statement with no gap marker	Insert *unquantified*, *unmeasured*, *not established*, or *it is unknown whether* — and then justify it
"There is a need for a better method"	Better by what measure, relative to what threshold, needed by whom?
Stating the inadequacy without evidence	Cite the papers whose limitations or protocols support your claim
Restating an engineering requirement	Apply the four-part test (§5.3)
"Apply X to Y" with no mechanism	Apply §5.5
Omitting scope, hoping nobody notices	State exclusions explicitly; reviewers respect it
Choosing the problem to fit a method already chosen	Reverse the order; reviewers detect retrofitted problems easily

Exercises

Exercise 5.1 Write your problem statement using the five blocks of Figure 5.1, one or two sentences per block. Mark which block is weakest.

Exercise 5.2 Apply the four-part test. For test 3, write one sentence describing what a negative result would look like and why it would be worth publishing.

Exercise 5.3 If your problem has the form "apply X to Y", rewrite it using §5.5: name the violated assumption, the predicted consequence, and the measurement.

Exercise 5.4 Give your problem statement to someone outside your immediate group. Ask them to tell you, in their own words, what is currently unknown. If they cannot, the statement has failed regardless of how clear it seems to you.

Chapter 6 — Research Questions

6.1 Definition and purpose

Definition. A research question is an interrogative statement that specifies exactly what the study will determine, in terms concrete enough that a reader can tell what evidence would answer it.

Purpose. The research question is the hinge of the whole project. It converts a problem (a statement about what is missing) into something that can be *answered*, and thereby determines the design, the data, the metrics, and the structure of the eventual paper. In a well-constructed paper, the results section has one subsection per research question, in order.

6.2 Properties of a good research question

Property	Test
Specific	Every noun is concrete; no unspecified “various” or “different”
Answerable	You can describe the experiment or argument that would settle it
Bounded	It contains one relationship, not three
Non-trivial	The answer is not already obvious or already published
Informative either way	Both outcomes advance knowledge
Appropriately scoped	Answerable with your data, compute, and time
Honest about type	Descriptive, comparative, or causal — and phrased accordingly

6.3 Question types and the designs they imply

Table 6.1 — Question types and the study designs they imply

Type	Template	Example *[HYPOTHETICAL]*	Design implied
Descriptive	How much / how many / what is the distribution of...?	How large is the AUC reduction from random to institution-disjoint evaluation?	Measurement across strata; confidence intervals
Comparative	Does A differ from B with respect to M?	Does site-adversarial training outperform empirical risk minimisation on worst-institution AUC?	Controlled comparison; equal budgets; paired test
Relational	How does M vary with X?	How does detection accuracy vary with illumination level?	Stratified measurement; regression or trend analysis
Causal / mechanistic	Why does A produce B? Which component is responsible?	Which component of the proposed method accounts for the improvement?	Ablation; intervention; one factor at a time
Conditional	Under what conditions does A hold?	Under what degree of distribution shift does the gain persist?	Systematic variation of the condition until the effect vanishes
Performance / feasibility	Can the target be met within constraint C?	Can worst-institution AUC exceed 0.85 within a 50 ms inference budget?	Constrained evaluation with cost measurement
Existence	Does there exist an X such that...?	Is there a label-free procedure attaining most of the label-supervised benefit?	Constructive demonstration plus upper-bound comparison
Synthesis	What does the evidence collectively show?	What evaluation protocols are used across the literature, and how do they affect reported performance?	Systematic review with structured extraction

Two practical notes. First, **ordering matters**: descriptive questions should precede comparative ones, which should precede mechanistic ones, because each supplies the context the next requires. Second,

causal phrasing carries obligations. If your question asks *why*, your design must isolate the cause; observational data plus a *why* question is a validity failure that reviewers reliably catch.

6.4 Bad and good research questions

Weak question	Diagnosis	Improved
Can deep learning be used for chest X-ray classification?	Already answered; answerable by yes	How much does classification performance degrade when random splits are replaced by institution-disjoint splits, across five published architectures?
Is our proposed model better?	Better than what, at what, under what conditions?	Does the proposed regulariser improve worst-institution AUC relative to the strongest of four baselines under an identical tuning budget?
What are the challenges in domain generalisation?	A literature-review prompt, not a question	Which of three documented failure modes accounts for the largest share of cross-institutional error in this task?
How can we improve accuracy?	Unbounded	Does inferring pseudo-domains by clustering embedding statistics recover a majority of the benefit obtained with true domain labels?
Does the model work well?	“Well” is undefined	Does the model attain sensitivity ≥ 0.90 at specificity 0.95 on each of three external institutions?
What is the effect of various hyperparameters on different datasets?	Two unspecified plurals; unbounded	How sensitive is worst-institution AUC to the invariance weight λ over the range 0.1–10 on three datasets?

6.5 Frameworks for constructing questions

Several disciplines have formalised question construction. All are variations on “name your components explicitly”, and any of them is better than none.

PICO (health sciences): **P**opulation, **I**ntervention, **C**omparison, **O**utcome. Widely used to frame systematic-review questions.

PICOC adds **C**ontext, which is useful in engineering and software research where setting dominates.

A computational adaptation. For machine-learning and data-science work, the following six slots map more directly onto what a paper must report:

Slot	Content	Example *[HYPOTHETICAL]*
D — Data	Population, datasets, split unit	CheXpert, MIMIC-CXR, ChestX-ray14; split by institution
M — Method	The intervention	Clustering-based invariance regulariser
B — Baseline	The comparison	ERM, Mixup, CORAL, IRM, at equal tuning budget
C — Condition	The circumstance varied	Random versus institution-disjoint evaluation
O — Outcome	Metrics, with the decision they serve	Macro AUC, worst-institution AUC, expected calibration error
T — Threshold	What would count as a meaningful effect	≥ 0.03 AUC, judged against seed variance

Filling all six slots produces a research question, an experimental design, and the skeleton of your results tables simultaneously. The **T** slot is the one most often omitted and the most valuable: deciding in advance what size of effect would matter prevents the common outcome of celebrating a statistically detectable but practically irrelevant difference.

6.6 From problem to questions: worked conversion

Take the inadequacy identified in §5.4.1. Each clause of the gap converts mechanically into a question.

Gap clause	Question
"…reported figures cannot be interpreted as cross-institutional performance"	**RQ1** (descriptive) How large is the difference between random-split and institution-disjoint performance across published architectures?
"…mitigation in the setting where institutional metadata is unavailable"	**RQ2** (comparative) Can a method that does not use institutional labels match methods that do?
"…a hospital has no basis to estimate accuracy it would obtain"	**RQ3** (conditional) Is worst-institution performance predictable from properties observable before deployment?
implied by any improvement claim	**RQ4** (mechanistic) Which component of the proposed method accounts for the observed improvement?

Notice that four questions of four different types emerged from one problem statement, and that together they specify a complete study. This is the normal relationship: one problem, two to four questions, each answerable by a distinct experiment.

6.7 Common mistakes

Mistake	Correction
Questions answerable by yes or no alone	Prefer *how much*, *under what conditions*, *which component*
Three relationships in one question	Split into separate numbered questions
Questions no experiment could settle	Specify the measurement, or discard the question
Causal phrasing on observational evidence	Change the phrasing or add an intervention
Questions written after the experiments	Reviewers detect this; pre-specify instead (§3.4)
No threshold for a meaningful effect	Fill the **T** slot before collecting data
Ten questions	Three or four; a paper cannot defend more

Exercises

Exercise 6.1 Convert your problem statement into two to four research questions using the mechanical procedure of §6.6, and label each with its type from Table 6.1.

Exercise 6.2 Fill the six DMBCOT slots (§6.5) for your principal question. Any slot you cannot fill is an unmade design decision.

Exercise 6.3 For each question, write the single sentence that would appear in your results section if the answer were *no*. If any of those sentences would be unpublishable, revisit the question.

Chapter 7 — Research Objectives

7.1 Five statements and how they differ

Beginners routinely conflate these five. Examiners and reviewers do not, and the distinctions are load-bearing.

Figure 7.1 — Aim, objectives, questions, hypotheses, and contributions

```
AIM             ONE sentence. The overall purpose. Broad but bounded.
 │              "To establish how much X degrades and whether Y mitigates it."
 │
 ├─ QUESTIONS   2–4 interrogatives. What will be determined.
 │              "RQ1: How large is the degradation?"
 │
 ├─ OBJECTIVES  3–5 commitments to work. "To + measurable verb + object + condition."
 │              "O1: To re-evaluate five architectures under both protocols…"
 │
 ├─ HYPOTHESES  Falsifiable predictions with a named test.
 │              "H1: institution-disjoint AUC is lower (paired Wilcoxon, α=0.05)."
 │
 └─ CONTRIBUTIONS  2–4 noun phrases. What the field gains afterwards.
                "C1: the first leakage-free multi-institution quantification of…"

TRACEABILITY REQUIREMENT
┌───────────┬───────────┬───────────────────┬────────────────┐
│ Objective │ Method    │ Results section   │ Contribution   │
├───────────┼───────────┼───────────────────┼────────────────┤
│ O1        │ §III-A    │ §V-A              │ C1             │
│ O2        │ §III-B    │ §V-B              │ C2             │
│ O3        │ §III-C    │ §V-C              │ C3             │
└───────────┴───────────┴───────────────────┴────────────────┘
Every row must be complete. An objective with no results section reads as an
abandoned promise; a results section with no objective reads as an afterthought.
```

Statement	Count	Grammatical form	Function
Aim	exactly 1	“To + purpose”	Orientation; the destination
Question	2–4	Interrogative	What will be determined
Objective	3–5	“To + measurable verb + object + condition”	Commitment to specific work
Hypothesis	1–4	Declarative prediction, with test and α	What you expect, refutably
Contribution	2–4	Calibrated noun phrase	What the field retains

7.2 Writing measurable objectives

Procedure.

1. Begin with **“To”** followed by a verb whose completion is *observable*.
2. Name the **object** — what exactly is acted upon.
3. Add the **condition or scope** — on what data, under what protocol, with what comparison.
4. Where possible, name the **outcome measure**.
5. Verify that an examiner could mark the objective *done* or *not done* without argument.

Verbs that work: quantify, measure, compare, characterise, derive, prove, construct, annotate, validate, determine, isolate, evaluate, establish.

Verbs and phrases that fail: study, analyse, investigate, explore, look into, understand, work on, develop a framework for, implement. These name activities with no completion criterion. (*Investigate* and *explore* are acceptable in genuinely exploratory studies — §2.5 — but then the objective must specify what will be produced.)

Table 7.1 — Weak and strong research objectives

Weak objective	Why it fails	Strong objective
To study deep learning for medical imaging	No boundary, no completion criterion	To quantify the change in macro AUC and worst-institution AUC of five published architectures when random splits are replaced by institution-disjoint splits, across three public datasets and ten seeds
To implement a CNN model	An activity completable in a day; no knowledge	To determine whether label-free pseudo-domain inference recovers at least half of the worst-institution improvement achieved with true domain labels
To improve accuracy	Unquantified; no comparison point	To reduce worst-institution AUC degradation by at least 0.03 relative to empirical risk minimisation at equal parameter count and inference latency
To analyse various algorithms	“Various” is unspecified	To compare ERM, Mixup, CORAL, and IRM under an identical 50-trial tuning budget on identical splits
To develop a novel framework	Unfalsifiable; “framework” hides the claim	To construct and evaluate an adaptation module requiring no institutional metadata, assessed on worst-institution AUC, calibration error, and inference latency
To achieve 99% accuracy	Fixates on a number that may be meaningless or unattainable	To characterise the accuracy–latency trade-off across four model scales on a specified edge device
To explore the potential of transformers	No output specified	To determine which of three transformer variants attains the best worst-group performance under a fixed 12 GB memory budget

7.3 Hypotheses

Not every discipline states hypotheses explicitly — much computational work does not — but **writing them anyway is a diagnostic**, because a hypothesis you cannot phrase is usually a hypothesis your experiment cannot test.

Form. State a null and an alternative, name the test, and fix the significance level *before* looking at results.

[HYPOTHETICAL] Example.

	Statement
H0	Macro AUC does not differ between random-split and institution-disjoint evaluation.
H1	Macro AUC is lower under institution-disjoint evaluation.
Test	Paired Wilcoxon signed-rank across architecture × dataset pairs; $\alpha = 0.05$; Holm correction across five architectures.
Effect size	Reported as the mean difference with a 95% bootstrap confidence interval.
H2	Label-free pseudo-domain inference improves worst-institution AUC by ≥ 0.03 over ERM.

Three rules that prevent the most common statistical failures:

1. **Pre-specify the test.** Choosing a test after seeing which one gives significance is p-hacking (§3.4, §49).
2. **Report an effect size, not only a p-value.** With enough seeds, a difference of 0.001 becomes statistically detectable and remains practically worthless.
3. **Correct for multiple comparisons.** Testing five architectures × three datasets is fifteen comparisons; uncorrected, roughly one spurious “significant” result is expected by chance at $\alpha = 0.05$.

7.4 Contributions

Contributions are covered fully in Chapter 21. Two points belong here because they constrain how objectives are written.

Contributions must map one-to-one onto objectives. If you have four objectives and two contributions, either two objectives are not producing knowledge, or you are under-claiming. If you have two objectives and five contributions, you are inflating.

Contributions must be calibrated. Compare:

> **[HYPOTHETICAL]** □ "We propose a novel state-of-the-art framework that solves cross-domain generalisation."
>
> **[HYPOTHETICAL]** □ "We provide the first leakage-free multi-institution quantification of degradation for five widely used architectures, with confidence intervals and paired tests; and a label-free adaptation module that recovers approximately 82% of the benefit obtained by methods using privileged institutional labels, at equal parameter count."

The second is longer, more specific, bounded, and — critically — much harder for a reviewer to refute.

7.5 Common mistakes

Mistake	Correction
Objectives beginning "Study of..." or "Analysis of..."	Use an observable verb and a completion criterion
Objectives that are really methods	An objective states what will be *determined*; the method states *how*
Seven or eight objectives	Three to five; more cannot be defended in one paper
Objectives with no corresponding results section	Complete the traceability table (Figure 7.1)
Hypotheses written after seeing results	Pre-specify and date them
p-values without effect sizes	Report both, always
No multiple-comparison correction	Apply Holm or an equivalent when testing many pairs
Contributions that restate the method three times	Distinct, auditable contributions only

7.6 Verification checklist for Part II

- [ ] My topic passes the one-breath test (§4.3.1).
- [ ] I have completed the feasibility audit with specifics (§4.6).
- [ ] My problem statement has all five blocks of Figure 5.1.
- [ ] My problem contains an explicit gap marker and passes the four-part test (§5.3).
- [ ] If my problem was "apply X to Y", I have added a mechanism (§5.5).
- [ ] I have two to four research questions, each typed (Table 6.1).
- [ ] I have filled all six DMBCOT slots, including the effect-size threshold (§6.5).
- [ ] Every objective starts with "To" plus an observable verb and can be marked done or not done.
- [ ] My hypotheses name a test and an α, and are dated before results.
- [ ] The traceability table in Figure 7.1 is complete, with no empty cells.

Exercises

Exercise 7.1 Write one aim, three to five objectives, and two to four contributions for your project. Check each objective against Table 7.1.

Exercise 7.2 Complete the traceability table. Any empty cell is either missing work or an unnecessary claim.

Exercise 7.3 Write H0 and H1 for your principal comparison, naming the test, α, and correction. Date the entry in your research journal.

Exercise 7.4 — Examiner simulation. Give your objectives to a colleague and ask them to mark each *done* or *not done* as though the work were finished. Any objective they cannot mark must be rewritten.

PART III — LITERATURE SEARCH

Part III turns literature searching from browsing into an instrument. Chapter 8 explains what the literature review is for and what specifically goes wrong when it is weak. Chapter 9 surveys the major databases with their strengths, limitations, and appropriate uses. Chapter 10 provides the mechanics: concept blocks, Boolean logic, field codes, portable search strings, and citation chaining.

The standard to aim for is stated once and applies throughout: **a search you cannot describe precisely enough for someone else to reproduce your result count is not a search — it is browsing.** Browsing is a legitimate way to become interested in a topic. It is not a legitimate basis for a claim that something has not been done.

Chapter 8 — Why Literature Review Is Necessary

8.1 The eight functions of a literature review

A literature review is not a chapter you write to demonstrate diligence. It performs eight distinct functions, each of which has a direct consequence if omitted.

#	Function	What it delivers	Consequence of omission
1	**Maps existing knowledge**	An accurate picture of what is settled, contested, and untested	You cannot tell whether your idea is new
2	**Prevents duplication**	Confidence that the specific question is open	Discovering mid-project that your result was published in 2023
3	**Identifies the gap**	The evidenced absence your work addresses	"No contribution" — the most common substantive rejection
4	**Informs method selection**	Knowledge of what has been tried and what failed	Repeating a known dead end; reinventing a solved subproblem
5	**Identifies datasets and benchmarks**	The resources your field considers standard	Using a non-standard dataset, making your results incomparable
6	**Identifies evaluation conventions**	The metrics and protocols reviewers expect	Reporting metrics nobody can compare, or inappropriate ones
7	**Establishes your baselines**	The specific methods you must beat or match	"The authors omit comparison with [X]" — a standard revision demand
8	**Supplies your motivation**	Cited evidence that the problem matters	An introduction that asserts significance without support

Functions 5, 6, and 7 are worth emphasising because beginners treat them as separable from the literature review. They are not. Your dataset, your metrics, and your baselines are all *findings of the literature review*, and discovering them late is one of the most common causes of wasted experimental effort.

8.2 What happens when the review is weak

Figure 8.1 — Consequences of a weak literature review, by stage

```
STAGE WHERE THE WEAKNESS SURFACES         WHAT IT COSTS
─────────────────────────────────────     ────────────────────────────────────────
Topic selection                           Months on a solved problem
  ↓ undetected
Method design                             Reinventing a known technique badly;
  ↓ undetected                            missing the standard trick everyone uses
Dataset choice                            Results incomparable to all prior work
  ↓ undetected
Experimental design                       Wrong metric; missing the obvious baseline
  ↓ undetected
Writing                                   Cannot articulate novelty; introduction
  ↓ undetected                            has no gap paragraph
Submission                                DESK REJECT: "insufficient novelty"
  ↓ or survives to
Peer review                               "The contribution over [ref] is unclear"
  ↓                                       "Comparison with [ref] is required"
                                          → major revision or rejection
  ↓ or survives to
Post-publication                          Someone shows your result was known.
                                          The paper stops being citable.
  ↓ or, in the worst case
Viva / defence                             "Why did you not compare with X?"
                                          — the question that cannot be answered late
```

The pattern is that the cost of a weak review **compounds with time**. A gap discovered in week two costs a week. The same gap discovered by Reviewer 2 in month nine costs a rewrite and a resubmission; discovered by an examiner, it costs far more.

8.3 How reviewers detect a weak review

It is useful to know the specific signals, because they are easy to eliminate once you know them.

Signal	What the reviewer infers
All citations from the last three years	The author does not know the field's foundations
All citations from one research group	The author found one cluster and stopped
Standard method used without citing its origin	The author learned it from a blog post
A well-known competing method absent	Either ignorance or avoidance; both damaging
"Few researchers have studied this" with no evidence	The author did not search systematically
Related work is a list of one-sentence summaries	The author read abstracts, not papers
Citation count far below the venue's norm	Under-engagement with the literature
The dataset is unusual with no justification	The author did not learn the field's conventions

8.4 How much is enough?

This is a recommendation, and norms vary by field and venue. As orientation for computational disciplines:

Purpose	Papers read at some depth	Papers cited
Deciding on a topic	10–20	—
Defining a defensible gap	20–40	—
A conference paper	30–60	25–40
A journal paper	50–100	35–70
A systematic review	300–1500 screened; 30–100 included	60–150
A doctoral thesis	200–500	120–300

More useful than any number is the **saturation criterion** (§10.9): you have read enough when new searches return only papers you have already seen, and when you can predict what a new paper in the area will say before reading it.

8.5 Common mistakes

Mistake	Correction
Reading only papers that support your idea	Actively seek work that would contradict or pre-empt it
Citing papers found through other papers' descriptions	Read the primary source; descriptions are frequently inaccurate
Treating the review as a one-off task	Set alerts (§10.9); your review must still be current at submission, possibly a year later
Reviewing only your own subfield	Adjacent fields often solved your problem under a different name
Ignoring older foundational work	Cite the origin of every method you use
Skipping the review because "the area is new"	New areas have precursors; find them

Exercises

Exercise 8.1 For each of the eight functions in §8.1, write one sentence stating what your current review delivers. Any function you cannot answer is an open task.

Exercise 8.2 Audit your current reference list against the eight signals in §8.3. Fix every one that applies.

Exercise 8.3 Name the three most likely candidates for “the paper that already did this”. Find them and read them properly. If they do not exist, record the searches that failed to find them — that record is evidence for your gap.

Chapter 9 — Research Databases and Search Platforms

9.1 A mental model

Before the individual platforms, the important distinction:

- **Discovery engines** maximise *recall* — they find everything, including material of unknown quality. Google Scholar and Semantic Scholar are of this kind.
- **Curated indexes** maximise *precision and reproducibility* — a defined corpus, controlled metadata, powerful query syntax, and analytics. Scopus and Web of Science are of this kind, and they are also the basis of most indexing and quality claims (Chapter 51).
- **Publisher platforms** provide *full text* for one publisher's output: IEEE Xplore, ScienceDirect, SpringerLink, ACM Digital Library.
- **Open bibliographic sources** provide free, programmable metadata at scale: OpenAlex, Crossref, and similar.

A rigorous search uses at least one curated index, at least two publisher platforms relevant to your field, one discovery engine, and one preprint source — and logs each.

[VERIFY] — an important caveat for this entire chapter. Platform interfaces, feature names, operator support, and coverage change, sometimes annually. Every workflow below should be checked against the platform's own current help documentation on the day you search. Where this handbook and the platform disagree, the platform is correct. Coverage figures are deliberately omitted or given only as orders of magnitude for the same reason.

9.2 Platform-by-platform

Table 9.1 — Research databases compared

Platform	Indexes	Access	Principal strength	Principal limitation
Google Scholar	Journals, conferences, preprints, theses, books, patents, some grey literature	Free	Widest recall; full-text indexing; “Cited by”	No quality control; limited field syntax; no bulk export of result sets; result counts not stable or reproducible
Scopus	Curated peer-reviewed sources	Subscription	Best query syntax and analytics; author and affiliation profiles; CiteScore	Coverage gaps for some CS conferences and books
Web of Science	Core Collection (multiple indexes including SCIE, SSCI, AHCI, ESCI)	Subscription	Strong curation; the basis of Journal Citation Reports and the Impact Factor	Narrower than Scopus in some fields; multiple editions cause confusion
IEEE Xplore	IEEE and partner journals, conferences, standards	Subscription	Essential for electrical engineering and much of CS; standards documents	Publisher-limited
ACM Digital Library	ACM journals, conferences, SIG proceedings	Subscription	Definitive for core computer science venues	Publisher-limited
ScienceDirect	Elsevier full text	Subscription	Full text of Elsevier journals	One publisher
SpringerLink	Springer/Nature journals, book series, LNCS proceedings	Subscription	Strong for LNCS conference proceedings and books	Weaker advanced query syntax
PubMed / PMC	Biomedical and life sciences	Free	**MeSH** controlled vocabulary — a genuine advantage for precision	Biomedical scope only
Semantic Scholar	Very broad, AI-enhanced metadata	Free	Machine-generated summaries, citation context, influential-citation signals, open API	Metadata noise; automated summaries are lossy
OpenAlex / Crossref	Open bibliographic metadata	Free	Programmable; excellent without a subscription	Less curation than Scopus or WoS
DOAJ	Vetted open-access journals	Free	A legitimacy signal for OA venues (Chapter 52)	Indexes journals, not articles

9.2.1 Google Scholar

When to use it. Early exploration; finding a paper you already know of; forward citation chasing; locating accessible copies; grey literature and theses.

Workflow.

1. Search with quoted phrases for multiword concepts.
2. Use the left-hand panel to restrict by year; sort by relevance and then re-sort by date, because the two orderings surface different papers.
3. On a relevant paper, use **Cited by** for forward chaining, and — importantly — **search within citing articles** to filter that citing set by keyword.
4. Use **Related articles** for sideways discovery.
5. Save items to *My Library*; set up alerts for a query or for citations to a key paper.

Limitations to respect. Result counts are estimates and are not stable, so they must not be reported as if reproducible. Inclusion is automated, so predatory-venue material appears alongside rigorous work. There is no reliable document-type filter. Do not build a systematic review’s search protocol on Scholar alone.

9.2.2 Scopus

When to use it. As your reference implementation for building a search string; for trend and venue analytics; for verifying indexing claims.

Workflow.

1. Advanced search with field codes, most usefully `TITLE-ABS-KEY( )`.
2. Restrict with year and document-type limits.
3. Inspect the analysis view for documents by year, source titles, authors, affiliations, and countries.
4. Refine iteratively, watching the result count change with each added concept block.
5. Export to RIS or BibTeX **including abstracts and keywords**, so your literature matrix can be partly populated from the export.

9.2.3 Web of Science

When to use it. Curated searching; citation reports; and — the distinctive function — verifying which index a journal belongs to and consulting Journal Citation Reports for the Impact Factor (Chapter 51).

Workflow. Field tags including `TS=` (topic), `TI=`, `AB=`, `AK=`, `SO=`, with `PY=` for the publication-year range and `DT=` for document type. The Analyze Results and Citation Report views provide the equivalent analytics to Scopus.

9.2.4 IEEE Xplore

When to use it. Electrical engineering, computer engineering, much of applied CS; and for standards, which are frequently the authoritative specification of something you are studying and are indexed nowhere else.

Workflow. The command-search interface accepts fielded queries with Boolean and proximity operators; fields include document title, abstract, index terms, and all metadata. Index terms are particularly useful for harvesting controlled vocabulary for your synonym table (§10.2).

9.2.5 ACM Digital Library

When to use it. Core computer science — systems, programming languages, HCI, software engineering, databases, theory. For many CS subfields the definitive venues are ACM conferences, and this is where they live.

9.2.6 ScienceDirect and SpringerLink

Both are primarily full-text access platforms for their respective publishers. ScienceDirect supports fielded and proximity searching, though with limits on the number of Boolean operators per field. SpringerLink's advanced syntax is comparatively limited, so it is best used to retrieve known items and to browse book series and LNCS proceedings rather than as a primary search instrument.

9.2.7 PubMed

When relevant. Any work touching health, biology, or medicine — including applied machine learning in those domains.

The distinctive advantage is MeSH, a curated controlled vocabulary. Searching a MeSH term retrieves papers indexed under that concept regardless of the authors' chosen wording, which solves the synonym problem that dominates searching elsewhere. If your work is biomedical, learning to use MeSH is the highest-return hour you can spend on searching.

9.2.8 Semantic Scholar

When to use it. As the strongest free substitute for a subscription index; for citation-context information; for programmatic access via its API when you want to build your own screening pipeline.

Verification requirement. Machine-generated summaries are convenient for triage and unreliable as evidence. Never characterise a paper's findings in your own writing on the basis of an automated summary

(§13.6, §45.4).

9.3 Working without subscriptions

Many researchers lack Scopus or Web of Science access. A rigorous search is still achievable.

Need	Free substitute
Curated searching	PubMed (if biomedical); Semantic Scholar; OpenAlex
Trend analysis	OpenAlex API aggregated by year; Semantic Scholar
Citation chaining	Google Scholar “Cited by”; Semantic Scholar; OpenAlex
Full text	Preprint servers; author copies; institutional repositories; interlibrary loan; polite email to the corresponding author
Venue legitimacy	DOAJ; publisher site; ISSN Portal; COPE membership (Chapter 52)
Bulk metadata	Crossref and OpenAlex APIs

Two practical notes. Many national and institutional consortia provide access that researchers do not realise they have — ask your librarian before concluding you have none. And emailing an author for a copy of their own paper is normal, accepted practice with a high response rate.

9.4 Common mistakes

Mistake	Correction
Using one platform only	Use at least three plus a preprint source
Reporting Google Scholar hit counts as reproducible	Use a curated index for reportable counts
Assuming a platform’s absence means a paper does not exist	Coverage differs; cross-check
Not exporting abstracts and keywords	Re-export; you will need them for screening and for the matrix
Never using controlled vocabulary	Use MeSH, index terms, or author keywords to build synonyms
Trusting automated summaries as evidence	Read the paper before characterising it

Exercises

Exercise 9.1 Determine which platforms in Table 9.1 you can actually access. Write the list down; it defines your search strategy.

Exercise 9.2 Run the same simple query on three platforms and compare the first twenty results. The differences in coverage will be larger than you expect.

Exercise 9.3 On one platform, locate the controlled vocabulary for your topic — MeSH terms, index terms, or author keywords — and record five terms you had not thought of.

Chapter 10 — Search Strategies

10.1 The objective

A good search is **reproducible**, **documented**, and **calibrated** — recall high enough not to miss the paper that pre-empts you, precision high enough to be readable. The practical target for a focused review is a set of roughly **80–300 records** before screening. Substantially more means your query is too loose; substantially fewer means it is too tight or your synonyms are incomplete.

10.2 Step one: concept blocks and synonyms

Procedure. Decompose your question into three to five *concepts*. Within each concept, list every way the literature might express it. Concepts are joined by `AND`; synonyms within a concept by `OR`.

Worked example. *Do deep models for chest-radiograph classification generalise across hospitals, and do domain-generalisation methods help?*

Block	Preferred term	Synonyms, variants, spellings	Narrower terms and proper nouns
C1 Data	chest radiograph	“chest X-ray”, CXR, “thoracic radiograph”, “chest radiography”	CheXpert, MIMIC-CXR, ChestX-ray14, PadChest
C2 Task	classification	detection, diagnosis, screening, “multi-label classification”	pneumonia, pneumothorax, cardiomegaly
C3 Method	“deep learning”	CNN, “convolutional neural network”, transformer, “vision transformer”, ViT	DenseNet, ResNet, EfficientNet, Swin
C4 Phenomenon	“domain generalisation”	“domain generalization”, “domain shift”, “distribution shift”, “out-of-distribution”, OOD, “external validation”, “cross-institution”, “site effect”, “dataset bias”	“site-wise split”, “hospital-level split”
C5 Outcome	AUC	AUROC, “area under the curve”, sensitivity, specificity, calibration	“worst-group accuracy”, ECE

Five rules that determine whether your search works:

1. **Include both -ise and -ize spellings.** “Domain generalisation” alone loses the majority of the literature, because most is written in US spelling. This single omission is the most common cause of a catastrophically incomplete search.
2. **Include acronyms *and* their expansions.** Neither alone is sufficient; and an acronym alone is often ambiguous (§10.5).
3. **Include British and American variants** of ordinary words where relevant (behaviour/behavior, modelling/modeling).
4. **Include proper nouns** — dataset names, benchmark names, tool names. These retrieve exactly the empirical papers you need, including papers that never use your abstract vocabulary.
5. **Harvest synonyms from the literature, not from memory.** Sources: the keyword lists of five papers you already trust; index terms on publisher platforms; MeSH; the reference titles of a seed paper.

10.3 Step two: Boolean operators and field codes

Table 10.1 — Boolean operators and field codes by platform

Function	Notation	Notes
Both terms	`AND`	Default on most platforms
Either term	`OR`	Always wrap `OR` groups in parentheses
Exclude	`NOT` / `AND NOT`	**Use sparingly** — see §10.5
Exact phrase	`" "`	Essential for multiword concepts
Grouping	`( )`	Precedence differs between platforms; parenthesise everything
Truncation	`*`	`generali*ation`, `cluster*`. Support and semantics vary; **[VERIFY]**
Single character	`?`	Supported on several curated indexes; **[VERIFY]**
Proximity	`W/n`, `NEAR/n`, `PRE/n`	Notation is platform-specific; **[VERIFY]**

Field codes, in outline. **[VERIFY] all of these against current platform documentation.**

Platform	Typical fielded syntax
Scopus	`TITLE-ABS-KEY( )`, `TITLE( )`, `AUTHKEY( )`, `SRCTITLE( )`, `DOCTYPE( )`, `PUBYEAR` with comparison operators
Web of Science	`TS=` topic, `TI=` title, `AB=` abstract, `AK=` author keywords, `SO=` source, `PY=` year range, `DT=` document type
IEEE Xplore	Command search with named metadata fields (document title, abstract, index terms, all metadata)
ScienceDirect	Title-abstract-keyword field search; a cap on Boolean operators per field
PubMed	`[MeSH Terms]`, `[tiab]`, `[au]`, `[dp]`
Google Scholar	`intitle:`, `author:`, `source:`, leading `-` to exclude; a length limit on queries

10.3.1 Proximity operators: the most underused tool

`AND` matches a paper in which your two concepts appear anywhere — perhaps in unrelated paragraphs. Proximity requires them to appear near each other, which usually means they are actually related.

```
"deep learning" AND "chest radiograph"
   → matches a paper whose introduction mentions deep learning
     and whose unrelated related-work paragraph mentions chest radiographs

"deep learning" W/5 "chest radiograph"          (Scopus notation)
"deep learning" NEAR/5 "chest radiograph"       (Web of Science notation)
   → matches papers where the two concepts occur within five words,
     which is far more likely to indicate a genuine topical match
```

Adopting proximity operators typically improves precision dramatically at little cost in recall.

10.4 Step three: assembling and porting the string

Figure 10.1 — Building a search string from concept blocks

```
      ┌─── C1: DATA ───┐   ┌── C3: METHOD ──┐   ┌─ C4: PHENOMENON ─┐
      │ "chest X-ray*"  │   │ "deep learning"│   │ "domain           │
      │ OR "chest       │   │ OR CNN         │   │  generali*ation"  │
      │    radiograph*" │AND│ OR transformer*│AND│ OR "domain shift" │
      │ OR CXR          │   │ OR ViT         │   │ OR "out-of-       │
      │ OR CheXpert     │   │                │   │    distribution"  │
      │ OR "MIMIC-CXR"  │   │                │   │ OR "external      │
      └─────────────────┘   └────────────────┘   │    validation"    │
                                                 └───────────────────┘
                                 │
              AND  year range  AND  document type  AND  language
                                 │
                                 ▼
             RESULT SET (target: 80–300 records)
                           │
                  ┌────────┴────────┐
                  ▼                 ▼
        too many (>500)      too few (<40)
        add a concept        drop a block;
        block; restrict      add synonyms;
        to title; tighten    check -ise/-ize;
        proximity            widen years
```

Reference implementation (Scopus-style syntax). Build here, then port.

```
TITLE-ABS-KEY(
   ("chest X-ray*" OR "chest radiograph*" OR CXR OR "thoracic radiograph*"
      OR CheXpert OR "MIMIC-CXR" OR "ChestX-ray14" OR PadChest)
   AND ("deep learning" OR CNN OR "convolutional neural network*"
      OR transformer* OR "vision transformer" OR ViT)
   AND ("domain generali*ation" OR "domain shift" OR "distribution shift"
      OR "out-of-distribution" OR OOD OR "external validation"
      OR "cross-institution*" OR "site effect*" OR "dataset bias")
)
AND PUBYEAR > 2021
```

…with document-type and language limits applied through the interface.

Ported to Web of Science:

```
TS=(("chest X-ray*" OR "chest radiograph*" OR CXR OR CheXpert OR "MIMIC-CXR")
  AND ("deep learning" OR CNN OR transformer*)
  AND ("domain generali?ation" OR "domain shift" OR "distribution shift"
      OR "out-of-distribution" OR "external validation"))
AND PY=(2022-2026)
```

Ported to a publisher platform (IEEE-style command search), then narrowed with the interface filters for year and content type:

```
("All Metadata":"chest X-ray" OR "All Metadata":CXR OR "All Metadata":CheXpert)
AND ("All Metadata":"deep learning" OR "All Metadata":transformer)
AND ("All Metadata":"domain shift" OR "All Metadata":"domain generalization"
     OR "All Metadata":"out-of-distribution")
```

Simplified for Google Scholar, which has a query-length limit and weaker field support — so use it for chaining, not as your primary protocol:

```
"chest radiograph" OR "chest x-ray" "domain shift" OR "external validation" "deep learning"
```

10.5 Ineffective and effective queries

Table 10.2 — Ineffective and effective search queries

Ineffective	Why it fails	Effective
`deep learning chest x ray`	No phrases, no synonyms, no fields — enormous noisy result set	`TITLE-ABS-KEY(("chest X-ray*" OR CXR) AND "deep learning")`
`how can I improve generalisation of CNNs across hospitals?`	A natural-language question in a Boolean system	Concept blocks joined by `AND` / `OR` (§10.4)
`"domain generalisation"`	One spelling — silently loses most of the literature	`"domain generali*ation" OR "domain shift" OR "out-of-distribution"`
`CXR AND DL AND DG`	Ambiguous acronyms: DL is also *downlink*, *Dice loss*, *description logic*	Each acronym `OR` its expansion
`(A OR B) AND (C OR D) NOT COVID NOT survey NOT review`	Over-exclusion removes methodology papers and the reviews you need	Retain everything; exclude at the screening stage, where it is documented
`chest x-ray classification 2026`	Year typed as a keyword rather than applied as a filter	`PUBYEAR > 2021 or PY=(2022-2026)`
Sorting by relevance only	Buries recent work	Sort by citations *and* separately by date
One database	Coverage bias; not reproducible	Three or more, each logged

On `NOT`. Exclusion inside a query is invisible to your reader and frequently removes relevant work. Excluding “COVID” also removes strong methodological papers that happen to validate on a COVID subset. The principle: **retrieve broadly, exclude during screening, and report the screening funnel** — where the exclusion is visible and auditable.

10.6 Filters and how to use them

Filter	Use	Caution
Year	Methods: a recent window. Foundations: no limit — you must cite origins	A recent-only window signals ignorance of the field (§8.3)
Document type	Search **reviews first** to harvest taxonomies and vocabulary, then primary studies	Excluding conference papers loses much of CS
Subject area	Removes cross-domain false positives	Can remove genuinely interdisciplinary work
Source title	Restrict to your candidate target journals to learn their conventions	Not for the main search
Open access	Only when you lack full-text access	Never a quality filter
Language	Practical necessity	If you restrict to English, **declare it as a limitation**

10.7 Citation counts and how to read them honestly

Citation counts are useful and routinely misinterpreted.

- They are **age-confounded**. A 2025 paper with eight citations may be more influential than a 2018 paper with sixty. Normalise by citations per year.
- They measure **attention, not correctness**. A heavily cited paper may be heavily *criticised*; check what the citing papers actually say. Citation-context features on some platforms help here.
- They are **field-relative**. Absolute counts across disciplines are meaningless.
- They can be **inflated** by self-citation clusters and coordinated citation practices (§49.6).

Practical use: sort by citations to find the canonical works you must cite; sort by date to find the frontier you must not be scooped by. Do both, always.

10.8 Citation chaining

This is how you find the papers your keyword search missed — and every keyword search misses papers, because authors do not share your vocabulary.

Figure 10.2 — Citation chaining: backward, forward, and sideways

```
                             ┌──────────────────────┐
     BACKWARD                │                      │         FORWARD
  (older foundations)        │   YOUR SEED PAPER    │   (newer work: extensions,
                             │                      │    improvements, refutations)
  read its reference    ◄────┤                      ├───►  "Cited by", then search
  list; find the origin      │                      │     WITHIN the citing set
  of every method used       └──────────┬───────────┘
                                        │
                                     SIDEWAYS
                      (parallel work that never cites your seed)
                      citation-graph tools; co-citation; shared references

  STOP when all three directions return only papers you have already seen.
```

Procedure. For each of your five to eight core papers:

1. **Backward** — read the reference list. Identify the origin paper for every method, dataset, and metric used. Cite origins, not only recent users.
2. **Forward** — use the "Cited by" function, then *filter within* the citing set by keyword to make a large citing list tractable. Forward chaining is how you find out whether your seed was later refuted, which is information you very much need.
3. **Sideways** — use a citation-graph tool to find prominent neighbours that share references with your seed but do not cite it. This is where independently developed parallel work hides.

10.9 Saturation, logging, and alerts

The stopping rule is saturation, not a target count: stop when new searches and all three chaining directions return only papers you have already seen, and when you can anticipate a new paper's content from its title and abstract.

The search log. Record this for every database, and reproduce it in your paper's method section if you are writing a review:

Field	Example
Database	Scopus
Date searched	12 March 2026
Exact string	*(verbatim, including all parentheses)*
Filters applied	2022–2026; article and conference paper; English
Records retrieved	412
After title/abstract screening	118
After full-text screening	34
Included in matrix	15

The date matters because databases are updated continuously: the same string run four months later returns a different count. A review without search dates cannot be reproduced.

Alerts. Set a saved-search alert on your principal string in at least two systems, plus a citation alert on your two most important papers. Your literature review must still be current at submission — which may be a year after you first searched — and again at revision. This costs five minutes once and prevents the specific disaster of a reviewer pointing out a paper published while yours was under review.

10.10 Verification checklist for Part III

- [ ] I have three to five concept blocks with at least three synonyms each.
- [ ] Both -ise and -ize spellings are present.
- [ ] Every acronym appears with its expansion.
- [ ] Dataset and benchmark proper nouns are included.
- [ ] Synonyms were harvested from the literature, not from memory.
- [ ] I have used proximity operators where the platform supports them.
- [ ] Year is applied as a filter, not typed as a keyword.
- [ ] I have used `NOT` sparingly or not at all, and exclude at screening instead.
- [ ] My result set is between roughly 80 and 300 records.
- [ ] I searched at least three databases plus one preprint source.
- [ ] I have completed backward, forward, and sideways chaining on my core papers.
- [ ] I have reached saturation, or know how far from it I am.
- [ ] My search log records database, date, exact string, filters, and counts.
- [ ] I have set at least two alerts.
- [ ] I have exported records with abstracts and keywords into a reference manager.

Exercises

Exercise 10.1 Build your concept-synonym table with at least three blocks and three synonyms per block, including dataset names.

Exercise 10.2 Compose a fielded string in Scopus or Web of Science syntax. Run it, then add one concept block at a time and record the result count at each step. Watching the count fall is how you learn to calibrate a query.

Exercise 10.3 Port your string to two other platforms and record all three counts in your search log.

Exercise 10.4 Pick your single most important paper. Perform all three chaining directions and record how many *new* relevant papers each direction produced. Most researchers find that chaining yields papers their keyword search missed entirely.

Exercise 10.5 Set two alerts. Record the date you set them.

PART IV — RESEARCH PAPER READING

Part IV addresses a skill that is almost never taught explicitly and that determines how fast a researcher can develop: reading papers efficiently and critically. Chapter 11 dissects the anatomy of a paper and specifies what to extract from each section. Chapter 12 gives a three-pass reading method with time budgets. Chapter 13 provides a sixteen-field extraction framework that converts reading into structured data you can analyse.

The governing insight is that efficient reading is not *faster* reading. It is **selective depth**: triaging everything, comprehending a core set, and reconstructing only the few papers your contribution actually stands on.

Chapter 11 — Anatomy of a Research Paper

11.1 Why structure knowledge helps you read

A research paper is a highly conventional document. Once you know what each section is *for*, you know where to look for what you need and — just as important — you can detect when something that should be present is missing. An absent ablation, an unstated split protocol, or an unreported variance is invisible if you read linearly and obvious if you read structurally.

11.2 Section by section

Table 11.1 — What to extract from each section of a paper

Section	Its function	What you should extract	Warning signs
Title	Identify and index the work	Method, problem, domain; whether it claims a finding or an artefact	Vague or buzzword-laden; claims not supported later
Abstract	Let a reader decide whether to read on	Problem, gap, method, data, headline result	No numbers; no dataset named; "results show effectiveness"
Keywords	Indexing and discovery	Controlled vocabulary for your own synonym table (§10.2)	Keywords absent from the paper's own content
Introduction	Establish importance and the gap	The claimed gap; the contribution list; the framing you may need to challenge	No gap paragraph; contributions that restate the method
Related work	Position the contribution	Their taxonomy of the field; who they consider competitors; **who they omit**	Paper-by-paper list; obvious competitor missing
Methodology	Enable reproduction	Architecture, algorithm, loss, the *novel component* isolated from the inherited parts	Cannot tell what is new; no rationale for design choices
Dataset	Define the evidence base	Names, versions, sizes, class balance, licence, **split protocol and split unit**	Unnamed or private data; split unit unstated
Experimental setup	Establish comparability	Baselines, tuning budget, seeds, hardware, framework versions	No seed count; no tuning budget; baselines from a paper rather than re-run
Results	Report observations	Numbers *with their comparison point*, variance, statistical tests	Single-run values; only favourable datasets reported
Discussion	Interpret	Mechanism offered; reconciliation with prior work; admitted limits	Results restated as interpretation; no limitations
Conclusion	State what is now known	The claim as the authors would defend it	New claims not supported by the results
References	Credit and traceability	Origin papers you must also read; the field's canon	Very recent only; single-group clustering; canon missing
Appendices / supplementary	Detail that would obstruct the main text	The honest details — full hyperparameters, extra datasets, failure cases	Material that contradicts the main text's emphasis
Declarations	Ethics, funding, data and code availability	Whether you can obtain the artefacts	"Available on request"; no ethics statement where required

11.3 Where the important information actually hides

Three practical observations that experienced readers rely on:

The tables tell you what was really compared. An introduction tells you what the authors want you to believe was compared; the tables tell you what was. Read the tables before the prose.

The appendix tells you what was really done. Page limits push inconvenient detail into supplementary material: the full hyperparameter grid, the datasets where the method did less well, the sensitivity analysis. Read it for any paper you intend to use as a baseline.

The limitations section is a gift. Authors are required to admit weaknesses and generally do so honestly. Those admissions, aggregated across many papers, are the raw material of your research gap (Chapter 18).

Exercises

Exercise 11.1 Take a paper in your area and, for each row of Table 11.1, write down what you extracted and any warning signs you found. Papers with three or more warning signs should not be used as baselines without a deep read.

Exercise 11.2 Find a paper whose abstract number does not appear in the same form in the results tables. This takes less searching than you would hope, and it permanently changes how you read abstracts.

Chapter 12 — How to Read a Research Paper

12.1 The three-pass method

The approach below is an adaptation of the widely used three-pass method described by Keshav (2007), extended with an extraction framework for empirical computational work. Keshav's original two-page article is worth reading directly; it takes ten minutes.

Figure 12.1 — The three-pass reading method

```
┌─ PASS 1 · TRIAGE ─────────────────────────── 5–10 min · 100% of papers ─┐
│  Read: title, venue, abstract, all figures and tables with captions,      │
│        section headings, conclusion, skim references                      │
│  Decide: DISCARD / CITE-ONLY / READ FULLY                                 │
│  Output: one line in your triage sheet                                    │
└───────────────────────────────────────────────────────────────────────────┘
                   │  (roughly 10–20% survive)
┌─ PASS 2 · COMPREHENSION ──────────────────── 45–60 min · core set ───────┐
│  Read: everything except heavy derivations; all tables carefully          │
│  Do:   redraw their pipeline yourself; interrogate every table            │
│  Output: a filled 16-field extraction template (Chapter 13)               │
└───────────────────────────────────────────────────────────────────────────┘
                   │  (roughly 3–6 papers per project)
┌─ PASS 3 · RECONSTRUCTION ─────────────────── 2–5 h+ · baselines only ────┐
│  Read: everything, including proofs, appendices, and THE CODE             │
│  Do:   re-derive equations; clone and run; reconcile code with paper      │
│  Output: a trustworthy baseline number; a list of hidden assumptions      │
└───────────────────────────────────────────────────────────────────────────┘
```

Table 12.1 — Reading passes: time, coverage, and output

	Pass 1	Pass 2	Pass 3
Time	5–10 min	45–60 min	2–5 h or more
Share of your set	100%	10–20%	2–5%
Governing question	Is this relevant to *my* question?	Do I believe the claim?	Can I build on, extend, or break this?
Output	Triage line	Extraction template + matrix row	Reimplementation; assumption list; new research ideas

The arithmetic that justifies this. Reading two hundred papers at forty-five minutes each is 150 hours. Triaging two hundred at seven minutes (23 hours), comprehending twenty-five at fifty minutes (21 hours), and deeply reading four (20 hours) is about 64 hours for *better* coverage, because you spend your depth where it matters. Efficiency here is not laziness; it is what makes a literature review finishable.

12.2 Pass 1: triage

Read in this order — and note that it is deliberately not the printed order:

1. **Title, venue, year.** Is it peer reviewed? Which community? Does the venue suggest a standard of evidence?
2. **Abstract.** Problem, method, headline claim.
3. **Figure 1** (usually the architecture or overview). The idea in one image.
4. **All tables.** What was compared, on what data, against what.
5. **Section headings.** The shape of the argument.
6. **Conclusion and limitations.** Their own account of what they did not achieve.
7. **Skim references.** Do you recognise the canon? Is anything conspicuously absent?

Reading figures and tables *before* the introduction is the key move: it protects you from adopting the authors' framing before you have assessed their evidence.

Record five items: problem; method in eight words or fewer; datasets; headline metric; relevance verdict.

Discard signals. Different task with no transferable mechanism; no baseline comparison at all; no named dataset and no access route; venue you cannot verify (Chapter 52); results implausible without explanation; superseded by a later version you already hold.

Cite-only signals. Provides a definition, statistic, or dataset you need; is the origin of a method you use; **contradicts another paper you are keeping** — flag contradictions explicitly, because they are the richest source of research gaps (§18.4).

12.3 Pass 2: comprehension

Procedure.

1. Re-read the abstract and write **their claim in your own words**. If you cannot, note it — that may be their writing failure rather than your comprehension failure, but you must resolve it either way.
2. **Redraw their pipeline as a block diagram yourself, without copying theirs.** This is the highest-value single habit in this chapter. Drawing forces you to make explicit what their prose left vague, and where you cannot draw, you have found either a gap in your understanding or an under-specified method. Both are useful.
3. Extract the experimental setup: datasets, splits and split unit, baselines, metrics, hyperparameters, tuning budget, seeds, hardware.
4. For **each table**, ask four questions:
    - What exactly is the comparison?
    - Was the baseline tuned with effort comparable to the proposed method?
    - Is the reported gain larger than the run-to-run variance?
    - Are *all* datasets and metrics reported, or only the favourable ones?
5. Mark unfamiliar terms and citations, and build a follow-up reading queue rather than breaking flow.
6. Fill the extraction template (Chapter 13).

Critical-reading questions that expose weak work.

Target	Question
Novelty	Which *exact* component is new? Which prior paper is closest, and how does this differ?
Fairness	Same data, splits, preprocessing, and tuning budget for every method?
Leakage	Any subject, site, or user appearing in both training and test? Any tuning on the test set?
Variance	Multiple seeds? Standard deviations? Confidence intervals? Any statistical test?
Selectivity	Are the omitted datasets or metrics the inconvenient ones?
Cost	Are parameters, FLOPs, latency, and memory reported?
Reproducibility	Are code and weights available, and do they match the paper?
Explanation	Do the authors explain *why* it works, or only *that* it wins?

The selectivity question requires care. Treat it as a hypothesis to check — compare the datasets named in the abstract with those in the appendix, and check whether the headline metric is used consistently throughout — not as an accusation to make.

12.4 Pass 3: reconstruction

When to invest this much. The paper is your principal baseline; you intend to extend its method; you suspect the result is overstated; or it is the theoretical foundation of your own contribution.

What you actually do.

1. Re-derive the key equations; check dimensions and assumptions.
2. Reconcile the notation in the paper with the variable names in the code. They frequently disagree, and the disagreement is informative.
3. Clone the repository. Run it **as published**, then on **your** data.
4. Record every discrepancy: undocumented preprocessing, hyperparameters in code differing from those in the paper, a metric implemented non-standardly, a different data split than described.
5. Read supplementary material and appendices in full.
6. Read public reviews and author responses if the venue provides them.
7. Email the authors with one specific, respectful question. Response rates are higher than early-career researchers expect.

What this produces: a baseline number you can defend to a reviewer; a list of hidden assumptions, each of which is a candidate research gap; and sometimes a reproducibility finding, which is itself publishable (§17.10).

On reporting reproduction failures. Write about the *method and protocol*, never about the authors' competence or honesty. A sentence naming a number, a protocol, a seed count, and a configuration is unanswerable:

> *"Using the released implementation with the described protocol, we obtained 0.86 ± 0.01 macro AUC over five seeds, compared with the reported 0.91. We were unable to identify the source of the discrepancy; the released configuration differs from the paper's description in the augmentation pipeline (Appendix C)."*

The most common causes of irreproducibility are undocumented preprocessing, different splits, unequal tuning budgets, and non-standard metric implementations — not misconduct. Documenting *which* one it was is a contribution.

12.5 Common mistakes

Mistake	Correction
Reading linearly from page one on first contact	Triage first (§12.2)
Highlighting instead of extracting	Fill fields; highlighting produces no reusable data
Believing the abstract's number without checking the table	Always verify against the table
Recording only the authors' stated limitations	Add your own independent critique (§13.4)
Never checking whether a paper was later refuted	Forward-chain every core paper (§10.8)
Letting an automated summary replace reading a core paper	Summaries are for triage only
Not recording the citation key at extraction time	You will lose hours reconstructing provenance

Exercises

Exercise 12.1 Triage five papers with a visible seven-minute timer. Write the five-item record for each. Most people are surprised how much is decidable in seven minutes.

Exercise 12.2 Pass-2 one paper fully. Redraw its pipeline without looking at their figure, then compare. Note every place your drawing was uncertain.

Exercise 12.3 For that paper, answer all eight critical-reading questions in §12.3. Then decide whether you would trust it as a baseline.

Exercise 12.4 Choose your most important paper, clone its code, and attempt to reproduce one reported number. Whatever happens, you will learn more in three hours than from a week of reading abstracts.

Chapter 13 — Research Paper Extraction

13.1 Purpose

Reading produces understanding; extraction produces **data**. The distinction matters because research gaps are found by *comparing across* papers (Chapter 18), and comparison requires structured fields, not prose notes. One filled template becomes one row of your literature matrix (Chapter 15) and, later, the raw material for your related-work section (Chapter 33).

Complete one during Pass 2. Save it as `<CitationKey>.md` — for example `Zech2018Confounding.md` — in one folder.

13.2 The sixteen-field framework

Table 13.1 — The sixteen-field extraction framework

#	Field	What to capture	Why it matters later
1	**Citation key, venue, year, indexing**	Full citation; DOI verified; venue type	Provenance; weighting evidence quality
2	**Research problem**	The unknown they address, **in your own words**	Grouping papers into themes
3	**Motivation**	Why it matters — application or theory	Your own introduction's ¶1
4	**Stated objective**	Their explicit aim or research question	Detecting mismatch between aim and evidence
5	**Claimed gap**	What they say prior work lacked, and whether they evidence it	Shows how gaps are argued in your field
6	**Dataset(s)**	Names, versions, sizes, classes, balance, licence, **split protocol and split unit**	**The most diagnostic field** — reveals leakage and incomparability
7	**Preprocessing**	Resizing, normalisation and where statistics were fitted, augmentation, resampling and whether before or after splitting	Leakage detection; reproduction
8	**Proposed method**	Architecture or algorithm; the *novel component* isolated; the stated mechanism	Building your method taxonomy
9	**Baselines**	What they compared against; whether tuned equally; number of baselines	Reveals weak-comparison papers
10	**Evaluation metrics**	Which, and whether appropriate to the problem	Reveals evaluation gaps
11	**Experimental setup**	Seeds, variance reporting, statistical test, cross-validation, optimiser, schedule, hardware, versions, tuning budget	Judging reliability
12	**Key results**	Two or three numbers, **each with its comparison point**, taken from the tables not the abstract	Your comparison tables
13	**Limitations — theirs**	Verbatim quotes with section and page	Citations for your gap paragraph
14	**Limitations — yours**	Your independent critique	**The field that produces research gaps**
15	**Future work**	Their suggestions, verbatim	Author-endorsed research directions
16	**Relevance and links**	Core / context / discard; which paper this contradicts; possible extension for you	Prioritisation; contradiction mining

13.3 Field 6 in detail: why the split protocol matters most

If you extract only one thing carefully, extract the **split unit**.

Data are frequently *grouped*: many images per patient, many commits per project, many utterances per speaker, many readings per sensor, many pupils per classroom. When such data are split randomly *by row*, records from the same group appear in both training and test partitions. The model can then succeed by recognising the group rather than by learning the target relationship, and the reported performance describes within-group recognition rather than the generalisation the paper claims.

This is not a hypothetical concern. Zech et al. (2018) demonstrated that chest-radiograph classifiers can exploit institution-specific signal, with the consequence that internal performance overstates external performance.

So field 6 must record not only “70/15/15” but **the unit**: split by image, by patient, by institution, by author, by project, by time period. When you later tally this field across fifteen papers and find that most used random row-level splits on grouped data, you have discovered a field-level methodological gap that a reviewer cannot dismiss — because the evidence is arithmetic.

13.4 Field 14 in detail: your own critique is mandatory

A template containing only the authors’ self-assessment reproduces their framing. That is summarising, not reviewing. Field 14 must contain *your* independent assessment, organised under headings:

- **Fairness of comparison** — equal tuning? official implementations? same preprocessing?
- **Statistical validity** — seeds, variance, tests, corrections?
- **Evaluation protocol** — appropriate metrics? leakage? split unit?
- **Generality** — how many datasets, populations, domains? What is excluded?
- **Cost and deployability** — reported at all?
- **Reproducibility** — code, weights, configurations, splits available?
- **Explanation** — is a mechanism offered and tested, or only a result reported?
- **Selectivity** — is anything conspicuously unreported?

13.5 Worked extraction

[HYPOTHETICAL] The following is an extraction of an invented paper, constructed for teaching. It is not a real work and must not be cited.

> **Paper P7** — “Site-Adversarial Training for Cross-Hospital Chest Radiograph Classification”, hypothetical journal, 2024.

Field	Content
2 Problem	CXR classifiers lose performance at unseen hospitals; the magnitude under leakage-free protocols is unquantified
3 Motivation	Deployment across institutions with differing equipment and protocols
4 Objective	Reduce cross-site AUC loss without harming in-domain AUC
5 Claimed gap	Prior domain-generalisation work evaluates on random splits that leak site identity
6 Datasets	CheXpert (train); ChestX-ray14 and MIMIC-CXR (external); **institution-disjoint** splits
7 Preprocessing	224×224, histogram equalisation, random crop and flip; normalisation statistics from training partition only; no lung segmentation
8 Method	DenseNet-121 with a gradient-reversal site-discriminator branch; invariance weight ramped over five epochs. Novel component: the ramping schedule. Mechanism claimed: site-invariant features transfer better
9 Baselines	ERM, IRM, CORAL, Mixup — four, from official implementations
10 Metrics	Macro AUC, worst-site AUC, expected calibration error
11 Setup	**3 seeds**; Adam 1e-4; 30 epochs; single GPU; code released; tuning budget not stated
12 Key results	Worst-site AUC 0.78 → 0.83; in-domain AUC 0.89 → 0.88; ECE 0.09 → 0.05
13 Limitations (theirs)	"Restricted to frontal adult radiographs; site labels are assumed known at training time." (§VI, p. 11)
14 **Limitations (mine)**	Only three seeds and **no statistical test**, so the 0.05 worst-site gain may lie within run-to-run variance. **Tuning budget unstated**, so fairness cannot be assessed. **Requires site labels**, which are typically stripped before data leaves an institution — this is the binding practical constraint. No subgroup analysis by age, sex, or device, so worst-*site* may conceal worst-*group*. No cost reporting. **No ablation on the ramping schedule**, which is the claimed novelty — so the paper does not establish that its own contribution is responsible for the gain.
15 Future work	Extend to lateral views and paediatric populations
16 Relevance	**Core** — principal baseline. Contradicts P4 on whether calibration improves. Extension: can pseudo-domains inferred *without* site labels recover most of this gain, and does the effect survive ten seeds with a paired test?

Observe what field 14 produced. Three candidate research directions emerged from critically reading one paper: a label-free variant (methodological gap), a statistical re-evaluation (evaluation and reliability gap), and a subgroup analysis (evaluation and fairness gap). None required an original idea from nothing; all came from asking *how do you know?* systematically.

13.6 Tool assistance and its limits

Software can pre-populate parts of this template, and doing so is legitimate and efficient — with a firm boundary.

Field	May be pre-filled by a tool	Must be yours
1, 6, 7, 9, 10, 11	□ Factual fields, then verified cell by cell against the PDF	
12 Key results	⚠ Extracted numbers **must** be checked against the actual table — this is where automated extraction most often errs, by attributing a number to the wrong condition	
2, 4, 5		Your restatement of their problem and gap
13		Verbatim quotes you located
14, 16		**Your critique and your judgement — never delegable**

The rule for the whole handbook, stated once and applied throughout: **tools may help you find, sort, translate, and pre-fill; you must read, verify, compare, judge, and claim.** Chapter 45 develops this.

13.7 Verification checklist for Part IV

- [ ] I triage every paper before reading it fully.
- [ ] My core set has a filled sixteen-field template each.
- [ ] Field 6 records the split **unit**, not only the ratio, for every paper.
- [ ] Field 12 numbers were taken from tables, not abstracts.
- [ ] Field 14 contains my own critique for every paper, under the eight headings.
- [ ] I have flagged every contradiction between papers in field 16.
- [ ] Every automated extraction has been verified against the PDF.
- [ ] I have deep-read and attempted to run at least one baseline.
- [ ] Citation keys were recorded at extraction time.

Exercises

Exercise 13.1 Complete the full sixteen-field template for three papers. Time yourself; the third will take half as long as the first.

Exercise 13.2 For each, write field 14 under all eight headings. Then list every candidate research direction that fell out. Expect two to four per paper.

Exercise 13.3 Tally the split unit across your extractions so far. If most are row-level random splits on grouped data, you have found a gap — proceed to Chapter 18.

Exercise 13.4 Take one paper, have a tool extract fields 6, 9, 10, and 12, then verify every cell against the PDF. Record how many were wrong. Whatever the answer, you now know the verification is necessary.

PART V — LITERATURE REVIEW

Part V converts extracted papers into an argument. Chapter 14 distinguishes synthesis from summary and demonstrates the difference on the same set of papers. Chapter 15 specifies the literature matrix — the single most useful artefact in this handbook — and shows how gaps are found by reading its *columns*. Chapter 16 covers the tools, with the verification each requires.

Chapter 14 — Literature Review Writing

14.1 Definition

Definition. A literature review is an argument about a body of work that (a) organises it into a structure of your making, (b) evaluates the quality of its evidence, and © demonstrates that a specific gap exists.

All three clauses are required. An organised account with no evaluation is a catalogue. An evaluation with no organisation is a series of opinions. Either without a terminating gap leaves the reader asking why they read it.

14.2 Review versus summary

	Summary (paper-by-paper)	Review (synthesis)
Unit of a paragraph	One paper	One idea, method family, or finding
Ordering principle	The order you happened to read them	Your taxonomy
Characteristic verbs	proposed, used, applied, achieved	converge on, diverge, in contrast, remains untested
Comparison	None	Explicit, along shared dimensions
Evaluation	None	Quality of evidence assessed with reasons
Ends with	The last paper	**The gap**
Citations per sentence	One	Frequently two to five, grouped

The diagnostic. Look at your draft. If most paragraphs begin with an author's name, you have written a summary. Run this test tonight on whatever you have.

Figure 14.1 — Paper-by-paper listing versus critical synthesis

```
SUMMARY (weak)

¶ Author A did X, got 0.90.
¶ Author B did Y, got 0.89.
¶ Author C did Z, got 0.91.
¶ Author D did W.
¶ Author E did V.

Reader learns: five facts.
Reader cannot tell what is KNOWN,
what is DISPUTED, or what is
MISSING.
```

```
SYNTHESIS (strong)

¶ Two families exist. FAMILY 1
  increases capacity [A,B,C];
  reports 0.89–0.91 BUT all three
  evaluate on random splits, so
  these are in-distribution numbers.
¶ FAMILY 2 targets invariance
  [D,E]; does evaluate across sites
  BUT both require site labels and
  report ≤3 seeds without tests.
¶ ACROSS ALL FIVE: worst-site
  performance and calibration are
  never reported.
¶ THEREFORE the gap is ...

Reader learns: the state of
knowledge, and what is absent.
```

14.3 Five organising structures

Structure	Organising principle	Use when	Risk
Chronological	Time; paradigm shifts	The field has genuine successive eras	Becomes a timeline with no argument
Thematic	Sub-problems or facets	Several distinct facets exist	Themes overlap; papers repeat
Methodological	Technique families	You will argue a *family* shares a weakness	Ignores differences in task and data
Comparative	Dimensions in a matrix	You have quantitative evidence across studies	Reads like a spreadsheet if not interpreted
Critical / argumentative	Your thesis about the field	You have deep mastery and a defensible position	Can appear biased; needs scrupulous fairness

Recommendation. Strong reviews are usually **hybrid**: a thematic top level, methodological groups within each theme, a comparison table per theme, and a critical synthesis paragraph closing each theme — with the section's final paragraph stating the gap.

A reusable skeleton for a related-work section:

```
2.1  Problem formulations, datasets, and evaluation conventions
2.2  Method family A          (+ comparison table, + critique paragraph)
2.3  Method family B          (+ comparison table, + critique paragraph)
2.4  Evaluation practices in this area        ← usually omitted; high value
2.5  Synthesis: what is settled, what is contested, what is untested
                                               ► THE GAP
```

Subsection 2.4 deserves comment. Almost no student paper contains a subsection on the *evaluation practices* of prior work, and it is where evaluation gaps become visible. If your contribution is a re-evaluation, 2.4 is the paragraph that justifies your entire paper.

A practical shortcut. Open the two most recent surveys in your area and examine their section structure. A taxonomy already exists. You may adopt it with citation, or deliberately improve on it — and explaining why the existing taxonomy is inadequate is itself a contribution.

14.4 Worked contrast on the same five papers

[HYPOTHETICAL] □ Weak — a summary chain

> "Zhang et al. [11] used a CNN with attention for chest X-ray classification and achieved 0.90 AUC. Patel et al. [12] used DenseNet-121 and reported 0.89. Lee et al. [13] applied a vision transformer, achieving 0.91. Gupta et al. [14] used CORAL for domain adaptation. Rao et al. [15] used IRM and showed improvement in external validation."

Five faults: chronological accident as structure; numbers from different datasets and splits presented as if comparable; no assessment of quality; no shared dimension; no gap at the end.

[HYPOTHETICAL] □ Strong — synthesis with a critical edge

> "**Capacity-scaling approaches.** One line of work treats cross-institutional robustness as a byproduct of stronger representations, progressing from attention-augmented convolutional backbones [11], [12] to vision transformers [13]. These report 0.89–0.91 AUC; however, all three evaluate on random splits in which radiographs from the same institution occur in both partitions (Table I), so their figures describe in-distribution performance and cannot be read as evidence of transfer.
>
> **Invariance-based approaches.** A second line optimises explicitly for domain invariance through feature alignment [14] or invariant risk minimisation [15], and does evaluate across institutions, reporting external gains of three to six AUC points. Both, however, require institutional labels during training, and neither reports variance over more than three seeds; since seed-to-seed variation of one to two AUC points is documented for these architectures [12], it is unclear whether the reported margins exceed run-to-run noise.
>
> **Evaluation practice.** Across the fifteen studies surveyed, none reports worst-institution AUC and only two report calibration, although a triage system's usability is governed by its weakest site and by the reliability of its confidence estimates rather than by mean discriminative performance.
>
> **Synthesis and gap.** The field has therefore established that distribution shift matters, but not how large it is under leakage-free protocols, nor whether invariance is attainable without institutional metadata, nor what happens to worst-site performance and calibration. This paper addresses these three questions."

Note the machinery: it groups; it names a shared dimension (evaluation protocol); it states what is credible and what is not, *with a reason*; it groups citations; it identifies what is missing across *all* studies; and its final sentence is a gap statement. No sentence begins with an author's name.

14.5 The synthesis phrasebook

Non-native English writers frequently find this the most immediately useful page in the handbook.

Agreement and convergence. "Consistently across [3]–[7], …" · "There is broad agreement that …, although the reported effect size ranges from X to Y." · "Both lines converge on …"

Contrast and contradiction. "In contrast to [4], who report …, [8] find the opposite when …" · "This discrepancy is plausibly attributable to differing split protocols rather than to the methods themselves." · "Whereas earlier work assumed …, more recent evidence suggests …"

Evaluating quality. "…, although the comparison employs an untuned baseline." · "…; the reported gain of 0.4 points is smaller than the seed-to-seed variance reported in [9]." · "…, on a single dataset, which limits generality." · "…, though the tuning budget is not stated, so fairness cannot be assessed."

Establishing absence — the gap moves. "**X** remains unquantified under **[condition]**." · "Existing evaluations do not report **[quantity]**, although it determines **[consequence]**." · "This assumption is unlikely to hold in **[setting]**, yet remains untested." · "We found no study that evaluates X under Y; the closest is [12], which does Z but not Y."

A warning about absolute claims. Never write "no work exists" casually — a specialist reviewer will find a counterexample, and the credibility damage extends to your whole paper. The safer and *stronger* formulation is the last one above: it demonstrates that you searched, names the nearest prior work, and survives contact with an expert.

14.6 Common mistakes

Mistake	Correction
Every paragraph begins with an author's name	Reorganise by idea, not by paper
Comparing numbers from different datasets and splits	State explicitly that they are not comparable; that statement is itself a finding
Listing without evaluating	Every group needs a critique sentence with a reason
No gap at the end of the section	The final paragraph must state the absence
"No work exists"	"We found no study that…; the closest is [x]"
Dismissing prior work rudely	Criticise the protocol, never the authors — they may be your reviewers
Omitting an evaluation-practices subsection	Add §2.4; it is where the cheapest defensible gaps live
Citing one paper per claim about the field	Claims about a field need grouped citations

Exercises

Exercise 14.1 Take four rows of your extraction set and write one synthesis paragraph of 120–180 words that groups them, evaluates their evidence, and ends with an absence statement.

Exercise 14.2 Have a colleague mark every sentence in that paragraph as *reports*, *groups*, *evaluates*, or *identifies absence*. Any sentence that only reports should be cut or upgraded.

Exercise 14.3 Examine the section structure of the two most recent surveys in your area. Write down their taxonomy, then write one sentence on how yours will differ and why.

Chapter 15 — The Literature Matrix

15.1 Purpose

The literature matrix is a table with one row per study and one column per attribute. Its purpose is to make patterns visible that no amount of reading prose can reveal, because **gaps are properties of columns, not of rows**.

This is the central mechanical insight of Part V. When you read papers one at a time, you see each study's contribution. When you tally a column across fifteen studies, you see what the *field* has and has not done. Research gaps are found by counting.

Build it in a spreadsheet, never in a word processor: you must be able to sort, filter, and count.

15.2 Fields

Table 15.1 — Literature matrix fields and why each earns its place

Col	Field	Example	Why it earns its place
A	Citation key	`Zech2018Confounding`	Links matrix to reference manager and bibliography
B	Year	2018	Chronology; locating the frontier
C	Venue and indexing	*PLOS Medicine*; indexed	Weighting evidence quality; identifying target journals
D	Research problem	Cross-site confounding	Grouping into themes
E	**Dataset(s) and split protocol**	CheXpert; **random** split	**The most diagnostic column** — reveals leakage and incomparability
F	**Split unit**	image / patient / **institution**	Separated from E because it is the single highest-value cell
G	Method	DenseNet-121 + CORAL	Building your method taxonomy
H	Novel component	The invariance term only	Distinguishes real from claimed novelty
I	Baselines and count	ERM only (1)	Reveals weak-comparison papers
J	Metrics	Accuracy, AUC	Reveals evaluation gaps
K	Key results	0.90 AUC in-domain	Comparable **only** within the same dataset and split
L	Seeds and variance	1 seed, no CI	Reveals reliability gaps
M	Statistical test	none	Almost always empty — and that emptiness is a finding
N	Cost reported	no	Reveals efficiency gaps
O	Code available	yes / on request / no	Reproducibility; feasibility as a baseline
P	Strengths	Public code; three datasets	Fair credit; tells you what to emulate
Q	Limitations (theirs)	Frontal views only	Raw material for the gap
R	**Limitations (yours)**	Random split; 1 seed; no ablation	**Where gaps are actually produced**
S	Gap it leaves	Leakage-free re-evaluation absent	Aggregates into your gap
T	Future work (verbatim)	"extend to lateral views"	Author-endorsed directions
U	Relevance	baseline / context / discard	Prioritisation
V	Contradicts	P4 on calibration	Contradictions are the richest gap signal

A critical warning about column K. Numbers across rows are usually **not comparable** — different datasets, splits, preprocessing, and metrics. The matrix's job is to *expose* incomparability, not to construct a leaderboard. Averaging column K across papers is a serious error that produces a meaningless number.

15.3 Reading the matrix by column

Figure 15.1 — Reading a literature matrix by column rather than by row

```
READING BY ROW (what beginners do)
────────────────────────────────────────────────────────────────
P1: ResNet-50, CheXpert, random split, 0.89 AUC, no baseline
P2: DenseNet, ChestX-ray14, random split, 0.90 AUC, 1 baseline
P3: ViT, site-wise, 0.84 external, 1 baseline
...
→ You learn what each paper did. You find no gap.

READING BY COLUMN (what produces gaps)
────────────────────────────────────────────────────────────────
Column F (split unit):     11/15 random at row level  ←── FINDING
Column I (baseline count):  9/15 have ≤1 baseline      ←── FINDING
Column J (metrics):        12/15 report only Acc/AUC   ←── FINDING
Column M (stat. test):      0/15 run any test          ←── FINDING
Column L (seeds):          13/15 report ≤3 seeds       ←── FINDING
Column N (cost):            2/15 report cost           ←── FINDING
Column O (code):            6/15 release none          ←── FINDING
→ Six field-level gaps, obtained by counting. No inspiration required.
```

Procedure for the tallies. In your spreadsheet, add a block below the data using count formulas — for example, in a spreadsheet where column F holds the split unit and your data occupy rows 2 to 30:

```
Total studies                =COUNTA(A2:A30)
Random row-level splits      =COUNTIF(F2:F30,"image")
Grouped splits               =COUNTA(F2:F30)-COUNTIF(F2:F30,"image")
Studies with ≤1 baseline     =COUNTIF(I2:I30,"<=1")
Studies reporting a test     =COUNTA(M2:M30)-COUNTIF(M2:M30,"none")
Studies releasing code       =COUNTIF(O2:O30,"yes")
Studies reporting metric X   =COUNTIF(J2:J30,"*ECE*")
```

Then write each tally as a sentence. **Those sentences are your gap paragraph** (Chapter 18) and, later, the evidence in your introduction's fourth paragraph (Chapter 32).

15.4 Worked extract

[HYPOTHETICAL] Papers P1–P10 are invented placeholders for teaching. They are not real works.

Paper	Yr	Dataset + split	Split unit	Method	Baselines	Metrics	Key result	Seeds	Test	Limitation (mine)
P1	2022	CheXpert, random	image	ResNet-50	none	Acc, AUC	0.89 AUC	1	none	Leakage; no comparison possible
P2	2022	ChestX-ray14, random	image	DenseNet-121	ResNet-50	AUC	0.90 AUC	1	none	Baseline untuned; no CI
P3	2023	CheXpert→CXR14, site-wise	institution	ViT-B/16	DenseNet	AUC	0.84 external	3	none	One external set; no worst-site
P4	2023	MIMIC-CXR, random	image	CNN + attention	2 CNNs	Acc, F1	0.91 Acc	1	none	Accuracy on imbalanced multi-label
P5	2023	CheXpert→CXR14	institution	CORAL	ERM	AUC	+0.03 external	3	none	Requires site labels
P6	2024	3 sets, site-wise	institution	IRM	ERM, CORAL	AUC, ECE	+0.04 worst-site	3	none	No significance test
P7	2024	3 sets, site-wise	institution	Site-adversarial	4 methods	AUC, worst-site, ECE	0.78→0.83	3	none	Site labels; no ablation on novelty
P8	2024	Private, undisclosed	unstated	Ensemble	1 CNN	Acc	0.95 Acc	1	none	Not reproducible; discard as baseline
P9	2025	CheXpert, random	image	Self-supervised	ERM	AUC	+0.02	5	none	No external validation
P10	2025	2 sets, site-wise	institution	Test-time adaptation	ERM, CORAL	AUC	+0.03	3	none	Latency unreported

Column tallies: 5/10 random row-level splits · 4/10 ≥1 baseline · **0/10 any statistical test** · 2/10 report calibration · 3/10 report ≥5 seeds · 1/10 reports cost · 0/10 operate without site labels.

Seven candidate gaps, produced by counting. Note in particular that **0/10 ran a statistical test** — an empty column is often the most valuable finding in the matrix.

15.5 Common mistakes

Mistake	Correction
Building the matrix in a word processor	Use a spreadsheet; you must sort, filter, and count
One column labelled “Notes”	Use specific fields; a notes column cannot be tallied
Averaging results across incomparable studies	Never; use the matrix to expose incomparability
Recording only the authors’ limitations	Column R is mandatory
Omitting the split unit	It is the highest-value cell in the table
Stopping at five papers	Patterns are invisible below roughly ten
Never computing the tallies	The tallies *are* the point
Filling cells from automated extraction without checking	Verify every cell against the PDF

Exercises

Exercise 15.1 Build a matrix of at least ten studies with columns A–V. Fill column F for every row.

Exercise 15.2 Compute at least six column tallies and write each as a sentence with its count.

Exercise 15.3 Identify every contradiction between studies (column V). Each is a candidate research question (§18.4).

Exercise 15.4 For non-computational fields, adapt the columns: dataset ▯ sample and population; method ▯ intervention or analytical approach; baselines ▯ comparison groups; metrics ▯ measures with their validity and reliability evidence. The logic is unchanged.

Chapter 16 — Literature Review Tools

16.1 Scope and a standing caution

[VERIFY] This chapter describes categories of tool and the workflows they support. Individual products change names, features, pricing, and availability frequently; some described here may have altered materially since writing. The *functional* classification is stable; the products are not. Verify current capability before relying on any of them, and prefer tools whose output you can export and audit.

Table 16.1 — Literature tools: purpose, workflow, limitations, verification

Tool	Purpose	Core workflow	Limitations	Verification you owe
Zotero	Reference library, PDFs, notes, citations, BibTeX	Install with browser connector and word-processor plugin → save from publisher page → add by DOI → organise with collections and colour tags → child notes using the extraction template → export CSV or BibTeX	Free storage tier is limited; publisher metadata is frequently wrong	**Check every metadata field.** Missing pages, ALL-CAPS titles, and wrong venue names are common
Mendeley	Same role, Elsevier ecosystem	Web importer → import RIS/BibTeX from Scopus or ScienceDirect → annotate → cite in word processor	Fewer extensions than Zotero; account-bound	Same metadata verification
EndNote	Institutional standard in many groups	Groups; cite-while-you-write; journal output styles	Licence cost; style files can lag journal requirements	Check output against the journal's current guide
Semantic Scholar	Discovery, citation context, alerts, API	Search → follow influential citations → set alerts → use API for bulk screening	Metadata noise; automated summaries lossy	Never characterise findings from a summary
Citation-graph tools (Connected Papers, Litmaps, ResearchRabbit)	Sideways discovery; finding what your matrix is missing	Seed with three to five core papers → inspect prominent unread nodes → export	Coverage depends on the underlying graph; incomplete	Treat as discovery, not as evidence of completeness
Elicit	Structured extraction across many papers	Upload or search a corpus → request columns (population, method, outcome) → export table	Extraction errors; mis-attributed numbers	**Verify every cell against the PDF**
Consensus	Evidence-oriented search over papers	Ask a question → read per-paper claim summaries → open the papers	Summaries compress nuance and can invert conditionals	Read the primary source before citing
NotebookLM	Grounded question-answering over *your own* uploaded documents	Upload your PDF set → ask cross-document questions → follow citations back to your sources	Limited to what you upload; still capable of error	Verify quotations and numbers against originals
Reference verification (DOI resolver, Crossref)	Confirming a reference exists and is correct	Resolve every DOI; compare fields against the publisher record	Authority for existence, not for page numbers	Do this for **every** reference before submission

16.2 A recommended concrete workflow

This is one workflow that works end to end; adapt freely.

1. **Search** in a curated index (Chapter 10). Export **RIS or BibTeX including abstracts and keywords**.
2. **Import** into Zotero. Create a collection per project.

3. **Fix metadata** on import — this is not optional; publisher exports contain errors that will surface as citation defects later.
4. **Triage** in Zotero with colour tags: red = core, yellow = context, grey = discard (§12.2).
5. **Extract** into child notes using the sixteen-field template (Chapter 13), one note per paper.
6. **Export** the collection to CSV; open in a spreadsheet; this seeds columns A–C, and partially E–J.
7. **Complete** the judgement columns (R, S, U, V) by hand. These cannot be automated.
8. **Tally** the columns (§15.3) and write the tally sentences.
9. **Chain** citations (§10.8) and use a citation-graph tool to check for prominent papers absent from your matrix.
10. **Maintain**: alerts feed new papers into the same pipeline monthly.
11. **Before submission**, resolve every DOI in your bibliography (§16.1, last row).

16.3 Where tool assistance is legitimate, and where it is not

Task	Tool may	You must
Finding candidate papers	Suggest, rank, expand queries	Run and log the searches yourself
Screening	Pre-rank by relevance	Make every include/exclude decision
Extracting factual fields	Pre-fill datasets, metrics, baselines	Verify every cell against the PDF
Extracting results	Propose numbers	**Check each against the actual table**
Summarising for triage	Produce a gist	Read core papers properly
Comparing papers	Suggest dimensions	Judge which comparisons are valid
Identifying gaps	Suggest gap *types* to check	Supply the evidence from your own tallies
Producing citations	Format existing verified records	**Never** accept a reference you have not resolved

The final row is absolute and is developed in §45.4 and §46.4: bibliographic details produced by a generative tool must be treated as unverified until resolved against the publisher record, because plausible-looking but non-existent references are a documented failure mode with severe consequences.

16.4 Verification checklist for Part V

- [ ] No paragraph of my review begins with an author's name.
- [ ] My review is organised by idea, with a stated taxonomy.
- [ ] Each group has a critique sentence giving a *reason*.
- [ ] I have a subsection on prior evaluation practice.
- [ ] The final paragraph states the gap.
- [ ] Absence claims are phrased as "we found no study that…", not "no work exists".
- [ ] My matrix has at least ten rows and includes the split unit for each.
- [ ] I have computed at least six column tallies and written them as sentences.
- [ ] Contradictions between studies are recorded.
- [ ] Every automated extraction has been verified against the source PDF.
- [ ] Every metadata field in my reference manager has been checked.

Exercises

Exercise 16.1 Set up the workflow in §16.2 end to end for ten papers. Time it. The second ten will take a third as long.

Exercise 16.2 Have a tool extract four factual columns for three papers, then verify every cell. Record the error rate; it calibrates how much verification you owe in future.

Exercise 16.3 Seed a citation-graph tool with your three core papers and list any prominent work absent from your matrix. Add them and re-tally.

PART VI — RESEARCH GAP

Part VI is the pivot of this handbook. Everything in Parts III to V was preparation for it, and everything in Parts VII to X depends on it. Chapter 17 defines what a research gap is, what it is not, and gives a thirteen-type taxonomy. Chapter 18 provides a systematic procedure for deriving a gap from a literature matrix. Chapter 19 contrasts weak and strong gap statements and specifies how a gap must be evidenced.

The central claim of this part: **a research gap is not discovered by inspiration. It is computed from a documented literature matrix, stated with counts, and stress-tested by a hostile reader before it reaches a reviewer.**

Chapter 17 — Understanding Research Gaps

17.1 Definition

Definition. A research gap is a specific, evidenced absence or inadequacy in the accessible literature, whose resolution would change what the field knows or can do.

Four components are obligatory. A statement missing any one of them is not yet a gap.

Component	Question it answers	Example *[HYPOTHETICAL]*
What is absent	Precisely what is missing?	No leakage-free, multi-institution quantification of CXR classifier degradation
Evidence of absence	How do you know?	11 of 15 surveyed studies use random row-level splits; 0 of 15 report worst-institution performance
Why it matters	Who is harmed by the absence?	Reported figures overstate deployable performance; a hospital cannot estimate the accuracy it would obtain
What resolution enables	What becomes possible?	A trustworthy degradation estimate, and a mitigation usable without privileged metadata

The second component is the one that separates a defensible gap from an assertion, and it is the one most often missing. "Evidence of absence" does not mean proving a negative; it means demonstrating that you looked systematically and reporting what the search found.

17.2 What is not a gap

Statement that is not a gap	Why not	How to rescue it
"Nobody has applied method X to dataset Y."	Absence of *activity*, not of *knowledge*. Doing something untried teaches nothing by itself	Identify which assumption of X is violated by Y, predict the consequence, measure it (§5.5)
"Accuracy can be improved."	True of every task, forever; unfalsifiable	Specify a threshold and why it matters: what decision changes at what level?
"The topic is new and trending."	Novelty of attention, not of knowledge	Find what is actually unresolved within the trend
"Existing methods have limitations."	Every method has limitations	Name the specific limitation, count how many studies share it, show what it costs
"No work exists in my country, language, or domain."	Geography is not a mechanism	Give a reason to expect different results here: what property differs?
"No one has combined A and B."	Combination is not knowledge	Predict *why* the combination should behave differently, and test the prediction
"Prior work is old; we use a newer model."	Recency is not a contribution	Ask what the newer model changes about the *conclusion*, not about the implementation
"There is limited research on this."	"Limited" is not a measurement	Count it. Report the count. Then say what the count implies

Nearly every first draft contains at least one of these. The rescue is always the same shape: **add a mechanism and a measurement.**

17.3 The thirteen-type taxonomy

Knowing which *type* of gap you have is practically useful, because each type demands a different kind of evidence and each carries a different cost to address.

Table 17.1 — Thirteen types of research gap

#	Type	Definition	Example [HYPOTHETICAL unless cited]	Evidence you need
1	**Knowledge**	A phenomenon is unexplained or unmeasured	Why does self-supervised pretraining help under distribution shift but not under label noise?	Prior work reports the effect but not its cause
2	**Methodological**	Existing methods rest on an assumption that fails in practice	All identified domain-generalisation methods require institutional labels; deployment strips them	The assumption, stated in each paper's setup
3	**Dataset / resource**	No suitable data exists for a population, language, modality, or annotation	No code-mixed paraphrase benchmark with graded obfuscation levels	A documented search of dataset registries returning nothing
4	**Performance**	The best known result is insufficient for the intended use	Worst-institution AUC of 0.83 against a clinically required 0.90	A requirement derived from the application domain
5	**Application / translation**	A method works in the laboratory and is unstudied under deployment constraints	Test-time adaptation requires target batches; latency and memory unreported for edge hardware	Absence of cost reporting in prior work
6	**Population / domain**	Evidence exists for one group and is untested for another, **with reason to expect difference**	Models trained on adult frontal radiographs untested on paediatric populations, where anatomy differs systematically	A domain argument *plus* absence of studies
7	**Evaluation**	Metrics or protocols do not measure what matters	Imbalanced multi-label classification evaluated by accuracy; calibration unreported	A column tally of metrics across studies
8	**Scalability**	Behaviour at the target scale is unknown	A graph method validated to 10^4 nodes; production graphs are 10^8	Complexity analysis plus absence of large-scale results
9	**Generalisation**	Results do not transfer across distributions	Chest-radiograph models can exploit institution-specific signal, so internal performance overstates external performance (Zech et al., 2018)	Direct re-evaluation under a leakage-free protocol
10	**Reproducibility**	Results cannot be independently obtained	Reported numbers not recoverable from released code under the described protocol	Your own documented reproduction attempt
11	**Computational efficiency**	Accuracy is achieved at unreported or unacceptable cost	A 0.02 AUC gain for four times the parameters and three times the latency, never reported	Cost measurements absent from prior work
12	**Explainability / trust**	Decisions are unexplained, or explanations are unvalidated	Saliency maps used as evidence of correctness without validation against expert annotation	Absence of explanation-validation studies
13	**Temporal**	Findings may not hold as the underlying distribution changes over time	A detector validated on 2019 data deployed against 2026 traffic patterns or 2026 malware	Evidence of drift plus absence of longitudinal evaluation

17.3.1 Which gaps are achievable with limited resources

This is a recommendation. Gaps of types **7, 9, 10, and 11** — evaluation, generalisation, reproducibility, and efficiency — are usually the most achievable for a researcher without large compute, for three reasons:

1. They require **careful experimental design** rather than new theory or enormous training budgets.
2. Their evidence is largely **arithmetic**, which makes them difficult for a reviewer to dismiss.
3. They frequently **change practice**, which makes them well cited.

The honest counterweight: work of this kind must be executed impeccably. If you are criticising the rigour of others, your own rigour must be beyond reproach — equal tuning budgets, released code, adequate seeds, appropriate statistics, and scrupulously neutral language about other authors.

17.4 Gap types often combine

Real gaps are rarely of a single type, and saying so is a strength rather than a hedge. The running example combines:

- Type 9 (generalisation) — degradation across institutions;
- Type 7 (evaluation) — worst-site performance and calibration unreported;
- Type 2 (methodological) — all mitigations assume metadata that deployment removes.

A gap statement that names two or three types with evidence for each is more convincing than one claiming a single dramatic absence.

Exercises

Exercise 17.1 Write your current gap statement. Check it against every row of §17.2. If it matches any, rescue it by adding a mechanism and a measurement.

Exercise 17.2 Classify your gap against Table 17.1. Expect two or three types. For each, write the specific evidence you hold.

Exercise 17.3 Consider deliberately whether a type 7, 9, 10, or 11 framing of your topic would be more achievable with your resources than your current plan.

Chapter 18 — How to Identify a Research Gap

18.1 The systematic procedure

Gaps are produced by a repeatable process, not by waiting for insight.

Figure 18.1 — From fifteen papers to a defensible research gap

```
STEP 1 · ASSEMBLE          15 studies in the matrix, from a logged search
                           (Chapters 10, 15)
      |
      ▼
STEP 2 · TALLY COLUMNS     Count every column. Write each count as a sentence.
                           • 11/15 random row-level splits on grouped data
                           • 12/15 report only accuracy or AUC
                           •  0/15 report worst-group performance
                           •  2/15 report calibration
                           •  0/15 run any statistical test
                           •  9/15 compare against ≤1 baseline
                           •  6/15 release no code
                           • 15/15 mitigations require domain labels
      |
      ▼
STEP 3 · CLUSTER INTO      L1  Evaluation protocol leaks group identity   (11/15)
         LIMITATIONS       L2  Reported gains may lie within seed noise   (0/15 tests)
                           L3  Domain labels assumed available            (15/15)
                           L4  Decision-relevant quantities unreported    (0/15, 2/15)
      |
      ▼
STEP 4 · UNRESOLVED        "The true cross-institutional degradation is unknown,
         PROBLEM            and no mitigation has been shown to work without
                            domain labels."
      |
      ▼
STEP 5 · GAP STATEMENT     Absence + evidence + significance + what resolution
                           enables.  (Template in §19.4)
      |
      ▼
STEP 6 · CONTRIBUTIONS     C1 ← answers L1, L2      C2 ← answers L3
         (1:1 mapping)     C3 ← answers L1, L4
```

Notice that **the study design fell out of the gap analysis**. This is the correct order. Most beginners design the method first and reverse-engineer a gap to justify it, which is why reviewers can smell it: the gap does not quite fit the method, and the contributions do not map onto the limitations of prior work.

18.2 The nine signals

Gaps announce themselves in nine recognisable places. Learn to read all nine.

Signal	How to detect it	Gap types it typically yields
Stated limitations	Read every limitations and threats-to-validity section; tally recurring items	2, 5, 6, 12
Future work	Collect verbatim; a suggestion repeated by three or more papers is a field priority	any
Contradictory results	Two studies, same task, opposite conclusions → find the confound	1, 7, 9
Missing datasets	Search dataset registries; note absent languages, populations, modalities	3, 6
Poor evaluation	Tally the metrics column: accuracy on imbalanced data, no CI, no test	7
Weak baselines	Tally the baseline column: "compared with ERM only", or with a five-year-old method	4, 7
Missing comparisons	Method families that have never been compared *under one protocol*	7, 9
Small samples, few seeds	*n* below field norms; one to three seeds; a single split	7, 10
Scope restrictions	"We consider only English / adults / frontal views / synthetic data"	6, 8, 9, 13

18.2.1 Mining limitations systematically

Procedure.

1. For each of ten to fifteen recent papers, open the limitations, threats-to-validity, discussion, and conclusion sections.
2. Copy each admitted weakness into a spreadsheet **verbatim**, with paper, section, and page.
3. Tag each with a category: `data`, `method`, `evaluation`, `scale`, `generality`, `efficiency`, `explainability`, `ethics`.
4. **Count the tags across papers.** A weakness admitted by six of fifteen studies is a field-level gap, not one author's excuse.
5. For each recurring weakness, ask the decisive question: *is it unsolved, or solved elsewhere and simply not applied here?*
    - **Unsolved** ▯ potential novel contribution.
    - **Solved elsewhere** ▯ potential transfer contribution. This is still publishable, but you must be honest in framing it, and you must explain why the transfer is non-obvious (§5.5).

Step 2 is not busywork. Those verbatim quotations, with page numbers, become the citations that justify the gap paragraph of your introduction (Chapter 32). Collecting them now saves reconstructing them later.

18.3 Reading "future work" correctly

Future-work sections are the most explicit, most freely given, and least used source of research directions in the literature.

Two cautions. First, a suggestion made once by one author is a hint; a suggestion made by five independent groups is a field priority and probably also a crowded space — check whether it has been done since. Second, authors sometimes propose future work they have already begun; forward-chain the paper (§10.8) to see whether the follow-up exists before investing.

18.4 Contradictions: the richest and least-used signal

When two studies on the same task reach opposite conclusions, the *explanation of the discrepancy* is a knowledge contribution — and one you can often produce with a controlled re-run rather than a new method.

Procedure.

1. Tabulate both studies side by side across every attribute in your matrix.
2. List every difference: dataset, split protocol, split unit, preprocessing, tuning budget, metric, seed count, model scale, evaluation code.
3. Form a hypothesis about which difference is the causal confound.
4. Design a single experiment that varies **only that factor**, holding everything else fixed.

This yields a clean, publishable study with a genuine knowledge claim, achievable in a semester, requiring no new theory. It is among the best first-paper designs available.

18.5 Worked derivation

[HYPOTHETICAL] Continuing the running example, with the tallies from Figure 18.1.

Step 3 — limitations with counts and citations.

ID	Limitation	Count	Evidence source
L1	Evaluation protocol permits group identity to leak between partitions	11/15	Matrix column F; mechanism documented by Zech et al. (2018)
L2	Reported improvements are not distinguished from run-to-run variance	15/15 report no test; 13/15 use ≤3 seeds	Matrix columns L, M
L3	Mitigation methods require domain labels at training time	15/15 of the mitigation subset	Each paper's setup section
L4	Decision-relevant quantities (worst-group performance, calibration) unreported	0/15 and 2/15	Matrix column J

Step 4 — unresolved problem. The magnitude of cross-institutional degradation is unknown under leakage-free protocols, and no mitigation has been demonstrated in the label-free regime that deployment actually imposes.

Step 5 — gap statement.

> "Across fifteen studies published between 2022 and 2026 and identified through a logged search of four databases, eleven evaluate chest-radiograph classifiers on random splits in which images from the same institution occur in both training and test partitions — a protocol shown to permit institution-specific confounding (Zech et al., 2018). Consequently the reported AUCs of 0.88 to 0.91 cannot be interpreted as cross-institutional performance, and the magnitude of degradation under leakage-free evaluation remains unquantified. Moreover, all eight mitigation methods we identified require institutional labels during training, whereas provenance metadata is routinely removed before data leaves an institution; and no study reports worst-institution performance or calibration, although these determine clinical usability. This study quantifies the degradation under institution-disjoint protocols with statistical validation, and evaluates a label-free mitigation on worst-institution AUC and calibration."

Step 6 — contributions mapped one-to-one.

Contribution	Answers	Requires new theory?	Compute
C1 Leakage-free multi-institution re-evaluation of five published models, ten seeds, paired tests, confidence intervals	L1, L2	No	Moderate
C2 A label-free adaptation method, with an ablation isolating the mechanism	L3	Modest	Moderate
C3 Public release of institution-disjoint splits, code, and weights	L1, L4	No	Negligible

Two of the three contributions require no new theory and modest compute. This is what a resource-constrained but well-framed project looks like.

18.6 Common mistakes

Mistake	Correction
Looking for a gap before reading	Read fifteen papers and build the matrix first
Looking for a gap in your head rather than in the tallies	Count the columns; the gap is usually audible when you read the tallies aloud
Finding the gap *after* choosing the method	Reverse the order (§18.1)
Treating one paper's limitation as a field gap	Count across papers; require a meaningful share
Ignoring contradictions	They are the richest signal (§18.4)
Not recording verbatim quotes with pages	You will need them as citations in your introduction
Choosing a gap you cannot address with your resources	Re-check the feasibility audit (§4.6)
Claiming a single dramatic gap	Two or three evidenced types are more convincing

Exercises

Exercise 18.1 Complete Steps 1–6 of Figure 18.1 on your own matrix. Produce the tallies, the limitation table with counts, the unresolved problem, the gap statement, and the contribution mapping.

Exercise 18.2 Mine limitations from fifteen papers using §18.2.1. Tag and count them. Report which weakness is most widely shared.

Exercise 18.3 Find one contradiction in your set and complete the four-step procedure of §18.4. Assess whether the resulting experiment is within your resources.

Exercise 18.4 For each contribution you propose, name the limitation it answers. Any contribution that answers none is unnecessary; any limitation with no contribution is an opportunity you are leaving unused.

Chapter 19 — Strong versus Weak Research Gaps

19.1 Why this chapter exists

The difference between a publishable and an unpublishable paper is very often a single paragraph. Reviewers form a judgement about the gap early and read the remainder of the paper in that light. A gap stated with counts and citations produces a sympathetic reading; a gap stated as “existing methods have limitations” produces a sceptical one that the rest of the paper must then overcome.

19.2 Side-by-side comparison

Table 19.1 — Weak and strong gap statements

Weak *[HYPOTHETICAL]*	**Diagnosis**	**Strong** *[HYPOTHETICAL]*
“Many researchers have worked on chest X-ray classification, but there is still room for improvement.”	Vague; no absence identified; unfalsifiable; true of everything	“Across fifteen studies (2022–2026), eleven evaluate on random splits in which images from the same institution appear in both partitions; consequently the reported AUCs of 0.88–0.91 cannot be interpreted as cross-institutional performance, and the degradation under leakage-free protocols remains unquantified.”
“Deep learning has not been applied to our hospital’s dataset.”	Local activity, not knowledge; no mechanism	“Our institution’s imaging protocol differs from those represented in public benchmarks in exposure settings and device mix; since models are known to exploit acquisition artefacts (Zech et al., 2018), it is unknown whether published performance transfers to this acquisition regime.”
“Existing methods have low accuracy.”	Unquantified; low relative to what threshold?	“Reported worst-institution AUC does not exceed 0.83 in any study we identified, whereas triage support has been argued to require sensitivity above 0.90 at fixed specificity; the shortfall is therefore approximately 0.07 AUC and its cause is unexamined.”
“No one has combined transformers with domain adaptation for radiographs.”	Untried combination; no reason to expect anything	“Transformer attention aggregates global context, which we hypothesise increases sensitivity to institution-level acquisition signal relative to convolutional inductive bias; if so, transformers should degrade *more* under institution shift, which is testable and has not been tested.”
“Research on explainability in medical imaging is limited.”	“Limited” is not evidence	“Of fifteen studies, twelve present saliency maps as evidence of clinical validity, and none validates them against expert region annotation; the agreement between saliency and expert attention, and whether it predicts model correctness, is therefore unestablished.”
“Prior work is old; we use a newer model.”	Recency is not a contribution	“Prior comparisons predate architectures with global attention; whether the *conclusion* that invariance methods help survives the change in inductive bias is unknown, since no study evaluates both families under one protocol.”

The pattern is consistent. Strong statements are **longer**, contain **counts**, carry **citations**, name a **mechanism**, specify a **threshold**, and are **falsifiable**.

19.3 The three rescue operations

Almost every weak gap can be repaired by one of three moves.

Rescue 1 — Add a mechanism. Convert “X has not been applied to Y” into “assumption A of X is violated by property P of Y, so we predict effect E.” Now there is a prediction that could fail.

Rescue 2 — Add a count. Convert "existing methods have limitations" into "n of N studies exhibit limitation L." Now the claim is evidenced and auditable.

Rescue 3 — Add a consequence. Convert "accuracy could be improved" into "performance is x against a required y, so decision D cannot currently be supported." Now significance is grounded in a decision rather than asserted with adjectives.

Most strong gap statements use all three.

19.4 The template

> "Across **[N]** studies identified through **[databases, search strings, date range]**, **[pattern, with counts]** holds. Consequently, **[specific quantity or mechanism]** remains **[unquantified / untested / unexplained]**, even though **[why it matters, and to whom]**. This study addresses that by **[action]**."

Every slot must be filled. If you cannot fill the first slot, you have not searched systematically; if you cannot fill the third, you have not identified an absence; if you cannot fill the fourth, you have not established significance.

19.5 How a gap must be evidenced

A gap is a claim about the literature, and like any claim it requires support. Four acceptable forms of evidence:

1. **Counts from a documented search.** The strongest form: "eleven of fifteen studies use protocol P". Requires the search log (§10.9) so a reviewer can audit it.
2. **Citations to authors' own admissions.** "Three of the studies we surveyed explicitly note this limitation [4], [9], [12]." Very strong, because the authors themselves concede it.
3. **A documented negative search.** "We searched four databases with the strings in Appendix A and identified no study that evaluates X under condition Y; the closest is [12], which evaluates X under Z." This is how absence is claimed responsibly.
4. **Your own measurement.** "In our reproduction, the reported result was not recoverable under the described protocol (§V-C)." Strongest of all for reproducibility gaps, but requires the work first.

Not acceptable: "to the best of our knowledge" with no search described; "few studies exist" with no count; an absence claim contradicted by a paper a reviewer finds in five minutes.

19.6 Stress-testing before submission

Before you commit, have a critical reader — supervisor, colleague, or reading group — attempt to break the gap with these six questions. This is the single highest-value review you can obtain, and it is free.

Question	What a failure reveals
Has this been done? Which paper is closest, and how exactly does yours differ?	If you cannot name the closest paper, you have not searched enough
How do you know the gap is real — which count supports it?	If there is no count, the gap is an assertion
Who benefits if you succeed, outside your institution?	If nobody, significance is not established
What is your riskiest assumption?	If you cannot name one, you have not examined your own design
Can you address it with your data, compute, and time?	If not, the gap is real but not *yours* — narrow it
What would a negative result look like, and would it be publishable?	If not, this is advocacy rather than research (§5.3)

Participants consistently want to skip this step. It is the step that prevents six months of work on an indefensible premise.

19.7 Common mistakes

Mistake	Correction
"Room for improvement" as a gap	Apply all three rescues (§19.3)
A gap with no numbers	Count your matrix columns
"No work exists" without a documented search	Use the negative-search form (§19.5, item 3)
A gap requiring resources you lack	Narrow it, or reframe to type 7, 9, 10, or 11 (§17.3.1)
Gap and contributions that do not correspond	Map one-to-one (§18.5)
A gap found after the method was chosen	Reviewers detect retrofitting; reverse the order
Claiming a single gap type when two or three apply	Name them all, with evidence for each
Never stress-testing	Run §19.6 before writing anything else

19.8 Verification checklist for Part VI

- [ ] My gap statement contains all four obligatory components (§17.1).
- [ ] It matches none of the eight non-gaps in §17.2.
- [ ] It is classified against the thirteen types, and I expect two or three.
- [ ] Every absence claim is supported by a count, an admission, a documented negative search, or my own measurement.
- [ ] I have written the tallies as sentences and they appear in the statement.
- [ ] I have collected verbatim limitation quotes with page numbers.
- [ ] I have examined every contradiction in my matrix.
- [ ] My contributions map one-to-one onto counted limitations.
- [ ] The gap is addressable with my actual data, compute, and time.
- [ ] A negative result would still be publishable, and I have written the sentence that would report it.
- [ ] A critical reader has attempted all six stress-test questions and the gap survived.

Exercises

Exercise 19.1 Rewrite your gap statement using the template in §19.4, filling every slot with specifics from your matrix.

Exercise 19.2 Check it against every row of Table 19.1 and apply whichever of the three rescues it needs.

Exercise 19.3 Identify which of the four evidence forms in §19.5 supports each absence claim you make. Any claim with no evidence form must be softened or removed.

Exercise 19.4 Run the six stress-test questions with a partner. Record which question was hardest; that is where your project is weakest, and it is where a reviewer will begin.

PART VII — NOVELTY AND CONTRIBUTION

> Part VII addresses the two questions a reviewer asks first and that authors answer worst: *what is new here?* and *what does the field gain?* Chapter 20 dissects novelty into nine recognisable kinds and confronts the most common misconception in applied research — that combining existing components constitutes novelty. Chapter 21 teaches contribution statements as a specific writing genre with its own rules.
>
> The gap you established in Part VI says what is missing. Novelty and contribution say what you are adding. These are different statements and both must be written explicitly.

Chapter 20 — Research Novelty

20.1 Definition

Definition. Novelty is the property of a contribution being new relative to the accessible published record — not new to the author, and not merely newly combined.

Purpose. Novelty is the price of entry to the permanent record. The record exists to accumulate knowledge; a paper that adds nothing new adds only volume. This is why "insufficient novelty" is the most common substantive rejection at selective venues, and why it is so often disputed: authors experience the *effort* of their work and assume it corresponds to novelty, while reviewers assess only the *increment*.

20.2 Novelty is a spectrum, not a binary

A great deal of unnecessary anxiety comes from treating novelty as an all-or-nothing property. It is better understood as degrees.

Figure 20.1 — Degrees of novelty

```
LOW ─────────────────────────────────────────────────────────────► HIGH

┌─ REPLICATION ─────────────────────────────────────────────────────┐
│ Same method, same data, independent execution.                    │
│ Publishable when the original is influential and unverified.      │
└───────────────────────────────────────────────────────────────────┘

┌─ INCREMENTAL IMPROVEMENT ─────────────────────────────────────────┐
│ Established problem, established route, measurably better result. │
│ Publishable if the gain survives equal tuning, seeds, and tests.  │
└───────────────────────────────────────────────────────────────────┘

┌─ SUBSTANTIVE CONTRIBUTION ────────────────────────────────────────┐
│ A new method, resource, evaluation protocol, or empirical finding.│
│ The category into which most good published research falls.      │
└───────────────────────────────────────────────────────────────────┘

┌─ REFRAMING ───────────────────────────────────────────────────────┐
│ Shows the community was measuring, assuming, or asking wrongly.   │
│ Rare; disproportionately cited.                                   │
└───────────────────────────────────────────────────────────────────┘

┌─ PARADIGMATIC ────────────────────────────────────────────────────┐
│ Changes what the field considers the problem to be.               │
│ A handful per decade per field. Not a reasonable PhD target.      │
└───────────────────────────────────────────────────────────────────┘
```

The practical point. A doctorate does not require reframing or paradigm change. It requires a defensible substantive contribution — and the expectation that it must be revolutionary is a documented cause of paralysis and delay. Aim for *substantive*, execute it impeccably, and let the field decide whether it turns out to be more.

20.3 The nine kinds of novelty

Each kind carries a distinct evidential obligation. Knowing which you are claiming tells you what experiments you owe.

#	Kind	You are claiming	Evidence obligation	Characteristic failure
1	**Algorithmic**	A new procedure for computing something	Correctness; complexity; comparison with the algorithm it replaces	Presenting a known algorithm under a new name
2	**Architectural**	A new arrangement of model components	Ablation isolating the new arrangement at matched parameter count	Concatenating two published blocks and calling it an architecture
3	**Dataset / resource**	New data, annotations, or a benchmark	Construction protocol, annotation agreement, quality audit, licence, baseline results	Releasing data with no documentation or baselines
4	**Application**	A method transferred to a genuinely different setting	A *mechanism* argument for why transfer is non-trivial, plus measurement of what changes	"Apply X to Y" with no mechanism (§5.5)
5	**Optimisation**	Better cost for equal quality, or better quality for equal cost	Cost measured honestly — parameters, FLOPs, latency, memory, energy — on stated hardware	Reporting accuracy gains while concealing added cost
6	**Evaluation**	A protocol or metric that measures what matters better	Demonstration that current evaluation is inadequate; validation that yours is better	Proposing a metric with no validation that it measures the construct
7	**Integration**	A combination whose behaviour is not predictable from its parts	A prediction about the interaction, and an experiment that could refute it	Combination without a prediction (§20.4)
8	**Theoretical**	A proof, bound, or formal characterisation	Correct derivation; assumptions stated; the bite of the assumptions acknowledged	A correct result whose assumptions exclude all practical cases
9	**Empirical**	A measured fact about the world or about existing methods	Careful measurement, controls, variance, replication across settings	Measurement without controls, presented as discovery

Kinds 3, 5, 6, and 9 are systematically undervalued by early-career researchers and systematically valued by editors, because they change what practitioners do. They are also the kinds most achievable without large compute.

20.4 Why combination is not automatically novelty

This is the most important section in the chapter, because "we combine A and B" is the single most common weak novelty claim in applied machine learning.

Consider the claim: *"We propose a novel hybrid CNN–LSTM–Attention model."* A reviewer's internal response is: CNNs, LSTMs, and attention are each a decade or more old, and their combination has been published many times across many tasks. What is new?

The problem is not that combinations are worthless. It is that a combination, by itself, makes **no claim that could be false**. If you assemble three components and report a number, the reader learns that this assembly produces that number on that data — which is a fact about your run, not knowledge about the world.

The conversion. A combination becomes a contribution when you state a **prediction about the interaction** and test it.

Combination as assembly	Combination as hypothesis
"We combine a CNN with an LSTM."	"Convolutional features capture local morphology but discard temporal ordering, while recurrent state captures ordering but degrades on long sequences. If the two failure modes are independent, the combination should be robust to both — which predicts that the gain over each component alone should be *largest* at intermediate sequence lengths and vanish at the extremes. We test this by crossing sequence length with noise level."
"We add attention to improve performance."	"If attention helps by suppressing background regions, then its benefit should scale with the proportion of background in the image, and masking the background should eliminate the benefit. We measure both."
"We use transfer learning with fine-tuning."	"Pretraining transfers low-level filters but not domain-specific texture statistics; we predict that freezing early layers preserves the benefit while freezing late layers destroys it, and we measure the crossover point."

Notice what each rewrite adds: a **mechanism**, a **prediction that could be wrong**, and a **measurement that would detect it being wrong**. That is the difference between engineering assembly and scientific contribution, and it usually costs one extra experiment rather than a new idea.

20.5 Establishing novelty in writing

Novelty is not asserted; it is *positioned*. Three moves, in order.

Move 1 — Name the closest prior work explicitly. Reviewers trust authors who identify their nearest competitor. Concealing it looks either ignorant or evasive, and a specialist will find it regardless.

> *"The closest prior work is [12], which also infers latent groupings for invariance training. It differs from ours in requiring a known group count and in clustering on raw inputs rather than on embedding statistics; §V-D compares the two directly."*

Move 2 — State the delta precisely. One sentence naming exactly what is different — a component, an assumption relaxed, a condition tested, a quantity measured.

Move 3 — Say why the delta matters. A difference that changes nothing is not a contribution. Connect the delta to a consequence: something now possible, cheaper, more reliable, or newly known.

On the phrase "to the best of our knowledge". This is acceptable *only* when preceded by a documented search, and it is stronger when paired with the nearest-neighbour formulation of §19.5. "To the best of our knowledge, no prior study evaluates X under condition Y; the closest is [12], which evaluates X under Z" is defensible. The same phrase with nothing behind it is a liability.

20.6 Common mistakes

Mistake	Why it fails	Correction
“Novel” as an adjective applied to your own work	Assertion, not evidence	Delete the word; demonstrate the delta instead
Combination without a prediction	Nothing could be false	Apply §20.4
Claiming architectural novelty for a concatenation	Precedent will be found	Reclassify honestly — often it is an application or empirical contribution
Hiding the nearest prior work	Reviewers find it; credibility collapses	Name it and differentiate (§20.5)
Claiming novelty of *effort*	Reviewers assess increment, not labour	Identify what is *learned*
Claiming optimisation novelty without cost measurement	The claim is unverifiable	Measure parameters, FLOPs, latency, memory
“First work in this area”	Almost always falsifiable	Use the bounded, scoped form
Novelty claimed in the abstract but never isolated experimentally	Ablation absent	Every novelty claim needs an experiment that isolates it

20.7 Verification checklist

- [ ] I can name which of the nine kinds of novelty I claim.
- [ ] I know the evidence obligation for that kind, and I have planned it.
- [ ] I can name the single closest prior work and state the delta in one sentence.
- [ ] If my contribution is a combination, I have stated a falsifiable prediction about the interaction.
- [ ] I have an experiment that isolates my novel component from everything inherited.
- [ ] If I claim efficiency, I measure cost on stated hardware.
- [ ] I have not used the word “novel” as a substitute for demonstrating novelty.
- [ ] Any absolute claim of priority is scoped and supported by a documented search.

Exercises

Exercise 20.1 Classify your contribution against the nine kinds in §20.3. If you cannot choose one, your novelty claim is not yet formed.

Exercise 20.2 Name your closest prior work. Write the delta in one sentence, and the consequence of the delta in a second. If you cannot name a closest work, return to Part III.

Exercise 20.3 If your work combines components, apply §20.4: write the mechanism, the prediction, and the experiment that could refute it.

Exercise 20.4 For your claimed novelty, name the specific experiment that isolates it. If no such experiment exists in your plan, add it now — a reviewer will demand it.

Chapter 21 — Research Contribution

21.1 Definition and purpose

Definition. A contribution is a transferable statement of what the field gains from your work, expressed so that its truth can be checked against your evidence.

Purpose. Contribution statements do heavy structural work in a paper. They appear as a numbered list at the end of the introduction, they determine the shape of the results section, they are what the conclusion restates, and they are what a reviewer scores. They are also what a hiring or promotion committee reads when they read only your abstract.

21.2 The five types, and what each requires

Type	Statement form	Evidence required	Example *[HYPOTHETICAL]*
Methodological	A procedure others can adopt	Fair comparison; ablation; released code	“A site-clustering regulariser that requires no institutional metadata at training time”
Dataset / resource	Data or a benchmark others can use	Construction protocol; annotation agreement; licence; baselines	“A benchmark of 4,000 code-mixed paraphrase pairs at four graded obfuscation levels, with inter-annotator agreement of κ = 0.81”
Experimental / empirical	A measured fact about the world	Controls; variance; replication across settings	“The first leakage-free multi-institution quantification of degradation for five widely used architectures”
Theoretical	A formal result	Proof; stated assumptions	“A bound on the excess risk of invariance training under bounded group imbalance”
Practical	An artefact or protocol with demonstrated utility	Deployment evidence; cost measurement; user evidence where relevant	“An inference configuration attaining worst-site AUC above 0.85 within a 50 ms budget on commodity edge hardware”

Most papers make two or three contributions of *different* types. A paper claiming four methodological contributions is usually claiming one and padding.

21.3 The anatomy of a contribution statement

A well-formed contribution has four elements. Missing any one produces a recognisable defect.

```
[ARTEFACT OR FINDING]  +  [SCOPE]  +  [QUANTIFICATION]  +  [SIGNIFICANCE]

"A label-free adaptation module    ← artefact
 for multi-institution chest        ← scope
 radiograph classification
 that recovers 0.041 ± 0.009        ← quantification (with uncertainty)
 worst-institution AUC over the
 strongest of four baselines at
 equal parameter count,
 thereby attaining most of the      ← significance
 benefit of methods requiring
 privileged site labels without
 using them."
```

Missing element	Resulting defect
Artefact	Vague — the reader cannot tell what they are being given
Scope	Overclaim — invites a counterexample outside your conditions
Quantification	Unverifiable — “improves performance” cannot be checked
Significance	Uninterpretable — the reader cannot tell why the number matters

21.4 Weak and strong contribution statements

Table 21.1 — Weak and strong contribution statements

▯ **Weak** *[HYPOTHETICAL]*	**Diagnosis**	▯ **Strong** *[HYPOTHETICAL]*
"We propose a novel deep learning framework for image classification."	"Framework" conceals the artefact; "novel" is asserted; no scope, quantification, or significance	"We propose a clustering-based invariance regulariser that operates without domain labels, and show it recovers 82% of the worst-group improvement obtained by label-supervised adversarial training on three public datasets."
"Our method outperforms state-of-the-art methods."	Unbounded; unquantified; invites refutation	"Our method improves worst-institution AUC by 0.041 (95% CI 0.032–0.050) over the strongest of four baselines tuned under an identical 50-trial budget."
"We conduct extensive experiments."	Effort, not knowledge	"We report the first evaluation of these five architectures under institution-disjoint splits across three datasets with ten seeds and paired significance testing, establishing a mean degradation of 0.116 AUC (95% CI 0.104–0.128)."
"We provide a comprehensive survey of the field."	"Comprehensive" is unverifiable	"We synthesise 68 studies retrieved by a logged four-database protocol, and show that 44 of them evaluate on splits that permit group leakage, which reconciles the apparent contradiction between [refs]."
"We release our code."	An artefact, not yet a contribution	"We release institution-disjoint split definitions, training code, and trained weights, enabling protocol-consistent comparison — currently impossible because no two of the surveyed studies share a split definition."
"This work has significant practical implications."	Asserted significance	"Because worst-site rather than mean performance governs deployment risk, the 0.116 gap implies that a model reported at 0.90 AUC may present as 0.78 at an unseen institution — a difference that changes the triage threshold a hospital should adopt."
"We solve the domain shift problem."	Overclaim	"We reduce but do not eliminate cross-institution degradation; approximately 0.075 AUC of the observed 0.116 gap remains unaddressed, and we characterise where it concentrates."

The pattern is consistent throughout this handbook: strong statements are longer, carry numbers with uncertainty, name their scope, and concede limits. **Calibration reads as competence.** Overclaiming invites a reviewer to hunt for the counterexample, and they will find one.

21.5 Writing the contribution list

Procedure.

1. Write one contribution per **objective** (Chapter 7). If an objective yields no contribution, either it is not producing knowledge or you are under-claiming.
2. Order them by **importance**, not chronology. The reader's attention is highest at the first item.
3. Give each a **label** — C1, C2, C3 — and use those labels consistently in the results section and the conclusion. This is a small formatting habit with a large effect on how coherent the paper reads.
4. Include the **artefact release** as a numbered contribution where you have one; it is genuinely valuable and reviewers credit it.
5. Verify each against the four-element anatomy of §21.3.
6. Verify that each maps to a **results subsection** that supplies its evidence.

A worked list. *[HYPOTHETICAL]*

> The contributions of this paper are:
>
> **(1)** the first leakage-free, multi-institution quantification of performance degradation for five widely used chest-radiograph architectures, with ten seeds, paired significance tests, and confidence intervals (§V-A);
>
> **(2)** CLUSTER-DG, an adaptation method that requires no institutional metadata and recovers 0.041 ± 0.009 worst-institution AUC over empirical risk minimisation at equal parameter count (§V-B);
>
> **(3)** an ablation isolating the contributions of cluster granularity, invariance weight, and augmentation, which shows that the ramping schedule accounts for approximately one third of the gain (§V-C);
>
> **(4)** a public release of institution-disjoint split definitions, code, and trained weights, enabling protocol-consistent comparison in future work (§VI).

Each item names an artefact or finding, bounds its scope, quantifies where quantification is possible, and points to the evidence.

21.6 Calibration: a short discipline

Three habits prevent nearly all overclaiming.

Bound every superlative. Not “the best”, but “the best of the four methods we evaluated under this protocol”. Not “the first”, but “to our knowledge the first under condition Y; the closest prior work is [12]”.

Report what you did not do. One sentence in the contribution list or the limitations paragraph — “we did not evaluate methods requiring paired multi-site supervision” — removes an entire category of objection, because the reviewer’s observation is already yours.

Prefer a smaller claim you can defend completely. A tightly bounded finding is more citable than a grand one that a later paper narrows. In the long run, calibration is also self-interested: the reputational cost of an overturned claim is much larger than the benefit of a bolder abstract.

21.7 Common mistakes

Mistake	Correction
Contributions that restate the method three times	Merge into one; find genuinely distinct gains
A contribution with no corresponding results subsection	Add the evidence or delete the claim
“Extensive experiments” as a contribution	Experiments are method; the *finding* is the contribution
Ordering contributions chronologically	Order by importance
Contribution list inconsistent with the abstract or conclusion	Make them substantively identical; reviewers check
Omitting artefact release from the list	Include it; it is a real contribution
Unbounded superlatives	Bound every one (§21.6)
No quantification anywhere in the list	Add numbers with uncertainty wherever available

21.8 Verification checklist for Part VII

- [ ] Each contribution names its type from §21.2.
- [ ] Each has all four elements of §21.3: artefact, scope, quantification, significance.
- [ ] Each maps one-to-one onto an objective and to a results subsection.
- [ ] Contributions are labelled C1…Cn and those labels recur in results and conclusion.
- [ ] Every superlative is bounded by the conditions actually tested.
- [ ] At least one sentence states what the work does *not* address.
- [ ] The list in the introduction matches the abstract and the conclusion in substance.

- [ ] I have not used “novel”, “extensive”, or “comprehensive” as evidence.

Exercises

Exercise 21.1 Write two to four contribution statements for your work, each containing all four elements of §21.3.

Exercise 21.2 For each, name the results subsection that will provide its evidence. Any contribution without one is currently unsupported.

Exercise 21.3 Rewrite each contribution twice: once deliberately overclaimed, once deliberately under-claimed. Choose the version you could defend under hostile questioning, and note how close it is to the under-claimed one.

Exercise 21.4 Write the one sentence stating what your work does not address. Keep it in the paper.

PART VIII — RESEARCH METHODOLOGY

Part VIII covers the production of trustworthy evidence. Chapter 22 treats methodology as the logical bridge from objectives to conclusions. Chapters 23 and 24 address data — selection, licensing, quality, and the preprocessing decisions that silently determine your results. Chapter 25 covers model selection and its scientific justification. Chapter 26 specifies experimental design: splits, tuning, baselines, ablations, and reproducibility controls.

A single theme runs through all five chapters: **the credibility of your result is determined by decisions made before you run anything.** No amount of careful writing repairs an experiment in which the baseline was under-tuned or the split leaked.

Chapter 22 — Designing the Methodology

22.1 What methodology is, and what it is not

Definition. The methodology is the defensible logical chain connecting your research questions to the evidence you will present.

It is emphatically **not** a list of software. "Methodology: Python, TensorFlow, Google Colab" describes an *environment*. A reviewer reading that sentence learns nothing about whether your conclusions follow from your procedure.

Figure 22.1 — Methodology as the bridge from objectives to evidence

```
OBJECTIVES  ─────────────────────────────────────────────►  CONCLUSIONS
 (Chapter 7)                                                  (Chapter 37)
     │                                                              ▲
     │                     THE METHODOLOGY                          │
     │                                                              │
     │   ┌──────────────────────────────────────────────────┐      │
     └──►│ 1  DESIGN     What kind of study? What is          │──────┘
         │               manipulated, what is held fixed?    │
         │ 2  DATA       Which datasets, why those, what     │
         │               splits, what known biases?          │
         │ 3  PROCEDURE  Preprocessing, model, training,     │
         │               tuning protocol                     │
         │ 4  CONTROLS   Baselines, seeds, equal budgets,    │
         │               blinding where applicable           │
         │ 5  ANALYSIS   Metrics, statistical tests,         │
         │               error analysis                      │
         │ 6  THREATS    Validity limits and mitigations     │
         └──────────────────────────────────────────────────┘

TEST OF ADEQUACY: for each objective, can you trace a path through all six
layers to a specific number or figure in your results? If not, the design
does not yet answer the objective.
```

22.2 The six layers, with the questions each must answer

Layer	Questions it must answer	Where it appears in the paper
Design	Comparative, ablative, observational, or proof-based? What is the independent variable? What is held constant?	Methodology, opening paragraph
Data	Which datasets and why? What is the split unit? What biases are documented?	Dataset subsection; Table I
Procedure	Exactly what happens to an input, from raw file to prediction?	Methodology, main body; Algorithm 1
Controls	Which baselines, tuned how? How many seeds? What is equalised?	Experimental setup
Analysis	Which metrics, why those, which statistical test, what effect size?	Experimental setup; Results
Threats	What could make this conclusion wrong, and what did you do about it?	Discussion; Threats to validity

22.3 Procedure: designing a methodology from objectives

Step 1 — Write each objective as a required comparison. An objective such as *"To quantify degradation when random splits are replaced by institution-disjoint splits"* implies a comparison in which the *only* thing that changes is the split protocol. Everything else — architecture, preprocessing, tuning budget, seeds — must be held fixed. Naming the comparison first prevents the common error of designing an experiment in which several things differ at once, making the result uninterpretable.

Step 2 — Identify the confounds the comparison must survive. For the example above: different tuning effort between conditions, different numbers of training images after re-splitting, different class balance across institutions. Each confound needs an explicit control, and each control belongs in the paper.

Step 3 — Specify the pipeline as a sequence of transformations. Write it as a chain, then check that each arrow is fully specified:

```
raw file → decode → filter (which records excluded, why) → resize/tokenise
→ normalise (statistics fitted on WHAT?) → augment (train only?) → model
→ output → threshold (chosen on WHAT?) → metric
```

Every arrow that you cannot describe in one sentence is a reproducibility hole and, frequently, a leakage risk.

Step 4 — Choose the controls before the results exist. Baselines, seed count, tuning budget, and statistical test are all design decisions, not reporting decisions. Choosing them after seeing outcomes is the mechanism of p-hacking (§3.4).

Step 5 — Write the threats-to-validity list. Do this *before* running experiments. It routinely reveals a design flaw while it is still cheap to fix.

22.4 System architecture, framework, and workflow

Three words are used loosely; distinguishing them helps you write clearly.

- A **system architecture** describes components and the data flowing between them. It answers *what is connected to what*.
- A **workflow** or **pipeline** describes an ordered sequence of operations. It answers *what happens in what order*.
- A **framework** — a word best used sparingly — implies a general structure into which different components can be substituted. Only use it if substitution is actually part of your contribution; otherwise name the thing precisely (a module, a loss, a protocol, a pipeline).

Recommendation. Present one overview figure early in the methodology (Figure 1 of the paper), with the novel component visually distinguished and identified as such in the caption. Reviewers form their understanding of your method from that figure. Chapter 38 covers how to draw it.

22.5 Design rationale: the sentences that separate research from engineering

For every non-obvious choice, the methodology should contain one sentence of the form:

> *"We use [choice] because [property of the problem] implies [expected consequence]."*

Compare:

> □ *"We add a clustering branch that assigns each sample to one of K groups."*
>
> □ *"We add a clustering branch over channel-wise embedding statistics, because acquisition artefacts are known to dominate low-order feature statistics; if that is so, clusters recovered from those statistics should approximate institutional identity closely enough to support an invariance penalty without any metadata. §V-C tests this by comparing inferred clusters against true site labels."*

The second version states a belief that could turn out to be false and names the experiment that checks it. That is the difference between describing a system and doing science, and it costs one sentence plus one experiment.

22.6 Common mistakes

Mistake	Correction
Methodology as a tool list	Describe design, controls, and analysis — not the environment
A comparison in which several factors differ	Isolate one factor per comparison
No design rationale	Add one “because” sentence per non-obvious choice
Controls chosen after seeing results	Pre-specify baselines, seeds, budget, and test
Pipeline arrows that cannot be described in a sentence	Specify them; each is a reproducibility hole
“Framework” used to conceal what the contribution is	Name the artefact precisely
Threats to validity written last	Write them first; they reveal design flaws cheaply

Exercises

Exercise 22.1 For each objective, write the comparison it requires and list what must be held constant.

Exercise 22.2 Write your pipeline as a chain of arrows and mark every arrow you cannot yet describe in one sentence.

Exercise 22.3 Write one design-rationale sentence for each non-obvious choice in your method, in the “because … implies …” form.

Exercise 22.4 Write your threats-to-validity list now, before running experiments. Note any threat with no mitigation.

Chapter 23 — Dataset Selection

23.1 Why this decision dominates your results

Dataset choice constrains everything downstream: what claims are possible, what baselines are comparable, which metrics are appropriate, and whether anyone can reproduce your work. It is also nearly irreversible — discovering in month six that your dataset cannot support your claim usually means restarting.

Table 23.1 — Dataset selection criteria

Criterion	What to check	Why it matters
Relevance	Does it contain the phenomenon you study?	A dataset lacking the condition cannot test your claim
Standard use	Is it the benchmark your field uses?	Non-standard data makes your results incomparable
Size	Enough for the model class and the effect size?	Under-powered studies detect nothing reliably
Label quality	Who labelled it, how, with what agreement?	Label noise caps achievable performance and can invert conclusions
Class distribution	Imbalance ratio per class	Determines metric choice (Chapter 27)
Grouping structure	Multiple records per subject, site, author, session?	Determines the split unit — the most common validity failure
Documentation	Is there a datasheet or paper describing construction?	Undocumented data cannot be reasoned about
Licence	Research use, redistribution, derivatives, commercial terms	Determines what you may publish and release
Access route	Open download, registration, data-use agreement, application	Determines your timeline
Ethics	Consent basis; identifiability; approval needed?	Determines whether the study is permissible at all
Known biases	Documented confounds and artefacts	Prevents rediscovering a known artefact as a finding

23.2 Public, private, and synthetic data

Public datasets are the default choice and should be your first option. They make results comparable, permit independent verification, and eliminate access risk. Their weakness is saturation — headline performance on a well-worn benchmark is usually near a ceiling, so a pure performance claim is hard. This is precisely why *evaluation* and *generalisation* contributions on public data are attractive (§17.3.1): the data are shared, so the contribution is the protocol rather than the access.

Private or proprietary datasets allow genuinely new questions and carry three costs that must be confronted honestly: nobody can verify your results; nobody can compare against them; and reviewers discount claims they cannot check. If you must use private data, mitigate: report descriptive statistics fully, release the code even when you cannot release the data, evaluate additionally on at least one public dataset, and state the limitation explicitly.

Synthetic data is legitimate for controlled study of a mechanism — you can vary one factor exactly, which real data rarely permits. Its threat is that conclusions may describe the generator rather than the world. The standard mitigation is to pair synthetic results with real-data validation, and to state which conclusions rest on which.

23.3 Data leakage: the failure that invalidates results silently

Definition. Leakage occurs when information from the evaluation partition influences training or model selection, so that measured performance overstates real performance.

Leakage is the most consequential technical error in applied machine learning because it is invisible: nothing crashes, and results *improve*.

Figure 23.1 — Data leakage pathways

```
□ GROUP LEAKAGE — the most common
   Records sharing a group (patient, site, author, session, project, device)
   appear in both train and test. The model recognises the group, not the target.
   FIX: split by the group, not by the row.

□ PREPROCESSING LEAKAGE
   Normalisation statistics, scalers, vocabularies, or feature selection
   fitted on all data before splitting.
   FIX: fit on training folds only; apply to validation and test.

□ RESAMPLING LEAKAGE
   Oversampling (e.g. synthetic minority generation) applied before splitting,
   so synthetic copies of test records appear in training.
   FIX: resample inside the training fold only, after the split.

□ TEMPORAL LEAKAGE
   Future records used to predict the past because the data were shuffled.
   FIX: split by time; train on earlier, test on later.

□ DUPLICATE LEAKAGE
   Exact or near-duplicate records straddling the split.
   FIX: deduplicate before splitting, including near-duplicates.

□ TARGET LEAKAGE
   A feature encodes the label (an identifier, a post-outcome measurement,
   a filename convention).
   FIX: audit features for post-hoc information.

□ SELECTION LEAKAGE (tuning on test)
   The test set consulted repeatedly during development.
   FIX: touch the test set once, at the end. Log every access.
```

Pathway □ deserves particular emphasis because it is both the most frequent and the most field-general. Zech et al. (2018) demonstrated the consequence concretely in medical imaging: models can key on institution-specific signal, so evaluation that mixes institutions across partitions measures something other than the transfer being claimed. The same structure occurs with multiple utterances per speaker, multiple commits per project, multiple pupils per classroom, and multiple readings per sensor.

The declarative sentence reviewers look for. Include it verbatim:

> *"All preprocessing statistics, resampling, and feature selection were fitted on training folds only. Splits are patient-disjoint and institution-disjoint: no patient and no institution appears in more than one partition."*

23.4 Licensing, ethics, and what you may publish

Licensing. Read the actual licence, not the summary on an aggregator. Distinguish: research-only versus commercial use; permission to redistribute the data; permission to publish derivatives such as split files, features, or trained weights; attribution requirements. **Recommendation:** if you cannot redistribute the data, you can almost always redistribute *split definitions* (lists of record identifiers), which is the single most valuable reproducibility artefact you can release.

Ethics. Human-subject data, patient data, and identifiable data generally require institutional review-board or ethics-committee involvement, and the requirement is determined by your institution and jurisdiction rather than by whether the data are already public. **[VERIFY] with your own committee.** Two points that beginners routinely miss: publicly available data are not automatically exempt from review, and web-scraped data may be subject to terms of service and data-protection law independently of any ethics approval.

Declarations. You will need statements on ethics approval, consent, data availability, and code availability at submission (§53.4). Draft them when you choose the dataset, not on submission day.

23.5 Common mistakes

Mistake	Correction
Splitting by row when data are grouped	Split by group; state the unit explicitly
Fitting scalers or vocabularies before splitting	Fit on training folds only
Oversampling before splitting	Resample inside the training fold
Shuffling time series	Split temporally
Using a non-standard dataset without justification	Add the field-standard benchmark, even if unfavourable
Not reporting the imbalance ratio	Report it; it determines metric validity
Discovering the licence at submission	Read it in week one
Reporting accuracy on data with known artefacts as a finding	Check documented biases first
“Data available on request”	Deposit in a repository with a DOI, or state a specific, justified restriction

Exercises

Exercise 23.1 Complete Table 23.1 for every dataset you intend to use, with specific answers.

Exercise 23.2 Identify your grouping structure and state the split unit. If records share any group, a row-level random split is invalid.

Exercise 23.3 Audit your pipeline against all seven leakage pathways in Figure 23.1. Write down which apply and what you did.

Exercise 23.4 Draft your data-availability and ethics statements now.

Chapter 24 — Data Preprocessing

24.1 Why preprocessing must be documented

Preprocessing decisions frequently affect results more than model choice, and they are the most under-reported part of most papers. Two consequences follow. First, undocumented preprocessing is the leading known cause of irreproducibility — when a reported number cannot be recovered from released code, a preprocessing difference is the most common explanation. Second, preprocessing is where leakage enters (§23.3).

Table 24.1 — Preprocessing decisions and their documentation requirements

Operation	Decisions to record	Leakage risk	Common error
Cleaning / filtering	Exclusion criteria; resulting counts at each step	Low	Filtering on a property correlated with the label
Missing values	Mechanism assumed; imputation method; whether an indicator was added	**High** — imputation fitted on all data	Mean imputation computed over train and test together
Normalisation / standardisation	Which statistics, computed on which partition	**High**	Statistics computed on the full dataset
Encoding	Scheme; how unseen categories are handled	Medium	Vocabulary built from all data
Resizing / resampling signals	Target size; interpolation; aspect-ratio handling	Low	Unreported interpolation, which changes results measurably
Tokenisation	Tokeniser identity and version; vocabulary source	Medium	Vocabulary fitted including test text
Augmentation	Exact transforms, parameters, probabilities; **train only**	Medium	Augmentation applied at test time
Class balancing	Method; applied **after** splitting, inside training folds	**Very high**	Synthetic oversampling before the split
Feature selection	Criterion; fitted on training folds only	**Very high**	Selection using labels of the whole dataset
Dimensionality reduction	Method; number of components; fitted on train only	**High**	Projection fitted on all data
Deduplication	Exact or near-duplicate; threshold	Low	Skipped entirely

The three rows marked *very high* or *high* account for most invalidated student results. The rule is uniform and simple: **anything that learns from data — a scaler, an imputer, a vocabulary, a selector, a projection, a resampler — is a model, and must be fitted on training data only.**

24.2 Documenting preprocessing: weak and strong

[HYPOTHETICAL] □ Weak. *“The images were preprocessed and resized. Data augmentation was applied to increase the dataset. The data was divided into training and testing sets.”*

Unanswerable questions: resized to what, with what interpolation? Which augmentations, with what probabilities? Was augmentation applied to test data? What split ratio, and split by what unit?

[HYPOTHETICAL] □ Strong. *“Images are resized to 224×224 using bilinear interpolation with aspect ratio preserved by centre cropping. Intensities are normalised using channel statistics computed on the training partition only ($\mu = 0.503$, $\sigma = 0.291$). Training augmentation comprises random resized crop (scale 0.8–1.0), horizontal flip ($p = 0.5$), and rotation ($\pm 10°$); no augmentation is applied at validation or test time. Records are deduplicated by perceptual hash before splitting (412 near-duplicates removed). Splits are patient- and institution-disjoint in the* *ratio 70/10/20 by patient count. Class prevalence ranges from 1.2% to 38.5%, which motivates the per-class metrics reported in §V."*

The strong version is reproducible, states where statistics were fitted, confines augmentation to training, names the split unit, and connects the class distribution to the metric choice.

24.3 The five sentences every data subsection needs

1. **Provenance** — dataset name, version, size, source, licence.
2. **Filtering** — what was excluded and why, with resulting counts.
3. **Transformations** — exact operations and parameters, and **which partition any fitted statistics came from**.
4. **Split protocol** — ratios *and the unit of splitting*.
5. **Distribution** — class balance, and its consequence for metric choice.

Exercises

Exercise 24.1 Write the five sentences of §24.3 for your own data.

Exercise 24.2 For every operation in your pipeline that *learns* anything from data, state which partition it was fitted on. Any answer other than "training only" is a defect.

Exercise 24.3 Take the weak paragraph in §24.2 and rewrite it for your dataset at the strong version's level of specificity.

Chapter 25 — Model Selection

25.1 Justifying model choice scientifically

The weakest sentence in applied papers is *"we use model M because it has achieved good results in recent literature."* That is an appeal to popularity. A scientific justification connects a property of the **problem** to a property of the **model**.

Table 25.1 — Model families and the conditions that justify them

Family	Justified when	Inductive bias	Characteristic weakness
Trivial / majority baseline	Always, as a floor	None	— (its purpose is to bound the task's difficulty)
Linear / logistic models	Few samples; interpretability required; near-linear structure	Linearity, additivity	Cannot represent interactions
Tree ensembles (random forests, gradient boosting)	Tabular, heterogeneous features; moderate sample size	Axis-aligned partitions	Poor on raw high-dimensional signals
Convolutional networks	Signals with local structure and translation-equivariant patterns	Locality, weight sharing	Limited long-range context
Recurrent networks (RNN, LSTM, GRU)	Sequential data with order dependence; streaming	Sequential state, recency	Degradation on long dependencies; limited parallelism
Transformers	Long-range dependence matters; data are plentiful; pretraining available	Global attention; weak locality prior	Data- and compute-hungry; quadratic attention cost in sequence length
Transfer learning / fine-tuning	Target data are scarce and the source domain is related	Inherited from pretraining	Negative transfer under domain mismatch; pretraining data may overlap your test set
Ensembles	Variance reduction; competition settings	Averaging	Multiplied inference cost; obscures mechanism
Hybrid architectures	Two failure modes are complementary **and you can predict the interaction**	Combined	Frequently claimed, rarely justified (§20.4)

25.2 Procedure for justifying a choice

1. **Name the property of your data** that dictates the requirement — long-range dependence, local texture, grouped structure, tabular heterogeneity, extreme scarcity.
2. **Name the model property** that addresses it.
3. **Name what you give up** — every inductive bias is a restriction, and stating the trade-off demonstrates understanding.
4. **Include a baseline from a different family**, so that the comparison tests your reasoning rather than assuming it.
5. **State the cost** in parameters, memory, and latency.

[HYPOTHETICAL] Worked example. *"Radiographic findings are characterised by localised texture at multiple scales, which motivates a convolutional backbone; we nonetheless include a vision transformer because global attention may capture the bilateral comparisons radiologists report using, and because it allows us to test whether stronger global context increases sensitivity to institution-level acquisition signal (§V-D). We additionally report a gradient-boosted baseline over handcrafted features as a non-deep reference point, since the literature we surveyed contains no such comparison."*

25.3 Why a strong classical baseline is not optional

Recommendation, with an evidential basis. Include a well-tuned classical baseline. Structured re-evaluation studies in several areas of machine learning have found that reported advantages of newer, more complex methods narrowed or disappeared when simpler baselines were tuned with comparable effort — reported for recommender systems by Dacrema et al. (2019), for language-model architectures by Melis et al. (2018), and for generative adversarial models by Lucic et al. (2018).

The practical consequence for you is twofold. If a reviewer suspects you avoided a strong simple baseline, your improvement claim is dead. And if the classical baseline *wins*, that is a genuine and publishable finding — not a failure.

25.4 Common mistakes

Mistake	Correction
Justifying by popularity	Connect a data property to a model property
No baseline from a different family	Add one; it tests your reasoning
Omitting the trivial baseline	Include it; it bounds task difficulty
Omitting a tuned classical baseline	Include it (§25.3)
Claiming a hybrid without predicting the interaction	Apply §20.4
Comparing models at different parameter counts	Match capacity, or report the mismatch prominently
Ignoring pretraining-data overlap with your test set	Check and disclose; it is a leakage pathway
Not reporting cost	Report parameters, FLOPs, latency, memory

Exercises

Exercise 25.1 Justify your model in the four-part form of §25.2, including what you give up.

Exercise 25.2 Identify a strong classical baseline for your task and commit to tuning it as hard as your own method.

Exercise 25.3 If you use pretrained weights, check whether the pretraining corpus plausibly overlaps your evaluation data, and write a sentence disclosing what you found.

Chapter 26 — Experimental Design

26.1 The seven design decisions

Decision	Standard to meet
Datasets	Two or more with differing characteristics; include the field-standard benchmark
Splits	Grouped and, where relevant, temporal; fixed; identical across all methods; published
Baselines	Four categories (§26.5); tuned with equal effort
Tuning	Identical search space size and trial budget for every method — and stated
Repetition	Five seeds minimum, ten preferred; report mean with standard deviation or confidence interval
Ablation	One factor removed at a time, plus a cumulative build-up
Cost	Parameters, FLOPs, training time, inference latency, memory

26.2 Training, validation, and testing

The three partitions have distinct and non-interchangeable roles:

- **Training** — parameters are fitted.
- **Validation** — hyperparameters, architecture variants, early-stopping points, and decision thresholds are selected.
- **Test** — the final estimate is computed, **once**.

The cardinal rule. Every additional look at the test set is a selection step, and selection on test invalidates the estimate. Keep a dated log of each time you evaluated on test and why (§3.5). Researchers who do this discover that they touch the test set far more often than they believed.

26.3 Splitting schemes

Figure 26.1 — Splitting schemes and when each applies

```
Is your data grouped (multiple records per subject/site/author/session)?
│
├─ NO ──── Is it time-dependent?
│          ├─ NO ──── Is the dataset large?
│          │          ├─ YES → fixed train/val/test (benchmark convention)
│          │          └─ NO  → k-fold CV (k=5 or 10); stratified if imbalanced
│          └─ YES → temporal split; rolling-origin evaluation. NEVER shuffle.
│
└─ YES ─── Split by the GROUP.
           ├─ Estimating in-domain performance → grouped k-fold
           ├─ Claiming cross-domain transfer   → leave-one-group-out
           │                                     (e.g. leave-one-site-out)
           └─ Small data + tuning required     → nested CV
                                                 (outer: estimation,
                                                  inner: hyperparameters)
```

Scheme	Use when	Note
Fixed train/val/test	Large data; established benchmark convention	Report the split source; use the official split if one exists
k-fold CV	Small to medium data; variance estimates needed	Report per-fold results, not only the mean
Stratified k-fold	Class imbalance	Preserves class proportions in each fold
Grouped k-fold	Repeated measures per unit	Prevents group leakage (§23.3, pathway ①)
Nested CV	Tuning *and* unbiased estimation on small data	Expensive but correct; the honest choice for small datasets
Leave-one-group-out	Cross-domain generalisation claims	The design that actually supports a transfer claim
Temporal / rolling-origin	Any time-dependent data	Shuffling time series is a common invalid setup
Repeated CV with different seeds	Reporting variance honestly	Combine with the above

26.4 Hyperparameter tuning and budget parity

The fairness principle. An improvement claim is only meaningful if every method received comparable optimisation effort. Otherwise you have measured your own tuning diligence.

Procedure.

1. Define one search space per method, of comparable dimensionality.
2. Fix a **trial budget** — the same number of trials for every method.
3. Use the same search strategy throughout (random search is a reasonable default and is easier to describe than manual tuning).
4. Select using the **validation** partition only.
5. Report the search space, the strategy, the budget, and the selected values.
6. Write the sentence: *"All methods received an identical budget of N trials over search spaces of comparable size, selected on validation data."*

That one sentence removes the most common reason reviewers distrust an improvement claim, and it costs nothing if you actually did it.

26.5 Baselines: four categories

Table 26.1 — Baseline categories

Category	Purpose	Example
Trivial	Bounds the task's difficulty; exposes degenerate metrics	Majority class; random; predict-the-mean
Strong classical	Tests whether complexity is necessary at all	Gradient boosting on features; TF-IDF with a linear model
Current state of the art	Positions the contribution	The best published method, from official code where possible
Your method minus its novelty	Isolates the contribution, holding everything else constant	The full pipeline with the new component removed

The fourth category is the one reviewers trust most, because it controls for every other difference. It is also the first row of your ablation table, so it costs nothing extra.

On reusing published numbers. Copying a baseline's number from its original paper is acceptable *only* if the dataset, split, and metric are provably identical. When they are not — which is usual — you must re-run the baseline. If you re-run and obtain a lower number than published, report both and say so plainly (§12.4).

26.6 Ablation studies

Definition. An ablation removes or replaces one component at a time to determine each component's contribution.

Purpose. Without ablation, you have shown *that* a pipeline works, not *which part* matters — so nobody, including you, knows what to reuse. An ablation is also the only experiment that tests a mechanistic claim (§1.5).

Design rules.

- **One factor per row.** Two simultaneous changes make the row uninterpretable.
- Report a **cumulative build-up** (baseline □ +A □ +A+B □ full) *and* **leave-one-out** rows where they differ informatively.
- Include an **upper bound (oracle)** where one exists — for instance a version using privileged information you claim not to need. This converts "we improved by 0.04" into "we recovered 82% of the oracle's benefit without its requirements", a much stronger scientific statement.
- **Ablate the hyperparameters of your novel component** in a sensitivity analysis. If performance depends critically on a value you tuned, say so.
- **Ablate the data**: what happens at 10%, 25%, 50% of labels?
- If a component contributes approximately nothing, **report that and remove it**. Honest negative ablations increase credibility more than they cost — a paper in which every component contributes exactly as hoped reads as suspicious.

26.7 Sensitivity and error analysis

Sensitivity analysis varies one hyperparameter or condition across a range and plots the outcome. It answers *how fragile is this?* — a question practitioners care about far more than the peak number.

Error analysis examines *which* cases fail. It is the highest-value-per-hour activity in empirical research and the most frequently skipped.

Procedure. Inspect the confusion matrix for semantically sensible confusions; stratify errors by subgroup (site, device, demographic, input length, illumination) and report the **worst** subgroup, not only the mean; sample thirty to fifty errors manually and categorise their causes into a taxonomy with counts; correlate errors with input properties; and test for shortcut learning by masking the suspected shortcut and observing whether performance collapses.

Manually examining thirty failures teaches more about a model than a week of hyperparameter search, and the resulting taxonomy is often the most cited figure in the paper.

26.8 Statistical validation

Situation	Appropriate test
Two models, same test set, per-item correctness	**McNemar's test**
Two models, paired scores across folds or seeds	Paired t-test (if normality is plausible) or **Wilcoxon signed-rank**
Two AUCs on the same sample	**DeLong's test**
More than two models across many datasets	**Friedman test** with a post-hoc procedure; critical-difference diagrams (Demšar, 2006)
Small data, tuning and estimation together	5×2-fold cross-validated paired t-test (Dietterich, 1998)
Any metric, distribution-free interval	**Bootstrap** confidence interval (≥1,000 resamples)
Any comparison	Report an **effect size** alongside the p-value

Three rules: pre-specify the test; report effect sizes because significance without magnitude is uninformative; and correct for multiple comparisons (Holm or an equivalent) when testing many model–dataset pairs.

McNemar's test deserves specific mention because it is the correct test for comparing two classifiers on a shared test set and is the one most often omitted from student papers.

26.9 Reproducibility controls

Control	Concrete action
Seeds	Fix and record seeds for the language runtime, the numerical library, the framework, and the accelerator; report ≥5 runs with mean and dispersion. Never report a single best run
Determinism	Enable deterministic kernels where available; document residual nondeterminism
Environment	Export a dependency manifest; record framework, accelerator library, and driver versions; note the hardware
Configuration over code edits	One configuration file per experiment; never tune by editing source
Data versioning	Record dataset version, download date, and a checksum of your split files
Release the splits	The cheapest high-impact reproducibility artefact; often permissible even when the data are not redistributable
Logging	Track every run with an experiment-tracking tool or structured logs; keep run identifiers for the appendix
Code release	Public repository, tagged release, archival DOI, licence, and a README with exact commands
Availability statement	A data-and-code statement with a resolvable link

Red flags reviewers look for: four-decimal accuracy from a single seed; "best of N runs"; no variance anywhere; hyperparameters "chosen empirically" with no search described; test-set numbers used for selection; and "code available on request".

26.10 Verification checklist for Part VIII

- [ ] Each objective traces through all six methodology layers to a specific result.
- [ ] Every non-obvious design choice has a "because" sentence.
- [ ] I have audited all seven leakage pathways and stated my controls.
- [ ] The split unit is the grouping unit of my data.

- [ ] Everything that learns from data is fitted on training folds only.
- [ ] All four baseline categories are present, including “mine minus its novelty”.
- [ ] Every method received an identical, stated tuning budget.
- [ ] I run at least five seeds and report dispersion.
- [ ] My ablation changes one factor per row and includes an upper bound where possible.
- [ ] I have performed stratified and manual error analysis.
- [ ] The statistical test was pre-specified, with effect size and multiple-comparison correction.
- [ ] Cost is measured on stated hardware.
- [ ] Seeds, environment, configurations, and split files are recorded and releasable.
- [ ] The test set was touched once, and I have the log to show it.

Exercises

Exercise 26.1 Write your experimental protocol: datasets, split scheme and unit, baselines in all four categories, tuning budget, seed count, metrics, statistical test. One page.

Exercise 26.2 Design your ablation table on paper, one factor per row, including an oracle row if one exists.

Exercise 26.3 Compute your total run count as configurations × seeds × datasets, then multiply by three (§4.6.1). Compare with your resources.

Exercise 26.4 Sample thirty errors from any model you have already trained and categorise their causes. Note how much you learn relative to the time spent.

PART IX — EVALUATION AND RESULTS

Part IX concerns measurement and its honest presentation. Chapter 27 treats evaluation metrics in depth — formula, meaning, interpretation, appropriate use, limitations, and worked example for each. Chapter 28 covers results presentation and the specific ways visualisations mislead. Chapter 29 draws the boundary between Results and Discussion, the most frequently violated structural convention in scientific writing.

The theme: **a metric is a claim about what matters.** Choosing accuracy for a rare-disease detector is not a technical shortcut but a substantive assertion that false negatives and false positives are equally costly — an assertion that is usually false and almost never defended.

Chapter 27 — Evaluation Metrics

27.1 The principle of metric selection

Definition. An evaluation metric is a function mapping predictions and ground truth to a scalar summarising performance along one dimension.

The principle. Choose metrics from the **decision the model supports**, not from convention or convenience. Then justify the choice in one sentence in the paper. That sentence — *"because prevalence is 1.2%, we report average precision and recall at fixed specificity rather than accuracy"* — pre-empts an entire class of reviewer objection and takes ten seconds to write.

27.2 The confusion matrix and its family

Nearly all classification metrics derive from four counts.

Figure 27.1 — The confusion matrix and the metrics derived from it

```
                         PREDICTED
                   Positive       Negative
                  ┌────────────┬────────────┐
        Pos       │     TP     │     FN     │  → Recall = TP/(TP+FN)
ACTUAL            │            │            │    (sensitivity, TPR)
        Neg       │     FP     │     TN     │  → Specificity = TN/(TN+FP)
                  └────────────┴────────────┘
                         |
                  Precision =
                  TP/(TP+FP)

Accuracy         = (TP + TN) / (TP + TN + FP + FN)
F1               = 2·Precision·Recall / (Precision + Recall)
Balanced accuracy= (Recall + Specificity) / 2
FPR              = FP / (FP + TN)  =  1 − Specificity
```

Always report the confusion matrix itself. A single scalar tells you how much error there is; the matrix tells you *which* errors occur, and that is what determines whether the model is usable.

27.3 Classification metrics in detail

Table 27.1 — Classification metrics: use and misuse

Metric	Formula	Meaning	Use when	Do not use when	Worked example *[HYPOTHETICAL]*
Accuracy	`(TP+TN)/N`	Proportion correct	Balanced classes; symmetric error costs	**Imbalanced data**	1,000 patients, 10 positive. Predict all negative → accuracy 99.0%, recall 0. Useless model, excellent accuracy
Precision	`TP/(TP+FP)`	Of flagged cases, how many are real	False positives are costly (referral, spam)	Alone — maximised by predicting almost nothing	Flag 5 cases, 4 correct → precision 0.80, but 6 of 10 cases missed
Recall (sensitivity, TPR)	`TP/(TP+FN)`	Of real cases, how many were found	Missing a case is costly (cancer, fraud, safety)	Alone — maximised by predicting everything positive	Flag all 1,000 → recall 1.00, precision 0.01
Specificity (TNR)	`TN/(TN+FP)`	Of negatives, how many correctly cleared	Screening; always paired with sensitivity	Alone	Sensitivity 0.90 at specificity 0.95 is an interpretable operating point
F1	`2PR/(P+R)`	Harmonic mean of precision and recall	One number needed under imbalance	Error costs are asymmetric — use Fβ	P = 0.80, R = 0.40 → F1 = 0.53 (the harmonic mean punishes imbalance between them)
Fβ	$(1+\beta^2)PR/(\beta^2P+R)$	Weighted harmonic mean	β > 1 favours recall; β < 1 favours precision	Without justifying β	F2 = 0.44 for the same values — recall weighted more heavily
Balanced accuracy	`(TPR+TNR)/2`	Mean of per-class recall	Imbalance; interpretable to non-specialists	When per-class detail matters	The all-negative predictor scores 0.50, correctly exposing it
MCC	correlation of predicted and actual, from all four cells	Single number using the whole matrix	Imbalance; a stricter summary than F1	When stakeholders cannot interpret it	The all-negative predictor scores 0 (Chicco and Jurman, 2020)
Cohen's κ	agreement corrected for chance	Agreement beyond chance	Inter-rater agreement; imbalanced classification	Across datasets with differing prevalence	κ = 0.81 indicates substantial annotator agreement
ROC-AUC	area under TPR vs FPR	Probability a random positive is ranked above a random negative	Threshold-free ranking quality; moderate imbalance	**Severe imbalance** — a large TN pool makes it look optimistic	AUC 0.90 can coexist with precision 0.10 at 1% prevalence
PR-AUC / average precision	area under precision vs recall	Ranking quality focused on the positive class	**Rare positives**: anomaly, rare disease, retrieval	Comparing across datasets with different prevalence — the baseline shifts	With 1% prevalence, a random ranker has PR-AUC ≈ 0.01 and ROC-AUC ≈ 0.50
Top-*k* accuracy	correct if truth is in top *k*	Ranked-output tolerance	Many classes; ranked suggestions acceptable	Binary or decision-critical tasks	Top-5 accuracy 0.95 with top-1 0.72
Calibration: ECE, Brier	binned gap between confidence and accuracy; mean squared probability error	Are the probabilities trustworthy?	Probabilities inform decisions or thresholds	— (under-reported; reporting it distinguishes your paper)	ECE 0.09 means predicted confidence is off by ~9 points on average (Guo et al., 2017)

27.3.1 ROC versus precision–recall: the mechanism

This distinction is the most consequential in the chapter and is worth understanding mechanically rather than as a rule.

Figure 27.2 — ROC versus precision–recall under class imbalance

```
Scenario: 10,000 samples, 100 positive (1% prevalence).
A model flags 200 samples; 80 are true positives.

TP = 80    FP = 120    FN = 20    TN = 9,780

ROC axis:  FPR = FP/(FP+TN) = 120/9,900 = 0.012      ← barely moves!
           TPR = 80/100     = 0.80
           → looks excellent

PR axis:   Precision = TP/(TP+FP) = 80/200 = 0.40    ← exposes the problem
           Recall    = 0.80
           → 60% of alerts are false alarms

WHY: FPR has the huge TN pool in its denominator, so hundreds of false
positives hardly register. Precision has only TP in its denominator
alongside FP, so the same errors are fully visible.

RULE: under severe imbalance, prefer PR-AUC / average precision.
CAVEAT: the PR-AUC of a random ranker equals the positive prevalence,
so PR-AUC is NOT comparable across datasets with different prevalence.
```

This behaviour is documented in the methodological literature; Saito and Rehmsmeier (2015) give a detailed treatment for imbalanced biological data.

27.3.2 Multi-class and multi-label reporting

For multi-class problems, report **per-class** precision, recall, and F1 alongside a **macro** average. Macro averaging weights every class equally and therefore exposes failure on rare classes; micro averaging weights by frequency and therefore conceals it. Reporting only a micro average on imbalanced data is a common way to hide the clinically or practically important failures.

For **multi-label** problems — where each instance may carry several labels — report per-label AUC and average precision plus the macro average. A single global accuracy over a multi-label problem is close to meaningless, because it is dominated by the prevalent labels.

27.4 Regression metrics

Table 27.2 — Regression metrics: use and misuse

Metric	Formula	Meaning	Use when	Do not use when	Note
MAE	`mean(	y − ŷ	)`	Average absolute error, in target units	Robust, directly interpretable
MSE	`mean((y − ŷ)²)`	Average squared error	Large errors matter disproportionately	Heavy outliers dominate	Units are squared, so hard to interpret
RMSE	`sqrt(MSE)`	Root mean squared error, in target units	Large errors matter; interpretable units wanted	Outlier-heavy data	Always ≥ MAE; the gap indicates error-variance spread
R^2	`1 − SS_res/SS_tot`	Proportion of variance explained	Familiar; linear-model contexts	**Comparing across datasets with different target variance**; non-linear or heteroscedastic fits	Can be negative; is *not* “accuracy”
MAPE	`mean(	y−ŷ	/	y	)·100`
MedAE / Huber	median absolute error; hybrid loss	Outlier-robust error	Outlier-heavy data	—	Huber transitions from squared to absolute
RMSLE	RMSE of log1p values	Error on a multiplicative scale	Targets spanning orders of magnitude	Negative targets	Penalises underprediction more

Two cautions. R^2 is routinely misreported as though it were accuracy; it is not, it can be negative, and it is not comparable across datasets whose targets have different variance. And whichever metric you report, **plot residuals against predictions**: a good RMSE with structured residuals indicates a mis-specified model, and the scalar alone will never reveal it.

27.5 Object detection and segmentation

Table 27.3 — Detection and segmentation metrics

Metric	Formula	Meaning	Notes and pitfalls
IoU (Jaccard)	`	A∩B	/
Precision / recall at IoU τ	as §27.2, with correctness defined by IoU ≥ τ	Detection quality at one strictness level	**Always state τ**
AP	area under the precision–recall curve for one class	Single-class detection quality	Interpolation convention varies between benchmarks
mAP	mean AP over classes	Overall detection quality	`mAP@0.5` and `mAP@[.5:.95]` are different numbers — often by 15–20 points. Quoting one against the other is a real and common error
AR	average recall over IoU thresholds	Coverage	Report with a small/medium/large size breakdown where the benchmark provides it
Dice (F1 per pixel)	`2	A∩B	/ (
Dice ↔ IoU	`Dice = 2·IoU/(1+IoU)`	Monotonic relation	Dice is always ≥ IoU for the same overlap; never compare one against the other
mIoU	mean IoU over classes	Semantic segmentation standard	Report per class as well as the mean
Pixel accuracy	correct pixels / total	Proportion of pixels correct	**Usually misleading** — background is often more than 95% of pixels
Hausdorff distance / ASSD	boundary distance measures	Boundary accuracy	Essential where shape or margin matters, e.g. surgical planning

Two rules. `mAP` without a stated IoU threshold and non-maximum-suppression settings is not comparable to anything. And pixel accuracy on data with dominant background is a vanity metric; report per-class Dice or IoU plus a boundary metric when shape matters.

27.6 Metric selection by situation

Table 27.4 — Metric selection by problem situation

Situation	Report these	Explicitly avoid
Binary, balanced	Accuracy, F1, ROC-AUC, confusion matrix	—
Binary, severe imbalance (<5% positive)	PR-AUC / AP, recall at fixed precision, F2, MCC, confusion matrix	Accuracy alone; ROC-AUC alone
Multi-class, imbalanced	Per-class P/R/F1, **macro** F1, confusion matrix, MCC	Micro average alone
Multi-label	Per-label AUC **and** AP, macro average	A single global accuracy
Screening / triage	Sensitivity at fixed specificity (or the converse), PR curve, calibration	A single operating point with no curve
Cost-sensitive	Expected cost, cost curves, decision-curve analysis	Symmetric metrics
Ranking / retrieval / recommendation	nDCG@k, MRR, MAP@k, Recall@k, coverage	Accuracy
Probabilistic forecasting	Brier score, ECE, reliability diagram, log loss	Hard-label accuracy alone
Text generation	Task-specific automatic metrics **plus human evaluation with inter-rater agreement**	Automatic overlap metrics alone as a quality claim
Deployment-constrained	The accuracy metric **plus** latency, memory, energy, parameters	Accuracy alone
Any comparison claim	Mean ± CI over ≥5 seeds **and** a paired statistical test	A single-run point estimate

27.7 A note for non-computational disciplines

The logic transfers directly even though the instruments differ. Where this chapter says *metric*, read *measure*; the requirement that the measure match the decision and the data structure is identical. The corresponding apparatus includes reliability (internal consistency, intraclass correlation), agreement (Cohen's or Fleiss' kappa), validity (construct, content, criterion), effect sizes (standardised mean differences, variance-explained measures), and, for qualitative work, trustworthiness criteria and saturation. The failure modes are also analogous: using a measure because it is conventional rather than because it captures the construct, and reporting a point estimate without any indication of precision.

27.8 Common mistakes

Mistake	Correction
Accuracy on imbalanced data	Use PR-AUC, macro F1, MCC; always show the confusion matrix
ROC-AUC alone under severe imbalance	Add PR-AUC; understand the mechanism (§27.3.1)
Comparing PR-AUC across datasets with different prevalence	Not comparable; the baseline differs
Reporting R^2 as accuracy	Report MAE and RMSE in target units alongside
MAPE with near-zero targets	Use MAE, MedAE, or a scaled alternative
`mAP` without IoU threshold and NMS settings	State both
Comparing Dice against IoU	Convert or report both
Pixel accuracy on background-dominated data	Per-class Dice or IoU plus a boundary metric
Micro average only on imbalanced multi-class	Add macro and per-class
Automatic generation metrics as a quality claim	Add human evaluation with agreement statistics
No calibration reported where probabilities drive decisions	Report ECE or Brier and a reliability diagram
Four decimal places from one seed	Mean ± CI over ≥5 seeds
No justification sentence for the metric choice	Add it; it costs one sentence

Exercises

Exercise 27.1 Compute the §27.3.1 example by hand for your own class prevalence. If your positive class is below 5%, check whether your current headline metric conceals your error profile.

Exercise 27.2 Write your metric set with a one-sentence justification per metric, tied to the decision the model supports.

Exercise 27.3 For each metric, write the sentence a reviewer would use to attack it. Then decide whether to change the metric or defend it in the paper.

Exercise 27.4 Identify the subgroup on which your model performs worst, and decide whether the paper will report it. (It should.)

Chapter 28 — Results Presentation

28.1 Purpose and the reading order of a reviewer

A reviewer typically reads the abstract, then the figures and tables, then decides how sceptically to read the prose. Your floats are therefore not illustrations of the argument — for many readers they *are* the argument.

Recommendation. Before writing prose, list the figures and tables the paper needs. A typical empirical paper has six to eight:

```
Fig. 1   System overview (novel component highlighted)
Table I  Dataset statistics (size, classes, balance, split unit, licence)
Table II Main comparison against baselines, with dispersion and significance
Fig. 2   The key effect (e.g. degradation under protocol change)
Table III Ablation study, one factor per row, with an oracle row
Fig. 3   Error analysis or qualitative failure cases
Table IV Cost: parameters, FLOPs, latency, memory
Fig. 4   Sensitivity to the principal hyperparameter
```

If you cannot produce this list, you are not ready to write — you are still doing experiments. If the list contains no baseline comparison, no ablation, and no cost table, the paper will struggle regardless of how well it is written.

28.2 Choosing the right display

Display	Use for	Avoid when
Table	Precise values; many conditions; anything readers will quote	Showing a trend — use a line plot
Line plot	A metric varying over a continuous variable (epochs, λ, sequence length)	Categories with no ordering
Bar chart	Comparing a few discrete categories	Many categories, or an implied continuum
Box or violin plot	Distributions across seeds or folds	Fewer than about five observations
Scatter plot	Relationships; trade-offs such as accuracy against latency	Heavy overplotting without transparency or binning
Confusion matrix heat map	Which errors occur	More than roughly 20 classes without aggregation
ROC curve	Threshold-free ranking, balanced data	Severe imbalance — add a PR curve
Precision-recall curve	Rare positives	Cross-dataset comparison at differing prevalence
Reliability diagram	Calibration	—
Training curves	Convergence behaviour; over-fitting evidence	As a substitute for test performance
Critical-difference diagram	Many methods across many datasets	Two methods only

28.3 How visualisations mislead

Most misleading figures are produced without any intention to mislead. Knowing the failure modes is the defence.

Figure 28.1 — Misleading and honest presentations of the same result

```
□ TRUNCATED AXIS                         □ FULL OR ANNOTATED AXIS
   0.86 ┤       █                            1.0 ┤
   0.85 ┤   █   █                                │   ███  ███  ███
   0.84 ┤   █   █                            0.5 ┤   ███  ███  ███
   0.83 ┤█  █   █                                │   ███  ███  ███
        └──A───B───C                          0.0 └───A────B────C
   "C is twice as good as A"                  0.831  0.848  0.862
   — the visual gap is an artifact            The real difference: 0.031
   of the axis, not the data.                 Now the reader judges it.

OTHER FAILURE MODES
□ No error bars → the reader cannot tell whether the difference is noise.
□ Error bars whose definition is unstated (std? SEM? 95% CI?) — these
   differ by large factors and are not interchangeable.
□ Dual y-axes → any desired correlation can be manufactured by rescaling.
□ Cherry-picked subset of datasets or metrics shown in the figure.
□ Colour as the only channel → illegible to colourblind readers and in
   greyscale print.
□ 3-D bars and perspective → distort area and hide values.
□ Connecting unordered categories with a line → implies a trend.
□ Different y-scales across panels compared side by side.
□ Log axis unlabelled as such.
```

The single most important rule: put dispersion on every mean. A bar chart of means without error bars is uninterpretable, and a reader is entitled to assume the omission is convenient.

28.4 Table construction

Rule	Detail
Horizontal rules only	Top, header, bottom. No vertical rules, no full grid
Bold the best per column	And state in the caption what bold denotes
Uniform decimal places	Two or three; aligned on the decimal point
Include dispersion	`0.822 ± 0.009`, with the definition given in the caption
Mark significance	A symbol with its meaning stated (`*p < 0.05`, Holm-corrected)
Group rows meaningfully	Trivial baselines → classical → prior state of the art → ours → oracle
Include cost columns	Parameters, FLOPs, latency in the main comparison table
Delete unused columns	If a column is never discussed, it is noise
Never screenshot	Typeset tables as tables

28.5 Captions, numbering, and cross-referencing

- **Figure captions below the figure; table captions above the table** — the standard convention in IEEE and most engineering styles. **[VERIFY]** against your target journal's guide.
- **Captions must be self-contained.** A reader who reads only figures and captions should understand what is shown, on what data, and what to notice. Name the dispersion measure and the number of runs in the caption.
- **Number in order of first mention**, and mention every float in the text. An unreferenced float will be queried.
- Use the journal's abbreviation conventions consistently (for example "Fig. 3" mid-sentence, "Figure 3" at the start of a sentence; tables often numbered in roman numerals).
- **Place floats near their first mention.**

28.6 Common mistakes

Mistake	Correction
Means with no error bars	Add dispersion to every mean
Dispersion measure unstated	Define it in the caption
Truncated axes exaggerating differences	Start at zero or annotate the truncation explicitly
Dual y-axes	Use two panels
Colour as the sole distinguishing channel	Add markers and line styles; use a colourblind-safe palette
Screenshots of plots or notebook output	Export vector graphics (PDF, SVG, EPS)
Unreadable font after scaling to column width	Set font sizes explicitly; check at final print size
Captions that merely name the figure	Make them self-contained and say what to notice
Tables with vertical rules and full grids	Booktabs style: horizontal rules only
A float never mentioned in the text	Reference it or remove it
Only favourable datasets plotted	Show all, or state clearly which are omitted and why

Exercises

Exercise 28.1 Write your figure and table list before writing any prose. Check it contains a baseline comparison, an ablation, an error analysis, and a cost table.

Exercise 28.2 Take one existing figure of yours and fix three things: font size, error bars with a stated definition, and vector export.

Exercise 28.3 Read one of your captions aloud to a colleague and ask what they should notice. If they cannot say, rewrite the caption.

Exercise 28.4 Audit your figures against all nine failure modes in §28.3.

Chapter 29 — Results versus Discussion

29.1 The boundary

Figure 29.1 — The boundary between Results and Discussion

```
┌─ RESULTS ──────────────────────┐    ┌─ DISCUSSION ───────────────────┐
│ Question: WHAT HAPPENED?       │    │ Question: WHY, AND SO WHAT?    │
│                                │    │                                │
│ Contains                       │    │ Contains                       │
│  • numbers, tables, figures    │    │  • mechanisms and explanations │
│  • statistical test outcomes   │    │  • comparison with literature  │
│  • observed patterns           │    │  • practical implications      │
│  • magnitudes and uncertainty  │    │  • limitations and threats     │
│                                │    │  • future directions           │
│ MUST NOT contain               │    │ MUST NOT contain               │
│  ✗ interpretation              │    │  ✗ new numbers                 │
│  ✗ speculation                 │    │  ✗ new experiments             │
│  ✗ "this proves…"              │    │  ✗ new tables or figures       │
│  ✗ comparison to other papers  │    │                                │
│    beyond stating the number   │    │ Tense: present for established │
│                                │    │  meaning ("this suggests")      │
│ Tense: past ("we observed")    │    │                                │
└────────────────────────────────┘    └────────────────────────────────┘

The two classic violations, both routinely flagged by reviewers:
 □ A Results section asserting "this proves our method is superior."
 □ A Discussion introducing a new table.
```

Some journals merge the two sections; check the template. When merged, the *functions* remain distinct even though the heading does not, and the discipline of separating observation from interpretation still applies.

29.2 Writing Results

Structure. One subsection per research question, in the order the questions were posed. This makes the traceability of Chapter 7 visible to the reader.

The four-part narration pattern. For each table or figure:

1. **Point** to the float.
2. State the **pattern**.
3. State the **magnitude**.
4. State the **uncertainty**.

Table 29.1 — Results language versus Discussion language

Belongs in Results	Belongs in Discussion
"Macro AUC fell from 0.897 ± 0.004 to 0.781 ± 0.011."	"This suggests that reported performance reflects within-institution recognition rather than transfer."
"The difference was significant (paired Wilcoxon, $p = 0.004$)."	"The effect is consistent with institution-level confounding reported by Zech et al. (2018)."
"Gains on the smallest institution were not significant ($p = 0.21$)."	"The absence of an effect at the smallest site may reflect limited statistical power rather than a genuine boundary."
"Expected calibration error improved from 0.094 to 0.052."	"Improved calibration matters here because triage thresholds are set on predicted probabilities."

[HYPOTHETICAL] □ Weak Results. *"Our model achieved 95% accuracy. Table 3 shows the results. From Table 3 it is clear that our model is better than the other models. The graph in Fig. 4 shows the comparison."*

Defects: no comparison point, no dispersion, no pattern, no magnitude; the reader must do all the work; and "it is clear that" is interpretation misplaced into Results — and is not evidence in any case.

[HYPOTHETICAL] □ Strong Results. *"Under institution-disjoint evaluation, macro AUC falls from 0.897 ± 0.004 to 0.781 ± 0.011 across the five architectures (Table II), a mean reduction of 0.116 (95% CI 0.104–0.128); the direction is consistent for every architecture and every seed. Degradation is largest for the transformer backbone (-0.142) and smallest for DenseNet-121 (-0.093). CLUSTER-DG raises worst-institution AUC from 0.781 ± 0.011 to 0.822 ± 0.009 (paired Wilcoxon signed-rank, p = 0.004, Cohen's d = 1.6), while in-domain AUC decreases by 0.008, which lies within seed variance. Expected calibration error improves from 0.094 to 0.052 (Fig. 3). On the smallest institution (n = 412) the improvement is not significant (p = 0.21)."*

The final sentence — an honestly reported null result — is the sentence that makes a reviewer trust the rest. Authors who report their own inconvenient findings are believed; authors whose every result is favourable are audited.

29.3 Writing Discussion

The six-move paragraph pattern. For each principal finding:

1. **Restate** the finding in one sentence, without numbers.
2. **Explain** the mechanism — why this happened.
3. **Reconcile** with prior literature, including work that disagrees.
4. **State the implication** — what someone should now do differently.
5. **State the limitation** on this specific finding.
6. **Point** to the next question.

[HYPOTHETICAL] □ Weak Discussion. *"Our proposed method achieved the best performance among all compared methods. This demonstrates the effectiveness and superiority of our approach. The results show that our method is suitable for real-world applications. In future we will extend the work to other datasets."*

Every sentence restates the result or asserts merit. No mechanism, no engagement with literature, no implication, no limitation. This paragraph could be attached to any paper on any topic.

[HYPOTHETICAL] □ Strong Discussion. *"Cross-institutional degradation is substantially larger than published within-institution figures imply, and most of it is recoverable without institutional metadata. The mechanism our ablation supports is that acquisition artefacts concentrate in low-order embedding statistics: clusters recovered from those statistics agree with true institutional labels for 78% of samples (§V-C), which is evidently sufficient to drive an invariance penalty. This is consistent with the confounding reported by Zech et al. (2018) and extends it by quantifying the gap under a leakage-free protocol. It also reconciles the apparent disagreement between [6] and [9] regarding calibration: both evaluated on random splits, where calibration is measured on the training distribution and therefore appears better than it is. The practical implication is that a hospital evaluating a published model should expect performance closer to the worst-site figure than to the reported mean, and that mitigation does not require the provenance metadata that privacy procedures remove. Two limitations bound this conclusion. First, our institutional partition is coarse: we treat each public dataset as one institution, whereas each aggregates several sites, so true site-level degradation may be larger. Second, the absence of a significant effect at the smallest institution leaves open whether the method helps where data are scarcest — the setting in which it would matter most. Resolving this requires multi-site data with genuine site labels for evaluation, which we identify as the next step."*

Note what it does: mechanism supported by a specific ablation number; reconciliation that *explains a contradiction in the literature*; a concrete implication for a named decision-maker; two specific limitations with their consequences; and a next question that follows from the limitation rather than being generic.

29.4 Interpreting rather than restating

The commonest Discussion failure is restatement. The test: if a sentence would still be true with all the numbers deleted and the method renamed, it is restatement.

Restatement	Interpretation
"Our model achieved 95% accuracy, which is higher than the baselines."	"The gain concentrates entirely in the two most prevalent classes; on the three rarest classes our method is indistinguishable from the baseline, which suggests the improvement comes from better calibration on abundant data rather than from improved representation of rare findings."
"The ablation shows all components contribute."	"Removing the ramping schedule costs 0.014 AUC while removing the clustering branch costs 0.031, indicating that the branch — not the schedule — carries the contribution; the schedule appears to act as an optimisation aid rather than a source of invariance."
"Results demonstrate effectiveness."	"The effect size (d = 1.6) exceeds the seed-to-seed variation by roughly an order of magnitude, so the difference is unlikely to be an artefact of initialisation; however, it was measured under a single augmentation policy and may not persist under stronger augmentation."

29.5 Reporting limitations well

Limitations are not a confession; they are a specification of where your claim stops being safe. A good limitation names the threat, its likely direction, and its consequence.

Vague	Specific
"Our study has some limitations."	"We treat each public dataset as a single institution, though each aggregates multiple sites; this makes our estimate of site-level degradation conservative — true degradation is likely larger, not smaller."
"More datasets could be used."	"We evaluate on three datasets, all from high-resource settings with digital radiography; performance under computed radiography or portable acquisition is untested and we would expect it to be worse."
"The method could be improved."	"The clustering step assumes the number of latent groups is roughly known; we show sensitivity to K over 2–16 (Fig. 4), but performance degrades beyond K = 20, which bounds applicability to settings with a moderate number of sites."

29.6 Common mistakes

Mistake	Correction
Interpretation in Results	Move it to Discussion
New numbers or tables in Discussion	Move them to Results
Results as a table dump with no narration	Apply the four-part pattern (§29.2)
Discussion restating results	Apply the interpretation test (§29.4)
No engagement with contradicting literature	Address it explicitly; it strengthens the paper
Only favourable findings reported	Report null and adverse findings; it is what earns trust
Generic limitations	Name the threat, its direction, and its consequence
Generic future work	Derive it from your stated limitations
Causal language on non-causal designs	Match claim strength to design (§2.5)

29.7 Verification checklist for Part IX

- [ ] Every metric is justified in one sentence by the decision it supports.
- [ ] Under imbalance I report PR-AUC or macro F1 or MCC, not accuracy alone.
- [ ] Confusion matrix or per-class breakdown is present.
- [ ] Calibration is reported where probabilities drive decisions.

- [ ] Detection metrics state the IoU threshold; segmentation reports per-class overlap plus a boundary metric where shape matters.
- [ ] Every mean carries dispersion, and the dispersion measure is defined.
- [ ] My float list includes baseline comparison, ablation, error analysis, and cost.
- [ ] No axis is truncated without explicit annotation; no dual y-axes.
- [ ] Figures are vector, colourblind-safe, and legible at print size.
- [ ] Results contains no interpretation; Discussion contains no new numbers.
- [ ] Each Results subsection corresponds to a research question.
- [ ] At least one inconvenient or null finding is reported.
- [ ] Every limitation names a threat, a direction, and a consequence.

Exercises

Exercise 29.1 Take your strongest table and narrate it in the four-part pattern: pointer, pattern, magnitude, uncertainty.

Exercise 29.2 Write one Discussion paragraph using all six moves of §29.3.

Exercise 29.3 Apply the restatement test to every sentence of your current Discussion. Delete or upgrade every sentence that survives with the numbers removed.

Exercise 29.4 Write your three most important limitations in the specific form of §29.5, each naming a direction of bias.

PART X — RESEARCH PAPER WRITING

Part X is the writing manual proper: one chapter for each section of a paper, in the order the reader encounters them. Chapters 30 to 32 cover the three sections that determine whether your paper is read at all — title, abstract, introduction. Chapters 33 to 37 cover related work, methodology, results, discussion, and conclusion.

Before beginning, one structural point that saves more time than any other piece of writing advice.

Write in this order, not in reading order.

Step	Section	Why here
1	Figures and tables	They *are* the paper; if the story is not in them, prose cannot rescue it
2	Methodology	Easiest — you did it; write while details are fresh
3	Experimental setup	Mechanical; forces you to notice missing controls
4	Results	Describe floats you have already made
5	Discussion	Interpreting is where you discover what the paper is *about*
6	Related work	Now you know what to contrast against; use your literature matrix
7	**Introduction**	Written late, so it promises exactly what you delivered
8	Conclusion	Compresses the introduction’s promise plus the results
9	Abstract	Compresses the whole paper
10	Title	Compresses the abstract

Introductions written first make promises the experiments never keep, and reviewers notice the mismatch between claimed and delivered contribution.

Chapter 30 — Title Writing

30.1 What a title must do

A title is read thousands of times more often than the paper. It must (a) contain the search terms a reader would type, (b) identify the method, problem, and domain, © be specific enough that a competitor could not reuse it, (d) claim nothing the paper does not deliver, and (e) be readable — typically 8 to 15 words.

Mechanism note. Indexing and search ranking weight title terms heavily. A title lacking your field's search vocabulary costs citations for the paper's entire life.

30.2 Patterns that work

Pattern	When to use	Example *[HYPOTHETICAL]*
`[Method] for [Problem] in [Domain]`	Method contributions	"Cluster-Based Invariance Regularisation for Cross-Institution Chest Radiograph Classification"
`[Finding]: [evidence or scope]`	Empirical contributions	"Site-Wise Evaluation Reduces Reported Chest Radiograph AUC by 0.12: A Multi-Institution Re-Analysis"
`[Question]?`	Reframing or investigative work	"Do Domain-Generalisation Methods Help When Site Labels Are Unavailable?"
`[Resource name]: A [type] for [purpose]`	Datasets and benchmarks	"CodeMixQA: A Benchmark for Extractive Question Answering in Code-Mixed Hindi–English"
`[Scope] ([years]): A Systematic Review of [dimension]`	Reviews	"Edge-Deployable Crop-Disease Models (2020–2025): A Systematic Review of Accuracy–Latency Trade-offs"

The colon construction is standard in strong venues because it allows both a memorable claim and a precise scope.

30.3 Words to delete

A Study of · An Investigation into · Novel · New · Efficient (when unmeasured) · Towards (when vague) · Using · Based on · An Approach for · A Framework for · Some Observations on · System

Each consumes title space without adding search value. "Novel" in particular signals inexperience: novelty is demonstrated in the paper, not asserted in the title.

30.4 Weak and improved titles

Table 30.1 — Weak and improved titles

□ **Weak** *[HYPOTHETICAL]*	**Diagnosis**	□ **Improved** *[HYPOTHETICAL]*
"A Study on Deep Learning for Medical Images"	No method, no problem, no domain; unsearchable	"Institution-Disjoint Evaluation of Chest Radiograph Classifiers: Quantifying Cross-Hospital Degradation"
"A Novel Efficient Framework for Text Classification Using Machine Learning"	Four dead words; "frame-work" hides the artefact; no domain	"Morphology-Aware Subword Tokenisation for Extractive Question Answering in Marathi and Kannada"
"Improved Accuracy in Plant Disease Detection"	Improved over what? Which crop? Which condition?	"Few-Shot Leaf-Disease Detection Under Field Illumination With Fewer Than 50 Labelled Images per Class"
"Deep Learning Based Plagiarism Detection System"	"System" signals a project; engineering title	"Script Normalisation Recovers Sentence-Embedding Plagiarism Detection on Code-Mixed Text"
"AI in Agriculture: A Review"	Scope unbounded	"Edge-Deployable Crop-Disease Models (2020–2025): A Systematic Review of Accuracy–Latency Trade-offs"
"Performance Analysis of Various Classifiers on Different Datasets"	Two unspecified plurals; no finding	"Tree Ensembles Match Deep Models on Tabular Clinical Data Under Equal Tuning Budgets"

30.5 The verification test

Paste your title into a scholarly search engine. If the top results are not your intended peers, your keywords are wrong. Then check: does every substantive word earn its place? Would a specialist know from the title alone whether to read it?

Exercises

Exercise 30.1 Write three candidate titles, at least one using the `[Finding]: [scope]` pattern — which forces you to know what your finding is.

Exercise 30.2 Apply the search test to your best candidate.

Exercise 30.3 Delete every word from §30.3 and check whether the title lost information.

Chapter 31 — Abstract Writing

31.1 The seven moves

Table 31.1 — The seven moves of an abstract

Move	Sentences	Content	Characteristic failure
1 **Background**	1–2	Why the problem matters. No history of the discipline	Three sentences on the history of AI
2 **Problem / gap**	1–2	The specific unknown, ideally with a number	“However, there are some challenges”
3 **Objective / approach**	1	What you did, named precisely	“We propose a novel framework”
4 **Method detail**	1–2	The mechanism, enough to be identifiable	Marketing adjectives instead of mechanism
5 **Data / experiment**	1	Datasets named, scale, protocol, baselines	Datasets never named
6 **Results**	1–2	Numbers **with comparison point and dispersion**	“Results show the effectiveness of our method”
7 **Conclusion / implication**	1	What this now means; scope of the claim	Repeating the results verbatim

Hard rules. No citations (in most venues); no undefined acronyms; no references to figures, tables, or sections; no claim absent from the paper; every number identical to the Results section. Respect the word limit exactly — submission systems truncate.

Structured abstracts. Some journals require explicit labels (Background, Objective, Methods, Results, Conclusions). **[VERIFY]** the target journal’s guide and use the labels verbatim if requested.

31.2 Weak and improved: the same study

[HYPOTHETICAL] □ Weak (118 words).

> “Deep learning has revolutionised medical image analysis in recent years. Many researchers have proposed various models for chest X-ray classification. However, there are still some challenges and room for improvement. In this paper, we propose a novel and efficient deep learning framework for chest X-ray classification. Our proposed model uses advanced techniques to extract better features. Experiments were conducted on a benchmark dataset and the results show that our proposed method outperforms existing state-of-the-art methods in terms of accuracy. The results demonstrate the effectiveness and superiority of the proposed approach. In future, we will extend our work to other medical imaging modalities.”

Diagnosis: move 2 is a cliché; moves 4–6 contain no mechanism, no dataset name, no number, no baseline; every claim is unverifiable. This abstract cannot attract a reader or survive a desk check.

[HYPOTHETICAL] □ Improved (247 words), with moves labelled.

> **[1]** Deep learning models for chest radiograph (CXR) classification report areas under the ROC curve (AUC) of 0.88 to 0.91 and are increasingly proposed for triage support. **[2]** However, in a review of 15 studies published between 2022 and 2026, 11 evaluate on random splits in which images from the same institution appear in both training and test partitions, and none report worst-institution performance or calibration; the degradation incurred when a model is deployed at an unseen hospital is therefore unquantified. **[3]** We re-evaluate five published architectures under leakage-free, institution-disjoint protocols and ask whether site-label-free adaptation can mitigate the resulting degradation. **[4]** We propose CLUSTER-DG, which replaces the site-supervised discriminator of adversarial domain generalisation with unsupervised clustering of channel-wise embedding statistics, requiring no provenance metadata. **[5]** Experiments use CheXpert, MIMIC-CXR, and ChestX-ray14 with institution-disjoint splits, ten random seeds, an identical 50-trial tuning budget for all methods, and four baselines (ERM, Mixup, CORAL, IRM). **[6]** Replacing random with institution-disjoint splits reduces macro AUC from 0.897 ± 0.004 to 0.781 ± 0.011 (mean ± 95% CI), a reduction of 0.116. CLUSTER-DG recovers 0.041 ± 0.009 of worst-institution AUC over empirical risk minimisation (paired Wilcoxon, $p = 0.004$) and reduces expected calibration error from 0.094 to 0.052, without site labels and at equal parameter count. **[7]** Reported CXR performance therefore substantially overstates cross-hospital behaviour, and most of the recoverable gap can be closed without provenance metadata. Code, splits, and trained weights are released.

Five professional signals: the gap carries a **count**; the mechanism is named in one clause; datasets and scale are **explicit**; results carry **dispersion and a statistical test**; the final sentence gives a **scope-limited implication** plus artefact release.

31.3 Procedure

1. Write **one sentence per move — seven sentences, nothing else.** Do not begin with prose.
2. Expand to the word limit.
3. Where you do not yet have a result, write the *shape* of the claim with an explicit `[TBD]` placeholder — **never** a fabricated number, not even in a draft. Placeholder numbers have been known to survive into submitted manuscripts.
4. Have a colleague label each sentence 1–7 and mark any move that is missing or doubled.
5. Verify every number against the Results section.

31.4 Checklist

- [ ] Within the word limit, counted not estimated
- [ ] All seven moves present, in order
- [ ] Gap sentence contains a specific unknown, ideally quantified
- [ ] Method mechanism identifiable in one sentence
- [ ] Datasets named with scale; baselines named
- [ ] At least two result numbers with comparison point and dispersion
- [ ] Statistical test named if a comparison is claimed
- [ ] Every number matches Results exactly
- [ ] No citations, no undefined acronyms, no float or section references
- [ ] No claim the paper does not deliver
- [ ] Final sentence states implication or scope
- [ ] Keywords chosen from the journal's or a standard thesaurus
- [ ] Read aloud once; no sentence longer than about 35 words

Exercises

Exercise 31.1 Write your abstract as exactly seven sentences, one per move. Then expand.

Exercise 31.2 Have a partner label your sentences 1–7. Fix any missing or doubled move.

Exercise 31.3 Check every number in your abstract against your results tables.

Chapter 32 — Introduction Writing

32.1 The six-paragraph blueprint

Figure 32.1 — The six-paragraph introduction, as a pressure curve

```
¶1 BACKGROUND        | Build   The domain problem; why it matters.
   stakes            | ↑       1–2 authoritative citations, a concrete number.
¶2 CURRENT STATE     | ↑       What is done now; what is unsatisfactory.
¶3 EXISTING          | ↑       2–3 grouped families of prior solutions.
   APPROACHES        | ↑
─────────────────────┼──────────────────────────────────────────────────
¶4 LIMITATIONS       | ◄ THE HINGE — maximum pressure.
   → GAP             |   Evidenced absence, with counts. Everything
                     |   before builds pressure; everything after
                     |   releases it.
─────────────────────┼──────────────────────────────────────────────────
¶5 PROPOSED          | ↓ Release  What you do, the mechanism, and why
   APPROACH          | ↓          it should work.
¶6 CONTRIBUTIONS     | ↓          3–4 numbered contributions + roadmap.
                     | ↓

LOGIC:  GENERAL → SPECIFIC → PROBLEM → GAP → SOLUTION → CONTRIBUTION
```

¶	Job	Length
1	Background and stakes	4–6 lines
2	Current state and its inadequacy	5–8 lines
3	Existing approaches, grouped into families	6–10 lines
4	**Limitations → gap (the hinge)**	6–10 lines
5	Proposed approach and its mechanism	6–10 lines
6	Numbered contributions and roadmap	bulleted

Rules. ¶4 is the hinge — weak introductions have no hinge and drift from background straight to "in this paper we propose", so the proposal answers no question. Each paragraph has exactly one job; if you cannot name a paragraph's job in six words, merge or delete it. Contributions in ¶6 must match the abstract, the results sections, and the conclusion in substance. Never put implementation detail (learning rates, library versions) in the introduction.

32.2 A complete worked introduction

[HYPOTHETICAL, grounded in the real finding of Zech et al. (2018).]

[¶1 Background] Chest radiography is the most frequently performed diagnostic imaging examination worldwide, and radiologist workload has motivated automated triage of abnormal studies. Deep classifiers now report areas under the ROC curve of 0.88 to 0.91 on public benchmarks such as CheXpert and MIMIC-CXR, a level often described as approaching expert agreement.

[¶2 Existing problem] These figures, however, are obtained almost entirely within a single institutional distribution. Acquisition hardware, exposure protocols, patient demographics, and post-processing pipelines differ substantially across hospitals, and models have been shown to exploit institution-specific signal — including markers, collimation borders, and department-level prevalence differences — as shortcuts for prediction (Zech et al., 2018). A model that is accurate on the distribution it was trained on may therefore be unsafe at the hospital that deploys it, which is precisely the setting that matters clinically.

[¶3 Existing approaches] Two families of work address this. The first increases representational capacity, progressing from convolutional backbones to vision transformers; these report in-distribution gains of one to three AUC points but do not target transfer. The second optimises explicitly for invariance across domains, using feature-distribution alignment (Sun and Saenko, 2016), invariant risk minimisation (Arjovsky et al., 2019), or adversarial domain discriminators; these report external gains of three to six AUC points.

[¶4 Limitations □ gap — the hinge] Both families rest on assumptions that our review of 15 studies published between 2022 and 2026 shows to be problematic. First, 11 of the 15 evaluate on random splits in which radiographs from the same institution — and frequently the same patient — occur in both training and test partitions, so the reported AUCs measure in-distribution performance rather than transfer. Second, all eight invariance methods we identified require institutional labels at training time, whereas provenance metadata is routinely stripped before data leaves a hospital for privacy reasons. Third, none of the 15 studies reports worst-institution AUC or calibration error, although these quantities, not the mean, determine whether a triage system is deployable. The magnitude of cross-hospital degradation under leakage-free protocols is therefore unknown, and no mitigation has been demonstrated in the label-free regime that deployment actually imposes.

[¶5 Proposed approach] We address both parts of this gap. We first re-evaluate five published architectures under institution-disjoint splits on three public datasets, with ten seeds and an identical tuning budget for every method, which isolates the effect of the evaluation protocol from the effect of the model. We then propose **CLUSTER-DG**, which replaces the site-supervised discriminator of adversarial domain generalisation with unsupervised clustering of channel-wise embedding statistics. The intuition is that acquisition artefacts dominate low-order feature statistics, so clusters recovered from those statistics should approximate institutional identity closely enough to support an invariance penalty without any metadata — a belief our ablation tests directly by comparing inferred clusters against true site labels.

[¶6 Contributions] Our contributions are: **(1)** the first leakage-free, multi-institution quantification of degradation for five widely used CXR architectures, with confidence intervals and paired significance tests; **(2)** CLUSTER-DG, a site-label-free adaptation method that recovers 0.041 ± 0.009 worst-institution AUC over empirical risk minimisation at equal parameter count; **(3)** an ablation isolating the contributions of cluster granularity, invariance weight, and augmentation; and **(4)** a public release of institution-disjoint splits, code, and trained weights, enabling protocol-consistent comparison in future work. Section II reviews related work, Section III describes the method, Sections IV and V present the setup and results, and Section VI discusses limitations.

What to notice. ¶2 ends on *stakes*, not a technicality. ¶3 groups into two families rather than one paper per sentence. ¶4 contains **three numbered evidence items with counts** — this is where the literature matrix of Chapter 15 cashes out. ¶5 explains *why the mechanism should work* and names the experiment that tests it; most student papers state what was built and never why it should work, and reviewers read that omission as the absence of an idea. ¶6 gives four auditable contributions, one of which is an artefact release.

32.3 The reverse-outline test

After drafting, write the job of each paragraph in the margin in six words or fewer. If two paragraphs share a job, or a job from §32.1 is missing, the structure is broken — fix structure before sentences. This test works on theses and grant proposals equally well.

32.4 Common mistakes

Mistake	Correction
Textbook history ("AI began in 1956…")	Start at the specific problem with a concrete number
Gap as "less work has been done"	Count it: "11 of 15 studies…"
Contributions restating the method three times	Three or four distinct, auditable contributions
Introduction promising more than results deliver	Write the introduction *after* results
Related work duplicated in full in the introduction	¶3 gives two or three grouped families; detail belongs in Section II
Hyperparameters in ¶5	Mechanism and rationale only
No roadmap sentence	One sentence at the end of ¶6
Citing only the last three years	Cite the origin *and* the frontier
"We solve the problem of…"	Bound the claim: "we quantify…", "we recover approximately…"

Takeaway. The introduction's job is to make your contribution feel *inevitable*. If a reader could finish ¶4 and predict ¶5, you have written it correctly.

Exercises

Exercise 32.1 Draft ¶4 and ¶6 for your paper — the hinge and the contributions. These two carry the paper.

Exercise 32.2 Verify ¶4 contains at least one count from your literature matrix.

Exercise 32.3 Run the reverse-outline test on your draft.

Exercise 32.4 Check that every ¶6 contribution maps to an objective from Chapter 7 and a results subsection.

Chapter 33 — Related Work Writing

33.1 Construction procedure

1. **Sort** your literature matrix by the method column; look for natural families.
2. **Name** two to four families with substantive labels — "capacity-scaling approaches", "invariance-based approaches", "test-time adaptation" — not "CNN-based" or "other methods".
3. **One subsection per family.** Within each: what the family assumes □ representative works with grouped citations □ what it achieves □ **the weakness it shares**.
4. **Insert a comparison table** with the columns your argument needs (typically: work, data, split protocol, baselines, metrics, key result, limitation).
5. **Add a subsection on evaluation practice** — unusual in student papers, and where evaluation gaps become visible.
6. **Close with a synthesis paragraph** stating what is settled, contested, and untested □ the gap, phrased identically to Introduction ¶4.

The worked weak/strong contrast for this section appears in §14.4 and is not repeated here; the material in Chapter 14 applies directly.

33.2 Division of labour between table and prose

The table carries facts; the prose carries judgement. Students frequently reverse this — narrating facts in prose and never interpreting. A comparison table lets you state fifteen studies' protocols compactly, freeing the prose to do the only thing prose can do: evaluate.

33.3 Citation mechanics

Table 33.1 — Citation practices

Practice	Do this
Grouping	`[3]–[5]` for a shared claim; individual citations only when the specific paper matters
Attribution of origin	Cite the paper that *introduced* a method, plus a recent user if useful
Density	A claim about a field needs two or more citations; one citation is an anecdote
Quotation	Rare in computational fields; if used, quotation marks plus citation plus page
Secondary citation	Avoid "as cited in"; read the primary source
Self-citation	Only where genuinely relevant; excessive self-citation is flagged by editors
Adversarial citation	If a paper contradicts you, **cite it and address it** — concealment is worse
Citation as noun	Write "Smith *et al.* [3] showed…", not "In [3], they showed…" repeatedly. `[3]` is not a noun

Absolute rule. Never cite a paper you have not opened, and never accept a reference produced by a generative tool without resolving it against the publisher record (§45.4).

Exercises

Exercise 33.1 Write one family subsection of 150–200 words plus its comparison table, ending with an absence sentence.

Exercise 33.2 Have a partner mark each sentence as *groups*, *evaluates*, or *identifies absence*. Strike any sentence that only reports.

Chapter 34 — Methodology Writing

34.1 Structure

1. **Overview paragraph** plus Figure 1: the pipeline in five to seven sentences, one per block.
2. **Formal problem statement**: notation, inputs, outputs, objective.
3. **Component subsections**, one per block, with the **novel component clearly isolated** from what is inherited.
4. **Algorithm box** with pseudocode and stated complexity.
5. **Loss function** and how components combine.
6. **Design-rationale sentences** — the "because" sentences of §22.5.

34.2 The reproducibility standard

> *Could a competent stranger, with this paper and the released code, obtain my numbers within noise?*

If the answer requires an email from you, the methodology is incomplete. Chapter 22's six layers and Chapter 26's controls specify what must be present; this chapter is about writing them down.

The leakage sentence — include it verbatim: *"All preprocessing statistics, resampling, and feature selection were fitted on training folds only. Splits are [unit]-disjoint."*

The fairness sentence — include it verbatim: *"All methods received an identical budget of N trials over search spaces of comparable size, selected on validation data."*

These two sentences remove the two most common reasons a reviewer distrusts an empirical result.

34.3 Notation discipline

Define every symbol at first use. Keep one convention throughout (for example bold lowercase for vectors, bold uppercase for matrices, calligraphic for sets). Never reuse a symbol for two meanings. Match your **code variable names** to the paper's symbols — this costs nothing during implementation and saves hours during revision, and it makes your released code checkable against the paper.

34.4 Common mistakes

Mistake	Correction
Methodology as a chronological narrative of what you tried	Present the final design; put exploration in an appendix if it is informative
Novel component not distinguishable from inherited parts	Say explicitly which is which, in prose and in Figure 1
No complexity or cost statement	State cost relative to the baseline
Symbols undefined or reused	One meaning per symbol, defined at first use
Missing leakage and fairness sentences	Add both verbatim
Method described so abstractly it cannot be implemented	Give shapes, dimensions, and defaults

Exercises

Exercise 34.1 Write your methodology overview paragraph — one sentence per pipeline block.

Exercise 34.2 Add the leakage sentence and the fairness sentence to your draft.

Exercise 34.3 Give your methodology to a colleague and ask them to list what they would still need in order to reimplement it. Every item is a defect.

Chapter 35 — Results Writing

Chapter 29 established the boundary between Results and Discussion and gave worked weak/strong examples. This chapter adds the mechanics.

35.1 Structure and procedure

1. **One subsection per research question**, in the order posed.
2. Open each subsection by naming the question it answers.
3. For each float: **point □ pattern □ magnitude □ uncertainty** (§29.2).
4. Report **inconvenient findings** explicitly. Reviewers find them anyway; finding them yourself is credibility.
5. **No interpretation.** None.

35.2 Objectivity in practice

□ Not objective	□ Objective
"Our method clearly outperforms all baselines."	"Our method attains the highest worst-institution AUC of the five methods evaluated (Table II); the margin over the second-best is 0.017 ± 0.011."
"The improvement is significant."	"The improvement is statistically significant (paired Wilcoxon, W = 3, p = 0.004, Holm-corrected across five architectures)."
"As expected, performance improved."	"Performance increased by 0.041 (95% CI 0.032–0.050)."
"Surprisingly, the baseline failed."	"The CORAL baseline attained 0.771 ± 0.014, below empirical risk minimisation (0.781 ± 0.011); this difference is not significant (p = 0.31)."

Note that "as expected" and "surprisingly" are interpretation and belong in the Discussion. So does "clearly".

35.3 Reporting conventions

- Report **dispersion with every mean**, and define the dispersion measure once.
- Report **effect size with every significance claim**.
- State the **number of runs** in the caption or the first sentence of the section.
- Use **consistent decimal places**, and do not report more precision than your dispersion justifies: `0.8221 ± 0.0090` should be `0.822 ± 0.009`.
- Report **negative and null results** in the same detail as positive ones.

Exercises

Exercise 35.1 Rewrite one paragraph of your results removing every interpretive word.

Exercise 35.2 Check that every mean in your results carries dispersion and every significance claim carries an effect size.

Chapter 36 — Discussion Writing

Chapter 29 gave the six-move paragraph pattern and worked examples. This chapter covers what else a Discussion must contain.

36.1 The four obligations

1. Interpretation. Explain *why* the observed pattern occurred, with reference to a mechanism your evidence supports. Apply the restatement test (§29.4).

2. Comparison with literature. Place your result against prior findings — including, especially, those that disagree. Reconciling a contradiction is one of the most valuable things a Discussion can do (§29.3, the worked example).

3. Implications. Name who should now do what differently. Implications are concrete: a threshold a practitioner should adopt, an experiment other researchers should stop running, a design decision that is now better informed.

4. Limitations. Name the threat, its **direction**, and its **consequence** (§29.5). Stating the direction — whether the bias makes your estimate conservative or optimistic — is what distinguishes a competent limitation from a ritual one.

36.2 Matching claim strength to design

This is where papers most often overreach. The permissible claim strength is set by the design, not by the size of the effect.

Design	Permissible language
Exploratory, small sample	“suggests”, “is consistent with”, “warrants investigation” — explicitly preliminary
Descriptive measurement	Factual about the studied sample; generalisation must be argued separately
Controlled comparison	“improves”, “outperforms” — bounded by the conditions tested
Ablation or intervention	“accounts for”, “is responsible for” — within the tested configuration
Observational, no intervention	“is associated with” — **not** “causes”, “improves”, or “leads to”

36.3 Significance without inflation

Significance is established by connecting your finding to a decision, not by adjectives. Compare:

> ▯ “This work has significant practical implications for the healthcare industry.”
>
> ▯ “Because deployment risk is governed by worst-site rather than mean performance, a 0.116 gap implies that a model reported at 0.90 AUC may present as 0.78 at an unseen institution — a difference large enough to change the operating threshold a hospital should adopt, and large enough that internal validation alone is insufficient evidence for procurement.”

Exercises

Exercise 36.1 Write one Discussion paragraph fulfilling all four obligations of §36.1.

Exercise 36.2 Check every claim verb in your Discussion against Table §36.2. Downgrade any that outrun the design.

Exercise 36.3 Rewrite your significance sentence to name a decision that changes.

Chapter 37 — Conclusion Writing

37.1 What a conclusion is for

The conclusion states what is now known. It is not a summary of what you did; it is the claim as you would defend it, compressed.

Structure — four to six sentences, or two short paragraphs.

1. **The problem and gap**, in one sentence.
2. **What you did**, in one sentence.
3. **What was found** — the two or three headline numbers.
4. **What it means** — the implication, scope-bounded.
5. **The principal limitation**, in one sentence.
6. **The next question**, derived from that limitation.

37.2 What must not appear

Do not include	Why
New results, numbers, or figures	Evidence belongs in Results
Claims stronger than the abstract’s	Reviewers compare the two
New citations introducing new argument	The argument is closed by now
A restatement of the entire methodology	The reader has just read it
Generic future work (“we will apply this to other datasets”)	Derive future work from your stated limitations
Anything not supported by your own evidence	This is the section most often over-claimed

37.3 Worked contrast

[HYPOTHETICAL] □ Weak. *“In this paper, we proposed a novel framework for chest X-ray classification. Extensive experiments demonstrated the effectiveness and superiority of our approach compared to state-of-the-art methods. The results show that our method is highly accurate and suitable for real-world clinical deployment. In future work, we will apply our method to other medical imaging modalities and explore other deep learning architectures.”*

Defects: no numbers; “effectiveness and superiority” asserted; “suitable for real-world clinical deployment” is a claim the study cannot support; future work is generic and unconnected to any limitation.

[HYPOTHETICAL] □ Strong. *“Reported performance for chest radiograph classifiers has been measured predominantly on splits that permit institutional leakage, leaving cross-hospital behaviour unquantified. We re-evaluated five published architectures under institution-disjoint protocols and introduced CLUSTER-DG, a label-free adaptation method. Macro AUC falls by 0.116 (95% CI 0.104–0.128) when leakage is removed, and CLUSTER-DG recovers 0.041 ± 0.009 of worst-institution AUC without using provenance metadata — approximately 82% of the benefit obtained by methods that require it. For practitioners, this implies that internal validation substantially overstates the performance a new institution should expect, and that mitigation does not require metadata that privacy procedures remove. Our institutional partition is coarse, treating each public dataset as one institution although each aggregates several sites, so true site-level degradation is likely larger than we report. Establishing that requires multi-site data with genuine site labels, which we identify as the necessary next step.”*

Every sentence does one job; the numbers match the abstract; the claim is bounded (“for practitioners”, “approximately 82%”); the limitation names its direction (“likely larger”); and the future work follows from the limitation rather than being decorative.

37.4 Verification checklist for Part X

- [ ] Written in writing order, not reading order.
- [ ] Title contains searchable terms and no dead words.
- [ ] Abstract has all seven moves; every number matches Results.
- [ ] Introduction ¶4 is a hinge containing at least one count.
- [ ] Introduction ¶5 states a mechanism and names the experiment testing it.
- [ ] ¶6 contributions match abstract, results subsections, and conclusion.
- [ ] Related work is grouped by family, with a comparison table and an evaluation-practice subsection.
- [ ] Methodology contains the leakage sentence and the fairness sentence.
- [ ] Results contains no interpretation; every mean has dispersion.
- [ ] Discussion fulfils all four obligations; claim verbs match the design.
- [ ] Conclusion introduces no new evidence and no claim stronger than the abstract.
- [ ] Future work is derived from a stated limitation.

Exercises

Exercise 37.1 Write your conclusion in the six-sentence structure of §37.1.

Exercise 37.2 Compare your conclusion's claims against your abstract's. Any escalation must be removed.

Exercise 37.3 Check that your future work follows from a limitation you actually stated.

PART XI — FIGURES, TABLES AND ALGORITHMS

Part XI covers the visual and formal apparatus of a paper. Chapter 38 addresses figures — architecture diagrams, flowcharts, and plots. Chapter 39 covers the four table types every empirical paper needs. Chapter 40 treats algorithms, pseudocode, and mathematical formulation.

Chapter 28 established *how results are displayed and how displays mislead*; this part concerns *how the artefacts are constructed*.

Chapter 38 — Research Figures

38.1 The five figure types

Type	Purpose	Typical position
System architecture	Show components and data flow; identify the novel block	Figure 1, in Methodology
Flowchart	Show an ordered procedure with decisions	Methodology, or the screening funnel in a review
Conceptual diagram	Illustrate an idea, relationship, or taxonomy	Introduction or Related Work
Result plot	Show a measured relationship	Results
Qualitative examples	Show representative inputs, outputs, or failures	Results or Discussion

38.2 Architecture diagrams

Figure 38.1 — Anatomy of a good architecture diagram

```
┌────────────────────────────────────────────────────────────────────────┐
│  □  ONE DIRECTION of flow: left→right or top→bottom, never both         │
│                                                                         │
│   Input          Backbone                          Head                 │
│   224×224×3  ──► DenseNet-121 ──► ║ CLUSTERING ║ ──► 14 logits          │
│      □           (7.0M params)    ║   BRANCH   ║      □                 │
│   shapes on                                                             │
│   every arrow                     □ novel block                         │
│                                     visually distinct                   │
│                                                                         │
│  □ every symbol/abbreviation either spelled out or defined in caption   │
│  □ font ≥ 8 pt AFTER scaling to column width                            │
│  □ legible in greyscale — never rely on colour alone                    │
│  □ caption is self-contained and says what to notice                    │
└────────────────────────────────────────────────────────────────────────┘
```

Rules.

1. **One direction of flow.** Diagrams that flow right, then down, then left are read wrongly.
2. **Label every arrow** with what flows along it: tensor shape, data type, or quantity.
3. **Distinguish the novel block** by outline weight, fill, or double border — and say so in the caption. This is the single most useful thing a Figure 1 can do.
4. **Include input and output shapes.** Reviewers use these to check the method is coherent.
5. **Font size □ 8 pt after scaling** to final column width. Check at print size, not on screen.
6. **Greyscale-legible.** Many readers print; some are colourblind. Use shape, hatching, or line style in addition to colour.
7. **Self-contained caption.**

38.3 Flowcharts

Use conventional shapes consistently: rectangles for processes, diamonds for decisions, parallelograms or cylinders for data. Label every branch out of a decision. Keep to one page-width column where possible; if a flowchart needs a full page, it is probably two flowcharts.

A specific and valuable use: the **screening funnel** for a systematic review, showing records retrieved, screened, excluded with reasons, and included. Reporting frameworks such as PRISMA (Page et al., 2021) specify this diagram, and it makes your search reproducible at a glance.

38.4 Tools

Tool	Best for	Notes
draw.io / diagrams.net	Architecture diagrams, flowcharts	Free; export **SVG or PDF** (vector)
Inkscape	Publication-grade vector editing	Free; full control over every element
PowerPoint / Keynote	Fast block diagrams	Acceptable — but export as PDF, never as a screenshot
Mermaid / Graphviz	Diagrams generated from text	Version-controllable and reproducible; excellent with Git
TikZ / PGFPlots	LaTeX-native figures	Steepest learning curve, best typographic integration
Matplotlib / seaborn / ggplot2	Result plots	Set font sizes explicitly; save as PDF or SVG
Neural-network diagram tools	Architecture drawings for deep models	Useful for conventional block diagrams

Absolute rule. Never paste a screenshot of a diagram, plot, or table into a paper. Use vector formats (PDF, SVG, EPS); reserve raster formats for photographs and heat maps at 300 dpi or higher. Screenshots are the most visible marker of an inexperienced submission.

38.5 Common mistakes

Mistake	Correction
Screenshot of a plot, including notebook cell prompts	Export vector from the plotting library
6 pt text after scaling	Set sizes explicitly; verify at print size
Colour as the only distinguishing channel	Add markers, line styles, or hatching
Unlabelled arrows	Label with what flows
Novel component indistinguishable	Highlight it and state so in the caption
Different colour meanings across figures	One palette, one meaning, throughout
Inconsistent figure styles	Same fonts, palette, and sizing everywhere
3-D bars, gradients, decorative elements	Remove; they distort and distract
Caption that merely names the figure	Say what the reader should notice

Exercises

Exercise 38.1 Draw your system architecture in a vector tool, highlight the novel block, label every arrow with shapes, and export as PDF.

Exercise 38.2 Give the diagram alone to a colleague and ask them to explain your pipeline back to you. Every point of confusion is a defect to fix.

Exercise 38.3 Take one existing plot and fix three things: explicit font sizes, dispersion on every mean, vector export.

Chapter 39 — Tables

39.1 The four table types

Table 39.1 — Table types and their required columns

Type	Purpose	Required columns
Dataset table (usually Table I)	Establish the evidence base	Dataset, records, groups (patients/sites/authors), classes, class balance, **split unit**, licence
Literature comparison table	Position the contribution	Work, year, data, **split protocol**, baselines, metrics, key result, limitation
Performance table (main result)	The central comparison	Method, per-dataset metric with dispersion, significance marker, **cost columns**
Ablation table	Attribute the gain to components	Configuration, one column per component toggled, metric with dispersion, Δ column

Most empirical papers need all four. A paper with a performance table but no dataset table, no literature comparison, and no ablation is missing three quarters of its evidential structure.

39.2 Construction rules

Rule	Detail
Horizontal rules only	Top rule, header rule, bottom rule. No vertical rules; no full grid
Bold the best per column	State in the caption what bold denotes
Uniform decimals, aligned	Two or three places; align on the decimal separator
Dispersion on every mean	`0.822 ± 0.009`, with the measure defined in the caption
Significance markers explained	`*p < 0.05, Holm-corrected`
Group rows	trivial → classical → prior state of the art → **ours** → oracle
Δ column in ablations	Makes the attribution explicit rather than leaving arithmetic to the reader
Cost columns in the main table	Parameters, FLOPs, latency — a win bought with 4× compute is a different claim
Caption above the table	IEEE and most engineering conventions. **[VERIFY]** your target journal
Delete unused columns	If it is never discussed, it is noise
Never screenshot	Typeset it

39.3 A worked ablation table

[HYPOTHETICAL]

Configuration	Cluster branch	λ ramp	Full augmentation	Worst-institution AUC	Δ
ERM (baseline)	–	–	–	0.781 ± 0.011	—
+ augmentation	–	–	✔	0.794 ± 0.010	+0.013
+ cluster branch (λ fixed)	✔	–	✔	0.808 ± 0.012	+0.014
Full (CLUSTER-DG)	✔	✔	✔	**0.822 ± 0.009**	+0.014
Site-supervised (oracle)	✔ (true labels)	✔	✔	0.831 ± 0.010	+0.009

What makes this table strong: a cumulative build-up; exactly one change per row; dispersion on every value; a Δ column attributing gain to components; and an **oracle row** that converts “we improved by 0.041” into “we recovered approximately 82% of the benefit of privileged information without using it” — a substantially more interesting scientific statement and one much harder to dismiss.

Exercises

Exercise 39.1 Draft all four table types for your paper, headers only.

Exercise 39.2 Add an oracle or upper-bound row to your ablation if one exists for your problem.

Exercise 39.3 Check every table against §39.2 and remove every column you do not discuss in the text.

Chapter 40 — Algorithms and Pseudocode

40.1 When to include an algorithm box

Include pseudocode when the procedure has non-obvious control flow, when the order of operations matters for correctness, or when your contribution *is* the procedure. Do not include it for a standard training loop with nothing distinctive — a paragraph is clearer.

40.2 From prose to pseudocode to implementation

Figure 40.1 — From prose description to pseudocode to implementation

```
PROSE (ambiguous)
  "We cluster the embeddings and add an invariance penalty, ramping its
   weight during early training."
  → Cluster what, exactly? Ramping over how long? Refit when?

PSEUDOCODE (unambiguous, 10–15 lines, novel lines marked)
  Algorithm 1  CLUSTER-DG: one training epoch
  Input : batches B, encoder f_θ, head g_φ, cluster count K,
          invariance weight schedule λ_t, refit interval E
  ─────────────────────────────────────────────────────────────
   1  for each batch b in B do
   2      z   ← f_θ(b.x)                        ▷ embeddings
   3 ⋆    μ,σ ← channel_statistics(z)           ▷ low-order stats
   4 ⋆    ĉ   ← kmeans_assign([μ;σ], K)         ▷ inferred domain
   5      L_task ← BCE(g_φ(z), b.y)
   6 ⋆    L_inv  ← domain_confusion(z, ĉ)
   7      L      ← L_task − λ_t · L_inv
   8      θ,φ ← optimiser_step(∇L)
   9  end for
  10  if epoch mod E = 0 then refit k-means centroids
  ─────────────────────────────────────────────────────────────
  Output: θ, φ
  Cost  : O(NKd) additional per epoch over the ERM baseline
  ⋆ marks lines constituting the contribution.

IMPLEMENTATION
  Variable names match the symbols above (z, mu, sigma, c_hat, lam_t),
  so a reader can check the code against the paper line by line.
```

40.3 Rules for algorithm boxes

1. **Ten to twenty lines.** Longer belongs in an appendix or the repository.
2. **State inputs and outputs explicitly**, with types or shapes.
3. **Mark the novel lines.** A reader should see your contribution at a glance.
4. **State the cost** relative to the baseline. Reviewers ask "what does this cost?" without exception.
5. **Number the lines** so the prose can refer to them ("line 4 assigns…").
6. **Use consistent notation** with the mathematical formulation and with your code.
7. **No language-specific syntax.** Pseudocode should be readable by someone who does not use your framework.

Typesetting. In LaTeX, the `algorithm` float combined with `algorithmic`, `algorithmicx`, or `algorithm2e` provides standard formatting. **[VERIFY]** which package your target template expects, since some publishers supply their own.

40.4 Mathematical formulation

Purpose. Formal statement removes ambiguity that prose cannot. It is not decoration, and equations that restate prose without adding precision should be deleted.

Procedure.

1. **Define the setting.** *"Let D = {(x_i, y_i, s_i)} for i = 1…N, where x_i is an image, y_i a label vector, and s_i an unobserved institution identifier."*
2. **Define every symbol at first use**, and keep one convention (bold lowercase vectors, bold uppercase matrices, calligraphic sets).
3. **State the objective.** For the running example, a task loss combined with an invariance term:

```
L(θ, φ)  =  (1/N) Σ_i  ℓ_BCE( g_φ(f_θ(x_i)), y_i )   −   λ_t · I( f_θ(x), ĉ(x) )
            └──────────── task loss ──────────────┘        └── invariance term ──┘

where  ĉ(x)  is the cluster assigned by k-means over channel-wise statistics
       λ_t   ramps linearly from 0 to 1 over the first 5 epochs
       I(·,·) denotes the domain-confusion objective of Eq. (3)
```

4. **Number equations you refer to**; do not number equations you never cite.
5. **Explain what the equation means** in one sentence after it. An unexplained equation is decoration.
6. **State assumptions and their bite** — where they hold and where they fail.

40.5 Common mistakes

Mistake	Correction
Pseudocode reproducing a standard training loop	Delete it; describe in prose
Fifty-line algorithm box	Ten to twenty lines; rest in appendix
Novel lines not distinguished	Mark them
No cost statement	State complexity relative to baseline
Symbols in equations differing from those in pseudocode and code	Unify all three
Equations never referenced	Remove the numbering or the equation
Undefined symbols	Define at first use, without exception
Framework-specific syntax in pseudocode	Write language-neutral pseudocode
Assumptions unstated	State them and acknowledge where they fail

40.6 Verification checklist for Part XI

- [ ] Figure 1 shows the architecture with the novel block highlighted and named in the caption.
- [ ] Every arrow is labelled; input and output shapes are given.
- [ ] All figures are vector, greyscale-legible, and ≥ 8 pt after scaling.
- [ ] All four table types are present.
- [ ] Every table uses horizontal rules only, with dispersion and a defined significance marker.
- [ ] Cost columns appear in the main comparison table.
- [ ] The ablation table changes one factor per row and includes an oracle row where possible.
- [ ] Pseudocode is 10–20 lines with inputs, outputs, marked novel lines, and a cost statement.
- [ ] Notation is identical across equations, pseudocode, and released code.
- [ ] Every numbered equation is referenced in the text and explained in prose.
- [ ] No screenshots anywhere in the manuscript.

Exercises

Exercise 40.1 Write your method as pseudocode in 10–20 lines, marking the novel lines and stating the cost.

Exercise 40.2 Write the formal problem statement and objective, defining every symbol.

Exercise 40.3 Check that your paper's symbols, your pseudocode's symbols, and your code's variable names agree. Fix the disagreements now, while it is cheap.

PART XII — RESEARCH WRITING TOOLS

Part XII is practical and deliberately brief. Chapter 41 covers producing an academic manuscript in a word processor; Chapter 42 covers LaTeX and Overleaf; Chapters 43 and 44 give reference-management workflows.

[VERIFY] Software interfaces, menu names, and features change. The workflows below describe stable functional patterns; confirm specifics against current documentation.

Chapter 41 — Microsoft Word

41.1 When a word processor is the right choice

Word remains the required or preferred format for many journals, especially outside computer science and mathematics, and it is usually the pragmatic choice when co-authors do not use LaTeX. It is well suited to text-heavy manuscripts with few equations. It becomes painful with heavy mathematics, large bibliographies managed by hand, and precise float placement.

41.2 The essential workflow

1. **Start from the journal's template**, not from a blank document. Download it from the publisher's author-resources page.
2. **Use named styles** (Heading 1, Heading 2, Body Text) rather than manual formatting. This is the single highest-value habit: it enables a navigable outline, automatic numbering, and a generated table of contents, and it prevents the formatting drift that makes long documents unmanageable.
3. **Use automatic caption numbering** via *Insert Caption*, and reference floats with *Cross-reference*. Manually typed float numbers become wrong the moment you insert a figure.
4. **Use a citation plugin** — Zotero or Mendeley integration (Chapters 43–44). Never number references by hand: inserting one reference mid-draft renumbers everything after it.
5. **Insert figures as PDF, EMF, or high-resolution images**, not as pasted screenshots, and set text wrapping to *in line with text* to keep positions stable.
6. **Use the equation editor** for all mathematics; do not paste images of equations.
7. **Enable Track Changes** for co-author review, and keep a clean copy for submission.
8. **Save versions with dates** (`paper_v3_2026-03-12.docx`), or keep the file in a versioned folder. Word's own version history is not a substitute for deliberate snapshots.

41.3 Common problems and fixes

Problem	Fix
Figure numbers wrong after insertion	Use *Insert Caption* and *Cross-reference*, then update fields
Reference list out of order	Use a citation plugin, not manual numbering
Formatting drift across sections	Apply named styles; avoid direct formatting
Figures moving unpredictably	Set wrapping to in-line; avoid floating anchors
File corruption on a large document	Split into chapter files; keep dated backups
Equations rendering differently on another machine	Use the built-in equation editor; embed fonts on export
Template compliance failures at submission	Compare against the template's own sample paper before submitting

Exercises

Exercise 41.1 Download your target journal's Word template and set up your manuscript's heading styles in it.

Exercise 41.2 Convert every manually numbered figure reference in your draft to a cross-reference.

Chapter 42 — LaTeX and Overleaf

42.1 Why LaTeX dominates computational fields

LaTeX separates content from presentation, handles mathematics, cross-references, and bibliographies automatically, and produces consistent typography. Overleaf provides a browser-based collaborative editor with version history, which removes the installation barrier that historically deterred beginners.

The cost is a real learning curve and cryptic error messages. **Recommendation:** learn it if you work in computer science, mathematics, physics, or engineering; the investment repays within two papers.

42.2 The minimum you need

Table 42.1 — LaTeX constructs for academic papers

Need	Construct	Note
Document class	`\documentclass[conference]{IEEEtran}`	Use the publisher's class file unchanged
Sections	`\section{}`, `\subsection{}`	Numbering is automatic
Cross-references	`\label{}` and `\ref{}`, `\eqref{}`	Never type a number manually
Figures	`figure` environment with `\includegraphics`	Include PDF or EPS; omit the file extension
Tables	`table` + `tabular`; `booktabs` for rules	`\toprule`, `\midrule`, `\bottomrule`
Equations	`equation`, `align`	Number only what you cite
Algorithms	`algorithm` + `algorithmic` or `algorithm2e`	Check what your template supplies
Bibliography	`\bibliographystyle{IEEEtran}`, `\bibliography{refs}`	Or BibLaTeX with `biber`
Citations	`\cite{key}`	Numbering and ordering are automatic
Units and numbers	`siunitx` package	Consistent formatting of quantities

42.3 A working Overleaf workflow

1. **New Project □ Templates**, and select the publisher's official template (IEEE, ACM, Springer LNCS, Elsevier as applicable).
2. **Upload `refs.bib`**, or link it to your reference manager's automatic export (§43.2).
3. **Split long documents** into one file per section with `\input{}`, so co-authors can work without conflicts.
4. **Use the version history** and label milestones before major revisions.
5. **Keep a local Git mirror.** Overleaf offers Git integration; a local clone protects against service interruption and gives you real version control.
6. **Compile early and often.** A missing brace found after ten lines is trivial; found after four hundred, it is not.

42.4 Common problems and fixes

Problem	Fix
Capitals lowercased in titles	Brace them in the `.bib` file: `{BERT}`, `{C}hina`
Undefined references or citations	Compile twice; BibTeX requires multiple passes
Figures in the wrong place	Use `[t]` placement; avoid `[h!]` fighting the algorithm; trust LaTeX
Overfull hbox warnings	Rephrase, or allow hyphenation; never reduce margins in a publisher template
Bibliography style not matching the journal	Use the publisher's `.bst`; do not hand-edit entries
Template modified to fit more text	Never do this — publishers check, and it causes rejection
Compile timeouts on the free tier	Split files; reduce image resolution; compile locally

Exercises

Exercise 42.1 Create an Overleaf project from your target publisher's official template and compile the unmodified sample.

Exercise 42.2 Move your abstract and introduction into it, with `\cite{}` commands drawn from a `.bib` file.

Chapter 43 — Zotero

43.1 Why a reference manager is not optional

Manual bibliography management fails predictably: renumbering errors, inconsistent formats, missing entries, and — most seriously — citations to papers that were never opened. A reference manager makes the correct behaviour the easy behaviour.

Zotero is free and open source, has the widest plugin ecosystem, and exports cleanly to BibTeX, which makes it the default recommendation for computational researchers.

43.2 Step-by-step workflow

1. **Install** Zotero, the browser connector, and the word-processor plugin.
2. **Create a collection** per project.
3. **Add items** three ways: the browser connector on a publisher page; *Add Item by Identifier* for a DOI, ISBN, or arXiv identifier; or by importing the RIS or BibTeX file exported from your database search (§10.9).
4. **Fix the metadata immediately.** Publisher exports routinely contain wrong or missing page numbers, ALL-CAPS titles, and incorrect venue names. This is the step everyone skips and everyone pays for at submission.
5. **Attach the PDF** and let Zotero index it, so full-text search works across your library.
6. **Tag by triage status** using colour tags — core, context, discard (§12.2).
7. **Add a child note per paper** using the sixteen-field extraction template (Chapter 13). Your related-work section will be written from these notes, not from the PDFs.
8. **Install Better BibTeX** for stable, human-readable citation keys and automatic export of a `.bib` file that stays synchronised with your library.
9. **Export the collection as CSV** to seed your literature matrix (§16.2).
10. **Before submission**, resolve every DOI in the exported bibliography and compare fields against the publisher record.

43.3 Limitations and verification

Limitation	Mitigation
Free cloud storage is limited	Store PDFs locally, or use institutional or alternative storage
Publisher metadata is often wrong	Verify every field manually — this is a genuine obligation, not a nicety
Automatically generated keys can collide	Better BibTeX with a deterministic key pattern
Group libraries can drift out of sync	Nominate one owner of the authoritative library

Exercises

Exercise 43.1 Set up a Zotero collection, import your search results, and fix the metadata on ten items. Record how many had errors.

Exercise 43.2 Install Better BibTeX and configure automatic export to a `.bib` file used by your Overleaf project.

Chapter 44 — Mendeley

44.1 Role and practical workflow

Mendeley occupies the same functional position as Zotero — library, PDF annotation, citation insertion — within the Elsevier ecosystem. It is a reasonable choice if your institution standardises on it, if your co-authors already use it, or if you work heavily with Scopus and ScienceDirect, where the import path is direct.

Workflow.

1. Install the desktop application, the web importer, and the citation plugin for your word processor.
2. Import RIS or BibTeX exports from Scopus or ScienceDirect, or capture items with the web importer.
3. **Verify and correct metadata** on import — the same obligation as with Zotero.
4. Organise into folders per project; use tags for triage status.
5. Annotate PDFs; annotations sync across devices.
6. Insert citations with the plugin and generate the reference list in the journal's style.
7. Export BibTeX for LaTeX workflows.

44.2 Choosing between managers

Consider	Zotero	Mendeley	EndNote
Cost	Free, open source	Free tier	Paid, often institutionally licensed
BibTeX and LaTeX integration	Strongest, via Better BibTeX	Adequate	Adequate
Plugin ecosystem	Largest	Smaller	Moderate
Institutional standardisation	Less common	Common	Common in some institutions
Best fit	Computational researchers; LaTeX users	Elsevier-centric workflows	Groups with an existing licence and shared libraries

Recommendation. Any of the three, used consistently and with metadata verified, is vastly better than none. Choose the one your collaborators use; the cost of two managers in one project exceeds the benefit of the better tool.

44.3 Verification checklist for Part XII

- [ ] I am using the publisher's official template, unmodified.
- [ ] Headings use named styles (Word) or sectioning commands (LaTeX).
- [ ] All float numbers and citations are automatic, never typed.
- [ ] Figures are vector or □ 300 dpi, embedded as files rather than screenshots.
- [ ] Equations are typeset, not images.
- [ ] A reference manager holds every source, with metadata verified field by field.
- [ ] Citation keys are stable and human-readable.
- [ ] A `.bib` file is exported automatically, not maintained by hand.
- [ ] Every DOI in the bibliography resolves and matches the publisher record.
- [ ] The document is versioned with dated snapshots or Git.
- [ ] Extraction notes are stored per paper and will be the source for related work.

Exercises

Exercise 44.1 Choose one reference manager and migrate every source into it. Delete the others.

Exercise 44.2 Resolve every DOI currently in your bibliography and record how many fields were wrong. For fifty references this takes about thirty minutes and prevents the most common avoidable reviewer complaint.

PART XIII — AI TOOLS FOR RESEARCH

Part XIII addresses the responsible use of generative and AI-assisted tools in research. Chapter 45 classifies the tools by function and specifies the verification each requires. Chapter 46 provides a prompt library covering fourteen research tasks, with the safeguards each prompt must carry.

The governing principle is stated once and applied throughout:

> **AI tools may help you find, sort, translate, structure, and pre-fill. You must read, verify, compare, judge, and claim.**

This is not a moral posture but a practical one. The researcher is accountable for every sentence, every citation, every number, and every claim in their work. No tool assumes that accountability, and no disclosure transfers it.

Chapter 45 — AI-Assisted Research

45.1 A functional classification

[VERIFY] Tool names, capabilities, and pricing change rapidly, and some products described here may have altered materially. The *functional* categories are stable; verify current capability before relying on any specific product.

Table 45.1 — AI tools: capability and verification burden

Category	Genuinely useful for	Verification burden
General assistants (ChatGPT, Claude, Gemini and similar)	Structuring sections; improving clarity and grammar; generating search-string variants; explaining unfamiliar methods; drafting and refactoring code; building checklists; adversarial self-review	**High.** No reliable literature grounding by default. May state confident falsehoods. **Must never be trusted for citations, numbers, or claims about what the literature says**
Search-grounded assistants (Perplexity and similar)	Rapid orientation on an unfamiliar topic, with visible sources	**Medium–high.** Open every source. A cited page may exist and yet not support the sentence attributed to it
Literature extraction tools (Elicit and similar)	Screening many papers; extracting structured columns into a table	**Medium.** Extraction errors and mis-attributed numbers occur; verify every cell against the PDF
Evidence-search tools (Consensus and similar)	“What does the literature say about X?” with per-paper claim summaries	**Medium.** Summaries compress nuance and can invert conditionals; read the primary paper before citing
Grounded document assistants (NotebookLM and similar)	Question-answering across *your own* uploaded corpus, with citations back to your files	**Lower** — grounded in documents you supplied — but still verify quotations and numbers
Academic search engines (Semantic Scholar and similar)	Metadata, citation context, influential-citation signals, alerts, APIs	**Low** for metadata; automated summaries are lossy
Language tools (grammar and style checkers, translators)	Grammar, register, academic phrasing, drafting in a first language then translating	**Low**, but check that technical meaning and hedging were not altered
Coding assistants	Boilerplate, unit tests, refactoring, plotting, debugging	**Medium.** Never trust unexecuted code; run the tests

A note on grounding. Tools that answer from documents *you* supply are structurally safer than tools answering from model memory, because the provenance of every claim is checkable. This reduces but does not eliminate error.

45.2 What AI assistance legitimately accelerates

Task	Legitimate assistance
Search-string construction	Generating synonym candidates and syntax variants — which you then run and log yourself
Comprehension	Explaining an unfamiliar method before you read the original
Structuring	Proposing a section outline for content you supply
Language	Grammar, clarity, register — particularly valuable for non-native English writers
Code	Boilerplate, tests, plotting, debugging assistance
Checklists	Generating a task-specific checklist from criteria you specify
Adversarial review	Producing the objections a hostile reviewer would raise (§46.5 — the highest-value application)
Response drafting	Tone and phrasing for a reviewer response whose substance you supply

45.3 The red lines

These are not matters of etiquette. Each carries a specific, documented risk of invalidating your work or constituting misconduct.

Never rely on AI to	Why it fails	Consequence
Produce references	Generative models can compose plausible author–title–venue–DOI combinations that do not exist, or attach a real identifier to a wrong title	Fabricated citations have led to desk rejections, corrections, retractions, and — in documented professional contexts — formal sanctions
State what a paper found	Output is generated from patterns, not from reading your PDF	Misattributed claims; a citation that does not support your sentence
Report experimental results	It has run no experiment	**Data fabrication** — research misconduct (§49.2)
Invent datasets or statistics	Fluent invention is indistinguishable from fact in the output	An unverifiable and invalid study
Assert novelty (“no prior work exists”)	No comprehensive index; no capacity to reason about absence	The gap collapses under review
Replace critical reading	Cannot judge whether a baseline was fairly tuned	You lose the ability to defend your own paper
Replace scientific reasoning	Correlates text; does not test hypotheses	Unsupported claims
Write text you submit unread	You are the author and are accountable for every word	Integrity violation; failure under examination

On the claim that AI-generated text is “plagiarism-free”. This claim is false and should be treated as a warning sign wherever it is made. Generated text may closely reproduce training material; it may reproduce another author’s argument structure without attribution; and low similarity-detector output is not evidence of originality (§48.3). Separately, presenting generated text as your own where disclosure is required is itself an integrity violation regardless of similarity score.

45.4 The verification protocol

Figure 46.1 — The verification loop for AI-assisted research

```
AI OUTPUT
   │
   ▼
□ PROVENANCE — For every factual claim, identify the primary source yourself.
   │            Did the tool name one? Does it exist?
   ▼
□ RESOLVE     — Open it. Resolve the DOI. Read at least the abstract and the
   │            relevant table.
   ▼
□ CHECK THE   — Compare against the actual table or figure, not the summary.
  NUMBER        This is where automated extraction most often errs.
   │
   ▼
□ CHECK THE   — Does the source actually SUPPORT your sentence, or merely
  DIRECTION      mention the topic? Does it state the opposite under a
   │             condition you omitted?
   ▼
□ RE-EXPRESS  — Rewrite in your own words, with your own citation.
   │
   ▼
□ LOG         — Record what was AI-assisted, so you can answer honestly
                 if asked.
   │
   ▼
ONLY NOW may it enter your manuscript.
```

45.5 Disclosure

Facts. Major publishers and editorial-ethics bodies have converged on two positions: AI tools **cannot be listed as authors**, because authorship entails accountability and consent that software cannot provide; and **substantive use should be disclosed**, typically in a methods or acknowledgements statement. Many venues additionally restrict AI use in **peer review**, on confidentiality grounds.

[VERIFY] Specific requirements differ by publisher and change frequently. Read your target journal's policy page and follow it exactly. Where a policy is silent, disclose anyway — disclosure has never harmed a submission, and undisclosed use can.

A defensible disclosure statement:

> *"AI assistance was used for language editing and for generating candidate search-query variants. All literature was independently retrieved and verified by the authors against publisher records; all experimental results were produced by the authors' own code and data. The authors are responsible for the content of the manuscript."*

Confidentiality warning for reviewers. If you review manuscripts, do not paste an unpublished manuscript into a public AI service. The manuscript is confidential material entrusted to you, and uploading it may breach that confidence irrespective of any policy on AI use.

45.6 Common mistakes

Mistake	Correction
Accepting a citation that a tool produced	Resolve the DOI and read the paper before citing
Letting a summary substitute for reading a core paper	Summaries are for triage only
Asking a tool whether your idea is novel	Novelty requires a documented search (Part III)
Using extracted numbers without checking the table	Verify every number
Submitting text you have not read closely	You are accountable for every sentence
Assuming low similarity means originality	These are different properties (§48.3)
Failing to disclose substantive use	Read and follow the venue's policy
Pasting a manuscript under review into a public tool	Breach of confidentiality

Exercises

Exercise 45.1 Ask a general assistant for five recent papers on your topic with DOIs. Attempt to resolve each identifier. Record how many resolve, and how many resolve to the stated title. Whatever the outcome, you now know why step □ of §45.4 exists.

Exercise 45.2 Take a paper you have read carefully. Ask a tool to summarise it, then list every inaccuracy or lost nuance. This calibrates how much you can delegate.

Exercise 45.3 Draft your own AI-use disclosure statement now, and keep it with your manuscript.

Chapter 46 — Prompt Engineering for Researchers

46.1 Principles

Four properties make a research prompt safe and useful.

1. **Supply the material.** Prompts that ask the model to *recall* facts invite fabrication. Prompts that ask it to *operate on text you provide* do not.
2. **Forbid fabrication explicitly**, and require uncertainty to be flagged.
3. **Require verifiable output** — reasoning you can check, changes you can review, code you can execute.
4. **Retain the judgement.** The prompt should produce candidates, options, or objections; you decide.

A standard safety clause to append to any research prompt:

> *"Use only the information I have supplied. Do not invent references, datasets, numbers, author names, or DOIs. If something is uncertain or you do not know, say so explicitly rather than guessing. Mark any statement I will need to verify independently."*

46.2 Prompt library index

Table 46.1 — Prompt library index

§	Task	Primary risk it manages
46.3	Finding and narrowing research topics	Vague or infeasible scope
46.3	Generating search queries	Incomplete synonym coverage
46.4	Understanding a paper	Misunderstanding, not fabrication
46.4	Extracting paper information	Mis-attributed numbers
46.4	Comparing papers	False equivalence across incomparable studies
46.4	Building a literature matrix	Unverified cells
46.5	Identifying candidate research gaps	Fabricated absence claims
46.5	Adversarial self-review	Blind spots in your own design
46.6	Improving academic writing	Altered technical meaning
46.6	Structuring an introduction	Structure divorced from evidence
46.7	Reviewing methodology	Undetected design flaws
46.7	Checking logical consistency	Claims unsupported by results
46.8	Generating code	Unexecuted, incorrect code
46.8	Debugging research code	Silent numerical errors
46.8	Explaining statistical results	Misinterpretation of tests

46.3 Topic and search prompts

Narrowing a topic.

> "I am interested in **[broad area]**. My resources are **[compute, data access, time, skills]**. Propose five candidate research problems at a scope completable in **[N months]** with these resources. For each, state: the specific unknown; why the answer would matter and to whom; the datasets that would be needed and whether public ones plausibly exist; the baselines required; and the principal feasibility risk. Do **not** cite papers or claim that anything is novel — I will establish novelty myself through a documented search. Flag any suggestion whose feasibility you are uncertain about."

Generating search strings.

> "I want to find work on **[research question]**. Produce three search strings of increasing precision in **[Scopus / Web of Science]** syntax, using field codes and concept blocks. Include for each concept: British and American spellings, acronyms paired with their expansions, and relevant dataset or benchmark proper nouns. Explain what each block is doing. Do **not** cite any papers. I will run each string myself and record the result counts."

46.4 Reading and extraction prompts

Understanding a method.

> "Explain **[method]** to someone who already understands **[prerequisite]**. State its assumptions, the conditions under which it fails, and the three things I should check in an implementation. Distinguish clearly between what is established and what is your inference. I will verify against the original paper."

Extracting from a paper you supply.

> "Here is the full text of a paper **[paste or attach]**. Fill this table using **only** this document: datasets with versions and sizes; split protocol and split unit; preprocessing steps and where any fitted statistics came from; baselines and whether equal tuning is stated; metrics; seed count; statistical test; reported headline numbers with the table they come from. For each cell, quote the sentence or table you took it from. Write 'NOT STATED' where the paper does not say — do not infer or fill gaps."

The requirement to quote the source sentence for every cell is what makes this prompt safe: it converts an unverifiable summary into a set of claims you can check in seconds.

Comparing papers.

> "Here are extraction tables for five papers **[paste]**. Identify: which studies are directly comparable and which are not, with the reason; the dimensions on which they differ; and any apparent contradictions between their findings. Do **not** average or rank results across studies with different datasets or split protocols — instead state explicitly why such comparison is invalid."

46.5 Gap and review prompts

Candidate gaps — note carefully what this prompt does and does not do.

> "Here are my literature matrix column tallies **[paste]** and the limitations sections of fifteen papers **[paste]**. Based **only** on this material: group the limitations into recurring themes with counts; identify which themes appear in a substantial share of studies; and for each, suggest which of the following gap types it corresponds to **[list the thirteen types]**. Do **not** assert that anything is novel or unstudied — I will establish absence through my own documented search. Flag any theme where my supplied evidence seems too thin to support a claim."

This prompt performs *aggregation over evidence you supply*. It does not, and must not, assert absence.

Adversarial self-review — the single highest-value application.

> "Act as a critical reviewer for **[target venue]**. Here are my abstract, methodology, and results **[paste]**. List the ten strongest objections a knowledgeable reviewer could raise, ranked by severity. For each, state the specific additional experiment, analysis, or clarification that would neutralise it. Be harsh; do not praise. Identify any place where my claims exceed my evidence."

Run this before submission. The objections you cannot currently answer are your remaining experiments.

46.6 Writing prompts

Language editing.

> "Improve the clarity and grammar of the paragraph below for **[target venue]**. Do not change technical content, do not add claims, do not remove hedging or qualifications, and do not strengthen any claim. List every change you made and why, so I can review each one."

The instruction not to remove hedging matters: language tools routinely "improve" cautious scientific prose into overclaims by deleting qualifications.

Structuring an introduction.

> "Here is my gap statement, my four contributions, and my results summary **[paste]**. Propose a six-paragraph introduction outline, stating in one sentence the job of each paragraph. Flag any contribution that has no supporting result, and any paragraph whose job duplicates another's."

46.7 Methodology and consistency prompts

Methodology review.

> "Here is my experimental protocol **[paste]**. Identify: any data-leakage pathway; any comparison in which more than one factor differs; any place where a fitted transformation may have been applied before splitting; any missing baseline category from **[trivial, strong classical, current state of the art, ours-minus-novelty]**; and whether the tuning budget is stated and equal. For each issue, state the specific fix. Do not comment on the merit of the idea."

Logical consistency.

> "Here are my abstract, contribution list, results section, and conclusion **[paste]**. Check for: claims in the abstract not supported by a result; contributions with no corresponding results subsection; numbers that differ between sections; claim strength escalating from results to conclusion; and causal language where the design is observational. Report each inconsistency with the two locations that conflict."

This is a mechanical consistency check that reviewers perform and authors rarely do. It catches a whole class of avoidable criticism.

46.8 Code and analysis prompts

Generating code.

> "Write a **[language]** function computing **[metric or procedure]**, with a clear docstring stating inputs, outputs, and edge-case behaviour. Include at least three unit tests covering **[edge cases]**. Explain any design choice that affects the numerical result."
>
> *Then: run the tests. Never trust unexecuted code.*

Debugging.

> "Here is my function and its unexpected output **[paste code, inputs, expected and actual results]**. Propose three hypotheses for the discrepancy, ranked by likelihood, and for each a specific diagnostic I can run to confirm or eliminate it. Do not rewrite the whole function."

Explaining statistical output.

> "Here is the output of **[test]** on my data **[paste output and describe the design]**. Explain what it does and does not license me to claim, whether the test is appropriate for this design, whether the assumptions are plausible given what I have described, and what effect size I should report alongside it. Note anything about my design that would make this test invalid."

46.9 Common mistakes

Mistake	Correction
Asking for facts rather than operations on supplied text	Always supply the material
Omitting the anti-fabrication clause	Append it to every research prompt
Accepting output without the source sentence	Require quotations for every extracted cell
Asking a tool to assert novelty	Novelty requires your documented search
Using edited prose without reviewing the change list	Require and read the list; check hedging survived
Running code that was never executed	Run it; test it; read it
Never using the adversarial review prompt	It is the most valuable one in the chapter

46.10 Verification checklist for Part XIII

- [ ] I can state which tool categories I use and the verification each requires.
- [ ] Every reference in my manuscript was resolved by me against the publisher record.
- [ ] No number in my manuscript came from an automated summary unchecked.
- [ ] No claim about what the literature says rests on generated text.
- [ ] All experimental results were produced by my own code and data.
- [ ] Every prompt I use includes the anti-fabrication clause.
- [ ] Extraction prompts require a quoted source sentence per cell.
- [ ] I have run the adversarial-review prompt and recorded the objections I cannot answer.
- [ ] Language-edited passages were reviewed change by change, with hedging intact.
- [ ] All generated code has been executed and tested.
- [ ] I have read my target venue's AI policy and drafted a disclosure statement.
- [ ] I have never uploaded a manuscript under my review to a public tool.
- [ ] I understand and can state that I remain responsible for the accuracy, originality, citations, methodology, data, and results of my work.

Exercises

Exercise 46.1 Run the adversarial-review prompt on your own draft. Write down the three objections you cannot currently answer; these are your next experiments.

Exercise 46.2 Run the logical-consistency prompt across your abstract, contributions, results, and conclusion. Fix every inconsistency it reports.

Exercise 46.3 Take one extraction prompt output and verify every cell against the source, using the required quotations. Record the error rate.

PART XIV — PLAGIARISM AND RESEARCH ETHICS

Part XIV treats research integrity as professional knowledge rather than exhortation. Chapter 47 classifies plagiarism and demonstrates the paraphrase boundary that most researchers get wrong. Chapter 48 explains similarity checking and why a percentage is the wrong target. Chapter 49 surveys research misconduct, authorship, and conflicts of interest.

Most integrity failures by early-career researchers arise from ignorance or supervisory pressure rather than intent. The purpose of this part is to remove the ignorance and to name the pressures.

Chapter 47 — Plagiarism

47.1 Definition

Definition. Plagiarism is the use of another person's words, ideas, structures, figures, data, or code without appropriate attribution, presented in a way that implies they are your own.

Note that plagiarism concerns **attribution**, not permission or similarity percentage. Correctly quoted and cited material is not plagiarism however much of it there is; unattributed reuse is plagiarism however little.

47.2 The forms

Table 47.1 — Forms of plagiarism

Form	What it is	Example
Direct / verbatim	Sentences copied without quotation marks and citation	Two paragraphs of related work lifted from a survey
Mosaic / patchwriting	Substituting synonyms while retaining another author's sentence structure and clause order	See §47.3 — **this is plagiarism even when a citation is present, and even when similarity software scores it low**
Paraphrase plagiarism	Restating a specific argument or finding without citation	Presenting another author's explanation as your own reasoning
Idea plagiarism	Using someone's research design, framing, or gap formulation without credit	Adopting a reviewer's or seminar speaker's unpublished idea
Self-plagiarism / text recycling	Reusing your own published text without disclosure or citation	Copying your conference paper's method section into the journal version without citing it
Image / figure plagiarism	Reusing a figure without permission **and** citation	A diagram taken from a paper and redrawn without credit
Data plagiarism	Reusing data without citing the source or honouring its licence	Using a dataset without the required citation or attribution
Code plagiarism	Copying code without honouring licence and attribution	Incorporating a repository's code with the licence header removed
Translation plagiarism	Translating work from another language and presenting it as new	A paper translated and resubmitted as original
Reference padding	Citing works you have not read to appear well-read	Copying a citation list from another paper's bibliography

47.3 The paraphrase boundary, demonstrated

This is the most misunderstood point in the chapter. Most researchers believe that changing the words and adding a citation is sufficient. It is not: sentence architecture is itself borrowed expression.

> **Source:** *"Deep networks trained on single-institution radiographs frequently exploit acquisition artefacts as predictive shortcuts, which limits transfer to unseen hospitals."*

□ Patchwriting — still plagiarism:

> *"Deep networks trained on single-hospital X-rays often use acquisition artefacts as predictive shortcuts, limiting transfer to new hospitals [6]."*

Identical structure, identical clause order, synonym substitution only. The citation acknowledges the *idea* but the *expression* remains the source's. This is the form that similarity software frequently misses and that experienced reviewers and examiners notice immediately.

□ Genuine paraphrase, integrated into your own argument:

> *"Shortcut learning is a documented failure mode in this setting: models can key on scanner-specific artefacts rather than on pathology, so in-distribution accuracy overstates what transfers to a new site [6]. We test this directly by masking image borders (§V-D)."*

The sentence is restructured, the idea is credited, and it is connected to the author's own contribution. The second version is also *better writing*, which is the usual case — genuine paraphrase forces you to understand the point well enough to redeploy it.

47.4 The procedure that makes plagiarism structurally unlikely

Exhortation does not work; process does.

```
1  READ the source.
2  CLOSE the PDF.                     ◄— the critical step
3  WRITE from your extraction template, in your own words,
   in the service of your own argument.
4  REOPEN the source only to check facts, numbers, and the citation.
5  CITE.
```

Closing the source removes the borrowed sentence from view at the moment of composition. This single habit prevents most accidental patchwriting, and it is why Chapter 13's extraction templates matter: you write from *your* structured notes, not from someone else's prose.

47.5 Figures, data, and code

Figures. Reusing a published figure generally requires **permission** from the copyright holder *in addition to* citation, unless the licence (for example a permissive Creative Commons licence) allows reuse with attribution. Publishers provide a permissions route. Redrawing a figure yourself still requires citing the source of the idea. Students routinely do not know this.

Data. Cite the dataset using the citation its creators request, and comply with the licence — including restrictions on redistribution and commercial use.

Code. Open-source licences impose conditions: attribution, licence retention, and sometimes share-alike terms. Removing a licence header is both a licence violation and plagiarism.

47.6 Self-plagiarism and text recycling

This is the form early-career researchers most often commit unknowingly, typically when extending a conference paper into a journal article.

What is required. Cite your own earlier paper explicitly; state in the manuscript and the cover letter what is new; and comply with both venues' policies. Many journals expect a substantial proportion of new material in an extended version — **[VERIFY]** the specific requirement, as thresholds vary by publisher and are sometimes stated explicitly.

Salami slicing — dividing one study into several thin papers to increase output — is a related problem. The test is whether each paper answers a distinct research question and stands as a contribution on its own. If the papers only make sense read together, they are one paper.

Reused method descriptions. Some limited recycling of your own methods text is widely tolerated when the method genuinely is the same, but it must be disclosed and cited. The safe practice is to rewrite and cite.

47.7 Common mistakes

Mistake	Correction
Believing synonym substitution plus citation is sufficient	It is patchwriting; restructure genuinely (§47.3)
Copying your own conference text silently	Cite it and state what is new
Reusing a figure with citation but no permission	Obtain permission, or use a licence that allows reuse
Removing a licence header from copied code	Retain licence and attribution
Citing papers found in another paper's bibliography	Read the primary source before citing
Writing with the source PDF open	Close it (§47.4)
Assuming a low similarity score means no plagiarism	Different properties (§48.3)

Exercises

Exercise 47.1 Take a paragraph from a paper in your area. Write a patchwritten version, then a genuine paraphrase integrated into your own argument. The contrast is instructive and takes ten minutes.

Exercise 47.2 Audit your draft: for every figure not created by you, record whether you have both citation and permission.

Exercise 47.3 If you are extending your own prior work, write the sentence that states what is new, and put it in both the manuscript and the cover letter.

Chapter 48 — Similarity Checking

48.1 What the tools do

Tool	Typically used by	What it does
iThenticate	Publishers and editors, on submitted manuscripts	Matches text against a large corpus of published literature and web content
Turnitin	Universities, on student work	Matches against student submissions, publications, and web content
Publisher screening	At the desk-check stage	An editor sees a similarity report before reviewers are invited

[VERIFY] Which service your institution provides, and whether your target publisher screens submissions, varies. Ask your library or research office.

48.2 How to use a report responsibly

Figure 48.1 — Interpreting a similarity report

```
Overall similarity: 19%
    |
    ├─► DO NOT act on this number. Open the report and examine every match.
    |
    ▼
For each highlighted match, classify it:
    |
    ├─ (a) Correctly quoted and cited       → no action needed
    ├─ (b) Unavoidable technical phrasing    → no action ("convolutional
    |       or standard nomenclature            neural network", metric names)
    ├─ (c) Reference list / bibliography     → no action (inflates the % harmlessly)
    ├─ (d) Method boilerplate you wrote      → consider rewriting for clarity
    |       in a prior paper                    AND cite the prior paper
    └─ (e) UNATTRIBUTED BORROWING            → ✗ FIX by rewriting from your
            (verbatim or patchwritten)           notes and citing.
                                                NOT by word-swapping.
    ▼
Also check the SELF-MATCH against your own prior papers and thesis.
Disclose and cite any legitimate overlap.
```

Procedure. Run your own check before submission if your institution provides access; read the report match by match; classify each; fix category (e) by genuine rewriting and citation; check self-matches; and keep the report, since some journals request it.

48.3 Why the percentage is the wrong target

This section is the point of the chapter.

A similarity score is not a plagiarism score. The tools detect *string overlap*. Plagiarism is a matter of *attribution and intent*. The two diverge in both directions:

- **10% can be misconduct.** One uncited copied paragraph from a single source is plagiarism regardless of the total percentage.
- **25% can be entirely clean.** Long reference lists, standard technical nomenclature, dataset descriptions, and correctly quoted definitions all inflate the number harmlessly.

What the tools cannot detect: mosaic plagiarism that has been sufficiently reworded; translated plagiarism; idea and design plagiarism; fabricated data; image manipulation; and inappropriate authorship.

Deliberate evasion is worse than the original overlap. Using synonym-substitution tools, character substitution, invisible text, or paraphrasing services to reduce a score constitutes *intent to deceive*, and it is treated far more severely than the overlap it conceals. The legitimate action is always the same: express the idea in your own words and cite the source — which reduces similarity as a side effect.

The correct standard, therefore, is not a number:

> Every idea, sentence, figure, dataset, and line of code that is not yours is attributed; and every sentence in your paper is one you could defend as your own expression.

Meet that standard and the percentage takes care of itself. If your institution enforces a threshold, you must satisfy both the threshold and the standard — the threshold is a floor, not a goal.

48.4 A note on AI-detection tools

[VERIFY] and treat with caution. The reliability of tools claiming to detect AI-generated text is contested, and false positives have been reported, with particular concern for non-native English writers whose prose may be flagged disproportionately. Do not treat a detector score as proof in either direction.

The practical protection for an honest researcher is a **documented writing trail**: dated drafts, extraction notes, a research journal (§3.5), and reference-manager history. This evidence is far more persuasive than any detector output.

48.5 Common mistakes

Mistake	Correction
Acting on the percentage without reading the report	Open and classify every match
Reducing a score by synonym substitution	This is evasion; rewrite genuinely and cite
Assuming a low score certifies originality	It does not (§48.3)
Ignoring self-matches	Disclose and cite legitimate overlap
Treating an AI-detector score as proof	It is not; keep a documented writing trail
Not running a check before submission	Run it if you have access; editors will

Exercises

Exercise 48.1 If you have access, run a similarity check on a current draft. Classify every match into the five categories of §48.2. Record how much of your total is categories (a) to ©.

Exercise 48.2 Write down your institution's similarity threshold, if it has one, and the ethical standard of §48.3. Note that satisfying the first does not satisfy the second.

Chapter 49 — Research Misconduct

49.1 Scope and framing

This chapter describes professional norms you are expected to know, much as a practitioner is expected to know the regulations of their field. Authoritative guidance is maintained by bodies including the Committee on Publication Ethics (COPE), the International Committee of Medical Journal Editors (ICMJE) for authorship criteria, and the CRediT contributor taxonomy for contribution statements; your institution will also have a research-integrity policy that takes precedence for you. **[VERIFY]** against those sources.

Table 49.1 — Categories of research misconduct

Category	What it is	Concrete example	Typical consequence
Fabrication	Inventing data or results	Reporting accuracy for an experiment never run; adding rows to a results table	Retraction; institutional sanction; degree revocation
Falsification	Manipulating data, images, or analysis to misrepresent findings	Removing outliers to reach significance; selectively adjusting one image panel; changing a number after a null result	Retraction; misconduct finding
Plagiarism	Unattributed use of others' work	Chapter 47	Desk rejection; retraction; institutional action
Duplicate publication	Publishing the same study twice	Submitting to two journals; republishing without disclosure	Retraction of the later paper
Redundant publication / salami slicing	Dividing one study into multiple thin papers	Four papers where the study supports one	Editorial sanction; reputational cost
Image manipulation	Altering images beyond permitted adjustment	Splicing panels; duplicating regions; selective enhancement	Retraction
Citation manipulation	Distorting the citation record	Coercive citation; citation cartels; padding with irrelevant citations	Editorial sanctions; journal delisting
Authorship violations	Credit not matching contribution	Gift or honorary authorship; ghost authorship; adding a name without consent	Correction; retraction; disputes
p-hacking / HARKing	Analytic choices made after seeing results and presented as pre-planned	Reporting only the significant comparison of six; presenting a post-hoc hypothesis as a prediction	Irreproducible literature; increasingly detected
Undisclosed AI use	Substantive generated content presented as your own where disclosure is required	Submitting generated discussion text unverified and undisclosed	Rejection; integrity investigation
Ethical breaches	Missing approval or consent; privacy or licence violations	Patient data without approval; scraping against terms of service	Legal liability; retraction
Undisclosed conflict of interest	Failing to declare interests that could bias the work	Funding from a party with a stake in the outcome, undeclared	Correction; loss of trust

49.2 The boundary cases that matter in practice

Three situations account for most genuine confusion.

Data cleaning versus falsification. Removing records according to criteria specified *in advance* and reported in the paper is legitimate preprocessing. Removing records *because* they are inconvenient for your hypothesis is falsification. The distinguishing feature is pre-specification and disclosure: state the criterion, state the resulting counts, and report what happens with and without exclusion if the choice is consequential.

Image adjustment versus manipulation. Uniform adjustment of brightness or contrast applied to a whole image is generally acceptable *if disclosed*. Selective adjustment of a region, splicing, or duplicating regions is manipulation. Retain original unprocessed files.

Exploratory analysis versus HARKing. Exploring your data is legitimate and valuable. Presenting a hypothesis discovered during exploration as though it had been predicted is not. The remedy is labelling: report exploratory findings *as exploratory*, and state clearly which analyses were pre-specified. A dated pre-specification entry (§3.4) resolves this permanently.

49.3 Authorship

The widely used ICMJE criteria require **all four** of the following for authorship: substantial contribution to conception or design, or to data acquisition, analysis, or interpretation; drafting or critically revising the work; approval of the final version; and **accountability** for the work's integrity. **[VERIFY]** — some fields apply different conventions, and your discipline may differ.

Everyone who contributed but does not meet all four belongs in the **Acknowledgements**, with their permission.

Bad practice	Why it is wrong
Gift or honorary authorship — adding a senior figure who did not contribute	Misrepresents accountability; dilutes the meaning of authorship
Ghost authorship — omitting a real contributor	Denies credit; can conceal conflicts of interest
Adding a name without consent	Every author must approve submission
Buying or selling authorship (paper mills)	Fraud; has led to mass retractions

Strong recommendation. Agree the author list and order **in writing, before writing begins**, including what happens if someone leaves the project. Use the **CRediT** taxonomy to record who did what — conceptualisation, methodology, software, validation, formal analysis, investigation, resources, data curation, writing (original draft), writing (review and editing), visualisation, supervision, project administration, funding acquisition. Authorship disputes are among the most common and most damaging conflicts in research groups, and they are almost entirely preventable by a five-minute conversation at the start.

49.4 Conflict of interest

A conflict of interest is any circumstance that could reasonably be perceived to bias your work: funding, employment, consultancy, patents, equity, advisory roles, or close personal relationships with those affected by the outcome.

The obligation is **disclosure**, not absence. Having a conflict is usually acceptable; concealing it is not. Declare it in the manuscript, and also declare it when you are asked to *review* a paper by a competitor or collaborator — the correct action there is to inform the editor and let them decide.

49.5 What to do under pressure

Early-career researchers sometimes face pressure to act improperly: to add an author who did not contribute, to remove an inconvenient result, to split a study, or to inflate a claim. Three things are worth knowing in advance.

1. **The pressure is not evidence that the action is normal.** Practices vary between groups, and some groups have drifted.
2. **Your institution has a research-integrity office or equivalent**, and consultation is normally confidential. Find out now who that is, before you need them.

3. **Documentation protects you.** A dated research journal recording decisions, data provenance, and analysis plans is the single most effective protection available (§3.5).

49.6 Common mistakes

Mistake	Correction
Excluding data after seeing results, undisclosed	Pre-specify criteria; report counts and sensitivity
Reporting only significant comparisons	Report all pre-specified comparisons with correction
Presenting a post-hoc hypothesis as predicted	Label exploratory findings as exploratory
Adding a supervisor who did not contribute	Apply the four criteria; use Acknowledgements instead
Not agreeing authorship in advance	Agree it in writing at project start
Extending a conference paper without disclosure	Cite it; state what is new; comply with both policies
Assuming public data need no ethics consideration	Check with your committee; public ≠ exempt
Not declaring funding or interests	Disclose; concealment is the violation

49.7 Verification checklist for Part XIV

- [ ] Every non-original sentence, figure, dataset, and code fragment is attributed.
- [ ] I have checked my draft for patchwriting, not merely for verbatim copying.
- [ ] Reused figures have both citation and permission.
- [ ] Any overlap with my own prior work is cited and disclosed.
- [ ] I have run a similarity check and classified every match, not just read the percentage.
- [ ] Data exclusions were pre-specified and are reported with counts.
- [ ] All pre-specified comparisons are reported, with multiple-comparison correction.
- [ ] Exploratory findings are labelled as exploratory.
- [ ] The author list satisfies all four authorship criteria, with a CRediT statement.
- [ ] Every author has seen and approved the submission.
- [ ] Ethics approval, consent, funding, conflicts, and data/code availability are declared.
- [ ] AI use is disclosed per the venue's policy.
- [ ] I maintain a dated research journal.
- [ ] I know who my institution's research-integrity contact is.

Exercises

Exercise 49.1 Write your full declarations block now: author list with CRediT roles, ethics status, consent, funding, conflicts, data availability, code availability, AI use, and prior presentation. Most researchers discover at least one unresolved issue — resolving it now is far cheaper than at submission.

Exercise 49.2 Identify your institution's research-integrity contact and record it.

Exercise 49.3 Write your pre-specification entry for your principal analysis, and date it.

PART XV — JOURNAL PUBLICATION

Part XV covers venue selection. Chapter 50 distinguishes the venue types and access models. Chapter 51 gives the criteria and metrics, with the definitions and caveats that are routinely confused. Chapter 52 addresses predatory and low-integrity venues, with a verification checklist.

[VERIFY] throughout. Metric definitions, indexing status, and publisher policies change. Every procedure in this part therefore tells you how to check the current position against an authoritative source rather than asking you to trust a printed statement.

Chapter 50 — Understanding Journals

50.1 Venue types

Table 50.1 — Venue types compared

Dimension	Conference	Journal
Review	Single round, fixed deadline, programme-committee based	Multiple rounds, rolling submission, editor plus referees
Decision	Accept or reject; sometimes a revision round	Major or minor revision cycles, possibly iterating for months
Time to decision	Typically 2–4 months, with a fixed presentation date	Typically 3–12 months to first decision
Length	Usually 6–10 pages, hard limit	8–35 pages, often with appendices
Depth expected	A crisp new idea with adequate validation	A complete study: exhaustive baselines, ablations, statistics, thorough related work
Standing in computer science and AI	**Very high** — the leading conferences are the primary archival venue	High; required by most non-CS fields and many degree regulations
Extension	Frequently extended into a journal version	Usually terminal
Cost	Registration and travel	Article processing charge may apply

Two facts that early-career researchers receive contradictory advice about.

First, in computer science and artificial intelligence, the top conferences are genuinely the primary archival venue and are often regarded as more selective than journals. In nearly every other discipline, the journal is primary.

Second, **your degree regulations may not reflect your field's norms.** Many universities require a specified number of papers in journals indexed in particular databases, regardless of how strong a conference record is. **[VERIFY]** your own institution's requirement early, and plan a deliberate mix.

Other venue types. Workshops (early-stage work, community building, usually non-archival or lightly archival); short communications and letters (rapid, concise results); registered reports (protocol reviewed before results, an increasingly available option that protects against publication bias); and preprint servers, which establish priority and visibility but are **not** peer review.

50.2 Extending a conference paper

This is a common and legitimate route that becomes misconduct if handled carelessly.

Requirements: cite the conference paper explicitly in the journal submission; state in both the manuscript and the cover letter what is new; and comply with both venues' policies. Many publishers expect a substantial proportion of new material. **[VERIFY]** the specific expectation, as it varies and is sometimes stated numerically.

New material that genuinely counts: additional datasets; substantially expanded experiments; ablations absent from the conference version; theoretical analysis; a new method component; and a materially deepened related-work and discussion treatment. Reformatting and rewording do not count.

50.3 Access models

Model	Author pays	Reader pays	Notes
Subscription	Usually nothing (page or colour charges possible)	Yes, via institutional subscription	The traditional model
Gold open access	Article processing charge (APC)	No	Immediately open at the publisher
Hybrid	Optional APC to make your article open	Yes, unless the APC is paid	A subscription journal offering an open-access option
Diamond / platinum	Nothing	Nothing	Funded by institutions or societies
Green	Nothing	Varies	You self-archive the accepted manuscript in a repository, subject to the publisher's policy and any embargo

Practical guidance. Before paying an APC personally, check three things: whether your funder mandates open access; whether your institution has a read-and-publish or transformative agreement with the publisher that covers the charge; and whether a waiver or discount applies for authors in certain countries or circumstances. Many researchers pay charges that an existing agreement would have covered.

Green open access is the free route to visibility. Most publishers permit deposit of the accepted manuscript in an institutional or subject repository after an embargo. **[VERIFY]** the policy for your specific journal — publisher policy pages and services that aggregate self-archiving policies are the authoritative sources.

50.4 Special issues

A special issue is a themed collection with a guest editor and a submission deadline. An open call is the most explicit statement of editorial demand available: an editor is publicly stating which topics they want and will handle.

Advantages: a receptive editor, a clear timeline, and a thematically coherent readership. **Cautions:** review quality and rigour vary; confirm that the special issue is hosted by a journal that passes the verification of Chapter 52 — special-issue invitations are also a common vector for predatory solicitation.

Exercises

Exercise 50.1 Write down your degree regulation's publication requirement, verbatim, and check whether your planned venue satisfies it.

Exercise 50.2 Determine whether your institution has a read-and-publish agreement covering your candidate publishers.

Exercise 50.3 If you are extending a conference paper, list what is new and quantify it.

Chapter 51 — Journal Selection

51.1 Criteria in priority order

#	Criterion	How to judge it
1	**Scope fit**	Read the Aims and Scope statement. Then check that **three to five papers you cite** were published there. Scope mismatch is the leading cause of desk rejection and is entirely preventable
2	**Indexing**	Presence in Scopus and/or the Web of Science Core Collection, verified on the **official** lists — not on the journal's own claim
3	**Audience**	Are the people you want to cite you reading this venue?
4	**Article type fit**	Full paper, letter, short communication; page and figure limits
5	**Quality tier**	Quartile from an appropriate source, with the category stated
6	**Speed**	The journal's reported submission-to-first-decision and time-to-publication figures
7	**Cost**	APC; waivers; institutional agreements
8	**Legitimacy**	Chapter 52
9	**Special issues**	An open call matching your topic
10	**Your regulations**	Does your institution accept this venue for your degree?

Criterion 1 deserves emphasis. The most reliable way to find candidate journals is to look at where the papers *in your own reference list* were published. If none of your cited work appears in a journal, it is probably the wrong journal — and an editor will reach the same conclusion in about ninety seconds.

51.2 Metrics: what they actually mean

Table 51.1 — Journal metrics: source, definition, and caveat

Metric	Source	Definition, in brief	Caveat
Journal Impact Factor (JIF)	Clarivate, Journal Citation Reports (based on Web of Science)	Citations in a given year to items published in the two preceding years, divided by the number of those citable items	Available only for journals in the relevant indexes; strongly discipline-dependent; **describes a journal, not your paper**
JIF quartile (Q1–Q4)	Journal Citation Reports, **per subject category**	The journal's rank within its category, divided into quarters	A journal may be Q1 in one category and Q2 in another — **always state the category and the source**
CiteScore	Scopus (Elsevier)	Citations over a four-year window per document	A **different** database, window, and document set from JIF; the two are **not interchangeable**
SJR (SCImago Journal Rank)	SCImago, using Scopus data; freely available	Prestige-weighted citations, source-normalised	A useful free proxy for tiering when you lack subscription access
SNIP	Scopus	Field-normalised citation impact per paper	Designed to enable cross-field comparison
h5-index	Google Scholar Metrics	Journal-level h-index over a five-year window	Includes venues with no editorial vetting
Acceptance rate	The journal or publisher, when published	Proportion of submissions accepted	Frequently unpublished; treat third-party figures sceptically

Three points that are routinely confused and that you should be able to state correctly.

1. **"SCI indexed" is loose usage.** The Web of Science Core Collection comprises several indexes, including the Science Citation Index Expanded (SCIE), the Social Sciences Citation Index (SSCI), the Arts and Humanities Citation Index (AHCI), and the Emerging Sources Citation Index (ESCI). A journal's own website claiming "SCI indexed" is *not* evidence. **[VERIFY]** on Clarivate's Master Journal List, and note *which* index.
2. **Impact Factor and CiteScore are not the same quantity** and cannot be compared with one another. They come from different databases, use different citation windows, and count different document types. Quote each with its source and year.
3. **Journal-level metrics do not measure your paper.** This is not merely a philosophical point: research-assessment reform initiatives, notably the San Francisco Declaration on Research Assessment (DORA), explicitly recommend against using journal metrics as a proxy for the quality of individual articles, and many institutions and funders have adopted that position. A strong paper in a well-matched specialist journal frequently serves a career better than a weak paper in a high-metric generalist one.

51.3 Verification procedure

Ten minutes per candidate journal, always in this order, and never starting from an email.

Step	Check	Source
1	The journal exists on the **publisher's own domain**	The publisher's site, reached independently
2	Web of Science Core Collection membership, and **which index**	Clarivate Master Journal List
3	**Active** Scopus coverage (not discontinued); CiteScore	Scopus source list
4	Quartile, SJR, subject category, trend	SCImago
5	For open-access journals: listing status	Directory of Open Access Journals (DOAJ)
6	The **ISSN belongs to this journal** — this catches cloned sites	The ISSN Portal
7	Publisher or journal is a COPE member	COPE membership list
8	Three editorial board members verified on their institutional pages	Institutional websites
9	Read two recent papers — is this the standard you want to be judged by?	The journal itself
10	The title is not a hijacked imitation of a legitimate journal	Chapter 52

51.4 Strategy

- Choose **one target and two backups before writing**, and format for the target from the first draft.
- Aim the target one tier above your honest self-assessment; use the backups without ego if rejected.
- **Never submit to two journals simultaneously.** Duplicate submission can be treated as misconduct (§49.1).
- If rejected, **use the reviews before resubmitting elsewhere.** A rejection with substantive reviews is free expert consultancy, and resubmitting unchanged wastes it.
- Check whether the publisher offers **transfer** to a sister journal — often faster than a fresh submission and it carries the reviews with it.

Exercises

Exercise 51.1 Complete the ten-step verification of §51.3 for three candidate journals.

Exercise 51.2 For each, quote the scope phrase that covers your topic and count how many of your cited papers appeared there.

Exercise 51.3 Record each journal's quartile *with its source and category*, and its APC with any applicable waiver or agreement.

Chapter 52 — Predatory Journals

52.1 What the problem is

Predatory or low-integrity venues charge publication fees while providing little or no genuine peer review, editorial oversight, or archiving. They are a serious hazard for early-career researchers, because the consequences are largely irreversible: the work may be effectively uncitable, your institution may refuse to count it towards a degree, the paper generally cannot be withdrawn and republished elsewhere because it has already been published, and the fee is not recoverable.

A related hazard is the **hijacked journal** — a cloned website imitating a legitimate journal, often at a slightly different address, which collects fees under the real journal's reputation.

52.2 Warning signs

Table 52.1 — Predatory-journal warning signs

Signal	Why it indicates a problem
Unsolicited flattering email inviting a submission	Legitimate journals rarely solicit individual manuscripts this way
Promise of publication in **days**, or "guaranteed acceptance"	Genuine peer review cannot be completed on such timescales
An **unrecognised metric** — variants such as "Global Impact Factor" or similar invented indicators	Fabricated authority; the recognised metrics come from the sources in Table 51.1
Claims of indexing that **fail verification** at §51.3 steps 2 and 3	The claim is false
Impossibly broad scope spanning unrelated disciplines	No editorial board can competently review such a range
APC **not disclosed before acceptance**, or payment to a personal account	Legitimate publishers disclose charges up front and invoice institutionally
Editorial board **without affiliations**, or listing members who did not consent	Fabricated credibility
Website closely **imitating a known journal** at a different address	Hijacked journal
No DOIs; no preservation or archiving policy	The work will not be reliably findable or preserved
Contact only through a **free email account**	No institutional infrastructure
Grammatical errors on the journal's own homepage	No editorial capacity
"Review" completed with **no substantive comments**	No peer review occurred

52.3 The verification checklist

Use the ten-step procedure in §51.3, and treat the following as decisive:

- [ ] Found from the publisher's own domain, reached independently — **not** from an email link
- [ ] Indexing claims verified on the **official** lists, with the specific index named
- [ ] ISSN confirmed as belonging to this journal
- [ ] If open access, listed in DOAJ
- [ ] APC amount disclosed before submission, payable to the publisher
- [ ] Editorial board members verified independently on institutional pages
- [ ] Two recent articles read and judged acceptable in quality
- [ ] Publisher or journal is a COPE member

- [ ] DOIs assigned and a preservation policy stated
- [ ] Accepted by your institution for your degree or assessment

Prefer positive verification over blacklists. Establishing that a journal *is* present in Scopus, the Web of Science Core Collection, or DOAJ is more reliable than checking whether it appears on an informal list of suspect venues, since such lists are incomplete, contested, and sometimes out of date. The community checklist **Think. Check. Submit.** is a widely recommended framework for this process.

52.4 Practical rules

1. **Never respond to solicitation emails.** Choose journals by starting from your own reference list (§51.1).
2. **Always type the publisher's address yourself** rather than following a link in an email.
3. **Be sceptical of speed.** If the promised timeline is implausible for genuine review, it is not genuine review.
4. **Check before submitting, not after acceptance.** After acceptance you have very little room to act.
5. **Ask your librarian or supervisor** when uncertain. Librarians deal with this constantly and will usually give you an answer in minutes.

52.5 Verification checklist for Part XV

- [ ] I can state the difference between JIF and CiteScore, including their sources and windows.
- [ ] I always quote a quartile with its source and subject category.
- [ ] I understand that journal metrics do not measure my paper's quality.
- [ ] I have verified indexing on official lists, not on journal claims.
- [ ] I have quoted the scope phrase covering my topic for my target journal.
- [ ] At least three papers I cite were published in my target journal.
- [ ] I have a target and two backups, scored against the same criteria.
- [ ] I am formatting for the target from the first draft.
- [ ] My APC position is known, including waivers and institutional agreements.
- [ ] My target passes every item of the §52.3 checklist.
- [ ] I have confirmed my institution accepts this venue for my degree.
- [ ] I am submitting to one journal only.

Exercises

Exercise 52.1 Take a real solicitation email from your inbox and run the §52.3 checklist until it passes or fails. Record where it failed.

Exercise 52.2 Verify your target journal's ISSN on the ISSN Portal and confirm it matches the site you intend to submit to.

Exercise 52.3 Find and record your institution's list of accepted or recognised venues, if it maintains one.

PART XVI — MANUSCRIPT SUBMISSION

> Part XVI covers the mechanics of submission and review. Chapter 53 provides the pre-submission checklist. Chapter 54 covers the cover letter. Chapter 55 explains peer review and its decision points. Chapter 56 gives detailed patterns for responding to reviewers, including how to disagree professionally.
>
> Most desk rejections that are within your control originate in this part: the wrong template, an over-length manuscript, missing declarations, broken links, or an anonymity breach. These are the cheapest failures in research to eliminate.

Chapter 53 — Preparing the Manuscript

53.1 The submission pipeline

```
FINAL MANUSCRIPT
   |  format to the journal template
FORMATTING AND COMPLIANCE CHECK
   |  write cover letter and declarations
COVER LETTER + SUPPLEMENTARY MATERIAL
   |  create account, upload, proof the system-generated PDF
SUBMISSION SYSTEM
   |
EDITORIAL TRIAGE ──────────► DESK REJECT
   |                         (scope, format, similarity, language, ethics)
REVIEWER ASSIGNMENT  (weeks — often the longest silent period)
   |
PEER REVIEW ─────────────► FIRST DECISION
   |
REVISION + RESPONSE LETTER ──► RE-REVIEW ──► (loop 1–3 times)
   |
ACCEPT ──► PROOFS ──► EARLY ACCESS ──► ISSUE ──► DOI, INDEXING
```

53.2 Manuscript checklist

Table 53.1 — Pre-submission checklist

Item	Detail
Template	The journal's own template, **unmodified**; correct column format
Length	Within page or word limits; figure and table counts within limits
Title, abstract, keywords	Finalised; keywords from the journal's or a standard thesaurus
Sections	All required sections present in the journal's expected order
Figures	Vector or ≥ 300 dpi; fonts embedded; greyscale-legible; **all referenced in text**
Tables	Captions positioned per the journal's convention; all referenced; units stated
Equations	Numbered where cited; every symbol defined
References	Journal style; sequential; **every DOI resolved**
Anonymisation	If double-blind: no author names, no identifying acknowledgements, no "our previous work [7]" where [7] is obviously yours, anonymised repository link
Language	Professionally checked if needed
Similarity	Report run and reviewed **match by match** (§48.2)
Declarations	Ethics, consent, funding, conflicts, data availability, code availability, AI use, author contributions, prior presentation

53.3 Files and system checklist

- [ ] **Proof-read the system-generated PDF, page by page.** Figures degrade, symbols vanish, and merged files reorder. This is the file reviewers actually see, and it is not the file you uploaded
- [ ] Source files (`.tex` with `.bib` and figures, or `.docx`)
- [ ] Supplementary material: appendices, extra results, demonstration video
- [ ] Highlighted and clean versions if requested
- [ ] Graphical abstract or highlights if required
- [ ] Cover letter (Chapter 54)
- [ ] Ethics approval documentation if required

- [ ] Data and code availability statements with **links tested in a private browser window**
- [ ] All authors' names, affiliations, emails, and persistent identifiers entered correctly
- [ ] The corresponding author's email is one that will be monitored for months
- [ ] Funding information entered in the system's structured fields, not only in the text
- [ ] Everything archived locally in a dated folder, with the submission identifier

53.4 Declarations

Declaration	What to write
Ethics approval	Committee name, approval number, date — **or** an explicit statement that approval was not required, and why
Informed consent	Obtained or waived; for identifiable images, consent to publish
Conflict of interest	Funding, employment, patents, advisory roles — or an explicit statement that none exist
Funding	Agencies and grant numbers
Data availability	Repository with a persistent identifier, **or** a specific, justified restriction
Code availability	Repository with a tagged release and a persistent identifier
AI use	Per the journal's policy (§45.5)
Author contributions	A CRediT statement (§49.3)
Prior presentation	"An earlier version appeared as [x]; this paper adds …"

On "data available on request". Studies of data-sharing statements have found that such requests frequently go unanswered, and a number of editors now regard the phrase as inadequate. Deposit in a repository with a persistent identifier, or state a specific and justified restriction such as a data-use agreement or participant privacy.

Budget a full day for submission mechanics. Researchers consistently underestimate this and rush, which is when errors enter.

Exercises

Exercise 53.1 Complete Table 53.1 against your current draft and list every failing item.

Exercise 53.2 Write all nine declarations now.

Exercise 53.3 Open your data and code links in a private browser window and confirm they resolve without your credentials.

Chapter 54 — Cover Letter

54.1 Purpose

The cover letter is read by the handling editor, who decides whether to invite reviewers at all. It must answer three questions in about thirty seconds: **is this in scope, is it new, and is it sound enough to justify reviewers' time?**

It is not a summary of the paper and not an abstract. It is an argument for editorial attention.

54.2 Structure and template

Approximately 250 to 350 words, in six moves.

> Dear Professor **[editor's name]**,
>
> **[1 — Submission]** We submit our manuscript, *"**[title]**"*, for consideration as a **[article type]** in **[journal]**.
>
> **[2 — Problem and gap, two sentences]** Across fifteen studies published between 2022 and 2026, chest-radiograph classifiers are evaluated predominantly on random splits that permit institutional leakage, and worst-institution performance is not reported. The magnitude of degradation under leakage-free evaluation is therefore unknown.
>
> **[3 — What you did and the headline result, three sentences]** We re-evaluate five published architectures under institution-disjoint protocols on three public datasets and observe a mean AUC reduction of 0.116 (95% CI 0.104–0.128). We then introduce a label-free adaptation method that recovers 0.041 ± 0.009 of worst-institution AUC without institutional metadata. All experiments use ten seeds with an identical tuning budget across methods.
>
> **[4 — Fit, one or two sentences]** This work falls within **[journal]**'s scope in **[quote the scope phrase]**, and extends work published in your journal **[refs]**.
>
> **[5 — Declarations]** The manuscript is original, is not under consideration elsewhere, and all authors have approved the submission. An earlier version was presented at **[venue]**; this submission adds **[what is new]**. We declare no competing interests. Code, split definitions, and trained weights are available at **[persistent identifier]**.
>
> **[6 — Optional: reviewers]** We suggest the following reviewers: **[names, affiliations, emails, and one line on why each is appropriate]**. We respectfully request exclusion of **[name]** because **[factual professional reason]**.
>
> Sincerely, **[corresponding author, affiliation, persistent identifier, email]**

54.3 Notes on the difficult parts

Move 4 — fit. Quote the journal's own scope language and cite papers the journal has published. This is the single most effective sentence in the letter, because scope mismatch is the editor's commonest reason to decline without review.

Move 5 — prior presentation. Disclose a conference version here rather than hoping nobody notices. Pre-empting a duplicate-publication concern costs one sentence; having it raised later costs credibility (§47.6).

Move 6 — suggested reviewers. Legitimate and often requested. Suggest people you cite, from different institutions and countries, who are not collaborators or recent co-authors. Exclusion requests are also legitimate when justified professionally — a direct competitor, or a documented prior dispute. Keep the reason factual and brief; never disparage anyone.

Exercises

Exercise 54.1 Write your cover letter using the six moves. Keep it under 350 words.

Exercise 54.2 Quote your target journal's scope phrase in move 4 and cite two papers it has published.

Exercise 54.3 Identify three suitable reviewers with a one-line justification each.

Chapter 55 — Peer Review

55.1 The process

Figure 55.1 — The peer-review process and its decision points

```
SUBMISSION
   │
   ▼
EDITOR-IN-CHIEF triage  ──► desk reject (scope, format, quality, ethics)
   │                          typically 3 days – 3 weeks
   ▼
ASSOCIATE / HANDLING EDITOR assigned
   │  invites reviewers — often several declines before acceptances
   │  THIS IS THE LONGEST SILENT PERIOD; weeks are normal
   ▼
2–4 REVIEWERS report independently
   │
   ▼
EDITOR SYNTHESISES  ──► FIRST DECISION
   │
   ├─ ACCEPT (rare on round one)
   ├─ MINOR REVISION      → likely acceptance
   ├─ MAJOR REVISION      → genuine interest; substantial work required
   ├─ REJECT AND RESUBMIT → often a major revision with a reset clock
   └─ REJECT              → wrong venue, or a fundamental flaw
   │
   ▼
REVISION + RESPONSE LETTER → RE-REVIEW → (1–3 rounds)
   │
   ▼
ACCEPT → PROOFS → EARLY ACCESS → ISSUE → DOI, indexing
```

55.2 Review models

Model	Description	Note
Single-anonymous	Reviewers know the authors; authors do not know reviewers	Historically the most common
Double-anonymous	Neither party is identified	Requires careful anonymisation (§53.2)
Open	Identities and sometimes reports are published	Increasingly available; useful for learning field standards
Post-publication	Commentary after publication	Supplementary rather than a substitute

55.3 Decoding decisions

Decision	What it means	What to do
Minor revision	Likely acceptance	Respond fully and promptly; do not expand scope
Major revision	The editor is genuinely interested; substantial work needed	**This is a success.** Do the experiments
Reject and resubmit	Often a major revision with a reset clock, used administratively	Treat as a major revision; retain reviewer goodwill
Reject	Wrong venue, or a fundamental flaw	Fix what the reviews exposed, then choose the next venue deliberately

Two things worth saying plainly. Rejection is the modal outcome at selective venues; many strong journals accept a small minority of submissions. And “major revision” is not a criticism of you — it is an invitation.

55.4 Timelines and what you control

Stage	Typical duration	What you control
Desk check	3 days – 3 weeks	Scope fit, formatting, similarity, language, declarations
Under review	1 – 6 months	Nothing — but you may send a polite status enquiry after the journal's stated period has elapsed
Revision	You are given 30–90 days	Response quality; request an extension **before** the deadline, not after
Production	2 – 8 weeks	Proof accuracy — your **last** opportunity to correct errors

On the proofs. Read them completely. Typesetting introduces errors: figures are re-rendered, symbols dropped, references renumbered, captions detached. Authors who skim the proofs discover the error after publication, when correcting it requires a formal corrigendum.

Exercises

Exercise 55.1 Find your target journal's stated time to first decision and record it, so you know when a status enquiry becomes reasonable.

Exercise 55.2 If your venue publishes reviews, read the reviews of two accepted and two rejected papers in your area. This teaches the field's evidentiary standards faster than any other activity.

Chapter 56 — Responding to Reviewers

56.1 Principles

1. **Every comment receives a response** — including positive ones.
2. Follow the pattern **thank → restate → act → locate → evidence**.
3. **Locate precisely:** "revised text on p. 7, lines 312–319; new Table IV."
4. **Quote your new text verbatim** in the letter where it is short, so the reviewer need not hunt for it.
5. **Be courteous and unemotional, always.** Reviewers volunteered their time, and the editor reads the tone.
6. **Answer the question behind the comment.** A confused reviewer usually indicates unclear writing — which is your defect to fix, not theirs.
7. **If reviewers conflict, say so explicitly** and explain your resolution; let the editor arbitrate.
8. **Never make undisclosed changes**, and never remove content silently.
9. **Meet the deadline**, or request an extension before it passes.
10. **Use a table format**, and supply a change-marked manuscript.

Wait 48 hours after receiving harsh reviews before drafting your response. Never reply on the day they arrive.

56.2 The response table

Table 56.1 — Reviewer response patterns

#	Reviewer comment	Response	Change and location
R1.1	"Only three seeds are used; the claimed gain may be noise."	We agree. All methods were rerun with **ten seeds**, and we added paired Wilcoxon tests with 95% bootstrap confidence intervals. The improvement remains significant (p = 0.004).	Table II updated; new §V-B *Statistical validation*; p. 8, ll. 355–372
R1.2	"The comparison to CORAL is unfair — no tuning details are given."	We have equalised and documented the budget: every method received the same 50-trial random search over search spaces of comparable size.	New Table VII (search spaces); §IV-C; p. 6, ll. 240–258
R2.1	"The novelty over [9] is unclear."	[9] requires site labels at training time, whereas our method infers pseudo-domains from embedding statistics and requires no metadata. We now contrast the two explicitly and add [9] as an oracle upper bound.	§II-C rewritten; Table V adds a site-supervised oracle row; p. 4, ll. 150–163
R2.2	"Add experiments on a fourth dataset."	We were unable to obtain a fourth institution-disjoint public dataset with the required annotations within the revision period; the two candidates we identified lack finding-level labels (now noted in §VI-B). To address the underlying concern about generality, we added **leave-one-site-out** evaluation across the three existing datasets, yielding five held-out configurations.	New §V-E; Fig. 5; residual limitation stated p. 11, ll. 502–511
R3.1	"The paper is well written and the experiments are thorough."	We thank the reviewer for this positive assessment.	—

Note R2.2 particularly. It is the model for a request you cannot fulfil: do not refuse, and do not pretend. Name the constraint factually, satisfy the *underlying concern* by an alternative experiment, and state the residual limitation in the paper. Editors accept this routinely.

Note R1.1. The response strengthened the paper. Reframe reviews as free expert consultancy: the reviewer spent hours identifying the weakness that would otherwise have discredited the work after publication.

56.3 Disagreeing professionally

The four-step pattern.

1. **Acknowledge** the concern as legitimate.
2. **State your position** plainly.
3. **Give evidence** — a citation, a new experiment, a definition, or a computation — not a preference.
4. **Concede something**, so the reviewer's concern is visibly served even where you disagree.

Worked example.

> "We appreciate this concern and have considered it carefully. Because pneumothorax prevalence in this dataset is 1.2%, accuracy is dominated by the negative class: a constant-negative predictor attains 98.8% accuracy with zero recall. We therefore retain average precision and recall at fixed specificity as our primary metrics. To address the reviewer's underlying interest in comparability with earlier work, we have **added accuracy to Table II** and justified the metric choice explicitly in §IV-D (p. 6, ll. 268–274)."

Phrases to avoid.

Never write	Write instead
"The reviewer clearly did not read the paper."	"We apologise that this was unclear; we have expanded §III-B."
"This comment is wrong."	"We respectfully retain our original choice, for the following reason: …"
"As already stated on page 4…"	"This was stated only briefly; we have now made it explicit in §III-B."
"This is beyond the scope of our work."	"This is an important question that we cannot address with the available data; we have added it as a stated limitation and future direction (§VI-B)."
"We have added a discussion."	"We have added §V-E and Fig. 5 (p. 9, ll. 402–431)."

56.4 Tracking revisions

Artefact	Purpose
Response letter (table format)	Point-by-point, with quoted new text and precise locations
Marked-up manuscript	Changes highlighted (a difference tool for LaTeX, or tracked changes)
Clean manuscript	For typesetting and re-review
Private change log	Your own list of every edit, keyed to comment identifiers
Version control	Tags such as `v1-submitted`, `v2-r1-response`, `v3-accepted`

Build the response table as you make each change, not afterwards. Reconstructing line numbers at the end wastes hours.

56.5 Verification checklist for Part XVI

- [ ] The journal's unmodified template is used; length and float limits respected.
- [ ] All nine declarations are written and complete.
- [ ] Every DOI in the bibliography resolves.
- [ ] Data and code links tested in a private browser window.

- [ ] Anonymisation complete if the venue is double-blind, including repository links.
- [ ] Similarity report reviewed match by match.
- [ ] The system-generated PDF has been proof-read page by page.
- [ ] Cover letter quotes the journal's scope phrase and cites its papers.
- [ ] Prior conference presentation disclosed in manuscript and cover letter.
- [ ] Submitted to one journal only.
- [ ] Every reviewer comment has a response, a change, and a precise location.
- [ ] Disagreements follow the four-step pattern with evidence and a concession.
- [ ] A marked-up and a clean manuscript are both prepared.
- [ ] Tone reviewed by a co-author who did not draft the response.
- [ ] Abstract numbers updated if any result changed during revision.
- [ ] Proofs read completely before publication.

Exercises

Exercise 56.1 Take the three practice comments below and write a full response-table entry for each.

> **A.** "The paper reports results from a single run. Given known seed variance in these architectures, the claimed improvement is not credible as presented."
>
> **B.** "The authors should evaluate on at least two additional datasets from different countries to support the generality claim."
>
> **C.** "Accuracy should be reported as the primary metric, as it is standard in this literature and allows comparison with earlier work."

Suggested handling: **A** — valid; fix it by rerunning with at least ten seeds and adding paired tests and confidence intervals. **B** — may be infeasible; name the constraint and satisfy the concern with leave-one-group-out evaluation plus a stated limitation. **C** — disagree using the four-step pattern; add accuracy for completeness while retaining the appropriate primary metric.

Exercise 56.2 Compare your response to comment C with a colleague's. Discuss the register, not the content.

PART XVII — COMPLETE CASE STUDY

Part XVII works one study through every stage of the handbook, from an unfocused interest to a submitted manuscript with a reviewer response. Its purpose is to *demonstrate* the process rather than describe it, so each stage shows the actual artefact produced — the search log, the matrix, the tallies, the gap statement, the objectives, the tables, the abstract — rather than an account of how one might be produced.

[HYPOTHETICAL] — READ THIS BEFORE PROCEEDING.

The study in this part is **invented for teaching**. The method named CLUSTER-DG does not exist. The papers labelled P1 to P15 are placeholders, not real works. **Every number in this part is fabricated for instructional purposes** and describes no experiment that was ever run.

The underlying *phenomenon* is real and is cited where relevant: Zech et al. (2018) demonstrated that convolutional networks trained on chest radiographs can exploit institution-specific confounders, so that internally measured performance may overstate performance at a new institution. The datasets named (CheXpert, MIMIC-CXR, ChestX-ray14) are real, and their published characteristics must be verified from their own documentation before use.

Nothing in this part may be cited. It exists to show the shape of the work.

Chapter 57 — The Case Study, End to End

Figure 57.1 — The case study, end to end

```
§57.1  Area          medical imaging + deep learning       wk 1
§57.2  Narrowing     → domain generalisation in CXR        wk 1–2
§57.3  Search        4 databases, 6 strings, logged        wk 2–3
§57.4  Reading       118 triaged, 34 extracted, 4 deep     wk 3–6
§57.5  Matrix        15 rows × 22 columns                  wk 5–6
§57.6  Tallies→gap   counted, then stated with evidence    wk 6
§57.7  Objectives    aim, O1–O4, H1–H3, C1–C4              wk 6–7
§57.8  Feasibility   licences, baselines, GPU-hours        wk 7
§57.9  Method        mechanism hypothesis first            wk 8–9
§57.10 Experiments   5 models × 2 protocols × 10 seeds     wk 9–18
§57.11 Results       degradation quantified; gain tested   wk 18–20
§57.12 Discussion    mechanism, reconciliation, limits     wk 20
§57.13 Writing       floats first, then §-by-§             wk 20–23
§57.14 Abstract      seven moves                           wk 23
§57.15 Ethics        similarity, declarations              wk 23
§57.16 Journal       target + 2 backups, verified          wk 24
§57.17 Submission    cover letter, system PDF proofed      wk 24
§57.18 Review        major revision, 21 comments           wk 35
§57.19 Response      2 new experiments, 1 disagreement     wk 35–41
§57.20 Publication   accepted wk 52; early access wk 56
```

57.1 Starting point: an interest, not a topic

Week 1. The researcher has a background in computer vision, access to one mid-range GPU, and an interest in medical applications. The initial framing is *"improve chest X-ray classification with deep learning."*

Applying §1.3.1 — the diagnostic question — *if my system works exactly as intended, what will the field know that it did not know before?* — the honest answer is "that my model reached some accuracy on a public benchmark", which is not knowledge. This is a **development framing** (Table 1.1) and must be converted.

The conversion arrives from reading rather than from thinking. In triaging papers, the researcher encounters Zech et al. (2018), which reports that CXR classifiers can key on institution-specific confounders. That gives a **reason to doubt what published numbers mean** — which, per §1.3.2, is exactly how an engineering task becomes a research question.

57.2 Narrowing (Chapter 4)

The funnel of Figure 4.1, applied honestly:

Level	Content
Discipline	Computer science
Broad area	Deep learning for medical image analysis
Sub-area	**Domain generalisation in chest radiograph classification**
Specific problem	Published CXR classifiers are evaluated on splits that mix institutions across partitions, so reported performance may not describe cross-hospital behaviour
Research question	How large is the degradation under institution-disjoint evaluation, and can it be mitigated without institutional metadata?
Objective	See §57.7

One-breath test (§4.3.1): *"I compare five published architectures on three public CXR datasets under random versus institution-disjoint splits, measured by macro AUC, worst-institution AUC, and calibration error, to find out how much reported performance overstates cross-hospital performance."* All five slots filled. The topic is adequately narrowed.

57.3 Literature search (Chapter 10)

The concept table (§10.2) yields three blocks: the data (chest radiograph and its synonyms plus dataset names), the method (deep learning and architecture families), and the phenomenon (domain generalisation, distribution shift, external validation, and their spelling variants).

The search log, as it appears in the manuscript's methods section:

Database	Date	Records	After title/abstract	After full text	Included
Scopus	12 Mar 2026	187	61	19	9
Web of Science	12 Mar 2026	143	44	12	3
IEEE Xplore	13 Mar 2026	58	21	8	2
ACM Digital Library	13 Mar 2026	24	9	3	1
Preprint server	13 Mar 2026	71	18	6	0 (all superseded)
Total (deduplicated)		**412**	**118**	**34**	**15**

Six string variants were logged verbatim. Citation chaining (§10.8) on the four most-cited papers produced eleven additional records, of which three entered the final set — papers the keyword search had missed because they used "external validation" rather than any variant of "domain generalisation". **This is the practical payoff of §10.8 and the reason chaining is not optional.**

Two alerts were set on the principal strings, which later surfaced two papers published during the review period (§57.19).

57.4 Reading (Chapters 11–13)

- **Pass 1 (triage):** all 118 screened records, roughly seven minutes each — about fourteen hours.
- **Pass 2 (comprehension):** 34 papers, with the sixteen-field extraction template completed for each.
- **Pass 3 (reconstruction):** 4 papers, code cloned and run.

The deep read produced a finding that shaped the whole project. For one intended baseline, the released implementation did not reproduce the paper's headline number: 0.86 ± 0.01 macro AUC over five seeds against a reported 0.91. Inspection showed the released configuration differed from the paper's description in its augmentation pipeline. Following §12.4, this was recorded factually — about the protocol, not the authors — and reported in a footnote of the eventual manuscript with both numbers.

57.5 The literature matrix (Chapter 15)

Fifteen rows by twenty-two columns, seeded from a Zotero CSV export and completed by hand for the judgement columns. The extract below shows nine of the twenty-two columns for ten of the fifteen rows.

[HYPOTHETICAL — P1–P15 are placeholders]

Paper	Yr	Dataset + split	Split unit	Method	Baselines	Metrics	Seeds	Test
P1	2022	CheXpert, random	image	ResNet-50	none	Acc, AUC	1	none
P2	2022	ChestX-ray14, random	image	DenseNet-121	1	AUC	1	none
P3	2023	CheXpert→CXR14	institu-tion	ViT-B/16	1	AUC	3	none
P4	2023	MIMIC-CXR, random	image	CNN + attention	2	Acc, F1	1	none
P5	2023	CheXpert→CXR14	institu-tion	CORAL	1	AUC	3	none
P6	2024	3 sets, site-wise	institu-tion	IRM	2	AUC, ECE	3	none
P7	2024	3 sets, site-wise	institu-tion	site-adversarial	4	AUC, worst-site, ECE	3	none
P8	2024	private, undisclosed	unstated	ensemble	1	Acc	1	none
P9	2025	CheXpert, random	image	self-supervised	1	AUC	5	none
P10	2025	2 sets, site-wise	institu-tion	test-time adaptation	2	AUC	3	none

57.6 From tallies to gap (Chapter 18)

Step 2 — the column tallies, computed with count formulas (§15.3):

- 11 of 15 use random, row-level splits on data grouped by patient and institution
- 12 of 15 report only accuracy or AUC
- **0 of 15 report worst-institution performance**
- 2 of 15 report calibration
- **0 of 15 run any statistical test**
- 9 of 15 compare against one baseline or none
- 6 of 15 release no code
- 8 of 8 mitigation methods require institutional labels at training time

Step 3 — clustering into limitations with counts:

ID	Limitation	Count	Evidence
L1	Evaluation permits institutional identity to leak across partitions	11/15	Matrix column; mechanism documented by Zech et al. (2018)
L2	Reported gains are not distinguished from run-to-run variance	0/15 tests; 13/15 use ≤3 seeds	Matrix columns
L3	Mitigations require institutional labels	8/8 of the mitigation subset	Each paper’s setup section
L4	Decision-relevant quantities unreported	0/15 worst-site; 2/15 calibration	Matrix column

Step 5 — the gap statement, written to the template of §19.4:

> "Across fifteen studies published between 2022 and 2026, identified through a logged search of four databases, eleven evaluate chest-radiograph classifiers on random splits in which images from the same institution — and frequently the same patient — appear in both training and test partitions, a protocol shown to permit institution-specific confounding (Zech et al., 2018). Consequently the reported areas under the curve of 0.88 to 0.91 cannot be interpreted as estimates of cross-institutional performance, and the magnitude of degradation under leakage-free evaluation remains unquantified. Furthermore, all eight mitigation methods identified require institutional labels during training, whereas provenance metadata is routinely removed before data leaves an institution; and no study reports worst-institution performance or calibration error, although these rather than the mean determine clinical usability."

Gap types (Table 17.1): 9 (generalisation), 7 (evaluation), 2 (methodological). Three types, each with its own evidence — which per §17.4 is more convincing than one dramatic claim.

Stress test (§19.6). A colleague ran all six questions. Five survived. The sixth — *can you do this with your resources?* — failed initially: the original plan required a fourth, private dataset for which no access route existed. The project was reframed to three public datasets with leave-one-institution-out evaluation. **This is the intended function of the stress test: it cost an afternoon in week six instead of a year in month eight.**

57.7 Objectives, hypotheses, contributions (Chapter 7)

Aim. To establish how much chest-radiograph classifier performance degrades across institutions under leakage-free evaluation, and whether that degradation can be mitigated without institutional metadata.

	Objective
O1	To re-evaluate five published architectures on three public datasets under both random and institution-disjoint protocols, over ten seeds, reporting macro AUC, worst-institution AUC, and expected calibration error with 95% bootstrap confidence intervals
O2	To develop and evaluate an adaptation method requiring no institutional metadata, compared against ERM, Mixup, CORAL, and site-adversarial training under an identical 50-trial tuning budget
O3	To isolate the contribution of each component of the proposed method by ablation over cluster granularity, invariance weight, and augmentation
O4	To release institution-disjoint split definitions, code, and trained weights

	Hypothesis	Test
H1	Macro AUC is lower under institution-disjoint than random-split evaluation	Paired Wilcoxon signed-rank across architecture × dataset pairs, $\alpha = 0.05$, Holm-corrected
H2	The proposed method improves worst-institution AUC by ≥ 0.03 over ERM	Paired Wilcoxon; effect size as mean difference with bootstrap CI
H3	Inferred clusters agree with true institutional labels above chance	Adjusted Rand index against a permutation baseline

Hypotheses were dated in the research journal in week seven, before any test-set evaluation (§3.4).

Contributions, mapped one-to-one onto the limitations they answer (§18.5): **C1** □ L1, L2; **C2** □ L3; **C3** □ the mechanistic claim in C2; **C4** □ L1, L4.

57.8 Feasibility audit (§4.6)

Dimension	Finding
Data	Three public datasets; one requires a data-use agreement, signed in week 7
Licence	Redistribution of images not permitted; **redistribution of split definitions permitted** — hence C4
Ethics	De-identified public data; institutional committee confirmed no approval required; confirmation documented
Compute	5 methods × 10 seeds × 3 datasets = 150 runs; ≈2 GPU-hours each = 300; plus tuning (≈250) and ablation (≈150) ⇒ **≈700 GPU-hours**, about 10 weeks on the available GPU
Baselines	Four located; three ran unmodified; one required a dependency downgrade; one did not reproduce (§57.4)
Killer risk	Institutional partition is coarse — each public dataset is treated as one institution although each aggregates several sites. Mitigation: state it as a limitation and note the direction of bias

57.9 Method, with the mechanism stated first (Chapter 22)

The mechanism hypothesis, written before the implementation: acquisition artefacts — exposure settings, device characteristics, post-processing — are expected to dominate *low-order* feature statistics rather than high-order semantic content. If so, clustering images by channel-wise embedding statistics should approximate institutional identity closely enough to drive an invariance penalty **without any metadata**.

This belief is testable, which is what makes the work research rather than engineering (§22.5). H3 tests it directly.

Method. CLUSTER-DG replaces the site-supervised discriminator of adversarial domain generalisation with unsupervised k-means clustering over concatenated channel-wise means and standard deviations of encoder embeddings, with an invariance weight ramped over the first five epochs. Pseudocode as in Figure 40.1; additional cost O(NKd) per epoch over the ERM baseline.

57.10 Experimental design (Chapter 26)

Element	Specification
Datasets	CheXpert, MIMIC-CXR, ChestX-ray14
Splits	**Patient- and institution-disjoint**; fixed; identical across all methods; released
Protocol comparison	Random row-level split versus institution-disjoint, all else held constant
Baselines	Majority-class (trivial); gradient boosting on handcrafted features (classical); ERM, Mixup, CORAL, site-adversarial (state of the art); CLUSTER-DG minus its clustering branch (ours-minus-novelty); site-supervised (oracle upper bound)
Tuning	Identical 50-trial random search per method, selected on validation
Seeds	10, all recorded
Statistics	Paired Wilcoxon signed-rank; Holm correction; 1,000-resample bootstrap CIs
Cost	Parameters, FLOPs, inference latency on stated hardware
Test-set policy	Evaluated once per configuration; every access logged

57.11 Results (Chapter 35)

[HYPOTHETICAL — all numbers invented]

Table (main comparison), abbreviated:

Method	Macro AUC (random)	Macro AUC (site-wise)	Worst-institution AUC	ECE	Params
Majority class	0.500	0.500	0.500	—	—
Gradient boosting	0.812 ± 0.006	0.744 ± 0.013	0.701 ± 0.018	0.121	—
ERM (DenseNet-121)	0.897 ± 0.004	0.781 ± 0.011	0.781 ± 0.011	0.094	7.0 M
Mixup	0.894 ± 0.005	0.789 ± 0.012	0.786 ± 0.013	0.088	7.0 M
CORAL	0.893 ± 0.006	0.798 ± 0.010	0.794 ± 0.012	0.079	7.0 M
Site-adversarial	0.891 ± 0.005	0.812 ± 0.011	0.808 ± 0.012	0.061	7.1 M
CLUSTER-DG (ours)	0.889 ± 0.005	**0.826 ± 0.008**	**0.822 ± 0.009**	**0.052**	7.0 M
Site-supervised (oracle)	0.890 ± 0.006	0.834 ± 0.009	0.831 ± 0.010	0.048	7.1 M

Results prose, written to the four-part pattern of §29.2:

> “Replacing random with institution-disjoint splits reduces macro AUC from 0.897 ± 0.004 to 0.781 ± 0.011 for the ERM baseline (Table II), a reduction of 0.116 (95% CI 0.104–0.128); the direction is consistent across all five architectures and all ten seeds (paired Wilcoxon, $p < 0.001$, Holm-corrected). Degradation is largest for the transformer backbone (-0.142) and smallest for DenseNet-121 (-0.093). CLUSTER-DG raises worst-institution AUC from 0.781 ± 0.011 to 0.822 ± 0.009 ($p = 0.004$, Cohen’s $d = 1.6$), recovering 82% of the improvement attained by the site-supervised oracle, while in-domain macro AUC decreases by 0.008, within seed variance. Expected calibration error improves from 0.094 to 0.052. Inferred clusters agree with true institutional labels at adjusted Rand index 0.61 against a permutation baseline of 0.00 ± 0.02, supporting H3. **On the smallest institution (n = 412) the improvement is not statistically significant (p = 0.21).** The well-tuned gradient-boosting baseline reaches 0.744 ± 0.013 under site-wise evaluation, below all deep models but above the degraded transformer on two of three datasets.”

Two features of this paragraph do the most work. The **honestly reported null result** on the smallest institution is what makes a reviewer trust the rest (§35.1). And reporting that the **classical baseline was competitive in places** costs nothing and demonstrates that §25.3 was taken seriously.

57.12 Discussion (Chapter 36)

Written to the six moves of §29.3, abbreviated here:

> “Cross-institutional degradation is substantially larger than published within-institution figures imply, and most of it is recoverable without institutional metadata. The mechanism our ablation supports is that acquisition artefacts concentrate in low-order embedding statistics: inferred clusters agree with true institutional labels at ARI 0.61, evidently sufficient to drive an invariance penalty. This is consistent with the confounding reported by Zech et al. (2018) and extends it by quantifying the gap under a leakage-free protocol across three datasets. It also reconciles an apparent disagreement in the literature: P4 reports improved calibration where P6 does not, and both evaluated on random splits, where calibration is measured on the training distribution and therefore appears better than it is. For practitioners, a model reported at 0.90 AUC may present nearer 0.78 at an unseen institution — a difference large enough to change the operating threshold a hospital should adopt, and large enough that internal validation alone is insufficient evidence for procurement. Three limitations bound this conclusion. First, our institutional partition is coarse: each public dataset is treated as one institution although each aggregates several sites, so **true site-level degradation is likely larger than we report**. Second, the absence of a significant effect at the smallest institution leaves open whether the method helps where data are scarcest — the setting in which it would matter most. Third, we evaluate frontal adult radiographs only.”

Note that each limitation names its **direction** (§29.5): the first states the bias makes the estimate conservative.

57.13 Writing order (Chapter 30 preamble)

Floats first (eight of them, listed before any prose), then methodology, setup, results, discussion, related work from the matrix, introduction, conclusion, abstract, title. Total drafting time: three weeks.

Title, using the `[Finding]: [scope]` pattern (§30.2): *"Institution-Disjoint Evaluation Reduces Reported Chest Radiograph AUC by 0.12: Quantification and Label-Free Mitigation."*

57.14 Abstract (Chapter 31)

The seven-move abstract for this study is the worked example already given in full at §31.2 and is not repeated here. Note that its every number matches §57.11 exactly — the check of §31.3 step 5.

57.15 Ethics and similarity (Part XIV)

Step	Outcome
Similarity check	19% overall. Read **match by match**: 11% references and standard nomenclature; 6% correctly quoted definitions; **2% patchwritten related-work sentences → rewritten from notes and cited** (§47.3). The percentage was never the target
Ethics	Public de-identified data; no approval required, stated explicitly with the confirmation on file
Licence	Data not redistributed; split definitions released, as the licence permits
Declarations	All nine written (§53.4), including a CRediT statement and an AI-use statement covering language editing only
Artefacts	Code, splits, and weights deposited with a persistent identifier

57.16 Journal selection (Chapter 51)

Three candidates, each run through the ten-step verification of §51.3.

	Target	Backup 1	Backup 2
Scope phrase quoted	✔	✔	✔
Papers cited from this venue	5	3	2
Indexing verified on official lists	✔	✔	✔
Quartile (with source and category)	✔	✔	✔
APC position	Covered by institutional agreement	Subscription	APC, waiver applied for
Accepted by degree regulations	✔	✔	✔

Formatting followed the target's template from the first draft.

57.17 Submission (Chapters 53–54)

Template unmodified; system-generated PDF proof-read page by page (two figure captions had detached in conversion and were fixed); cover letter written to the six moves of §54.2, quoting the journal's scope phrase; three suggested reviewers with justifications; all links tested in a private browser window.

57.18 First decision (Chapter 55)

Week 35: major revision. Three reviewers, 21 comments. Per §55.3 this is a success, not a criticism.

57.19 Revision and response (Chapter 56)

Six weeks. Two new experiments, one reasoned disagreement, and two papers added from the alerts set in week 2.

#	Comment	Response	Change
R1.3	"Institutional partition is too coarse to support the claim."	We agree this is the principal limitation and had stated it; we now additionally report **leave-one-dataset-out** evaluation, giving three held-out configurations, and quantify how the estimate changes.	New §V-E, Fig. 5; limitation expanded p. 11
R2.4	"A fourth dataset is required for the generality claim."	We could not obtain a fourth institution-disjoint public dataset with finding-level annotations within the revision period (§VI-B). To address the underlying concern we added leave-one-dataset-out evaluation and bounded the claim to three datasets.	§V-E; abstract claim narrowed
R3.2	"Report accuracy as the primary metric for comparability."	Declined, with evidence: prevalence ranges from 1.2% to 38.5%, so accuracy is dominated by prevalent labels; a constant-negative predictor attains 98.8% on the rarest. We **added accuracy to Table II** for comparability and justified the primary metrics in §IV-D.	Table II; §IV-D
R1.7	"Two relevant papers published since submission are not discussed."	Added and discussed; one supports our finding, one reports a smaller gap, which we attribute to its within-country dataset composition.	§II-C

Note R3.2: the four-step disagreement pattern of §56.3 — acknowledge, state position, give evidence, concede something.

57.20 Outcome

Week 47: minor revision, four comments. Week 52: accepted. Week 56: early access, persistent identifier assigned, artefacts linked.

Elapsed: approximately thirteen months from first reading to early access, of which roughly six were spent waiting. Researchers planning "a paper this semester" should calibrate against this, and it is the strongest argument for beginning the search and matrix immediately rather than after the experiments.

57.21 What made this study work

Feature	Which chapter it came from
The gap came from **counting**, not from intuition	Ch. 15, 18
The **evaluation protocol itself** was part of the contribution	Ch. 17 (type 7)
Equal tuning budgets, stated explicitly	§26.4
Ten seeds, paired tests, confidence intervals	§26.8
An **oracle upper bound**, turning "+0.041" into "82% of privileged-information benefit"	§26.6
A **mechanism hypothesis stated before implementation** and tested by H3	§22.5
Negative sub-results reported (smallest institution; classical baseline competitive)	§35.1
Artefacts released, making it the protocol reference for later work	§26.9
Feasibility failure caught in **week 6, not month 8**	§19.6

57.22 Where it nearly failed

Week	Problem	Resolution
6	The design required a private fourth dataset with no access route	Reframed to three public datasets with leave-one-out evaluation
12	A baseline's released code did not reproduce its published number	Documented both numbers factually; contacted the authors; reported in a footnote
16	A gain visible at 3 seeds vanished at 10 for one architecture	**Reported it** rather than dropping the architecture

The third row is the ethical centre of the case study. The temptation was to retain three seeds; the decision not to is what makes the rest of the paper trustworthy, and it is precisely the decision that §3.4 and Chapter 49 exist to make easier when it arrives.

57.23 The reusable order of operations

```
1  Narrow until you can name datasets, baselines, and metrics in one breath
2  Search reproducibly; log database, date, string, filters, counts
3  Triage → extract → matrix
4  COUNT the columns → limitations → gap
5  Gap → questions → objectives → hypotheses → contributions (1:1)
6  Feasibility audit BEFORE experiments; stress-test the gap
7  State the mechanism before implementing
8  Baselines in four categories, equal budgets, ≥5 seeds, statistics
9  Ablate one factor per row; include an oracle
10 Error analysis on 30+ failures
11 Floats first, prose second, abstract last
12 Verify every DOI
13 Read the similarity report match by match
14 Declarations before submission day
15 Verify the journal on official lists
16 Respond to every comment with a precise location
17 Read the proofs completely
```

Notice what is absent from this list: no step at which anything was invented, no step at which generated text entered the manuscript unverified, and no step at which an inconvenient result was concealed.

Exercises

Exercise 57.1 Produce §57.3 (search log) and §57.6 (tallies, limitations, gap statement) for **your own** topic. These two artefacts are the foundation of everything else.

Exercise 57.2 Write your §57.9 — the mechanism hypothesis — *before* you implement anything. Date it.

Exercise 57.3 Complete §57.8 (feasibility audit) and identify your killer risk.

Exercise 57.4 Run the six stress-test questions (§19.6) on your gap with a colleague, as was done in week 6 here. Record which question was hardest.

Exercise 57.5 Write your §57.22 in advance: name the three ways this project could nearly fail, and what you would do in each case. Deciding in advance is far easier than deciding under pressure.

PART XVIII — RESEARCHER CHECKLISTS

Part XVIII collects the handbook's verification requirements into nine stage-gated checklists. Each is designed to be run *before* committing to the next stage, because every item is cheaper to satisfy now than to repair later.

How to use them. Treat each checklist as a gate, not a form. An unticked item is not a formality to be waived; it is a specific, named risk you are choosing to accept. Where a checklist item is not applicable to your discipline, record *why* rather than skipping it silently.

Section references point to the chapter where the item is explained.

Chapter 58 — Stage-Gated Checklists

58.1 Before starting research

Run this before committing time to a project.

- [ ] I can state in one sentence what the field will know if my study succeeds (§1.3.1)
- [ ] That statement is about a class of situations, not only about my system (§1.4)
- [ ] I can name which of the seven contribution types I am attempting (§1.8.1)
- [ ] I can name the evidence that contribution type requires (§1.8.1)
- [ ] I can state a mechanism — a reason my approach should work — that could turn out to be wrong (§1.7, §5.5)
- [ ] I can name an outcome other than “my method wins” that would be worth reporting (§5.3)
- [ ] I know my research type and its principal validity threat (Ch. 2)
- [ ] My area passes all five constraints, including **supervisory capability** (Table 4.1)
- [ ] I have plotted publication volume by year and can describe the curve (§4.2)
- [ ] I have listed the top five journals and top five authors in my area (§4.2)
- [ ] My topic passes the one-breath test with all five slots filled (§4.3.1)
- [ ] I can name at least one real dataset, with its licence and access route (§4.6)
- [ ] I have located at least one baseline implementation and confirmed it runs (§4.6)
- [ ] I have completed the compute arithmetic and multiplied by three (§4.6.1)
- [ ] I have identified my single greatest risk and written one sentence on mitigation
- [ ] I have started a dated research journal (§3.5)
- [ ] I have allocated at least four weeks to framing before beginning experiments (§3.3)

58.2 Before the literature review

- [ ] I have three to five concept blocks with at least three synonyms each (§10.2)
- [ ] Both -ise and -ize spellings are present (§10.2)
- [ ] Every acronym appears with its expansion (§10.2)
- [ ] Dataset and benchmark proper nouns are included (§10.2)
- [ ] Synonyms were harvested from the literature, not from memory (§10.2)
- [ ] I have used proximity operators where the platform supports them (§10.3.1)
- [ ] Year is applied as a filter, not typed as a keyword (§10.5)
- [ ] I have used `NOT` sparingly or not at all, excluding at screening instead (§10.5)
- [ ] My result set is between roughly 80 and 300 records (§10.1)
- [ ] I searched at least three databases plus one preprint source (§9.1)
- [ ] I have completed backward, forward, and sideways chaining on my core papers (§10.8)
- [ ] My search log records database, date, exact string, filters, and counts (§10.9)
- [ ] I have set at least two alerts (§10.9)
- [ ] Records are exported with abstracts and keywords into a reference manager (§16.2)
- [ ] Every metadata field in my reference manager has been verified (§43.2)
- [ ] I know the three most likely candidates for “the paper that already did this” and have read them (§8.5)

58.3 Before selecting a research gap

- [ ] I triage every paper before reading it fully (§12.2)
- [ ] My core set has a completed sixteen-field extraction template each (Ch. 13)
- [ ] Field 6 records the split **unit**, not only the ratio, for every paper (§13.3)
- [ ] Field 12 numbers were taken from tables, not abstracts (§13.2)
- [ ] Field 14 contains **my own** critique for every paper (§13.4)
- [ ] Every contradiction between papers is flagged (§18.4)
- [ ] Every automated extraction has been verified against the PDF (§13.6)
- [ ] I have deep-read and attempted to run at least one baseline (§12.4)
- [ ] My matrix has at least ten rows and includes the split unit for each (Ch. 15)
- [ ] I have computed at least six column tallies and written each as a sentence (§15.3)
- [ ] My gap statement contains all four obligatory components (§17.1)
- [ ] It matches none of the eight non-gaps (§17.2)
- [ ] It is classified against the thirteen gap types; I expect two or three (Table 17.1)
- [ ] Every absence claim is supported by a count, an author's admission, a documented negative search, or my own measurement (§19.5)
- [ ] The gap is addressable with my actual data, compute, and time (§4.6)
- [ ] A negative result would still be publishable, and I have written the sentence reporting it (§5.3)
- [ ] **A critical reader has attempted all six stress-test questions and the gap survived** (§19.6)

58.4 Before experiments

- [ ] Each objective starts with "To" plus an observable verb and can be marked done or not done (§7.2)
- [ ] The traceability table is complete, with no empty cells (Figure 7.1)
- [ ] Contributions map one-to-one onto counted limitations (§18.5, §21.5)
- [ ] Hypotheses name a test and an α, and are **dated before any test-set evaluation** (§7.3, §3.4)
- [ ] Each objective traces through all six methodology layers to a specific result (§22.1)
- [ ] Every non-obvious design choice has a "because" sentence (§22.5)
- [ ] Each comparison isolates one factor; everything else is held constant (§22.3)
- [ ] I have audited all seven leakage pathways and stated my controls (§23.3)
- [ ] The split unit is the grouping unit of my data (§23.3, §26.3)
- [ ] Everything that learns from data is fitted on training folds only (§24.1)
- [ ] Dataset licences permit my intended use and any artefact release (§23.4)
- [ ] The ethics route is confirmed, or non-applicability is documented (§23.4)
- [ ] All four baseline categories are present, including "mine minus its novelty" (§26.5)
- [ ] A well-tuned classical baseline is included (§25.3)
- [ ] Every method will receive an identical, stated tuning budget (§26.4)
- [ ] I will run at least five seeds and report dispersion (§26.9)
- [ ] The ablation changes one factor per row and includes an oracle where possible (§26.6)
- [ ] The statistical test is pre-specified, with effect size and multiple-comparison correction (§26.8)
- [ ] Seeds, environment, configurations, and split files are recorded and releasable (§26.9)
- [ ] I have a written policy for touching the test set once, and a log to record it (§26.2)

58.5 Before writing

- [ ] The test set was touched once per configuration, and the log shows it (§26.2)
- [ ] I have performed stratified and manual error analysis on at least thirty failures (§26.7)
- [ ] Every metric is justified in one sentence by the decision it supports (§27.1)
- [ ] Under imbalance I report PR-AUC, macro F1, or MCC — not accuracy alone (§27.6)
- [ ] A confusion matrix or per-class breakdown exists (§27.2)
- [ ] Calibration is reported where probabilities drive decisions (§27.3)
- [ ] Detection metrics state the IoU threshold; segmentation reports per-class overlap (§27.5)
- [ ] Every mean carries dispersion, and the measure is defined (§28.3)
- [ ] My float list includes baseline comparison, ablation, error analysis, and cost (§28.1)
- [ ] All four table types exist (§39.1)
- [ ] Figures are vector, colourblind-safe, and legible at print size (§38.2)
- [ ] No axis is truncated without annotation; there are no dual y-axes (§28.3)
- [ ] Pseudocode is 10–20 lines with inputs, outputs, marked novel lines, and a cost statement (§40.3)
- [ ] Notation is identical across equations, pseudocode, and released code (§40.4)
- [ ] I can state my novelty type and name the closest prior work and the delta (§20.5)
- [ ] Each contribution has artefact, scope, quantification, and significance (§21.3)

58.6 Before submission — manuscript content

- [ ] Written in writing order, not reading order (Part X preamble)
- [ ] Title contains searchable terms and no dead words (§30.3)
- [ ] Abstract has all seven moves; every number matches Results exactly (§31.1, §31.4)
- [ ] Introduction ¶4 is a hinge containing at least one count (§32.1)
- [ ] Introduction ¶5 states a mechanism and names the experiment testing it (§32.2)
- [ ] ¶6 contributions match the abstract, results subsections, and conclusion (§32.1)
- [ ] Related work is grouped by family, with a comparison table and an evaluation-practice subsection (§33.1)
- [ ] Methodology contains the **leakage sentence** and the **fairness sentence** verbatim (§34.2)
- [ ] Results contains no interpretation (§29.1)
- [ ] At least one inconvenient or null finding is reported (§35.1)
- [ ] Discussion fulfils all four obligations; claim verbs match the design (§36.1, §36.2)
- [ ] Every limitation names a threat, a direction, and a consequence (§29.5)
- [ ] Conclusion introduces no new evidence and no claim stronger than the abstract (§37.2)
- [ ] Future work is derived from a stated limitation (§37.2)
- [ ] Every superlative is bounded by the conditions actually tested (§21.6)
- [ ] I have run the logical-consistency check across abstract, contributions, results, and conclusion (§46.7)
- [ ] I have run the adversarial-review prompt and addressed or acknowledged each objection (§46.5)

58.7 Before journal selection

- [ ] I can state the difference between the Impact Factor and CiteScore, with sources and windows (§51.2)
- [ ] I always quote a quartile with its source and subject category (§51.2)

- [ ] I understand that journal metrics do not measure my paper's quality (§51.2)
- [ ] Indexing is verified on the **official** lists, not on the journal's claim (§51.3)
- [ ] I have quoted the scope phrase that covers my topic (§51.1)
- [ ] At least three papers I cite were published in the target journal (§51.1)
- [ ] The ISSN is confirmed as belonging to this journal (§51.3)
- [ ] If open access, the journal is listed in DOAJ (§51.3)
- [ ] Three editorial board members are independently verified (§51.3)
- [ ] I have read two recent articles and judged the quality acceptable (§51.3)
- [ ] The target passes every item of the predatory-venue checklist (§52.3)
- [ ] My APC position is known, including waivers and institutional agreements (§50.3)
- [ ] My institution accepts this venue for my degree or assessment (§50.1)
- [ ] I have one target and two backups, scored against the same criteria (§51.4)
- [ ] I am formatting for the target from the first draft (§51.4)

58.8 Before submission — mechanics and ethics

- [ ] The journal's unmodified template is used; length and float limits respected (§53.2)
- [ ] All nine declarations are written and complete (§53.4)
- [ ] **Every DOI in the bibliography resolves and matches the publisher record** (§16.1)
- [ ] No reference is cited that I have not opened (§33.3)
- [ ] Load-bearing references checked for retraction notices
- [ ] Foundational works cited, not only recent ones (§32.4)
- [ ] Data and code links tested in a private browser window (§53.3)
- [ ] Anonymisation complete if the venue is double-blind, including repository links (§53.2)
- [ ] Similarity report reviewed **match by match**, not by percentage (§48.2)
- [ ] Patchwriting checked for specifically, not only verbatim copying (§47.3)
- [ ] Reused figures have both citation and permission (§47.5)
- [ ] Any overlap with my own prior work is cited and disclosed (§47.6)
- [ ] Data exclusions were pre-specified and are reported with counts (§49.2)
- [ ] All pre-specified comparisons are reported, with correction (§49.2)
- [ ] Exploratory findings are labelled as exploratory (§49.2)
- [ ] The author list satisfies all four authorship criteria, with a CRediT statement (§49.3)
- [ ] Every author has seen and approved the submission (§49.3)
- [ ] AI use is disclosed per the venue's policy, and I can state that I remain responsible for the accuracy, originality, citations, methodology, data, and results (§45.5)
- [ ] The **system-generated PDF** has been proof-read page by page (§53.3)
- [ ] Cover letter written to the six moves, quoting the journal's scope phrase (§54.2)
- [ ] Prior conference presentation disclosed in manuscript and cover letter (§54.3)
- [ ] Submitting to **one** journal only (§51.4)
- [ ] Everything archived locally in a dated folder with the submission identifier (§53.3)

58.9 After reviewer comments

- [ ] I waited 48 hours before drafting the response (§56.1)
- [ ] Every comment has a response — including positive ones (§56.1)

- [ ] Each response follows thank □ restate □ act □ locate □ evidence (§56.1)
- [ ] Every response gives a precise location: section, page, lines, float number (§56.1)
- [ ] New text is quoted verbatim in the letter where short (§56.2)
- [ ] Requests I could not fulfil name the constraint and satisfy the underlying concern by an alternative (§56.2)
- [ ] Disagreements follow the four-step pattern with evidence and a concession (§56.3)
- [ ] No phrase from the “never write” list appears (§56.3)
- [ ] Conflicting reviewer requests are acknowledged explicitly and a resolution proposed (§56.1)
- [ ] No undisclosed changes; nothing silently deleted (§56.1)
- [ ] A marked-up and a clean manuscript are both prepared (§56.4)
- [ ] Version tagged in version control (§56.4)
- [ ] **Abstract numbers updated if any result changed during revision** (§56.5)
- [ ] Every new claim is supported by new evidence *in the manuscript*, not only in the letter (§56.4)
- [ ] Tone reviewed by a co-author who did not draft the response (§56.5)
- [ ] Deadline met, or an extension requested **before** it passed (§56.1)
- [ ] Newly published relevant work added, from alerts set at §58.2 (§10.9)
- [ ] Proofs read completely — this is the last opportunity to correct anything (§55.4)

58.10 A note on using these checklists

Three failure modes are worth naming.

Ticking without checking. A checklist run at speed the night before submission provides no protection. Run each one at the *stage gate* it belongs to, when there is still time to act on a failure.

Treating non-applicability as exemption. Many items in §58.4 and §58.5 are phrased for computational research. If you work qualitatively, the corresponding requirement still exists in a different form — inter-coder agreement rather than seed variance, saturation rather than sample size, audit trail rather than released configurations. Translate the item; do not delete it.

Missing the two that matter most. If you retain nothing else from Part XVIII, retain these:

> **Every absence claim is supported by a count, an admission, a documented negative search, or your own measurement (§19.5).**
>
> **Every reference in your manuscript has been opened by you and its identifier resolved against the publisher record (§16.1).**

The first determines whether your paper has a contribution. The second determines whether it can be trusted at all.

APPENDICES

> Sixteen reusable templates. These are intended to be copied into your own working documents and filled. Each names the chapter that explains it, so you can return to the reasoning when a field is unclear.
>
> A machine-readable version of the literature matrix (Appendix 4) as a spreadsheet is more useful than a printed one; build it in a spreadsheet application so you can sort, filter, and count (§15.3).

Appendix 1 — Research Problem Template

(Chapter 5)

Block	Your text
Context — domain and why anyone should care; one or two sentences, with a concrete number	
Current state — what is done now and what it achieves; cited	
The inadequacy — what specifically is missing, wrong, unmeasured, or assumed; **cite the evidence**	
Consequence — what cannot currently be done, decided, or trusted	
Scope — what you will and will not address	

Gap marker present? ☐ unquantified ☐ unmeasured ☐ not established ☐ unexplained ☐ it is unknown whether

Four-part test (§5.3)

#	Test	Pass?	Evidence
1	The answer is currently unknown in the accessible literature	☐	
2	It can be measured or proven	☐	
3	A negative result would still be publishable	☐	
4	It matters to someone outside my institution	☐	

If my problem has the form “apply X to Y” (§5.5): Assumption of X violated by Y: ___________ Predicted consequence: ___________ Measurement that would detect it: ___________

Appendix 2 — Research Question Template

(Chapter 6)

Question	Type (Table 6.1)
RQ1	
RQ2	
RQ3	

DMBCOT slots for the principal question (§6.5)

Slot	Content
D — Data: population, datasets, split unit	
M — Method: the intervention	
B — Baseline: the comparison, and its tuning budget	
C — Condition: the circumstance varied	
O — Outcome: metrics, and the decision each serves	
T — Threshold: what size of effect would matter, and why	

Quality check — for each question: ☐ One relationship only ☐ Answerable by an experiment I can run ☐ Not answerable by yes/no alone ☐ Informative whichever way it resolves ☐ Claim type matches the design

The negative-result sentence — write the sentence that would appear in your results if the answer were *no*: ____________

Appendix 3 — Research Objective Template

(Chapter 7)

Aim (exactly one sentence, "To + purpose"): ____________

	Objective ("To + measurable verb + object + condition")	Measurable outcome	Maps to contribution
O1			
O2			
O3			
O4			

Verb check. Acceptable: quantify, measure, compare, characterise, derive, prove, construct, annotate, validate, determine, isolate, establish. **Reject:** study, analyse, investigate, explore, understand, work on, develop a framework for, implement.

Hypotheses (§7.3)

	H0	H1	Test	α	Correction	Effect size reported as
H1						
H2						

Date pre-specified: ____________ *(before any test-set evaluation)*

Traceability table (Figure 7.1) — no cell may be empty

Objective	Method section	Results section	Contribution
O1			C1
O2			C2
O3			C3

Examiner simulation: could a colleague mark each objective *done* or *not done* without argument? ☐

Appendix 4 — Literature Matrix

(Chapter 15 — build this in a spreadsheet)

Columns

Col	Field	Col	Field
A	Citation key	L	Seeds and variance reported
B	Year	M	Statistical test
C	Venue and indexing	N	Cost reported
D	Research problem	O	Code available
E	Dataset(s) and split protocol	P	Strengths
F	**Split unit**	Q	Limitations (theirs)
G	Method	R	**Limitations (mine)**
H	Novel component	S	Gap it leaves
I	Baselines and count	T	Future work (verbatim)
J	Metrics	U	Relevance: core / context / discard
K	Key results	V	Contradicts which paper

Tally block (§15.3) — adapt ranges to your sheet

Tally	Formula
Total studies	`=COUNTA(A2:A30)`
Row-level random splits	`=COUNTIF(F2:F30,"image")`
Studies with ≤1 baseline	`=COUNTIF(I2:I30,"<=1")`
Studies running any test	`=COUNTA(M2:M30)-COUNTIF(M2:M30,"none")`
Studies releasing code	`=COUNTIF(O2:O30,"yes")`
Studies reporting metric X	`=COUNTIF(J2:J30,"*X*")`

Five tally sentences

1. Of ___ studies, ___ use ___________
2. Of ___ studies, ___ report ___________
3. ___ of ___ compare against ___________
4. No study (0 of ___) reports ___________
5. All ___ studies assume ___________

> **Warning.** Column K is comparable only within identical dataset and split. Never average results across studies.

Appendix 5 — Paper Reading Sheet

(Chapter 13 — one per paper; save as `<CitationKey>.md`)

#	Field	Content
1	Citation key, venue, year, indexing; **DOI verified** ☐	
2	Research problem, in **my** words	
3	Motivation	
4	Stated objective or research question	
5	Claimed gap, and whether evidenced	
6	Dataset(s): names, versions, sizes, classes, balance, licence, **split protocol and split unit**	
7	Preprocessing; **where fitted statistics came from**; resampling before or after split	
8	Proposed method; the **novel component isolated**; stated mechanism	
9	Baselines; count; tuned equally?	
10	Metrics; appropriate to the problem?	
11	Setup: seeds, variance, test, CV, optimiser, hardware, versions, **tuning budget**	
12	Key results — from **tables**, each with its comparison point	
13	Limitations — **theirs**, verbatim with section and page	
14	Limitations — **mine** (mandatory; see headings below)	
15	Future work, verbatim	
16	Relevance: core / context / discard · contradicts ___ · possible extension for me	

Field 14 headings (§13.4): fairness of comparison · statistical validity · evaluation protocol and leakage · generality · cost and deployability · reproducibility · explanation offered · anything selectively unreported

Reading passes completed: ☐ Pass 1 ☐ Pass 2 ☐ Pass 3

Tool-assistance log: fields pre-filled by a tool ___________ · verified against the PDF ☐ · corrections made ___________

Appendix 6 — Research Gap Analysis Sheet

(Chapters 17–19)

Step 1 — tallies: see Appendix 4.

Step 2 — limitations with counts and citations

ID	Limitation	Count (n of N)	Supporting papers	Evidence source
L1				
L2				
L3				
L4				
L5				

Step 3 — unresolved problem (one sentence): ____________

Step 4 — gap statement (§19.4 template)

> "Across **[N]** studies identified through **[databases, strings, date range]**, **[pattern with counts]** holds. Consequently, **[specific quantity or mechanism]** remains **[unquantified / untested / unexplained]**, even though **[why it matters, and to whom]**. This study addresses that by **[action]**."

Gap types (Table 17.1) — tick all that genuinely apply, expect two or three: ☐ 1 Knowledge ☐ 2 Methodological ☐ 3 Dataset ☐ 4 Performance ☐ 5 Application ☐ 6 Population/domain ☐ 7 Evaluation ☐ 8 Scalability ☐ 9 Generalisation ☐ 10 Reproducibility ☐ 11 Efficiency ☐ 12 Explainability ☐ 13 Temporal

Weak-gap self-check — my statement is **none** of these (§17.2): ☐ "room for improvement" ☐ "nobody has applied X to Y" ☐ "accuracy is low" ☐ "no one has combined A and B" ☐ "topic is trending" ☐ "no work in my country/language" ☐ "prior work is old" ☐ "limited research exists" ☐ contains no numbers

Evidence form for each absence claim (§19.5): ☐ count from a documented search ☐ authors' own admissions ☐ documented negative search ☐ my own measurement

Step 5 — stress test (§19.6) — a critical reader must ask all six

Question	My answer	Survived
Has this been done? Which paper is closest, and how does mine differ?		☐
How do you know the gap is real — which count supports it?		☐
Who benefits, outside my institution?		☐
What is my riskiest assumption?		☐
Can I do this with my data, compute, and time?		☐
What would a negative result look like — and would it be publishable?		☐

Appendix 7 — Research Contribution Template

(Chapter 21)

	Contribution	Type	Answers which limitation	Results section supplying evidence
C1				
C2				
C3				
C4				

Types: methodological · dataset/resource · experimental/empirical · theoretical · practical

Four-element check for each (§21.3)

	Artefact or finding	Scope	Quantification (with uncertainty)	Significance
C1	☐	☐	☐	☐
C2	☐	☐	☐	☐
C3	☐	☐	☐	☐

Novelty positioning (§20.5) Novelty kind (of nine): ____________ Closest prior work: ____________ The delta, in one sentence: ____________ Why the delta matters: ____________ Experiment that isolates my novel component: ____________

Calibration (§21.6) ☐ Every superlative is bounded by conditions actually tested ☐ One sentence states what the work does **not** address: ____________

Appendix 8 — Methodology Template

(Chapter 22, Chapter 34)

Six layers (§22.2)

Layer	Content
Design — study type; independent variable; what is held constant	
Data — datasets, why these, split unit, documented biases	
Procedure — preprocessing, model, training, tuning protocol	
Controls — baselines, seeds, equal budgets	
Analysis — metrics, statistical test, error analysis	
Threats — validity limits and mitigations	

Pipeline as a chain — every arrow must be describable in one sentence

`raw → ______ → ______ → ______ → ______ → ______ → output → metric`

Arrows I cannot yet specify: ____________

Design-rationale sentences (§22.5) — one per non-obvious choice, in the form *"We use [choice] because [property of the problem] implies [expected consequence]."*

1.
2.
3.

Two mandatory sentences (§34.2)

> Leakage: *"All preprocessing statistics, resampling, and feature selection were fitted on training folds only. Splits are _____-disjoint."*
>
> Fairness: *"All methods received an identical budget of ___ trials over search spaces of comparable size, selected on validation data."*

Attachments: ☐ architecture diagram, vector, novel block highlighted ☐ Algorithm 1, 10–20 lines, novel lines marked, cost stated ☐ formal problem statement with every symbol defined

Appendix 9 — Experimental Design Template

(Chapter 26)

Element	Specification
Datasets (≥2, including the field-standard benchmark)	
Split scheme and **unit**	
Protocol comparison (what varies, what is fixed)	
Tuning: search space, strategy, **trial budget per method**	
Seeds (≥5, 10 preferred)	
Metrics, primary and secondary	
Statistical test, α, correction, effect size	
Cost measured (params, FLOPs, latency, memory) and hardware	
Test-set access policy and log location	

Baselines (§26.5)

Category	Specific baseline	Source / repository	Runs?	Equal budget?
Trivial		—	—	—
Strong classical			☐	☐
Current state of the art			☐	☐
Mine minus its novelty		—	☐	☐
Oracle / upper bound			☐	☐

Leakage audit (§23.3) — record the control for each pathway

▯ ▯ group ▯ ▯ preprocessing ▯ ▯ resampling ▯ ▯ temporal ▯ ▯ duplicate ▯ ▯ target ▯ ▯ tuning on test

Ablation plan (§26.6) — one factor per row

Configuration	Component A	Component B	Component C	Metric	Δ
Baseline	–	–	–		—
+A	✔	–	–		
+A+B	✔	✔	–		
Full	✔	✔	✔		
Oracle					

Compute arithmetic (§4.6.1): configs ___ × seeds ___ × datasets ___ = ___ runs × ___ h = ___ GPU-h × 3 = ___ **GPU-h**

Appendix 10 — Results Table Template

(Chapters 28, 39)

Main comparison table

Method	Dataset 1	Dataset 2	Dataset 3	Worst-group	Params	Latency
Trivial baseline					—	—
Strong classical						
Prior SOTA [ref]						
Ours						
Oracle						

Caption must state: what the dispersion measure is (std, SEM, or CI) · the number of runs · what bold denotes · the meaning of any significance marker

Float list before writing (§28.1)

	Float	Purpose	Done
Fig. 1	System overview, novel block highlighted		☐
Table I	Dataset statistics incl. split unit and licence		☐
Table II	Main comparison with dispersion and significance		☐
Fig. 2	The key effect		☐
Table III	Ablation, one factor per row, with oracle		☐
Fig. 3	Error analysis or qualitative failures		☐
Table IV	Cost: params, FLOPs, latency, memory		☐
Fig. 4	Sensitivity to the principal hyperparameter		☐

Figure audit (§28.3): ☐ dispersion on every mean ☐ no truncated axis without annotation ☐ no dual y-axes ☐ colourblind-safe and greyscale-legible ☐ vector export ☐ ☐8 pt after scaling ☐ self-contained caption ☐ every float referenced in text

Appendix 11 — Journal Selection Checklist

(Chapters 51–52)

Verification sequence — never start from an email

#	Check	Source	J1	J2	J3
1	Journal found on the publisher's own domain	publisher site	☐	☐	☐
2	Web of Science Core Collection; **which index**	Master Journal List	☐	☐	☐
3	**Active** Scopus coverage; CiteScore	Scopus source list	☐	☐	☐
4	Quartile, SJR, subject category	SCImago	☐	☐	☐
5	If open access: DOAJ listing	DOAJ	☐	☐	☐
6	ISSN belongs to this journal	ISSN Portal	☐	☐	☐
7	COPE membership	COPE	☐	☐	☐
8	Three board members verified	institutional pages	☐	☐	☐
9	Two recent articles read and judged	the journal	☐	☐	☐
10	Not a hijacked imitation	Ch. 52	☐	☐	☐

Fit and cost

Item	J1	J2	J3
Journal / publisher			
Scope phrase covering my topic (quote it)			
Papers **I cite** published here (target ≥3)			
Index (SCIE / SSCI / ESCI / Scopus only / neither)			
Quartile **with source and category**			
Article type; page and float limits			
Reported time to first decision			
APC; waiver or institutional agreement			
Access model			
Template available			
Special issue open + deadline			
Accepted by my degree regulations			
Decision: TARGET / BACKUP / REJECT			

Predatory warning signs (Table 52.1) — one tick warrants investigation; two or more, walk away ☐ unsolicited flattering invitation ☐ publication promised in days ☐ unrecognised metric claimed ☐ indexing claim fails steps 2–3 ☐ impossibly broad scope ☐ APC undisclosed until acceptance or payable to an individual ☐ board without affiliations ☐ site imitates a known journal ☐ no DOIs or preservation policy ☐ free-email contact only ☐ errors on the homepage ☐ "review" with no substantive comments

Appendix 12 — Manuscript Checklist

(Chapter 53)

Content ☐ Journal template, unmodified ☐ within page/word and float limits ☐ title, abstract, keywords final ☐ all required sections in the expected order ☐ figures vector or ☐300 dpi, fonts embedded, greyscale-legible, all referenced ☐ tables captioned per convention, all referenced, units stated ☐ equations numbered where cited, symbols defined ☐ references in journal style, sequential ☐ language checked ☐ anonymised if double-blind, including repository links

References ☐ every in-text citation in the list and vice versa ☐ **every DOI resolves and matches the publisher record** ☐ consistent author formats and venue abbreviations ☐ no duplicates ☐ preprints labelled as preprints ☐ retraction check on load-bearing references ☐ foundational works cited ☐ **every reference opened and read** ☐ reference count appropriate for the venue

Declarations ☐ ethics approval or documented non-applicability ☐ informed consent ☐ conflict of interest ☐ funding with grant numbers ☐ data availability with a persistent identifier or a justified restriction ☐ code availability ☐ AI use per venue policy ☐ author contributions (CRediT) ☐ prior presentation disclosed

Files and system ☐ system-generated PDF proof-read page by page ☐ source files ☐ supplementary material ☐ marked-up and clean versions if requested ☐ graphical abstract if required ☐ cover letter ☐ ethics documentation ☐ links tested in a private window ☐ all author details and identifiers correct ☐ corresponding email monitored for months ☐ funding entered in structured fields ☐ archived locally with the submission identifier

Integrity ☐ similarity report reviewed match by match ☐ patchwriting specifically checked ☐ reused figures have citation **and** permission ☐ self-overlap cited and disclosed ☐ submitting to one journal only

Appendix 13 — Reviewer Response Template

(Chapter 56)

Cover note to the editor

> Dear Professor **[editor]**,
>
> Thank you for the opportunity to revise manuscript **[ID]**, *"**[title]**"*. We are grateful to the reviewers for their careful reading; the revision is substantially stronger as a result.
>
> We have addressed all **[N]** comments. The principal changes are: **(1)** ___________; **(2)** ___________; **(3)** ___________.
>
> A point-by-point response follows. New and modified text is highlighted in the marked-up manuscript; page and line numbers refer to that file. We have retained our original position on one point (**[comment ID]**) and explain our reasoning there, together with the additional material added to address the underlying concern.
>
> Sincerely, **[corresponding author]** on behalf of all authors

Response table

#	Reviewer comment (verbatim, abbreviated)	Our response	Change and exact location
R1.1		We thank the reviewer… We agree that…	Revised §__, p. __, ll. **–**; new Table __
R1.2			
R2.1			
R2.2			
R3.1			

Response patterns (§56.2–56.3)

Situation	Pattern
Valid criticism you can fix	Agree → what you changed → where → the new evidence
Request you cannot fulfil	Name the constraint factually → satisfy the underlying concern by an alternative → state the residual limitation in the paper
Misunderstanding	"We apologise that this was unclear" → what you clarified → where. **Never** blame the reviewer
Legitimate disagreement	Acknowledge → state position → give **evidence → concede something**
Conflicting reviewers	State the conflict → explain your resolution → invite the editor's guidance
Positive comment	"We thank the reviewer for this positive assessment."

Never write: "the reviewer clearly did not read the paper" · "this comment is wrong" · "as already stated on page 4" · "this is beyond the scope" · "we have added a discussion" *(vague)*

Tracking ☐ waited 48 hours ☐ every comment answered ☐ precise locations throughout ☐ new text quoted where short ☐ marked-up version ☐ clean version ☐ private change log keyed to comment IDs ☐ version tagged ☐ abstract numbers updated if results changed ☐ tone reviewed by a co-author who did not draft it ☐ deadline met or extension requested in advance

Appendix 14 — AI Research Prompt Library

(Chapters 45–46)

The standard safety clause — append to every research prompt

> *"Use only the information I have supplied. Do not invent references, datasets, numbers, author names, or identifiers. If something is uncertain or you do not know, say so explicitly rather than guessing. Mark any statement I will need to verify independently."*

Task	Prompt skeleton	Verification you owe
Narrowing a topic	"I am interested in [area]. My resources are [compute, data, time, skills]. Propose five candidate research problems completable in [N months]. For each state: the specific unknown; why it matters and to whom; datasets needed; baselines required; the principal feasibility risk. Do not cite papers or claim novelty."	Run the feasibility audit yourself
Search strings	"Produce three search strings of increasing precision in [Scopus/WoS] syntax with concept blocks and field codes. Include -ise/-ize variants, acronyms with expansions, and dataset proper nouns. Explain each block. Do not cite papers."	Run and log every string yourself
Understanding a method	"Explain [method] to someone who knows [prerequisite]. State its assumptions, when it fails, and three things to check in an implementation. Distinguish established facts from your inference."	Verify against the original paper
Extracting from a paper	"Here is the full text [attach]. Fill this table using only this document: datasets, split protocol and unit, preprocessing and where statistics were fitted, baselines and tuning parity, metrics, seeds, statistical test, headline numbers with their table. **Quote the sentence or table you took each cell from.** Write NOT STATED where absent."	Check every quotation
Comparing papers	"Here are extraction tables for five papers. State which are directly comparable and which are not, and why; the dimensions on which they differ; any contradictions. Do not average or rank across different datasets or split protocols."	Judge validity yourself
Literature matrix	"Convert these extraction sheets into a table with columns [list]. Leave cells blank rather than inferring."	Verify every cell against the PDF
Candidate gaps	"Here are my matrix tallies and fifteen limitations sections. Group the limitations into themes with counts; identify which appear in a substantial share; map each to one of these gap types [list]. **Do not assert that anything is novel or unstudied.**"	Absence claims are yours to evidence
Adversarial self-review	"Act as a critical reviewer for [venue]. Here are my abstract, method, and results. List the ten strongest objections, ranked by severity, and the specific experiment or clarification that would neutralise each. Be harsh; do not praise. Identify where my claims exceed my evidence."	These become your next experiments
Language editing	"Improve clarity and grammar for [venue]. Do not change technical content, add claims, remove hedging, or strengthen any claim. List every change and why."	Read the change list; confirm hedging survived
Structuring an introduction	"Here is my gap statement, contributions, and results summary. Propose a six-paragraph outline stating each paragraph's job in one sentence. Flag any contribution with no supporting result."	Structure must match your evidence
Methodology review	"Here is my protocol. Identify any leakage pathway; any comparison where more than one factor differs; any transformation possibly fitted before splitting; any missing baseline category; whether the tuning budget is stated and equal. Give the specific fix for each."	Fix them yourself
Logical consistency	"Here are my abstract, contribution list, results, and conclusion. Check for: abstract claims unsupported by a result; contributions with no results subsection; numbers differing between sections; claim strength escalating; causal language on observational design. Report each with the two conflicting locations."	Resolve each conflict
Generating code	"Write a [language] function computing [X], with a docstring stating inputs, outputs, and edge-case behaviour, plus three unit tests covering [cases]. Explain any choice affecting the numerical result."	**Run the tests**
Debugging	"Here is my function and its unexpected output [paste code, inputs, expected, actual]. Give three hypotheses ranked by likelihood and a specific diagnostic for each. Do not rewrite the whole function."	Run the diagnostics
Explaining statistics	"Here is the output of [test] and my design. Explain what it does and does not license me to claim, whether the test suits this design, whether the assumptions are plausible, and what effect size to report. Note anything that would invalidate it."	Confirm against a statistics reference

Prompts never to use: “give me ten citations with DOIs” · “write my related work” · “fill in the missing numbers” · “rewrite this so plagiarism software will not detect it” · “summarise this paper so I need not read it” *(for a core paper)*

Disclosure statement to adapt (§45.5)

> *“AI assistance was used for [language editing / search-query generation / code scaffolding]. All literature was independently retrieved and verified by the authors against publisher records; all experimental results were produced by the authors’ own code and data. The authors are responsible for the content of the manuscript.”*

Appendix 15 — IEEE Paper Structure Template

(Chapters 9, 30–37, 42)

```
TITLE                         8–15 words; method + problem + domain; no dead words
AUTHORS, AFFILIATIONS, persistent identifiers
ABSTRACT                      150–300 words; seven moves; no citations; numbers
                              matching Results exactly
KEYWORDS                      4–8, from the journal's or a standard thesaurus

I.   INTRODUCTION             ¶1 background and stakes
                              ¶2 current state and its inadequacy
                              ¶3 existing approaches, 2–3 grouped families
                              ¶4 limitations → GAP  ◄ the hinge; include counts
                              ¶5 proposed approach + mechanism
                              ¶6 numbered contributions + roadmap sentence

II.  RELATED WORK             A. problem formulations, datasets, evaluation
                              B. method family 1  (+ table, + critique ¶)
                              C. method family 2  (+ table, + critique ¶)
                              D. evaluation practices in prior work   ◄ high value
                              E. synthesis: settled / contested / untested → gap

III. METHODOLOGY              A. overview + Fig. 1 (novel block highlighted)
                              B. formal problem statement, all symbols defined
                              C. component subsections; novelty isolated
                              D. Algorithm 1 + complexity
                              E. loss / objective
                              (design-rationale "because" sentences throughout)

IV.  EXPERIMENTAL SETUP       A. datasets + Table I (incl. SPLIT UNIT, licence)
                              B. preprocessing  ◄ include the LEAKAGE sentence
                              C. baselines + tuning ◄ include the FAIRNESS sentence
                              D. metrics + one-sentence justification each
                              E. implementation: seeds, hardware, versions

V.   RESULTS                  A. RQ1 …  B. RQ2 …  C. ablation  D. error analysis
                              E. cost
                              (point → pattern → magnitude → uncertainty;
                               no interpretation; report null findings)

VI.  DISCUSSION               interpretation with mechanism · comparison with
                              literature incl. disagreements · implications ·
                              limitations with DIRECTION · threats to validity

VII. CONCLUSION               problem → what you did → what was found →
                              what it means → principal limitation → next question
                              (no new evidence; no claim beyond the abstract)

ACKNOWLEDGEMENTS
DECLARATIONS                  ethics · consent · conflicts · funding ·
                              data availability · code availability · AI use ·
                              author contributions (CRediT) · prior presentation
REFERENCES                    numbered by first appearance; every DOI resolved
APPENDICES / SUPPLEMENTARY    full hyperparameters · additional results ·
                              run identifiers
```

LaTeX skeleton

```
\documentclass[journal]{IEEEtran}
\usepackage{graphicx,amsmath,booktabs,siunitx,algorithm,algorithmic,hyperref}
\begin{document}
\title{...}
\author{...}
\maketitle
\begin{abstract} ... \end{abstract}
\begin{IEEEkeywords} ... \end{IEEEkeywords}
\section{Introduction} \label{sec:intro}
...
\bibliographystyle{IEEEtran}
\bibliography{refs}
\end{document}
```

Use the publisher's class file **unmodified**. Never adjust margins or spacing to fit more text; publishers check, and it causes rejection.

Appendix 16 — Research Workflow Checklist

(the whole handbook, on one page)

#	Stage	Key artefact	Gate
1	Research area	Sub-area with a readable literature	One-breath test passes (§4.3.1)
2	Research problem	Five-block problem statement	Four-part test passes (§5.3)
3	Research questions	2–4 typed questions with DMBCOT slots	Each answerable by an experiment you can run
4	Literature search	Search log: database, date, string, filters, counts	≥3 databases; saturation reached (§10.9)
5	Paper reading	16-field extraction per core paper	Split unit recorded for every paper
6	Literature matrix	≥10 rows × 22 columns	≥6 column tallies computed
7	Critical review	Synthesis grouped by family, ending in the gap	No paragraph begins with an author's name
8	Research gap	Evidenced gap statement, 2–3 types	Survives the six-question stress test (§19.6)
9	Objectives	Aim, O1–O4, H1–H3, C1–C4	Traceability table complete
10	Novelty and contribution	Closest prior work named; delta stated	Every contribution has all four elements (§21.3)
11	Methodology	Six layers; pipeline fully specified	Leakage and fairness sentences written
12	Dataset	Licences, ethics route, split unit	All seven leakage pathways audited
13	Experimental design	Protocol with four baseline categories	Equal tuning budget; ≥5 seeds; test pre-specified
14	Execution	Logged runs; configs; seeds; environment	Test set touched once, with a log
15	Evaluation	Metrics justified by the decision served	Dispersion and effect sizes throughout
16	Results	6–8 floats; null findings reported	No interpretation in Results
17	Discussion	Mechanism, reconciliation, implications	Limitations name a direction
18	Manuscript	Written in writing order	Consistency check across all sections
19	References	Verified bibliography	**Every DOI resolved**
20	Ethics	Similarity report classified; declarations	Patchwriting specifically checked
21	Journal selection	Target + 2 backups	Ten-step verification passed
22	Submission	Cover letter; system PDF proofed	One journal only
23	Peer review	—	Wait; enquire only after the stated period
24	Revision	Response table with locations	Every comment answered
25	Publication	Proofs corrected; artefacts deposited	Proofs read completely

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A note on this list, and on how to use it.

Every entry below is a **real, published work**. They are the sources actually referred to in the text, together with a small number of foundational works that define metrics or reporting standards discussed in the handbook.

Two deliberate omissions, and the reason for them.

First, **no digital object identifiers or page ranges are given** where the author could not confirm them character-for-character. Fabricating or mis-transcribing an identifier is precisely the failure this handbook warns against (§45.3, §16.1), and it would be incoherent to commit it here. Author, title, venue, and year are given so that you can locate each work in seconds using any of the databases in Chapter 9.

Second, **no entry should be cited from this list.** Retrieve each work yourself, resolve its identifier against the publisher record, and confirm that it says what you believe it says (§45.4). That is the standard the handbook asks of you throughout, and it applies to its own bibliography.

Works appearing in the extended case study of Part XVII as "P1" to "P15", and the method named CLUSTER-DG, are **hypothetical placeholders** and therefore do not appear here. They do not exist.

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> **Note.** Dataset sizes, class counts, and licence terms quoted anywhere in this handbook are indicative only and **must be verified against each dataset’s own current documentation** before use. Access conditions change, and several of these datasets are governed by data-use agreements.

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Organisational and standards sources consulted for policy discussion

The following maintain the guidance referred to in Parts XIV to XVI. These are **living documents**: consult the current version rather than any description of it, including the descriptions in this handbook.

Body or resource	What it provides	Referred to in
Committee on Publication Ethics (COPE)	Guidance and flowcharts on publication ethics, misconduct, authorship disputes	Ch. 47–49
International Committee of Medical Journal Editors (ICMJE)	Authorship criteria; recommendations for scholarly work in medical journals	§49.3
CRediT contributor taxonomy	Standardised contributor roles	§49.3
San Francisco Declaration on Research Assessment (DORA)	Position on the use of journal metrics in research assessment	§51.2
Clarivate Master Journal List	Authoritative record of Web of Science Core Collection coverage	§51.3
Scopus source list	Authoritative record of Scopus coverage and CiteScore	§51.3
SCImago Journal Rank	Freely available journal quartiles and prestige metrics	§51.2–51.3
Directory of Open Access Journals (DOAJ)	Vetted index of open-access journals	§51.3, §52.3
ISSN Portal	Authoritative ISSN records; detects hijacked titles	§51.3, §52.3
Think. Check. Submit.	Community checklist for assessing a venue	§52.3
Crossref	DOI registration and metadata; reference verification	§16.1
Retraction Watch Database	Record of retractions and expressions of concern	§53.2

GLOSSARY

Ablation study. An experiment removing or replacing one component at a time to attribute the effect to individual components. The only design that tests a mechanistic claim. §26.6

Accuracy. Proportion of predictions that are correct. Misleading under class imbalance. §27.3

Aim. The single overall purpose of a study, in one sentence. Distinct from objectives, which are the completable steps toward it. §7.1

Average precision (AP). Area under the precision–recall curve for one class; averaged over classes it gives mean average precision (mAP). §27.5

Baseline. A comparison method establishing what performance is achievable without your contribution. Four categories are required: trivial, strong classical, current state of the art, and "yours minus its novelty". §26.5

Calibration. The agreement between a model's stated confidence and its observed accuracy. Measured by expected calibration error or the Brier score. §27.3

CiteScore. A journal metric published by Scopus, computed over a four-year citation window. Not interchangeable with the Impact Factor. §51.2

Confusion matrix. The table of true and false positives and negatives from which most classification metrics derive. Should always be reported. §27.2

Construct validity. Whether a measurement corresponds to the concept it is claimed to measure. §1.7.1

Contribution. A transferable statement of what the field gains from your work, containing an artefact or finding, a scope, a quantification, and a significance. §21.3

CRediT. A standardised taxonomy of contributor roles used to state who did what on a paper. §49.3

Data leakage. Any pathway by which information from the evaluation partition influences training or model selection, causing measured performance to overstate real performance. Seven pathways are identified in §23.3.

Desk rejection. Rejection by an editor without external review, usually for scope mismatch, formatting, language, or ethics failures. Largely preventable. §53.1

Dice coefficient. An overlap measure for segmentation, equal to `2·IoU/(1+IoU)`. Never comparable directly with IoU. §27.5

DMBCOT. A six-slot framework for specifying a research question: Data, Method, Baseline, Condition, Outcome, Threshold. §6.5

DOAJ. Directory of Open Access Journals; a vetted index used as a legitimacy signal for open-access venues. §51.3

Effect size. The magnitude of a difference, reported alongside a *p*-value because statistical significance alone does not indicate practical importance. §26.8

External validity. Whether a result transfers beyond the studied sample, population, or setting. §1.7.1

Gap (research gap). A specific, evidenced absence in the accessible literature whose resolution would change what the field knows or can do. Requires four components: what is absent, evidence of absence, why it matters, and what resolution enables. §17.1

Grouped split. A partitioning that keeps all records sharing a group — patient, site, author, session — on one side of the split. Required whenever data are grouped. §26.3

HARKing. Hypothesising after the results are known, then presenting the hypothesis as though it had been predicted. §49.2

Hijacked journal. A cloned website imitating a legitimate journal in order to collect fees under its reputation. §52.1

Impact Factor (JIF). A journal-level metric from Journal Citation Reports, based on a two-year citation window. Describes a journal, not a paper. §51.2

Internal validity. Whether an observed relationship within a study is attributable to the manipulated factor rather than to a confound. §1.7.1

IoU (intersection over union, Jaccard index). Overlap between predicted and true regions; the threshold defining a correct detection. §27.5

Literature matrix. A table with one row per study and one column per attribute, from which research gaps are found by tallying columns rather than reading rows. Ch. 15

Macro average. An average over classes weighting each equally, thereby exposing failure on rare classes. Contrast micro average, which weights by frequency and conceals it. §27.3.2

MCC (Matthews correlation coefficient). A single-number classification metric using all four confusion-matrix cells; more robust than F1 under imbalance. §27.3

McNemar's test. The appropriate paired test for comparing two classifiers on the same test set. §26.8

Mosaic plagiarism (patchwriting). Substituting synonyms while retaining another author's sentence structure. Plagiarism even when a citation is present and even when similarity software scores it low. §47.3

Novelty. The property of a contribution being new relative to the accessible published record. A spectrum, not a binary; nine kinds are distinguished in §20.3.

Objective. A completable commitment to specific work, phrased "To + measurable verb + object + condition", markable done or not done. §7.2

Oracle (upper bound). A comparison configuration using privileged information you claim not to need, allowing you to report what fraction of its benefit you recovered without it. §26.6

PR-AUC. Area under the precision–recall curve. Preferred over ROC-AUC under severe class imbalance; not comparable across datasets of differing prevalence. §27.3.1

Pre-specification. Recording predictions, measurements, statistical tests, and significance levels before examining results. The practical safeguard against p-hacking and HARKing. §3.4

PRISMA. A widely used reporting framework for systematic reviews. §2.6

Reliability. Whether repeating a measurement yields a consistent answer. Addressed by running multiple seeds and reporting dispersion. §1.7.3

Reproducibility. Whether a different team can obtain the same result using the original artefacts. Distinguished from repeatability and replicability in §1.7.2.

ROC-AUC. Area under the receiver operating characteristic curve; the probability a random positive is ranked above a random negative. Optimistic under severe imbalance. §27.3.1

Salami slicing. Dividing one study into several thin papers to increase output. §47.6

Saturation. The stopping rule for a literature search: new searches and all three chaining directions return only papers already seen. §10.9

Self-plagiarism (text recycling). Reusing your own published text without disclosure or citation. §47.6

Similarity score. The proportion of text matching a tool's corpus. **Not** a plagiarism score: 10% can be misconduct and 25% can be clean. §48.3

Split unit. The entity by which data are partitioned — image, patient, institution, author, time period. The single most diagnostic field in a literature matrix and the commonest source of invalid evaluation. §13.3

Threats to validity. An explicit account of what could make a conclusion wrong, and what was done about it. Best written before data collection. §22.3

Three-pass reading. Triage, comprehension, and reconstruction, applied to progressively fewer papers at progressively greater depth. Ch. 12

Tuning budget parity. Giving every compared method an identical number of hyperparameter search trials over search spaces of comparable size, and stating that you did. The single most effective way to make an improvement claim credible. §26.4